

CASE BOOK

Industry Overviews

Industry Overview

The following slides offer a high-level overview of the industries commonly covered in case interviews.

Please do not memorize these industry facts. These slides are only meant to provide a foundational understanding for those who may not be familiar with these sectors, and the overviews are by no means exhaustive in nature.

It's also important to note that, in interview settings, cases sometimes centre around esoteric problems—for instance, evaluating a fox population's decline in a national forest.

Focus on mastering problem-solving frameworks that apply across industries and problem types, rather than memorizing industry facts and trends.

With that said, stay calm and confident, and good luck!

- From your friendly MCA Executive Committee of 2025-2026!

Industry Overview – Airlines

Industry Overview

- The global airline industry comprises several regions, including- Asia-Pacific, Europe, North America, Middle East, Latin America and Africa, with 33% of the revenue generated from APAC region in 2019.
- The key customers are individual consumers, corporations/small businesses and travel websites/retailers.
- Key sales channels are through airlines/travel websites, telephone, travel agents, over-the-counter at airports.

Key Trends

- **Low-Cost Carriers:** No-frills carriers competing for lowest base price but charging customers for Value-Added Services.
- **Fuel Efficiency:** Airline companies have been investing heavily in upgrading their fleet to more fuel-efficient aircrafts to reduce their biggest cost driver.
- **Changing Revenue mix:** Inadvertently rising fuel prices, dull economic conditions and increasing competition are realities that are biting into the revenue generating potential of the global airlines business today. Some of these strategies include tapping alternate revenue generating streams such as selling ancillary products and services across the value chain.
- **Rising Sustainability concerns:** Airlines will face growing pressure to address environmental sustainability over the next years. We have seen this topic gain significant momentum in just the past few months, and a number of airlines have recently announced ambitious targets to tackle emissions.
- **Usage of Analytics:** There are use cases for usage of advanced analytics spanning the entire value chain from a customer's initial travel inspiration and research through the ticket purchase and flight all the way to postflight services.

Terminology and Calculations

- **Revenue:** Ticket revenues, excess/oversize baggage fees, food and beverage sales.
- **Cost:** VC: fuel, food and beverage, ground crew/hourly employees FC: aircraft leases, airport gate leases, IT/admin costs, salaried employees (i.e., pilots)
- **Load Factor:** Measures the capacity utilization of transportation services and is equal to the actual utilization divided by the maximum capacity
- **Available Seat Miles (ASM):** Available seat miles refers to how many seat miles are actually available for purchase on an airline. Seat miles are calculated by multiplying the available seats for a given plane by the number of miles that plane will be flying for a given flight.
- **Revenue Passenger Miles (RPM):** Revenue passenger miles are calculated by multiplying the number of paying passengers by the distance traveled. For example, an airplane with 100 passengers that flies 250 miles has generated 25,000 RPM.
- **Cost Per Available Seat Miles (CASM):** CASM is a common unit of measurement used to compare the efficiency of various airlines. It is obtained by dividing the operating costs of an airline by available seat miles (ASM). Generally, the lower the CASM, the more profitable and efficient the airline.

Industry Overview – Financial Services (Retail Banks/Insurances)

Industry Overview

- Retail banking provides services include checking and savings accounts, mortgages, credit cards, and auto loans. The broad function of a bank is to collect money from those who have it, through savings, and disburse to those to need it, through loans.
- The insurance sector is made up of companies that offer risk management in the form of insurance contracts. The basic concept of insurance is that one party, the insurer, will guarantee payment for an uncertain future event. Customers make regular payments (premiums) to the insurer for coverage when unforeseen events, e.g. car crash; fire damage occur

Key Trends

- **Payment Modernization:** There is a 5-year plan undertaken by Payments Canada to modernize Canadian Interbank payment structure to enable rapid funds transfer and make same-day settlements
- **Evolution of Physical Branch:** The bank branch is no longer the only or primary way customers can interact with their bank. The everyday customer is just as interested in the quality of your website, mobile app, and ATM locations as the locations of your physical branches.
- **Rise of Blockchain Technology:** Blockchain's digital ledger has the potential of eliminating a lot of the paper-intensive process of traditional banking. It could also help banks with compliance due to its 'immutability' feature.
- **AI and RPA in Insurance:** Various elements of AI could eliminate brokers and paperwork, while its pattern recognition capabilities could minimize fraudulent claims. Chatbots are also on the rise as a 1st line of customer service.
- **Personalized Insurance Offerings:** With the rise of wearables and IoT-based devices, Insurance companies could monitor client behavior and tailor the offerings according to the risk levels of the individuals. Application Programming Interfaces (APIs) will enable the creation of insights-driven offerings as they integrate data from multiple sources.

Terminology and Calculations

- **Revenue (Banks):** Net revenue is the spread between bank's borrowing cost and the rates charged to borrowers; fees.
- **Revenue (Insurance):** Net revenue is the spread between premiums collected and claims/payments made over time.
- **Cost (Banks):** Overhead (branches, administration, compliance); Salaries; Bad Debt Expense
- **Cost (Insurance):** Overhead (administration, compliance); Salaries; Sales Commissions; Marketing
- **Know Your Customer:** The know your customer or know your client guidelines in financial services requires that professionals make an effort to verify the identity, suitability, and risks involved with maintaining a business relationship.
- Insurance Sales Channels are through the Insurance Agents who manage the front-end sales, online sales through websites and apps, along with direct marketing to employees via in-office demonstrations.
- Net Present Value (NPV) calculation, where NPV is the summation of all the future cash flows, discounted by a certain rate.
- **Private Equity:** Composed of investors and funds that invest directly into private companies or convert public companies to private companies to improve the target company's operations and financials with the goal of extracting a financial return from the company and reselling it another firm or the public at a profit
- **Assets Under Management:** Measures the total market value of all the financial assets which a financial institution manages on behalf of its clients and themselves.

Industry Overview – Manufacturing

Industry Overview

- Manufacturing is the process of converting raw materials and/or parts into finished goods that can be sold in wholesale or retail markets or exported for sale in other countries. Manufacturing represents 10% of Canadian GDP, employing 1.7M people.
- It covers a wide range of industries, from food and beverages to pharmaceuticals, iron and steel to textiles, as well as lumber, tobacco, automobiles, aerospace, and petrochemicals. There are 21 industry groups that represents Canadian Manufacturing.

Key Trends

- **Digitization through Industry v4:** Digital “muscle building” can be one of the leverage points to increase flexibility in global supply chains. Applying artificial intelligence, cloud computing, advanced analytics, robotics, and additive manufacturing to the value chain can increase visibility and transparency, allowing manufacturers to make faster changes to operations to respond to market-based threats or opportunities.
- **Trade wars and tariffs:** US imports from China were down 12.7 percent in the first eight months of 2019 versus the same period in 2018. Meanwhile, US imports from Mexico were up 5.9 percent, and US imports from Vietnam were up 37.4 percent. In a matter of months, manufacturers have shifted both sourcing and production to different geographies, seeking tariff-friendly combinations.
- **Use of renewable energy sources:** The commitment toward green and clean energy continues in 2020 as numerous manufacturers experiment with multiple renewable energy resources for current and future energy procurement, including solar, wind, hydro, and geothermal.
- **Sustainable Food Systems:** Vertical farming has been a growing trend in urban locations to minimize environmental foot-prints and bring produce to major cities

Terminology and Calculations

- **Revenue:** Customer segmentation (high value adding vs commodity), high volume vs high variation, emerging markets, consumer market demand, consumer confidence index
- **Cost:** Outsourcing (quality impact), supply chain optimization, labor cost, raw materials procurement, sales and marketing, fixed costs, depreciation
- **Bottleneck:** A bottleneck is a point of congestion in a production system (such as an assembly line or a computer network) that occurs when workloads arrive too quickly for the production process to handle.
- **Just-in-Time (JIT):** “Pull demand” inventory system in which assembly materials and support items are delivered as needed to minimize raw material inventory.
- **Lean Manufacturing:** Lean manufacturing is a methodology that focuses on minimizing waste within manufacturing systems while simultaneously maximizing productivity.

Industry Overview – Healthcare & Pharmaceuticals

Industry Overview

- The healthcare sector consists of businesses that provide medical services, manufacture medical equipment or drugs, provide medical insurance, or otherwise facilitate the provision of healthcare to patients.
- The pharmaceutical industry consists of branded/originator drug producers produce original patent-protected (for a certain period of time) drugs for human and animal diseases. Generic drug producers produce 'copy-cat' drugs (with the same medical result) at a lower development cost when the originator drug's patent expires

Key Trends

- **Precision Medicine:** Precision medicine is an innovative approach that takes into consideration individual differences in people's genes, environments, and lifestyles.
- **Wearable Medical Devices:** Activity trackers help patients stay more active and healthier on their own while also monitoring health metrics reducing the need to visit doctors frequently.
- **AR and VR in Healthcare:** Doctors could see the visual effect of a drug on an organ or conduct surgery on a VR platform
- **Patient Data Analytics:** It's driven by the belief that critical decisions affecting treatment should be based on reliable, real-time data. In order to bring fairness to healthcare, transparency in inpatient data is also vital.
- **Empowered Patients:** Medical professionals of every field have to get ready to provide answers and to treat patients as partners in their treatment.
- **Rise in Global Collaboration:** The COVID pandemic accelerated the joint ventures among different companies and governments in a race to find the most effect vaccine and preventative medications.

Terminology and Calculations

- **Revenue Drivers:** Size of specific treatment area / level of competition; Buy-in from doctors that will prescribe; Speed to market/ expertise in difficult products (for generics)
- **Cost Drivers:** VC: sales and marketing (doctor visits, sponsored studies); FC: R&D (drug discovery, formulation, clinical trials; a lot of this is now outsourced; generic companies only need to perform clinical trials)
- **Market Sizing:** Total Population → Population in the potential demographics → Rate of prevalence in demographics → Market share of Drug → Dosage / time period X price per dose = Market Size
- **Generics:** A prescription drug that has the same active-ingredient formula as a brand-name drug but sold at a cheaper cost. Typically occurs when name branded drugs lose patents
- **Biologics:** Any pharmaceutical drug product manufactured in, extracted from, or semi-synthesized from biological sources.
- **Orphan Drug:** An orphan drug is a pharmaceutical agent developed to treat medical conditions which, because they are so rare, would not be profitable to produce without government assistance.
- **Considerations:** Regulations – FDA Approvals, Patent Rights, Government laws. Competition. Effectiveness of Drug, Number of Dosages required. Manufacturing Capabilities. Pricing – varying by region (developed vs underdeveloped countries)

Industry Overview – Information Technology

Industry Overview

- The Information Technology Sector covers the following general areas: firstly, Technology Software & Services, including companies that primarily develop software in various fields such as the Internet, applications, systems, databases management and/or home entertainment, and companies that provide information technology consulting and services, as well as data processing and outsourced services; secondly Technology Hardware & Equipment, including manufacturers and distributors of communications equipment, computers & peripherals, electronic equipment and related instruments; and thirdly, Semiconductors & Semiconductor Equipment Manufacturers.

Key Trends

- **Artificial Intelligence:** Refers to the simulation of human intelligence in machines that are programmed to think like humans and mimic their actions. Has a huge number of implications in the rise of self-driven cars, smart homes, digital assistants.
- **Cloud Computing:** Is the practice of using a network of remote servers hosted on the Internet to store, manage, and process data, rather than a local server. Has implications for cost savings, security. Key players are AWS, Azure, Google Cloud.
- **Internet of Things (IoT):** A system of interrelated computing devices, mechanical and digital machines provided with unique identifiers and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction.
- **Blockchain:** a system in which a record of transactions made in bitcoin or another cryptocurrency are maintained across several computers that are linked in a peer-to-peer network.
- **Increase in Mobile Apps:** Brands and industries all over the world are trying to find ways in which one can improve their work using mobile apps and through the implementation of new resources that can make working on the go more efficient.
- **Increase in VR/AR usage:** Virtual reality is a simulated experience that can be similar to or completely different from the real world. Applications of virtual reality can include entertainment and educational purposes. Augmented reality is an interactive experience of a real-world environment where the objects that reside in the real world are enhanced by computer-generated perceptual information

Terminology and Calculations

- **Profit Drivers: Systems-** Lower margin (COGS management key to profitability). **Semiconductors-** High fixed costs (capex) and highly cyclical; manufacturing utilization. **Comms Equipment-** Manufacturing utilization. **Software-** license/maintenance versus subscription service model; renewal rates; high gross margins, but high R&D expenses. **IT Services-** staff util, revenues per employee. **Internet-** revenue per click
- **Unicorn:** A privately held startup company valued at over \$1 billion. Examples – DiDi, JUUL Labs, AirBnB
- **Software as a Service (SaaS):** Software as a service is a software licensing and delivery model in which software is licensed on a subscription basis and is centrally hosted.
- **Market Sizing:** Total Population → Number of users → Market Share → # of units per user X Price/Unit or subscription fees
- **Freemium:** A pricing model by which a product or service is provided free of cost, but money is charged for additional features

Industry Overview – Consumer Packaged Goods (CPG)

Industry Overview

- Consumer packaged goods (CPG) are items used daily by average consumers that require routine replacement or replenishment, such as food, beverages, clothes, tobacco, makeup, and household products.
- Customers could range from individuals to discount wholesalers (Sam's Club, Costco) to Large Box retailers (Walmart, Target) to Convenience Retailers (7-11, Rabba). Key CPG brands are Proctor and Gamble (P&G); Unilever, Clorox, Kellogg's, Campbell's, Frito Lay, ConAgra Foods, Colgate-Palmolive, L'Oréal, Estee Lauder

Key Trends

- **Private Label and Amazon effect:** Private label consumer products are eroding market share of large name brand products. This is partially driven by "the Amazon effect" of quick and cheap replacement. Brand loyalty is getting harder to win.
- **Enhanced In-store shopping experience:** Prepared foods and eating spaces, Individualization of experience, smartphone-integration for customized offerings. Potential advancement of mobile pay.
- **Retail Omnichannel:** Stores are offering a 'order online, pick-up at store' mix, in addition to enhancing the online buying experience. There is an effort to reduce store foot-print to store reduced level of inventory.
- **Data-Driven Social Media Engagement:** Data analytics can provide predictive insights into consumer preferences, helping companies of all sizes adapt to changing realities in the market. CPGs are also analyzing social media content to distinguish emerging trends and inform future product development, also tapping into social media influencers to advertise products.
- **Emergence of Subscription Model:** Amazon Dash buttons, Dollar Shave club, and HP instant ink are some examples of how the traditional transactional model of buying products when needed is being replaced by newer, stickier subscription models with more predictable demand signals and revenue streams.
- **Rising Environmentalism:** Recycling and reusing while reducing the global carbon footprint has become the norm.

Terminology and Calculations

- **Revenue:** Volume of goods sold; Price premium on branded goods
- **Cost:** Branding, Sales and Marketing, COGS (commodity costs – raw & packaging material)
- **Turnover** = (Sales/Inventory) | **Gross Margin** = (Revenue – COGS) / Revenue | **Contribution Margin** = Revenue – Var Cost
- **Mark-up** is the difference between the selling price of a good or service and cost. It is often expressed as a % over the cost
- **Velocity** describes how quickly a product is selling compared to similar products, or across markets and outlets.
- **Stock Keeping Unit (SKU):** A stock keeping unit is a distinct type of item for sale, such as a product or service, and all attributes associated with the item type that distinguish it from other item types.
- **Private Label Brands:** Private label products are goods and services created by one company to be sold and branded by another company. Popular examples of private label products include Walmart's Great Value brand, Target's Mainstays, and Amazon's Amazon Essentials.

Industry Overview – Telecommunications

Industry Overview

- The telecommunication sector is made up of companies that make communication possible on a global scale, whether it is through the phone or the Internet, through airwaves or cables, through wires or wirelessly.
- These companies created the infrastructure that allows data in words, voice, audio or video to be sent anywhere in the world. The largest companies in the sector are telephone (both wired and wireless) operators, satellite companies, cable companies, and internet service providers.

Key Trends

- **5G spectrum auction and investment:** Gartner predicts that 5G infrastructure revenue will reach \$4.2 billion in 2020, an 89% increase from 2019. 5G encompasses use cases including IoT, fixed wireless networks, automation, and home networks.
- **Increase in IoT devices:** A McKinsey report says that investments in IoT technology are projected to grow at 13.6 percent per year through 2022. IoT devices range from wearables to smart fridges to self-driven cars.
- **Increase in Edge Computing:** Edge computing just means processing and delivering information closer to the user. By enabling data aggregation and processing at the edge, companies can achieve bandwidth savings while also reducing latency and improving reliability.
- **Content Integration:** High profile acquisition like AT&T of Time Warner and Verizon of Yahoo illustrate a push to either get into the content creation game or to build out their advertising network. A Canadian example would be Crave from Roger.
- **Data Privacy and Security:** A recent Deloitte report shows that consumers still fear identity theft, financial loss, and unauthorized use of sensitive data—largely because many have experienced these threats directly.

Terminology and Calculations

- **Revenue Drivers:** Data services (SMS, email and internet access on cell phones), mobile advertising, app stores.
- **Cost Drivers:** VC: marketing & advertising, salaries; FC: capital costs (equipment, infrastructure – cell towers, network maintenance, stores); overhead
- **5G:** Fifth-generation wireless (5G) is the latest iteration of cellular technology, engineered to greatly increase the speed and responsiveness of wireless networks.
- **Carrier:** A company that is authorized by regulatory agencies to operate a telecommunications service system: Key mobile network operators in Canada are Roger, TELUS and Bell
- **Original Equipment Manufacturer (OEM):** A company whose goods are used as components in the product of another company that sells the finished goods to users
- **ROI:** $(\text{Future Profits} - \text{Investment Cost}) / (\text{Investment Cost})$ | **Client Acquisition Cost:** $\text{Marketing Expenses} / \text{New Clients}$
- **Considerations:** Regional competition, New entrants, spectrum sharing, Infrastructure support, 5G upgradation, marketing

Feedback Rubric



MCA Case Interview Feedback Rubric

Interviewer	Case
Candidate	Date

Case Recap: _____

1 (low)	2	3	4 (high)
Didn't recap or began recap with questions	Long rambling recap; repeated most of prompt	Succinct recap but didn't highlight business problem	Succinct recap; highlighted business problem; no questions inserted midway

Clarifying Questions: _____

1 (low)	2	3	4 (high)
Asked no questions	Asked too many questions (4+); questions were rambling, unfocused, and unexplained	Asked 1-4 questions; questions were unfocused but defended with rationale; questions were too focused on looking for new data	Asked 1-3 questions that were strongly framed, focused, and concise; added rationale to questions; questions sought to understand background and businesses vs. collecting data; < 2 minutes total

Case Structure: _____

Give interviewee time (if they ask for it) to structure thoughts.

Time how long it takes to structure, should be ~2min; penalize for < 1 and 2.5+ minutes.

1 (low)	2	3	4 (high)
Proposed at least 2 buckets but lacks depth in ideas under each bucket (<3); walkthrough of ideas is too brief (<1 min) or too long (>3 min)	Proposed at least 2 buckets; each bucket has ~3 ideas; buckets and ideas are not MECE; walkthrough of ideas is too brief (<1 min) or too long (>3 min)	Proposed at least 2 buckets; each bucket has ~2-3 ideas that focus on data; buckets and ideas are MECE; walkthrough of ideas takes 2-3 minutes and is fairly easy to follow	Proposed at least 2; each bucket has ~2-3 ideas that focus on data and may have another layer of ideas; buckets and ideas are MECE and prioritized; walkthrough of ideas takes 2-3 minutes, is clear and easy to follow, and hypothesis-driven

Exhibit / Qualitative: _____

1 (low)	2	3	4 (high)
Appeared confused; failed to describe the exhibit; failed to extract any insights.	Described the exhibit; Struggled to extract insights; Required interviewer to step in and help.	Correctly described the exhibit; Extracted good insights that are relevant to the overall objectives.	Accurately described the exhibit in a concise way; Extracted good insights and tied it back to the overall objectives; Suggested next steps based on the exhibits.

Calculation / Quantitative: _____

1 (low)	2	3	4 (high)
Took too long to solve; appeared confused; no verbal cues	Wrong answer; verbally talked through process; logic and calculations okay	Correct answer and approach quickly; verbally easy to follow; no insights or only Level 1 insights	Correct answer and approach quickly; good verbal and written structure; Provided insights

Brainstorming Question: _____

Give interviewee time (if they ask for it) to structure thoughts.

Time how long it takes to brainstorm, should be less than a minute.

1 (low)	2	3	4 (high)
Unclear and hard to follow (no structure)	Didn't take time to structure; Acceptable response but no structure	Generic or not MECE structure; lacks depth of ideas or rationale	MECE buckets or clear rationale; calm when asked "what else?"

Recommendation: _____

Give interviewee time (if they ask for it) to structure thoughts.

Time how long it takes to prepare final recommendation, should be ~1min.

1 (low)	2	3	4 (high)
No recommendation other than to collect "more data"; no or very limited next steps	Rambling recommendation; not tied to or supported by case findings; no or very limited risks or next steps	Clear recommendation supported by data from the case; missing risks or next steps	Concise recommendation supported by data from the case addressing business problem; risks and ~2-3 next steps; <2 min

Additional Comments:

Sample Behavioural Question Bank

Behavioural Question Bank

The following questions are sample interview questions designed to give you an idea of the types of questions that may be asked during the interview process. You **do not** need to prepare answers to all the listed questions. Instead, use there as a reference to understand the variety of topics and scenarios that could be discussed. Focus on developing your ability to answer questions **thoughtfully**, rather than memorizing specific answers. Remember adaptability is key during an interview!

Career Story or Personal Questions

1. Tell me about yourself.
2. Take a minute to run me through your resume.
3. Why did you choose to do an MBA?
4. Why was Rotman the right school for you?
5. What is an important issue facing your previous industry?
6. Tell me about your biggest strength/weakness?
7. What do you do for fun?
8. What skills can you bring to this position.
9. How would your teammates describe working with you?
10. What are a couple of the best and worst decisions you have made in the past year?
11. What do you do for fun outside of school?
12. Tell me about an important role model in your life.

Consulting-related Questions

1. Why do you want to get into consulting?
2. Why do you think you'd be a good consultant?
3. Why are you interested in working at Firm X?
4. What are you most excited about as you prepare for a role in consulting?
5. What skills can you bring to this position?
6. What do you see as the most challenging aspect of this job?

Situational Questions

1. Tell me about a time you had to make a difficult decision.
2. Tell me about a time when you had to convince a difficult client to follow your recommendations
3. Tell me about a time you faced conflict in the workplace.
4. Tell me about a time you failed.
5. Tell me about a time you needed to adjust a project schedule because you didn't have all the resources you needed.
6. Describe a positive/negative experience working with a customer.
7. Tell me about a time you pushed back on a project timeline.
8. Describe a time you had to change your communication style to influence different stakeholders.
9. Give me an example of a tough business problem you previously solved.
10. Describe the last time you disagreed with a supervisor?
11. Tell me about a time you pushed back on a project schedule.
12. What are the most difficult or challenging decisions you have had to make recently?
13. Tell me about a time you had a conflict with a teammate.
14. Tell me about the best piece of feedback you ever received.
15. Tell me about a piece of feedback you received that you disagreed with. How did you handle it?
16. Tell me about a time when you led a team through a challenge.
17. Give me an example of a time when you had to motivate others who did not report to you.
18. Describe a recent unpopular decision you made. What was the result?
19. Tell me about a successful business relationship you built with a client, boss, or peer in your previous job.
20. Tell me about a time when you exhibited leadership
21. Tell me about a time you faced opposition in the workplace. What did you do and what was the outcome?
22. What is your proudest accomplishment?
23. Tell me about a problem in your professional life and how you were able to overcome it

Cases

Guide to this Casebook

There are 3 categories of cases in the following casebook:

1. Essential Cases:

- These cases are foundational and recommended for MCA members to complete. They are representative of cases commonly encountered in interviews. Completing these cases will help you form a solid basis for your case-solving skills.

2. Legacy Cases:

- These cases are recommended for MCA members looking for extra practice.

3. Specialized:

- These cases focus on more varied, unconventional, or specialized industries, complete with a diverse range of exhibit types. MCA members are encouraged to complete these cases when they have more experience and want to diversify their practice. They offer a unique opportunity to tackle complex and unique scenarios.

Special thanks to Zenon Sirko, Majeet Jandu, Emma Wilson, Raghav Gupta from the 2024-2025 MCA Exec team as well as Georgio Homsy, Kimia Khoee, Sirun Wang, Christie Kwan, Emmanuel Olawoye, Kieran Grekul-Somerville, and Nishtha Taneja from the 2023-2024 MCA Exec Team for their revisions.

List of Cases (I of III)

#	Name of Case	Type	Difficulty level	Concepts	Industry	MCA Case Type
1	Gift Wrapping Paper	Interviewee	Easy	Opportunity Assessment	Consumer Products	Essential
2	Candy Manufacturer	Interviewer	Easy	Profitability	Consumer Products	Essential
3	Multipurpose Tool	Interviewee	Easy	Opportunity Assessment	Consumer Products	Legacy
4	Recreational Aircraft	Interviewer	Easy	Profitability	Manufacturing	Legacy
5	Sheep Auction	Interviewee	Medium	Opportunity Assessment	Other	Essential
6	Scotch Bar	Interviewee	Medium	Merger & Acquisition	Travel & Leisure	Essential
7	American Beauty Company	Interviewer	Medium	Profitability	Consumer Products	Essential
8	Gas Station	Interviewee	Medium	Opportunity Assessment	Consumer Products	Essential
9	Zoo Co	Interviewer	Medium	M&A	Financial Services	Essential
10	Brazilian Manufacturer	Interviewee	Medium	Profitability/Growth	Manufacturing	Essential
11	SPLAT!	Interviewer	Medium	M&A	Consumer Goods	Essential
12	HVAC Service Provider	Interviewee	Medium	Merger & Acquisition	Energy	Legacy
13	Vidi-Games	Interviewee	Medium	Opportunity Assessment	Telecom, Media & Tech	Legacy
14	Molds R Us	Interviewee	Medium	Opportunity Assessment	Consumer Products	Legacy
15	The Coffee Grind	Interviewer	Medium	Profitability	Consumer Products	Legacy
16	Winter Olympics Bidding	Interviewee	Medium	Valuation	Telecom / Media & Ent.	Legacy
17	Butcher Shop	Interviewee	Medium	Operations/Supply Chain	Manufacturing	Legacy

List of Cases (II of III)

#	Name of Case	Type	Difficulty level	Concepts	Industry	MCA Case Type
18	Everlasting Light Bulb	Interviewee	Medium	Valuation	Consumer Goods	Legacy
19	Vie Tire	Interviewee	Medium	Competitive Dynamics	Manufacturing	Legacy
20	Megabank Under-Penetration	Interviewer	Medium	Marketing	Financial Services	Legacy
21	Trains in Spain	Interviewer	Medium	Profitability	Transportation	Legacy
22	Money Bank Call Centre	Interviewee	Medium	Operations	Financial Services	Legacy
23	Upscale Restaurant	Interviewer	Hard	Profitability	Travel & Leisure	Essential
24	Jamaican Land Investment	Interviewee	Hard	Opportunity Assessment	Agriculture and Real Estate	Essential
25	Grape Control Systems	Interviewee	Hard	Profitability	Industrial Goods	Essential
26	New Rubber Plant Investment	Interviewee	Hard	Value Chain	Industrial Goods	Essential
27	Wealth Management	Interviewee	Hard	Operations	Financial Services	Essential
28	A+ Airline Co.	Interviewee	Hard	Opportunity Assessment	Airline	Essential
29	Old Pharma	Interviewer	Hard	M&A	Pharmaceuticals	Essential
30	Grocer Co.	Interviewee	Medium	Profitability	Consumer Goods	Essential
31	Electronics Retailer	Interviewee	Medium	Opportunity Assessment	Retail	Legacy
32	Frontier Adventure Racing Inc.	Interviewee	Hard	Profitability	Travel & Leisure	Legacy
33	Competitive Threat in Cookies	Interviewer	Medium	Profitability	Consumer Products	Legacy
34	Bell Computer Co	Interviewee	Hard	Profitability	Technology	Essential

List of Cases (III of III)

#	Name of Case	Type	Difficulty level	Concepts	Industry	MCA Case Type
35	Sheep of Wall Street Co	Interviewee	Medium	M&A - Financial Buyer (PE)	Private Equity	Specialized
36	Webflix	Interviewee	Medium	Pricing Strategy	Tech	Specialized
37	Death Valley Energy	Interviewee	Medium	Market Study - Strategy	ESG	Specialized
38	Beer Burger	Interviewee	Medium	Market Study - Strategy	ESG/Logistics	Specialized
39	Mayor of Maine	Interviewee	Hard	Market Study - Growth	Non-business/government	Specialized
40	Doctors Within Borders	Interviewee	Hard	Market Study - Growth	Non-profit	Specialized
41	TikTac Co.	Interviewee	Hard	M&A - Systems transformation	Tech	Specialized
42	Whatcha Say Research	Interviewee	Hard	Systems transformation	Professional Services	Specialized
43	Phuture	Interviewee	Hard	Internal Strategy	Tech	Specialized
44	Cloudforce	Interviewee	Hard	Pricing Strategy	Tech	Specialized

#1 – Gift Wrapping Paper

Boston Consulting Group

Analysis	Structure	Overall
Easy	Easy	Easy

Behavioural Prep

Tell me about yourself.
Tell me about a time when you exhibited leadership.

Case Tracker		Case Stem
Industry	Consumer Products	Your client is a gift-wrapping paper manufacturer in the U.S. They are considering a proposal to outsource their manufacturing to mainland China. You have been called in to assist in the go/no-go decision-making process. They would like to know your thoughts and recommendation.
Type	Opportunity Assessment	
Leader	Interviewee	
Concepts	Cost Analysis	

Interview Guidance
• This case involves some number crunching but overall is quite straight-forward. • Case is mainly interviewee-led but interviewer can provide some upfront guidance (see Steps 1-3 on next page) • The candidate should be able to recognize that this requires a cost-benefit analysis for the 2 options: manufacturing in U.S. vs. in China • Candidate can make the assumption to prioritize costs, and that revenues should not be greatly impacted by outsourcing manufacturing in the gift-wrapping paper industry.

#1 – Gift Wrapping Paper (II of VI)

Interviewer Guidance

Interviewer guidance:

STEP 1: Ask the candidate to begin by establishing cost buckets.

Variable costs to be provided upon request and fixed cost comparison provided on candidate handout slide (Exhibit 1). Do not provide until candidate outlines potential differences and asks for specifics between options.

STEP 2: After cost buckets are established, ask how they may differ in China (A: Lower labor costs...). A strong candidate will note early on that distribution costs may have a big impact.

STEP 3: Let the candidate steer the interview from here...

#1 – Gift Wrapping Paper (III of VI)

Information to be provided upon request

Only provide additional information after being specifically asked by the candidate.

Client info: The client manufactures printed gift-wrapping paper and sells to wholesalers and retailers in the U.S. Currently their entire operation is in the U.S.

Objective: Client would want to save at least 10% on their current cost to move the manufacturing to China.

Ask the candidate to begin by establishing cost buckets

Provide the costs information below after candidate identifies the cost buckets

Fixed costs include:

- Plant & machinery *Provide candidate exhibit on the following page*
- Employees *Provide candidate exhibit on the following page*

Variable costs include:

- Raw paper material \$20 per ream (same for U.S. and China)
- Ink \$100 per ream U.S.
\$50 per ream in China

Ink is special wrapping paper ink and an unavoidable cost

Shipping cost from China to US is \$150 per ream

No need to consider customs and taxes

#1 – Gift Wrapping Paper (IV of VI)

Sample Framework

While the case progression mainly focuses on the cost comparison between manufacturing in China and in the U.S., the candidate should try to structure a robust framework that covers more than just the financial considerations.

Basic structure:

Should our client outsource the manufacturing to China?

- Financial considerations
 - Comparing costs
 - Fix costs: facility, employees, etc.
 - Variable costs: raw material, ink, shipping, etc.
- Operational considerations
 - Vetting of Chinese manufacturers
 - Sourcing of raw material

Advanced structure:

Should our client outsource the manufacturing to China?

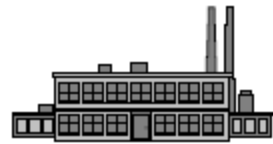
- Financial considerations
 - Basic cost buckets
 - Fix costs: facility, employees, etc.
 - Variable costs: raw material, ink, shipping, etc.
 - Additional costs
 - Shutdown cost for the U.S. operations (e.g. severance package, etc.)
 - Import licensing and additional documentation required to import
- Operational considerations
 - Vetting of Chinese manufacturers
 - Sourcing of raw material
 - Key talents
- Potential risks
 - Political tension between U.S. and China
 - Brand image
 - Product quality

#1 – Gift Wrapping Paper (V of VI)

Exhibit 1

Gift Wrapping Paper Production Comparability Between Markets

United States



1 Unit Parts & Machines
Cost: \$100 each

+

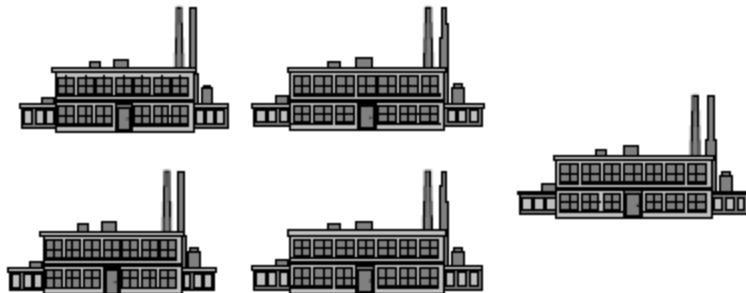


=



1 Ream Paper

China



5 Units Parts & Machines
Cost: \$20 each

+



25 Employees
Cost: \$2 each

=



1 Ream Paper

#1 – Gift Wrapping Paper (VI of VI)

Establish base <u>US</u> costs	\$100 Parts & machines \$100 Labor \$20 Paper \$100 Ink Expensive US based labor Commodity – difficult to lower Specialized product TOTAL COST : \$320 per ream
Generate comparable <u>China</u> costs	\$ 100 Parts & machines \$50 Labor \$20 Paper \$50 Ink \$150 Shipping More equipment may increase repair cost (5 * \$20) Lower per employee, but hiring/firing costs may increase No change Closer to supplier, but still expensive TOTAL COST : \$370 per ream
Compare two fully loaded costs for options	• Shipping is the deal-breaker for China • Lower shipping costs would increase attractiveness - What might some alternatives be? Bulk shipments? Sheets rather than rolls? • More variables to manage in China, not very labor-intensive product in the US
Generate Recommendation	• Conclusion: Moving manufacturing to China does not look viable at this time • Non-examined issues: US shutdown costs • Next steps: Consider ways to lower shipping costs in China; track ink, paper, employee costs in China to address long-term potential for move

#2 – Candy Manufacturer

Analysis	Structure	Overall
Medium	Easy	Easy

Behavioural Prep

Take a minute to run me through your resume.

Tell me about a time when you had to convince a difficult client to follow your recommendations

Case Tracker	
Industry	Consumer Products
Type	Profitability Improvement
Leader	Interviewer
Concepts	Cost Analysis

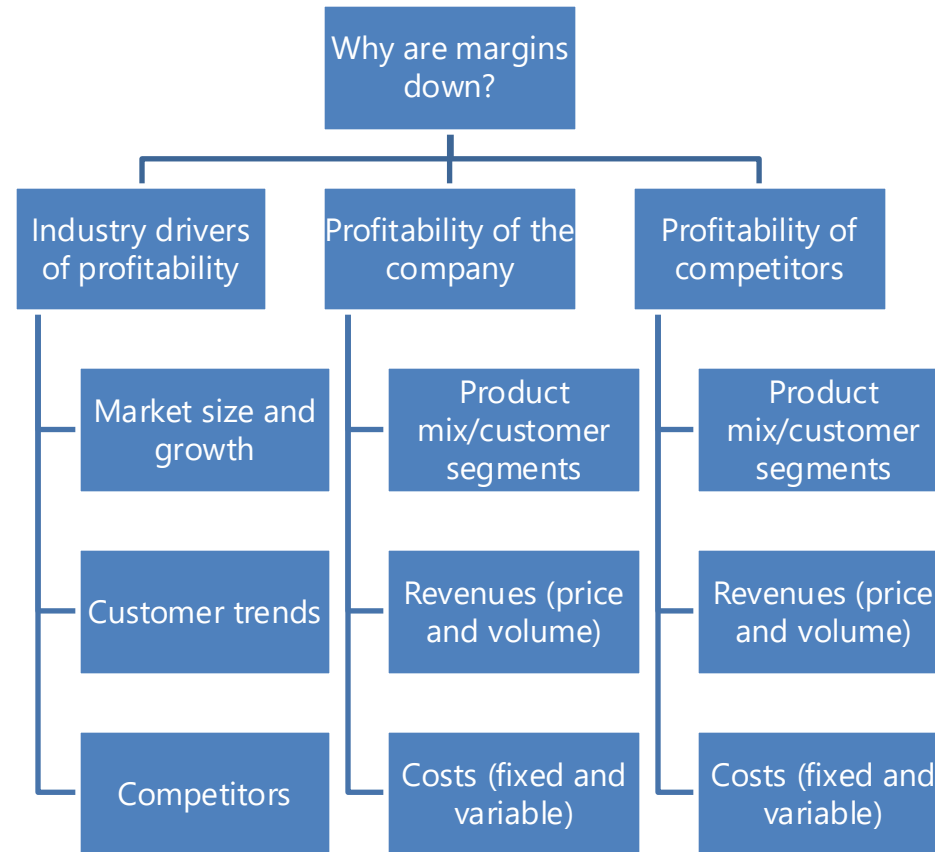
Case Stem
Your client is a candy manufacturing company that has been facing declining profit margins. The CEO has reached out to us to figure out why and what they should do about it (Question 1).

Interview Guidance

- This is a relatively straight-forward case, which includes the following: discussing profitability drivers, calculating profitability of the client versus competitors, discussing opportunities to increase revenues and decrease costs and a conclusion.
- If they ask as a follow-up to the question to the problem statement whether declining profitability is an industry-specific issue or a company-specific issue, highlight that it is a company-specific issue.
- This is an interviewer-led case. Interviewer-led cases are very good to practice specific skills (e.g., quant). Therefore, you will find these cases very helpful even if you might not get exposed to interviewer-led cases during recruiting. As an interviewer, you lead the case, asking questions, receiving answers, engaging in short discussions of the interviewee's answers and eventually closing the question and leading to the next question.
- Note: This is a long case and interviewee may not go through all the questions within 25-30 minutes. Gauge how the interviewee is progressing and adjust as appropriate to stay within time limit.

#2 – Candy Manufacturer (II of IX)

Sample Framework



#2 – Candy Manufacturer (III of IX)

Sample Framework

External:

1-Market:

- Size and growth
- Trends (switch to healthy food, more awareness regarding candies)

2-Competition:

- Market share and concentration
- Product mix and differentiators
- Distribution channels
- substitutes

3-Customers:

- Segments (size, growth)
- Change in preference
- Price sensitivity

4-Suppliers:

- Concentration
- Prices

Internal:

1-Revenue:

- # of customers
- Spending /customer
- Price

2-Cost:

FC

- Capex
- Rent
- Utilities
- Overhead

VC:

- Raw material
- Labor
- Shipping

3-Qualitative:

- Taste and quality of products
- Product mix
- Channel mix
- Reputation

#2 – Candy Manufacturer (IV of IX)

Question 2a: Industry trends

What do you think are the general trends in the industry?

Interviewer guidance:

- General brainstorming to test business intuition of the interviewer, i.e., this is a mature industry; candy manufacturer probably steadily growing at inflation, if anything probably declining due to more health-conscious customers/products;
- There are many big players like Hershey in this space, so it is likely a highly competitive/saturated market
- At the end of the discussion (if they haven't asked already), clarify that the industry itself has sustained higher profit margins than our client

Note:

- Interviewers in actual case interviews rarely ask such questions about industry knowledge directly and it is not required to have specific industry knowledge; however, it is beneficial to share one's business intuition as an interviewee to motivate assumptions, justify prioritization of certain elements of one's framework, etc.
- The purpose of this question is thus to practice sharing big picture business knowledge

Question 2b: Profitability drivers

1) What do you think are the main drivers of revenues in this industry?

Interviewer guidance:

Again, test for general brainstorming skills and business intuition.

In the discussion, provide details about our client's revenues:

- Pricing has been relatively steady on so has been sales
- Our client plays in a more niche part of the candy segment which has not been impacted by the rise in the more health-conscious consumer as it is particularly liked by children
- In fact, our client commands a price premium, most of our competitors have prices that are 10% lower than our client's.

2) What do you think are the main drivers of cost?

Interviewer guidance:

Again, test for general brainstorming skills and business intuition. Keep in mind the amount of time left and transition to next questions appropriately.

#2 – Candy Manufacturer (V of IX)

Question 2c: Competitive cost benchmarking

We will look at costs in a moment. However, we have an interim meeting with the client, and he has asked us a very specific question. Our client's profit margins are 10%. He believes the competitors' profit margins are higher and wants us to calculate the difference in profit margins. How would you go about doing that?

Interviewer guidance: (provide these answers when prompted)

- On the revenue side, you have already mentioned that our client commands a price premium, as competitors' prices are 10% lower. If the interviewee asks again, give it to them.
- From industry reports, we also know that competitors' costs are 20% lower
- This information should be sufficient to calculate the competitors' profit margins; if the interviewee asks for prices say that prices vary depending on the distribution channel and the package size
- If your interviewee is struggling, propose the trick to assume the price is \$1.00. This will make the math simpler.
- The answer should be the competitors' profit margin is 20%, 2X higher than our client

Sample solution for question 2c

Profitability of our client:

- Assume any price, e.g. \$1.00
- Hence, our client's profitability is: $\$1.00 \times 10\% = \0.10
- This means that our client's cost is: \$0.90

Profitability of our competitors:

- Competitors' price is 10% lower than our client: $\$1.00 \times 90\% = \0.90
- Competitors' cost is 20% lower than our client: $\$0.90 \times 80\% = \0.72
- Competitors' profits: $\$0.90 - \$0.72 = \$0.18$
- Competitors' profit margin: $\$0.18 / \$0.90 = 2/10 = 20\% \rightarrow$ Difference of profit margins: $20\% - 10\% = 10\%$

#2 – Candy Manufacturer (VI of IX)

Question 3a: Competitor profitability

What do you think is the main reason that competitors are more profitable?

Interviewer guidance:

- It is a difference in costs as their revenues are lower

Follow-up: Show the interviewee the exhibit on the next page and ask what it means in context of the overarching questions (of improving profit margins)

Interviewer guidance:

Key things the interviewee should notice from the graph:

- Raw materials are the highest percent of costs; this could be primarily what is driving the difference between the client and competitors; also, this would be where there is most leverage for our client to reduce costs; an excellent candidate would point out that doubling profit margins to match competitors would require cutting costs by \$0.10, e.g., cutting raw material costs by $\$0.10/\$0.63 = \sim 16\%$ ($\$0.63 = \$0.90 \times 70\%$)
- Equipment is the next highest, unless there are significant differences between the client and its competitors' equipment this won't drive down our client's costs sufficiently
- Labor costs are the smallest part of our client's costs; even if the client could reduce it, impact would be limited
- Summary: Focus on reducing raw materials costs

Question 3b: Math teaser

By approximately what percentage would raw materials cost have to be decreased to match our client's competitors' profit margins?

Interviewer guidance:

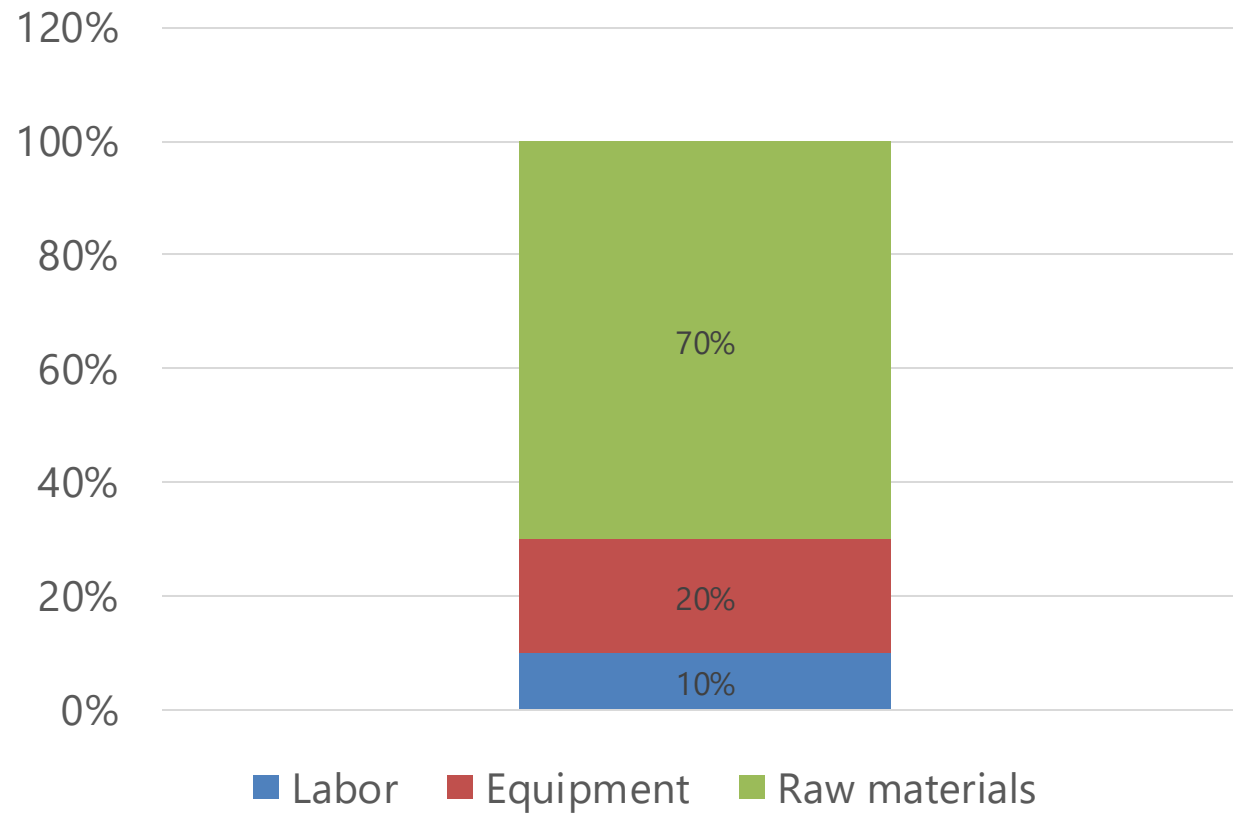
An excellent candidate will have calculated this proactively as part of the previous question.

- Competitors' profit margins are 20%
- Assuming a price of \$1.00 that means that the new profits needs to be \$0.20 and hence, costs \$0.80
- Thus, current cost of \$0.90 need to be reduced by \$0.10
- Raw materials cost is 70% of total current cost: $\$0.90 \times 70\% = \0.63
- This is a $\$0.10/\$0.63 = \sim 16\%$ decrease

A good candidate will gut-check this number and point out that a ~16% decrease seems to be a reasonable target

#2 – Candy Manufacturer (VII of IX)

Exhibit 1 – Client's cost structure



#2 – Candy Manufacturer (VIII of IX)

Question 3d: Raw materials

What could be the reasons why raw material costs could be higher for our client than its competitors?

Interviewer guidance:

Pose this as a general brainstorming question. Continue the discussion until these key points are addressed:

- Supplier bargaining power
 - supplier concentration relative to number customers
 - \$ purchased per given supplier
 - function of production volume, number of components per supplier, number of suppliers per component, purchasing frequency
- Types of raw materials used in production
- Amount of raw materials used in production (also amount of raw materials lost in production)

Interviewee should dive further into these points to extract the following information, which interviewer should provide if asked:

- Competitors are slightly larger than our client
- There a lot of supplier options for a given raw material
- Given our client is the result of a lot of mergers, they have a lot of local suppliers for the same raw materials

Good candidates will proactively explain how this new information changes their perspective. Prompt them if they don't.

#2 – Candy Manufacturer (IX of IX)

Wrap-Up and Recommendation

Your project manager comes in and asks for an update of your progress which he wants to share with the client in a phone call in about 5 minutes?

Recommendation:

- Declining profit margins are mainly due to a higher cost structure compared to competitors.
- We propose cutting raw material costs by 16%, aligning our profit margins with industry standards.
- To achieve this, we will conduct an assessment of available suppliers and select 2-3 that provide optimal pricing, service quality, financing terms etc..
(this is a possible answer. The recommendation would depend on the discussion in 3d. Other answers can be related to increasing efficiency, changing the type of raw material without affecting quality, etc..)

Risks: (also depend on the recommendation given)

- Transition Costs: Negotiating contracts, establishing new relationships, and adjusting production processes
- Quality Control: New suppliers may not meet the same quality standards as the previous supplier.
- Supply Disruption: production issues, quality problems, or logistical challenges could disrupt the flow of raw materials, leading to production delays or even shortages.
- Supplier Dependence: While diversifying suppliers is a prudent strategy, relying on only two suppliers may still leave the company vulnerable.

Next steps:

- Supplier Evaluation and Selection
- Pilot program with the new suppliers before a full-scale transition
- Risk Assessment and Contingency Planning specific to the transition

#3 – Multipurpose Tool

Boston Consulting Group

Analysis	Structure	Overall
Easy	Medium	Easy

Behavioural Prep

What skills can you bring to this position.
How would your teammates describe working with you?

Case Tracker		Case Stem
Industry	Consumer Products	Your client is a diversified hardware manufacturer who produces a multi-purpose hand tool. For several decades, your client was the only company to make the tool. Over the past two years, the company has seen a decline in revenue. You have been brought in to determine what is driving the decline, and what you can recommend as a solution.
Type	Opportunity Assessment	
Leader	Interviewee	
Concepts	Revenue drivers, price elasticity	

Interview Guidance
• This case tests the candidate’s brainstorming skills and business intuition • This case is more qualitative than quantitative, and interviewer should remain as passive as possible to allow candidate to demonstrate they can drive the case forward • After candidate realizes that increasing price by 10% will increase revenues, interviewer should push candidate to brainstorm next steps to investigate for the long-term • <i>Bonus:</i> Interviewer can also push candidate to qualitatively brainstorm other methods of growth (aside from price) and their implications – e.g., organic growth (new products, new markets) and inorganic growth (acquisitions, joint ventures)

#3 – Multipurpose Tool (II of V)

Interviewer Guidance

Only provide additional information after being specifically asked by the candidate.

- Price: \$50, constant over time – no recent increases
- Current volume: 100M units / year
- Channel: Hardware retailer – can not break contracts
- Price elasticity of demand: -0.5 (-10% change in price will raise demand + 5% and vice-versa)
- Market demand: Do-it-yourself market (client's market) in higher than ever
- Competition: several new competitors in past 2 years
 - Selling similar product for \$30
 - Channel: discount retailers (Wal-Mart, Target, etc.)

#3 – Multipurpose Tool (III of V)

Sample structure

External reasons:

1. Competition
 - Number of competitors and market share
 - Substitutes
 - Differentiations
2. Demand
 - Customer segments and size
 - Change in preferences
 - Price sensitivity

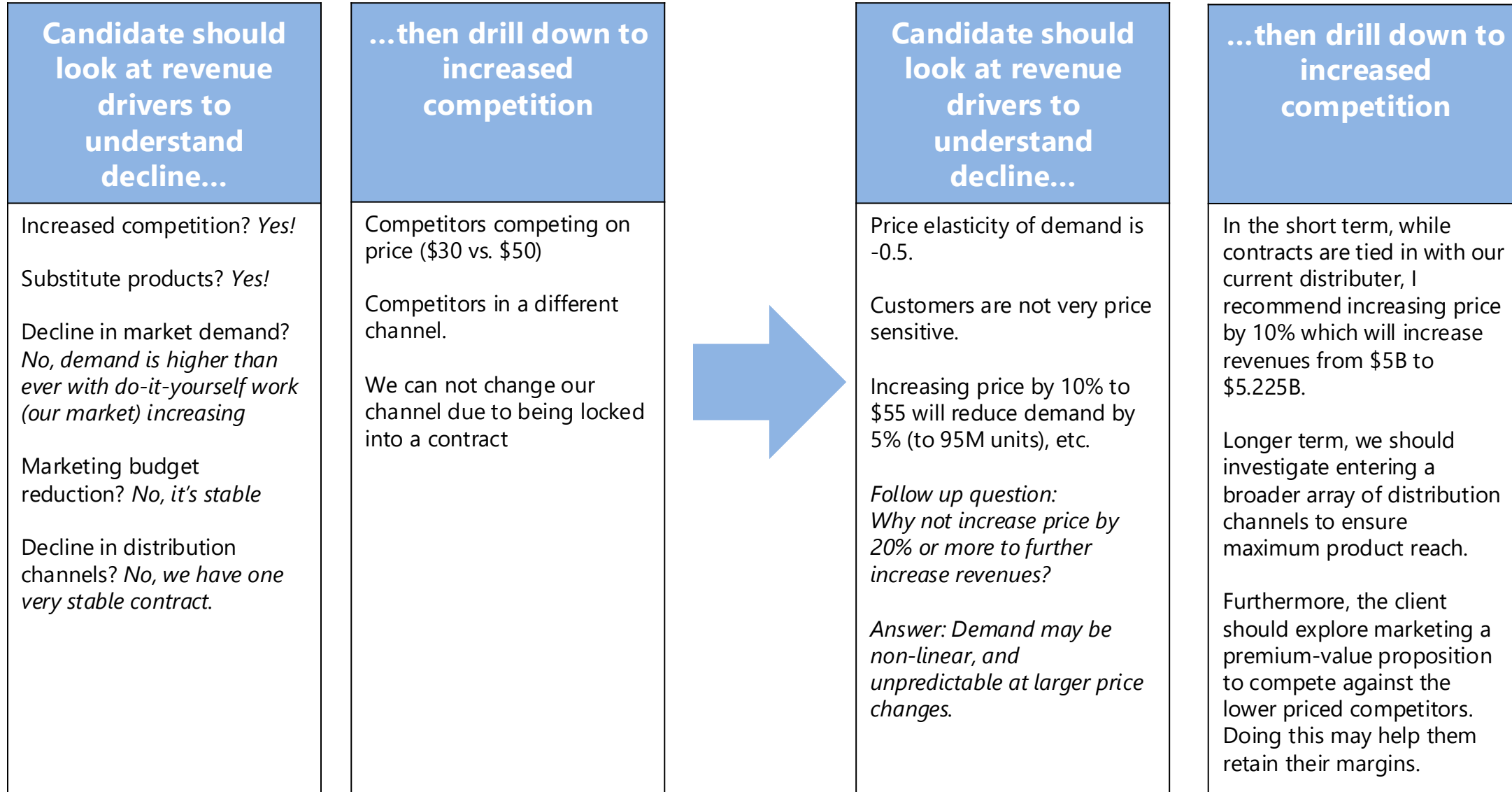
Internal reasons:

1. Revenue
 - Quantity
 - Price
2. Qualitative
 - Customer service
 - Marketing
 - Quality
3. Distribution Channels

Solutions:

1. Increase prices
2. Increase marketing
3. Explore new market
4. Explore new distribution channel

#3 – Multipurpose Tool (IV of V)



#3 – Multipurpose Tool (V of V)

Recommendations

Recommendation:

- In the short term, while contracts are tied in with our current distributor, I recommend increasing price by 10% which will increase revenues from \$5B to \$5.225B.
- Longer term, we should investigate entering a broader array of distribution channels to ensure maximum product reach.
- Furthermore, the client should explore marketing a premium-value proposition to compete against the lower priced competitors. Doing this may help them retain their margins

Risks:

- Customer Backlash: Increasing prices in the short term may lead to customer dissatisfaction and potential loss of market share.
- Market Saturation: Broader distribution channels might already be saturated, posing challenges for market entry.

Next Steps:

- Market Research: Conduct customer surveys to assess their willingness to accept a price increase, and evaluate the potential impact on market share.
- Market Entry Analysis: Investigate new distribution channels and conduct a comprehensive market entry analysis to identify viable opportunities.
- Competitive Analysis: Analyze the competitive landscape to develop a compelling premium-value proposition that resonates with customers.

#4 – Recreational Aircraft

Bain & Company

Analysis	Structure	Overall
Easy	Medium	Easy

Behavioural Prep

Why do you want to get into consulting?
 Tell me about a piece of feedback you received that you disagreed with. How did you handle it?

Case Tracker		Case Stem
Industry	Aircraft manufacturing	- Our client is a large industrial equipment company with four business units: - Airplane engines - Industrial cranes - Oil drilling equipment - Recreational aircraft - Question 1: What issues do you think are important to consider in assessing key drivers, opportunities, and risks for this company?
Type	Market-sizing, profitability	
Leader	Interviewer	
Concepts	Break-even analysis	

Interview Guidance
This is an atypical case because there is a two-part stem. If the candidate starts by asking to make a structure, tell them there will be time for that later. Potential response to question 1: Key Drivers - General economic factors (GDP growth, energy prices, construction market, etc.) - Availability of natural resources - Economic policies regarding oil exploration, airline industry regulation, etc. Opportunities - Trends across markets (client is 70% domestic; other 30% includes emerging markets) - Synergies across business units (aircraft engines with recreational aircraft, oil drilling equipment with industrial cranes, etc.) Risks - Issues in the suppliers’ industries; competition; customer concentration; price volatility

#4 – Recreational Aircraft (II of III)

Question 2

Recreational aircraft are small planes like Cessnas that people fly for fun. The client's recreational aircraft division was doing well in the 1980s but started declining in the 1990s. The division was shut down in 2000. The people were moved to other divisions and the plant was closed.

It is now 2009, and the client is wondering whether it should get back into the recreational aircraft market or sell off the assets and get out for good.

Question 2: What do you think the client should consider in this decision?

Potential response to Question 2

The candidate should now make his/her structure, which may include:

- Recreational plane industry overview (current market opportunity, competitive landscape, etc.)
- Financials (start-up costs, profit potential, etc.)
- Business fit (internal capabilities, product resurrection vs. new product development, rationale for original closing vs. rationale for re-entry, potential synergies, existing plant assets)
- Risks

Market Sizing

The candidate should begin by estimating the current market size:

Potential Responses

- Top-down requires determining how many players there are in the market and their sizes. It is probably only a few (Cessna and a few others) so we could determine their revenues and add them to get the overall market size.
- A bottom-up approach would start by estimating the number of pilots and planes in the US (make sure they get this before moving on)

Interviewer: 1% of the US population has a pilot's license. Of those, 1/3 has a high enough income to actually own a plane or a share of a plane. Of those, 10% buy their own plane and the remaining 90% belong to flying clubs where they share ownership of a plane with other pilots. We can assume there are 10 pilots/plane in these clubs.

Doing the Math:

- $300\text{M people} * 1\% = 3\text{M pilots}$
- $1/3 \text{ of } 3\text{M} = 1\text{M pilots with high incomes}$
- $10\% \text{ of } 1\text{M} = 100\text{K planes bought by individuals}$
- So, $900\text{K flying club members} / 10 \text{ ppl per plane} = 90\text{K planes bought by clubs}$
- $100\text{K} + 90\text{K} = 190\text{K planes total}$

Important: Interviewee should ask how often planes are replaced. The answer is every 20 years.

- $190\text{K planes} / 20 \text{ years} = 9.5\text{K new planes bought per year}$

#4 – Recreational Aircraft (III of III)

Profitability analysis

You should now direct the candidate to figure out how many planes the client would need to sell to make a profit.

Information to provide candidate, upon request:

- 10 years ago, fixed costs were \$450M per year and variable costs were \$1.8M per plane
- At that time, the selling price per plane was \$2M
- Assume those numbers are 10% higher today

Break-even analysis:

- Current price = $\$2M \times 1.1 = \$2.2M$
- VC = $\$1.8M \times 1.1 = \sim \$2M$
- So, \$200K profit per plane
- FC = $\$450M \times 1.1 = \sim \$500M$
- Total planes to break-even = $\$500M / \$200K = 2500$ planes

Given that the market is about 10K planes our client would need to capture 25% market share to break even. A good candidate should share this insight on their own without prompting.

Interviewer: Ask the candidate if they think it's realistic for our client to achieve 25% market share. Any response is okay, as long the candidate supports their response with good reasoning

Risks and recommendations

Interviewer: You have a meeting with the client's CEO tomorrow – what would you tell her about your findings?

- Recommendation should be concise and direct, and it should correspond to the previous points raised in the case (meaning it should be well-reasoned)
- He/she should also make note of any risks associated with the recommendation

Potential Risks

- Technology in plant could be obsolete (8 years is a long time for it to sit unused)
- Could take too long to restart operations in recreational aircraft
- Need to find trained personnel
- Economy could take turn for the worse
- Fuel prices could skyrocket
- Safety approval and/or flight testing could hold up process of restarting operations
- This could divert focus from other product areas
- Should also consider opportunity costs (i.e. how much client would receive from selling the plant)

#5 – Sheep Auction

Boston Consulting Group

Analysis	Structure	Overall
Medium	Medium	Medium

Behavioural Prep

Why are you interested in working at Firm X?
Tell me about an important role model in your life.

Case Tracker		Case Stem
Industry	Other	It is early 2006. Your client is looking at investing a significant amount of money to create an online auction company that facilitates sheep sales from producers to large customers. They will only do this if they could make roughly \$10M annual profit in 5 years, and they have enlisted your help in determining their go/no-go decision.
Type	Opportunity Assessment	
Leader	Interviewee	
Concepts	Market sizing, profitability	

Interview Guidance
• This is an exhibit heavy case. Candidate should recognize that they need to size the market to determine if the profitability objective can be met • Interviewer should push the candidate to think about what data they need to drive the case forward, and should show an exhibit only after candidate asks for the information (show exhibits one at a time as opposed to all at once) • When candidate calculates the profitability in 5 years time, they should recognize that they have not taken into account our client’s estimated market share (assumes 100% penetration). As the total profits calculated (\$6.6M) already falls short on the \$10M target, an insightful candidate can note that in actuality that the gap between estimated profits and target profits may be larger than calculated.

#5 – Sheep Auction (II of VIII)

Interviewer Guidance

Only provide additional information after being specifically asked by the candidate.

Current sheep sales model:

- There are two main sheep sales channel, in-person auction and contract. In-person auction means that customers travel to sheep farms to auction the sheep while under contracts, sheep are sold under a fixed price and a fixed quantity.
- Auction component includes both online and traditional selling models.
- Sales via auction and contract will not migrate – there is no way to steal share between channels.

Online auction: The company that operates the online auction platform would charge a service fee on a per unit basis

Competition: There is currently no other online auction platforms

Buyers: All large buyers use computers

Exhibits:

1. Overall market size (in lbs. of sheep)
2. Sheep prices (in \$/lb.)
3. Sheep sold at auction vs. contract
4. Farmers (producers) who use computers

Follow up questions for candidate upon completion of the calculation (which should in total fall short of \$10M)

- *What would you do to achieve the \$10M level?*
- *If launched, how would you market this product?*

#5 – Sheep Auction (III of VIII)

Sample Frameworks

Basic structure:

Should our client invest in online sheep auction?

- Market attractiveness
 - Market size and growth
- Financial considerations
 - Revenue
 - Number of sheep auctioned
 - Revenue per sheep auctioned
 - Cost
 - Online auction setup cost

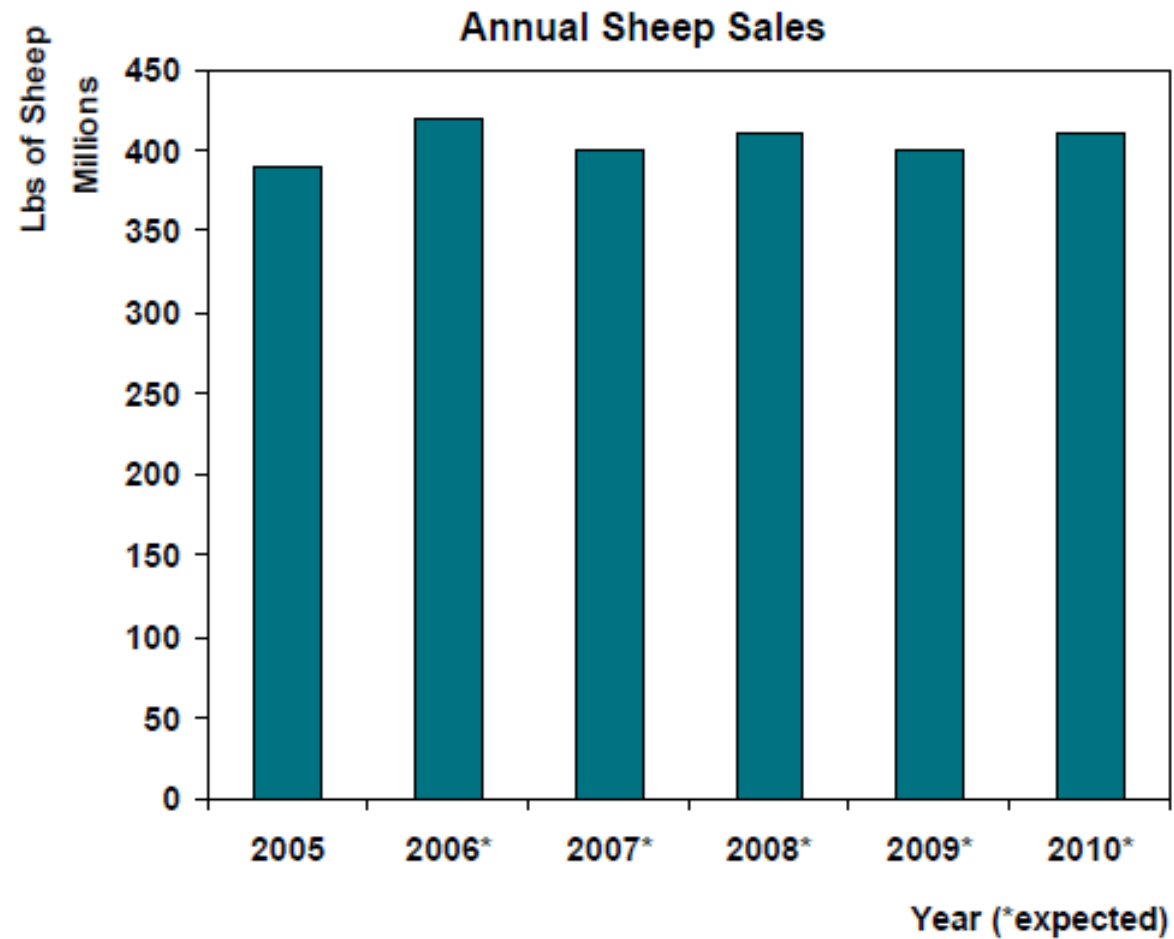
Advanced structure:

Should our client invest in online sheep auction?

- Market attractiveness
 - Market size and growth
 - Customer preference, both sheep farmers and end customers
 - Competitors
- Financial considerations
 - Revenue
 - Number of sheep auctioned
 - Revenue per sheep auctioned
 - Cost
 - Online auction setup cost
- Other considerations
 - Technological feasibility (given it's 2006)
 - Customer conversion

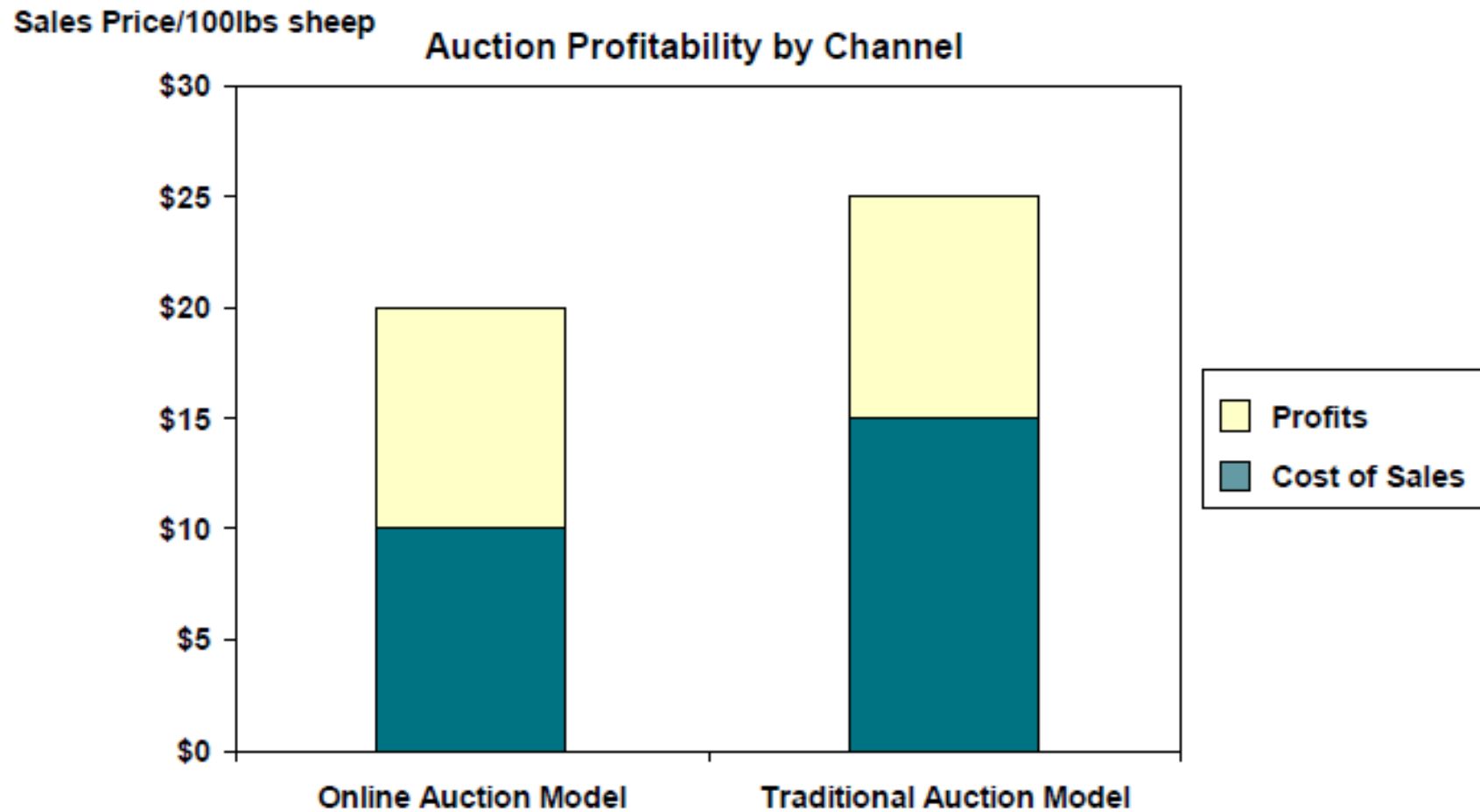
#5 – Sheep Auction (IV of VIII)

Exhibit 1



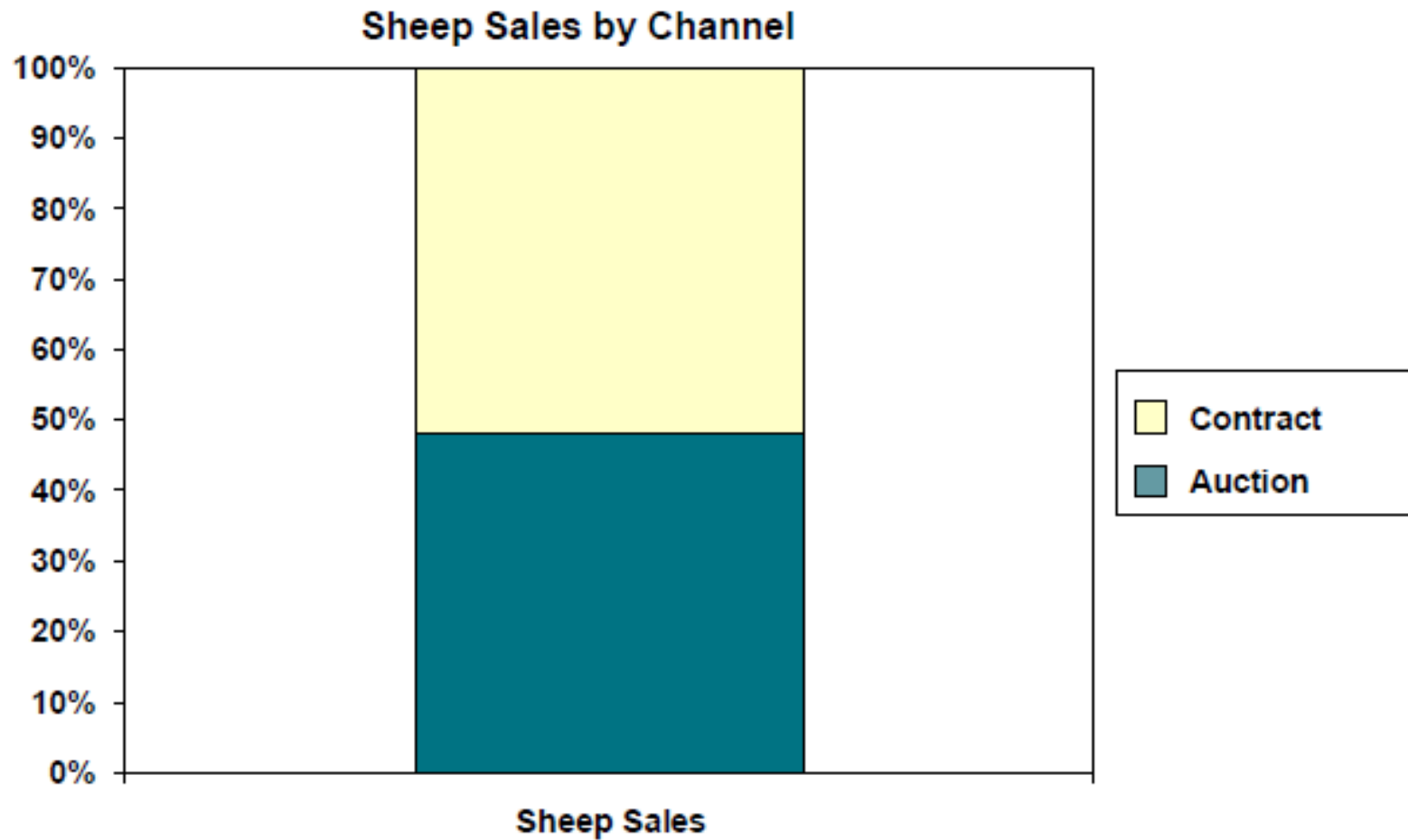
#5 – Sheep Auction (V of VIII)

Exhibit 2



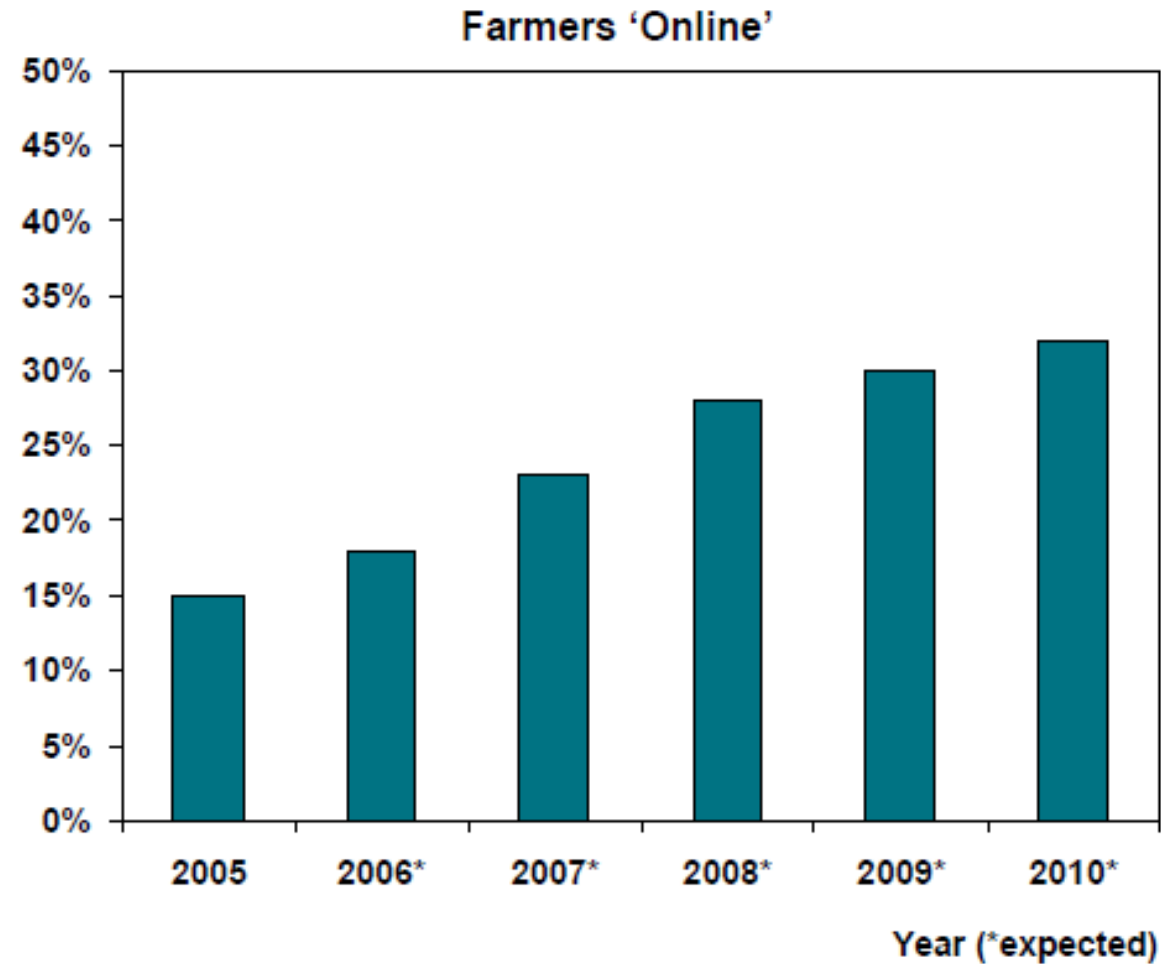
#5 – Sheep Auction (VI of VIII)

Exhibit 3



#5 – Sheep Auction (VII of VIII)

Exhibit 4



#5 – Sheep Auction (VIII of VIII)

Expected calculation (approximate)

Use 2010 numbers to show 5-year 'maturity' and steady-state for profitability. This model assumes a 100% penetration. Candidate should recognize that penetration is irrelevant given overall profitability

400M	x	50%	x	33%	x	\$10/100	=	\$6.66M Profit
lbs sheep (exhibit 1)		Auctioned (exhibit 3)		online farmers (exhibit 4)		\$10 is profit <u>per 100lbs</u> sold, so need to divide by 100 (exhibit 2)		<i>Less than \$10M</i>

Recommendation

Based on the 2010 figures, I would not recommend that the client move forward with the investment as it would not provide the required \$10M annual profit return 5 years from now. Excellent candidate will recognize that at least 50% of farmers must be online in order to obtain \$10M profit.

Additional questions for the candidate, time permitting:

- 1. If they're really determined to make this investment work, what would you do to reach the \$10M profit?*
- 2. If this product were already launched, how would you choose to market it?*

Sample answers:

- Train farmers and sheep producers on computer use
 - Provide central computer locations near farm sites to facilitate farmer interactions
 - Expand the auction tool to other animals
- Trade magazine advertisements
 - Door to door sales & training reps
 - Commission farmers as representatives
 - Relationships with sheep processors

#6 – Scotch Bar

Analysis	Structure	Overall
Medium	Medium	Medium

Behavioural Prep

Give me an example of a tough business problem you previously solved.
Describe the last time you disagreed with a supervisor?

Case Tracker		Case Stem
Industry	Travel & Leisure	You are sitting in one of Chicago's oldest scotch bars with a fellow intern. It is a Friday night after a busy week at your summer internship. The weather is mild—a perfect summer evening. While enjoying one of the bar's finest stogies and sipping an 18-year-old whiskey, your friend asks you how much you think the bar is worth. Using a back-of-the-envelope calculation, how would you go about determining the value of this bar?
Type	Merger & Acquisition	
Leader	Interviewee	
Concepts	Valuation and DCF	

Interview Guidance
• This case tests the candidate’s ability to estimate the value of the scotch bar using a discounted cash flow analysis (one method of valuation). <i>An excellent candidate will suggest a discounted cashflow during their framework.</i> • A candidate may mention other methods of valuation apart from DCF (e.g., benchmark with recent sales of comparable bars)– interviewer should guide candidate to DCF approach if candidate commits to another approach. • An excellent candidate will lead the case and make assumptions on possible revenue and cost estimates, before being provided the information through exhibits.

#6 – Scotch Bar (II of VI)

Information to provide if requested

Information to provide throughout the case:

- Product Mix and Pricing: The bar sells two things, liquor and cigars. The average cost of a cigar is \$9, and the average cost of a drink is \$12. (Note: these average cost numbers will prove irrelevant, but in cases one is sometimes given irrelevant info.)
- Location: The bar is located on one of Chicago's busier streets for foot traffic.
- Staff: A bartender, a waiter, and a waitress. All three were there the entire evening.
- Capacity: The maximum capacity is 100 people.
- Hours: The bar is open Tuesday thru Sunday from 5 pm until 2 am.
- Tax Rate: 40%
- Discount Rate: 13%
- Annual Growth Rate of Cash Flows: 3%

The candidate will most likely not ask for all this information upfront. Allow the candidate to make some assumptions about revenues. One way to project revenues is to estimate the number of customers per day or per week and multiply that by the average expenditure of each customer. Watch for realistic assumptions and logical thought progression.

#6 – Scotch Bar (III of VI)

Once the candidate has developed a structure...

Once you've given the candidate some time to go through their structure and they have not discussed the following already, prompt the candidate with the following question: **"What might drive variation in revenue and cost projections?"**

*The answer is days of the week (Fridays and Saturdays are typically busier than other days) and seasonality (people tend to be out more during summer than winter). While the candidate talks you through their approach, but before the candidate does a substantial amount of calculation, hand the candidate **Exhibit 1**.*

Calculation Explanations

- If the candidate does the math correctly, the revenue should add up to **\$568,000 (420,000+148,000)**.

Calculations	Fridays and Saturdays	Rest of Week
	\$4,800*2 Days = \$9,600 for Friday & Sat.	*16.667 is 1/6. Use fractions whenever possible
Spring and Summer (Equals 25 weeks. 50 weeks in a year)	\$9,600*25 Weeks = \$240,000 Total \$240,000 Total	\$4,800*1/6* 4 days (Monday is closed) = \$3,200 \$3,200 * 25weeks = \$80,000 Total
Fall and Winter	3/4* \$240,000 = \$180,000 \$180,000 Total	0.85*\$80,000 = \$68,000 Total
Total	\$420,000	\$148,000

Costs

- There are two main components: fixed costs and variable costs
- Have the candidate brainstorm possible fixed costs and variable costs. Under fixed costs the candidate might consider rent, general maintenance, management, insurance, liquor license, and possibly employees. The only real variable cost is the cost of goods sold.
- After the candidate has brainstormed costs, provide them with the following data:
 - Variable costs = 20% of total Revenue
 - Fixed costs = \$120,000
 - Note: Guide the candidate to remove variable costs before fixed costs in order to find correct annual income:*
 - Annual Income: (568,000*0.8) - 120,000 = \$334,400**

#6 – Scotch Bar (IV of VI)

Bonus Question

The **net income is \$334,400. The candidate can round to \$334,000 for ease of calculations.**

The candidate should ask for income tax rate (40%).

The after-tax cash flow number (approximately **$\$334,000 * (1-40\%) = \$200,400$** (or \$200,000 for ease of calculation in next prompt). You now have the annual cash flows generated by the bar.

At this point a great candidate will drive the process forward and recognize that they need to figure out a stream of cash flows going forward. The interviewer may have to nudge less-savvy candidates toward the next step (discounted cash flow analysis) by asking **“How does one perform a valuation of the business?”**

DCF Analysis

To perform a valuation in this case, the candidate must estimate the cash flows from the business and discount them back using a perpetuity formula. **The discount rate typically used for bars of this genre is 13%.** When the candidate inquires about growth rates, **say the bar’s cash flow is growing at 3%**-- the rate of inflation.

Thus, whatever numerator the candidate arrives at should be divided by $.13 - .03 = .10$, an easy calculation.

Perpetuity Calculation: **$CF / (r - g) = \$200,000 / .10$** , or \$2 million.

#6 – Scotch Bar (V of VI)

[Optional: time permitting]

Brainstorming question: Let's say your friend wants to buy the bar. In what ways could they improve its financial situation?

Sample structure:

Increasing Revenue:

- Attract additional customers: e.g., adding live music, promotions or deals (e.g., happy hours, promotions during popular sporting events, etc.)
- Increasing spend per customer: increase prices of existing revenue streams (cigars + alcohol), add additional products (e.g., start selling food)

Decreasing costs:

- Reducing COGS

Note: this provides the opportunity for the candidate to show creativity.

Recommendation: After our discussion today, what is the final value of the bar?

Sample Recommendation:

- Based on our DCP analysis, the value of the bar is \$2M. This was based on differences in sales in summer/spring compared to winter/fall, and weekends vs weekdays.

Risks

- Assumption of discount and growth rates
- Valuation using perpetuity, cash flows will vary
- Other relevant non-financial risks, e.g., competitors entering to steal market share will decrease cash flows

Next Steps:

- Conduct sensitivity analysis to test assumptions

#6 – Scotch Bar (VI of VI)

Exhibit 1: Daily Sales

	Fridays and Saturdays	Rest of Week
Spring and Summer	\$4,800	16 2/3% of Fri and Sat, (Spring and Summer)
Fall and Winter	3/4 of Spring and Summer, (Fridays and Saturdays)	85% of Spring and Summer (Rest of week)

#7 – American Beauty Company

Analysis	Structure	Overall
Medium	Medium	Medium

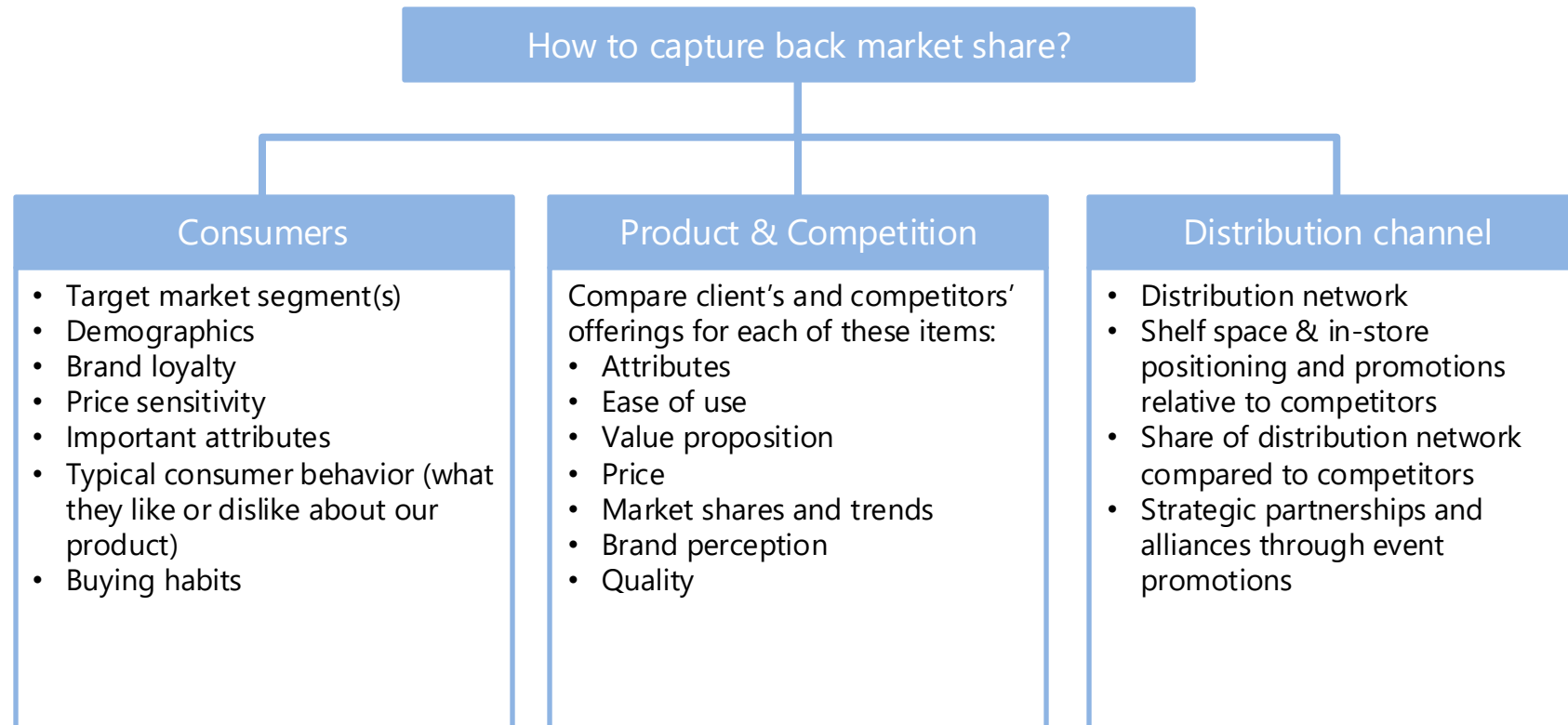
Behavioural Prep

Tell me about a time when you led a team through a challenge.
Give me an example of a time when you had to motivate others who did not report to you.

Case Tracker		Case Stem
Industry	Consumer Products	American Beauty Company is, as the name suggests, a high-quality beauty products company. They have done very well both in the US and global market and enjoy great brand recognition. ABC manufactures high quality 'use at home' hair color products, which is one of their major products. They sell through retail and drugstores, with all manufacturing done in-house. They have a 1-800 number for customer support. Recently, they have been experiencing declining revenues and market shares. The retailers have complained about their products and the competition Bell International is taking over. The firm has been called in to advise ABC on what to do. Question 1: How would you start thinking about this problem?
Type	Profitability Improvement	
Leader	Interviewer	
Concepts	Marketing	
Interview Guidance		
• The case contains qualitative as well as quantitative questions at a moderate difficulty level. • The initial question is open ended. There can be several ways to approach this problem. A crucial step is to look at the big picture and come up with three or four major areas that the candidate would like to explore given this specific product, industry and the situation. The interviewee should not get caught in the profitability trap due to the mention of declining revenues. If the candidate's structure is focused on profitability push him to think about other factors as well. • This is an interviewer-led case. As an interviewer, you lead through the case, asking questions, receiving answers, engaging in short discussions of the interviewees answers and eventually closing the question and leading to the next question. Since you are leading the case try to make sure the case lasts only about 25-30 minutes.		

#7 – American Beauty Company (II of V)

Sample framework (w/o profitability branch)



#7 – American Beauty Company (III of V)

Declining shares in segment of woman aged 18-55

Question 2: One of their biggest market segments is women aged 18-55, but the client's share in this segment has been declining recently. Why do you think this might be happening? How would you approach this issue?

Sample answers:

There might be several issues here:

- Candidate could suggest doing market research to break down the issue into brand awareness, customer preferences, trial and re-trial %, availability % (distribution) and identify which one of these may be critical for ABC
- The candidate could also suggest benchmarking against competitor products

Math teaser

Question 3: Using the following data, what sales are required for ABC to have 50% of the women's market in 2 years?

Interviewer guidance: Prompt candidate to write down the information in the following table. A good candidate will organize the data in a table.

Segment	Size (\$M)	Annual growth rate
Women	800	5%
Men	200	20%
Teens	100	10%

- It will be good to point out that based on this data it looks like their biggest segment, women, is maturing fast
- Total women's market in 2 years = $\$800M \times (1.05)^2 = \$882M$ (candidate can't round for this calculation)

Answer: Sales required for ABC to have 50% share of this segment = \$441M

#7 – American Beauty Company (IV of V)

Overall ABC market share

Question 4: What is the dollar market share for ABC currently if the company has 50% share in Women, 10% share in Men and 30% share in Teens? What will the market share be in 2 years if segment share remain constant?

Sample answer:

Segment	Size (\$M)	Segment share	ABC current sales (\$M)	Annual Growth rate	Size in 2 yrs (\$M)	ABC in 2 yrs (\$M)
Women	800	50%	400	5%	882	441
Men	200	10%	20	20%	288	28.8
Teens	100	30%	30	10%	121	36.3
Total	1,100		450		1,291	506.1
Market share			40.9%			39.2%

- While it is not required to calculate % market share data (unless the interviewer wants to give the candidate some extra practice doing complicated divisions without rounding), a strong candidate would point out that the relative market share would decrease because of small presence in high growth segments.

#7 – American Beauty Company (V of V)

Insights from brand awareness and perception surveys

Key insights: You can see from the results of both surveys that despite a high level of quality and brand recognition, the competitor Bell fare better relatively amongst customers in both dimensions whether users or non-users.

Implication: ABC should focus on improving its brand awareness and perception of quality. For brand awareness, they should focus on advertising to the major target customers. To improve perception of quality, they should invest in promotions, joint marketing efforts with retailers to push their product and trials. Customer testimonials around quality is useful too. If the product is truly performing worse, ABC might have to think about investing in R&D to boost product performance along dimensions important to consumers which can be understood through market analysis.

Recommendation

A good recommendation will include the following major points:

- Based on the analysis so far, it seems like the main reason for declining revenues and market shares is that the competitor Bell has achieved better brand awareness and perception of quality in the market compared to ABC. In the short term, ABC should focus on improving these through advertising and aggressive promotions and marketing to the target segment, women aged 18-55.
- Going forward, it seems like ABC has very low penetration in the men's and teen's segment. They should target these segments for long-term growth opportunities since the women's segment seems to be maturing.
- Also, the total market size of teens seems quite low. There may be opportunities there to expand the total market size through innovative products, increased usage and acceptance of hair color products.
- **Risks:** A) Opposite effects of branding (customers may further think poorly of our product due to aggressive branding). B) Competitive backlash – promotions on both end may lead to decreasing margins
- **Next steps:** Conduct market analysis and curate targeted branding strategies

#8 – Gas Station

Analysis	Structure	Overall
Medium	Medium	Medium

Behavioural Prep

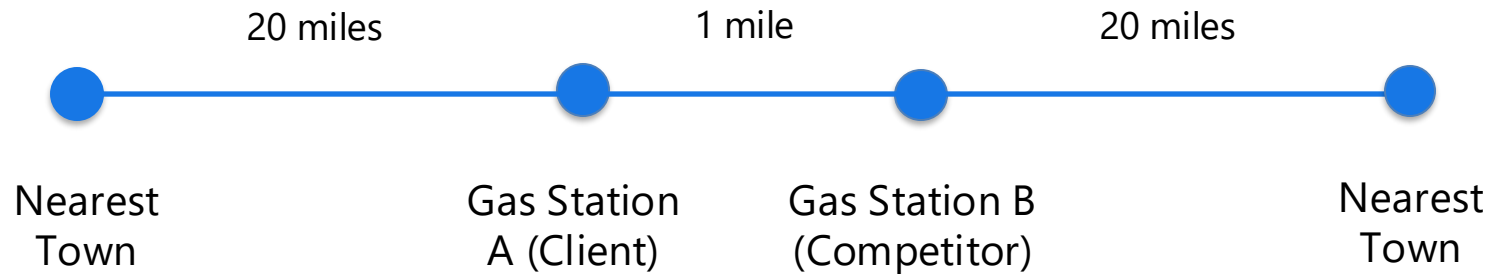
Describe a recent unpopular decision you made. What was the result?
Tell me about a successful business relationship you built with a client, boss, or peer in your previous job.

Case Tracker		Case Stem
Industry	Consumer Products	Your client is a gas company that operates in a town with a population of 1,000. There is only one other gas company in this town, and it is 1 mile away. The other nearest gas stations are outside town, and they are 20 miles away (show Exhibit on next page). Recently, our client was approached by a supermarket with the idea of selling groceries in the gas station. Our client is a simple businessman and has hired us to evaluate this proposal. What should our client consider?
Type	Opportunity Assessment	
Leader	Interviewee	
Concepts	Game Theory, Competitive Response	

Interview Guidance
The standard opportunity assessment component of this case is pretty straight-forward with a minor twist. What makes this case interesting is the game theoretical competitive dynamics. Try to push the candidate into a discussion about competitive responses in case they don't proactively uncover the respective insights (which a strong candidate should do).

#8 – Gas Station (II of VI)

Case Prompt Exhibit – Map of Area (not to scale)



#8 – Gas Station (III of VI)

Information to be provided upon request

1. What are the proposed groceries the gas station would now sell? (Turn around and ask the candidate.)
 - Likely answers include cigarettes, milk snack foods, but probably not fresh produce or healthy foods
2. Currently, the gas station is barely scraping by. Profits are essentially 0.
3. If asked anything about the other gas station, the answer is “we don’t know, but assume they are identical”.

The candidate should recognize that each gas station serves 500 people.

Sample Frameworks

Basic Structure:

- Revenue:
 - 1) Groceries: Increased revenue from selling groceries in store;
 - 2) New gas customers: more people coming to buy gas from this station instead of station B.
- Costs: Up front investment costs such as freezer, shelves, etc.
 - Recurring costs such as labour will be minimal since same staff can handle gas and groceries

Advanced Structure:

- Revenue:
 - 1) Groceries: Increased revenue from selling groceries in store;
 - 2) New gas customers: more people coming to buy gas from this station instead of station B.
- Costs: Up front investment costs such as freezer, shelves, etc.
 - Recurring costs such as labour will be minimal since same staff can handle gas and groceries
 - Will there be any profit-sharing with the grocery store?
- Other considerations:
 - Will there be any profit-sharing with the grocery store?
 - Competition: What is keeping the grocery store from going to gas station B as well?

#8 – Gas Station (IV of VI)

After presenting their structure, the candidate should start walking you through the components of their structure in more detail. Interviewer can prompt the candidate with the questions below if needed, and provide the information if the candidate does not think of it. but an excellent candidate will be leading.

Cost Analysis

What kind of costs could there be?

- Upfront costs: Investment in freezers, shelves, utilities

Information to be provided upon request:

- No increase in labour expected
- Total increase in costs: **one-time cost of \$1.25 M**

Revenue Analysis

Groceries:

- What kind of people will shop for groceries here?
 - People who shop here are probably not health conscious, since gas station groceries are not healthy. They are probably also in a rush, and will make small purchases (milk, cigarettes, jerky, etc.)
- How much do they spend per year?
 - Out of 1000 people, assume 500 will buy groceries here *[note: candidate should realize this from first exhibit]*
 - Information to provide upon request: average spend: \$200 per year per person
 - Additional revenue = \$200 / person / year * 500 people = \$100,000 per year

New customers:

- Increase in gas sales: very important distinction: people won't buy more gas, but they are stealing customers from station B
 - We can steal 50% of station B's customers (250 people), who purchase \$1,000 in gas per year
 - Additional Revenue: \$250,000 per year (250*\$1000)

Total increase in sales per year: \$350,000 (\$100,000+\$250,000)

The gas station owner is a simple man and wants to look at this like a perpetuity. What does this work out to?

Information to provide upon request: Discount rate= 10%, g = 0%

As a perpetuity: $\$350,000 / 10\% = \$3,500,000$

#8 – Gas Station (V of VI)

Competitive analysis

How will the competitor react?

- This is a prisoner's dilemma
- If station A [client] does it and B [competitor] won't, it is beneficial, because station A will steal \$250k in gas revenues per year from station B and achieve a positive NPV.
- But what is keeping station B from retaliating and doing the same thing? Remember, station A is barely scraping by, so station B is in a similarly bad situation. If station B responds, we could go out of business since revenue would decrease and the NPV of the investment would be negative (bad).
- If A does not do it, station B might do it. Then station A would lose \$250k in revenue per year and it could go out of business (bad).

Interviewer Guidance:

- No matter which scenarios the candidate chooses, push hard for the other option and play devil's advocate.
- An excellent candidate will create a matrix with payoffs (expressed in NPVs or as "Out of business"/"better, neutral, worse")

Sample Payoff Matrix:

		Station B	
		Sell groceries	Don't Sell groceries
Station A	Sell groceries	Negative NPV for station A	Positive NPV for Station A
	Don't Sell groceries	Negative NPV for station A	Status quo

#8 – Gas Station (VI of VI)

Recommendation

Interviewer guidance:

The recommendation could go two ways depending on what the interviewer decides.

Recommendation 1: Do it and try to push station B out of business.

Reasons: station a has a dominant strategy of offering groceries; if station A offers groceries, station B will go out of business and won't be able to finance the initial cost of launching groceries because no bank will fund an expected to be (less) unprofitable business

Risks: Station B might respond and bring both stations down.

Next steps: Try to form an exclusive contract with the grocery store.

Recommendation 2: Don't do it, and hope station B doesn't do it either.

Reasons: Station A and B have highest payoffs in the current state; if A offers groceries, B will too and both will risk going out of business; somehow the candidate should show that it is possible for the owners to reach an agreement even though their dominant strategies are to offer groceries

Risks: Station A might miss out on the opportunity and lose first mover advantage.

Next steps: Communicate to station B owner that you will offer groceries as soon as they do. To make this a credible threat station A owner could provide a business case showing that station A owner is better off launching grocery services if station B does so and showing an agreement with the local bank to provide appropriate financing.

#9 – Zoo Co

Analysis

Structure

Overall

Medium

Medium

Medium

Behavioural Prep

Tell me about a time you had a conflict with a teammate.

Tell me about the best piece of feedback you ever received.

Case Tracker

Industry	Financial services
Type	M&A
Leader	Interviewer
Concepts	Investment, break even, basic NPV

Case Stem

Our client is a zoo that is thinking about acquiring a famous zebra from an African preserve. It's a huge investment, but they believe the new zebra would be a great contribution to their animal community. You have been engaged to help decide whether this is a good idea. What would you consider when trying to help your client make this decision?

Interview Guidance

Even though the client is a Zoo, we're undertaking a similar process to what is done when underwriting an insurance policy. The case evaluates basic concepts, but involves many calculations and use of financial and assessment techniques.

Key case objectives:

1. Investment Valuation – Walk through the valuation process for an asset
2. Breakeven Analysis – Determine the revenue increase needed for a positive NPV
3. Risk Assessment – Should the zoo should use an insurance contract to hedge downside risk?

Rounding numbers is generally okay but should not be done to the extreme as it will alter the results

Clarifying information only provided when inquired by the candidate

Objective of the Zoo: they are primarily concerned with financial benefits – whether the zoo will earn higher profit.

Business model: main revenue stream of the zoo is ticket sales

Timeline: ASAP

#9 – Zoo Co (II of VII)

Sample Structures

Basic structure:

- Financial impact
 - Increased revenue
 - Increased costs (fixed and variable)
- Operational considerations
 - Transportation
 - Living space and condition

Advanced structure:

- Understanding Zebra
 - Popularity among visitors
 - Quantity at other competing zoos
- Financial impact
 - Increased revenue
 - Increased costs (fixed and variable)
- Operational considerations
 - Transportation
 - Living space and condition
 - Expertise
- Other considerations
 - Non-financial benefits (research, reputation, etc)
 - Government regulation
 - Risks (health and safety)

#9 – Zoo Co (III of VII)

Question 1: NPV

Zoo Co would like to know how much value bringing in the Zebra would add to the zoo, and would like you to determine this.

Notes:

- Use the data on the right to calculate benefit to zoo from acquisition
 - Determine whether or not this zebra purchase makes financial sense for the zoo, using the NPV value
- For simplicity, tell the candidate to use a perpetual NPV model, even though the zebra won't live perpetually
- Assume that attendance benefits are realized immediately and maintained thereafter

Calculations:

- Annual benefits = $(300K) * (\$15) * (0.08) = \$360K$
- Upfront costs = $\$235K + \$850K + \$110K = \$1.195M$
- Annual costs = $\$90K$
- NPV = $-\$1,195K + ((\$360K - \$90K) / 0.20) = \mathbf{\$155K}$

Revenue/Cost Data

Data to provide when asked

- 300K people visit the zoo yearly
- Admission is \$15
- Benefits from acquisition could lead to increased attendance. Another zoo that acquired a similar zebra had an 8% increase

Costs from zebra acquisition

- Immediate costs: acquisition fees, transportation costs, and new facilities
- Food, health costs and additional trainers are part of annual maintenance costs
- Acquisition cost: \$235K
- New facilities: \$850K
- Transportation: \$110K
- Annual maintenance: \$90K
- Discount rate = 20%
- Assume that immediate cost are paid today, and annual costs and benefits are realized beginning next year and sustained into perpetuity

#9 – Zoo Co (IV of VII)

Question 2: Break-even analysis

Zoo Co. is concerned about using the other zoo's attendance benefits as a proxy. They think that attendance could increase by less than 8%. What analysis could you perform to address their concerns? What is the breakeven attendance increase required?

Note to interviewer

The interviewee should determine that a sensitivity / breakeven analysis of the NPV calculation with lower attendance increases will help confirm that the project still makes sense

Calculations

Break-even:

$$= -\$1,195,000 + ((\text{revenue} - \$90,000) / 0.20)$$

$$= (\$1,195,000) \times .20 = \text{revenue} - \$90,000$$

$$\text{Revenue} = \$239,000 + \$90,000 = \$329,000 \text{ (*required additional revenue to break even)}$$

$$\$329,000 = (300,000) \times (15) \times (\% \text{ increase})$$

$$\% \text{ increase} = (\$329,000 / \$4.5\text{M}) = 7.3\%$$

#9 – Zoo Co (V of VII)

Question 3: Risk assessment

Since the zoo is very risk-averse, they're interested in hedging some of their downside risk.

An insurance company has offered to provide the Zoo with a constant revenue to increase revenue to \$250,000 per year if attendance increases are less than or equal to 5% (if revenue is \$135K, the insurance will give the Zoo, \$115K.

In exchange, the insurance company wants the zoo to pay 1% of the zoo's total annual revenues as a premium. What might you do to determine if this was a good deal?

Note to interviewer

The interviewee should recognize that additional information is needed, and that a market research study could aid in this process

Hand out Exhibit 1 after the interviewee identifies this notion.

The interviewee should use the market research to determine the probable attendance increase.

Calculations

After handing over exhibit 1

Annual cost to zoo:

$$1\% \text{ of total zoo revenues} = (0.01) * (\$4,752,000) = \$47,520$$

Annual expected benefit to zoo:

$$(\$250,000 - \$225,000) * (0.40) + (\$250,000 - 135,000) * (0.20) = \$33,000$$

Costs > Benefits, so this is not a good deal

#9 – Zoo Co (VI of VII)

Exhibit 1: Market research findings

Possible attendance	Annual revenue	Probability
3% increase	\$135k	20%
5% increase	\$225k	40%
7% increase	\$315k	30%
9% increase	\$405k	10%
Expected additional annual revenue	\$252k	
Plus: Current annual revenue	\$4,500k	
Expected total annual revenue	\$4,752k	

#9 – Zoo Co (VII of VII)

Solution and recommendation

- It is unlikely that the zebra acquisition is a good idea for the zoo to undertake given the information provided. At other zoos, attendance has gone up substantially due to a new zebra; however, based upon our market research, it seems less likely that we can breakeven on the investment through increased attendance. We have received an insurance contract to help mitigate some of the downside risk; however, it is too expensive to create value.
- In order to make the investment more palatable, we may consider negotiating with the insurance company to either increase the revenue benefits provided or decrease the premium cost.

Bonus guide to an excellent case

Excellent cases will:

- Identify that we can use another zoo's attendance increases as a proxy for estimating our own attendance increases
- Notice in Exhibit 1 that it is unlikely that attendance will increase sufficiently enough for the zoo to break even
- Notice that the insurance company's premiums and benefits are both impacted by attendance increases; so if attendance increases are always greater than 5%, the zoo will be paying even more but getting no benefit
- Notice that the insurance company's contract is essentially an option; so a different structure to the contract may be more suitable for the zoo

#10 – Brazilian Manufacturer

Analysis	Structure	Overall
Medium	Medium	Medium

Behavioural Prep

What is your proudest accomplishment?
Tell me about a problem in your professional life and how you were able to overcome it.

Case Tracker		Case Stem
Industry	Manufacturing	A widget manufacturer in Brazil has a 60% national market share and has a leadership position in South America. However, it only has a 5% international market share. The top 20 clients buy 80% of its products and the manufacturer is concerned with increasing margins and exploiting new sources of revenue. Their CEO has asked you to gather data and brainstorm potential ideas for increasing margins and market share.
Type	Growth Strategy; Profitability Improvement	
Leader	Interviewee	
Concepts	Market Share, Profit Margin	

Interview Guidance

Clarifying information only provided when inquired by the candidate

Market Information:

- The market is growing at 5%
- No significant competitors in the region
- Greatest threat comes from the US, Chinese, and Indian imports
- Premium pricing for national customers

Company Information:

- Plants are operating at 100% capacity
- Plants cannot shift from producing one product to another
- Currently, the company sells products directly to consumers

The key of the case is to focus on profit margins and breaking down the global and national markets. A strong candidate should lead the case and ensure they are providing insights throughout.

#10 – Brazilian Manufacturer (II of VII)

Sample Structure

Market Landscape (in both National and Global)

- Market share and growth rate
- Competitors (*highlight imports since clarifying info said no competitors in regions*)
- Customer Preferences

Manufacturers Capabilities:

- Current profit margin in both markets
 - Prices
 - Costs (production, COGS, distribution)
- Product mix
- [Capacity and product expansion]

Other Considerations:

- Regulatory considerations (tariffs, political considerations)
- Cultural congruencies affecting growth and ease of doing business in certain areas

Note: clarifying questions help guide. For example, if the candidate asks about capacity in the clarifying questions, then they would not focus on it here

#10 – Brazilian Manufacturer (III of VII)

Successful Case Roadmap and Interviewer Guidance

After presenting their structure, over the course of the case, the interviewee should try to understand the market dynamics, which products are more profitable than others and in what markets, and then should brainstorm ways to exploit this information to drive up profit margins and market share for the client.

The interviewer may ask to understand the products and profitability of each. Remind them that the company sells its products in local and international markets

***Use Exhibit A when asked**

The Interviewee may ask about the breakdown of international and national markets

***Use Exhibit B when asked**

- Ask them to calculate profit margins (will need to use unit costs in Exhibit A)
- Assume no freight costs, taxes, or any other extraneous costs

#10 – Brazilian Manufacturer (IV of VII)

Exhibit A

	Units	Average Price	Profit Margin %	Unit Cost
Product A	400	\$9.00	50%	\$4.50
Product B	200	\$5.50	40%	\$3.30
Product C	100	\$4.00	30%	\$2.80

#10 – Brazilian Manufacturer (V of VII)

Exhibit B

	Units	National Price	<i>Profit Margin %</i>		Units	Export Price	<i>Profit Margin %</i>
Product A	200	\$10.00		Product A	200	\$8.00	
Product B	150	\$5.00		Product B	50	\$6.00	
Product C	75	\$4.00		Product C	25	\$4.00	

#10 – Brazilian Manufacturer (VI of VII)

Exhibit B – Answer Sheet

	Units	National Price	Profit Margin %	Unit Cost
Product A	200	\$10.00	55%	\$4.50
Product B	150	\$5.00	34%	\$3.30
Product C	75	\$4.00	30%	\$2.80

	Units	Export Price	Profit Margin %	Unit Cost
Product A	200	\$8.00	44%	\$4.50
Product B	50	\$6.00	45%	\$3.30
Product C	25	\$4.00	30%	\$2.80

Use for interviewer reference only – or if the interviewee gets stuck

Prompt interviewee to explain insights as described below and on next slide:

- Product A is more profitable nationally
- Product B is more profitable internationally
- Product C has the same margins for both markets

#10 – Brazilian Manufacturer (VII of VII)

Possible Conclusions

Product A is more profitable nationally while product B is more profitable internationally.

For *National Market*:

- For national markets, the client should consider reducing prices to gain market share from India and China and the US, to maintain their leadership position
- The client could consider developing new products in partnership with local clients and establishing new/better distribution channels to better serve smaller national clients

For *International Market*:

- The client could seek support from the government to aid the export of products, i.e. the creation of commercial partnerships at the government level
- The client could consider acquiring companies in priority international markets to ease entry
- The client could look to establish more robust distribution networks in order to reach international customers

Sample Recommendation

- Note: could include any of the above points with supporting evidence and reasoning.

Recommendation:

National: The client should consider reducing prices for *national markets* to gain market share and maintain their leadership position, and developing new products.

International: They should consider expanding their *international* reach through acquiring companies to ease entry in priority international markets and establishing better distribution channels.

Risks:

National: reducing prices may result in lower profit margin, developing new products may be costly

International: acquiring companies require significant capital investment

Next Steps:

National: conduct sensitivity analysis to examine affect on decreasing price on profit margin, conduct market research on customer preferences for new products

International: examine markets for potential target firms to acquire, examine company financials for availability of capital

#11 – SPLAT!

Analysis	Structure	Overall
Medium	Medium	Medium

Behavioural Prep

How would your teammates describe working with you?
What is the riskiest decision you have made in your life?

Case Tracker		Case Stem
Industry	Consumer Products	We have a private equity client that is considering purchasing a small company that makes paintball guns (Splat). Sales last year were \$50M, net profit was \$20M, and they have been growing at 50% per year. The client has asked us for help in evaluating this investment. You & I are the team members who will do the bulk of the thinking and analysis. The client has asked us to help them to evaluate whether or not this paintball gun company (Splat) is a good investment?
Type	Acquisition	
Leader	Interviewer	
Concepts	3 C's; Market Sizing	

Interview Guidance

Even though the case has three distinct questions to evaluate, the candidate should still drive the case to some extent. It's important that the candidate not only deliver and 'answer' but also provide an 'insight' i.e., given what we know about (the company; the competitors; the customers; the market), here's my sense of the meaning/impact of this finding/number OR if they don't have enough information, telling you what information they would need to make a better or more complete assessment. They should also tie the questions together, applying learnings from one question to their approach/insights in another – this demonstrates that they're thinking holistically about the problem.

The ideal structure should cover: (a) **Market assessment**, including size, growth, and competitive dynamics; (b) **Company evaluation**, focusing on financial performance (revenues, costs), capabilities such as brand perception, product differentiation, and distribution channels; and (c) **Potential synergies or strategic opportunities** considering the PE firm's portfolio

#11 – SPLAT! (II of V)

Question 1

The first question that the client would like us to think about is: Is the (paintball) market attractive?

Additional information and solutions

Prompt is necessary: There are no secondary research reports on the paintball gun market. We're going to have to estimate the size and attractiveness of the market ourselves.

Market size: the interviewee should identify that the market can be sized by making assumptions about the following (prompt if necessary):

- Population size: US = 300MM, assume US only
- Paintball participation: 10% of the population plays paintball in a given year
- Gun purchasers: 10% of players actually purchase (rather than rent) a gun
- # of guns purchased per year: Average purchaser buys 1 per year
- Average price of a gun: \$100

*For simplicity, exclude the rental market.

Solution

Annual market size is $300M \times 10\% \times 10\% \times 1 \times \$100 = \$300M$

The company had \$50M of sales, which means they would have 1/6th or ~15-20% of the market.

Market growth: Ask candidates how they would estimate the growth rate.

- Look at the same drivers of market size to see if they are growing via:
 - Survey of end customers to find out about participation, purchasing habits
 - Calling stores that sell paintball equipment and ask them how fast sales are growing, and what is causing growth
 - Tracking the number of new entrants, store openings, or product launches in the paintball space to gauge momentum
- We found that the market is growing ~20% per year and will continue to do so for the next several years
 - This implies that at 50% growth Splat is gaining share

#11 – SPLAT! (III of V)

Question 2

The second aspect that the client would like us to think about is: What position does Splat have relative to its competitors? How do customers view their products?

Additional information and solutions

Provide only if asked:

- SPLAT! Sells both directly to consumers, through online sales, and through sporting goods retailers. They sell through two types of sporting goods retailers:
 - Specialty Paintball Retailers
 - Large Chain Retailers (Wal-Mart; Sport Check)
- Splat's sales have been steady through specialty channel, but growing ~60% per year in large chains.
 - Specialty stores sell all paintball gun brands; other brands experience roughly 20% sales growth YoY, while Splat's sales grow at roughly 25% YoY
 - The main specialty chain has been steadily adding Splat products to its many outlets over the last few years, which has supported the larger-than-average YoY sales growth. They are now in nearly all of the chain's stores.
- When surveyed, end customers indicated they buy based on reputation for quality, and Splat is known as having the best paintball guns on the market.
- There are several other paintball manufacturers competing in the market, however only two other players are viewed quite favorably, and not as highly ranked as Splat

#11 – SPLAT! (IV of V)

Question 3

The final thing that the client would like us to think about is: What, if any, are the risks they should consider when assessing the possible acquisition?

Additional information and solutions

What we know – The company has been growing fast for a few reasons:

- High market growth
- Quality product
- Adding stores in the specialty segment

However, adding stores appears to be done, so growth will only be from market growth and share gain vs. competitors, estimated at 25%. So, risks that may impact that growth will have significant repercussions for SPLAT!, for instance:

- **Participation risk:** Maybe paintball is a fad, and participation will stop increasing or even decline
- **Brand risk:** Maybe brand perceptions change quickly, and we will lose share to other competitors
- **Liability risk:** Maybe someone will get killed or injured playing paintball. If it's our gun, it's a legal issue. If it's another gun, it may result in decreased participation.

There's also a risk that the owners of the business won't agree to sell the company for a fair price (not rational agents, or may ascribe additional, non-financial value, to keeping the business for themselves) and thus may demand a higher price.

It's possible that Splat may not meet the required IRR of the PE firm

- Depending on the price of the deal, the company should buy Splat if the return meets their needs with the company growing 20-25% per year over the next few years

End case with a final recommendation based on analysis conducted.

#11 – SPLAT! (V of V)

Final evaluation

A good answer includes:

- Structuring an approach to answering the overarching question (i.e., identifying the need to understand the market, competitors, and customers)
- Understanding the different components that make up the market sizing

A better answer includes:

- Recognizes that Splat growth is much greater than market growth and quickly trying to identify the source of the discrepancy
- Realizing that roll-out into the retail channel/supply chain could be the current source of extraordinary growth for Splat

A superior answer includes:

- Move quickly through market sizing, uncover the difference between market and Splat growth, and understand the difference between the two
- Understanding the importance of end consumers and retailers in order to solve case
- Identifying that there are risks to every investment worth considering

#12 – HVAC Service Provider

Boston Consulting Group

Analysis	Structure	Overall
Medium	Medium	Medium

Behavioural Prep

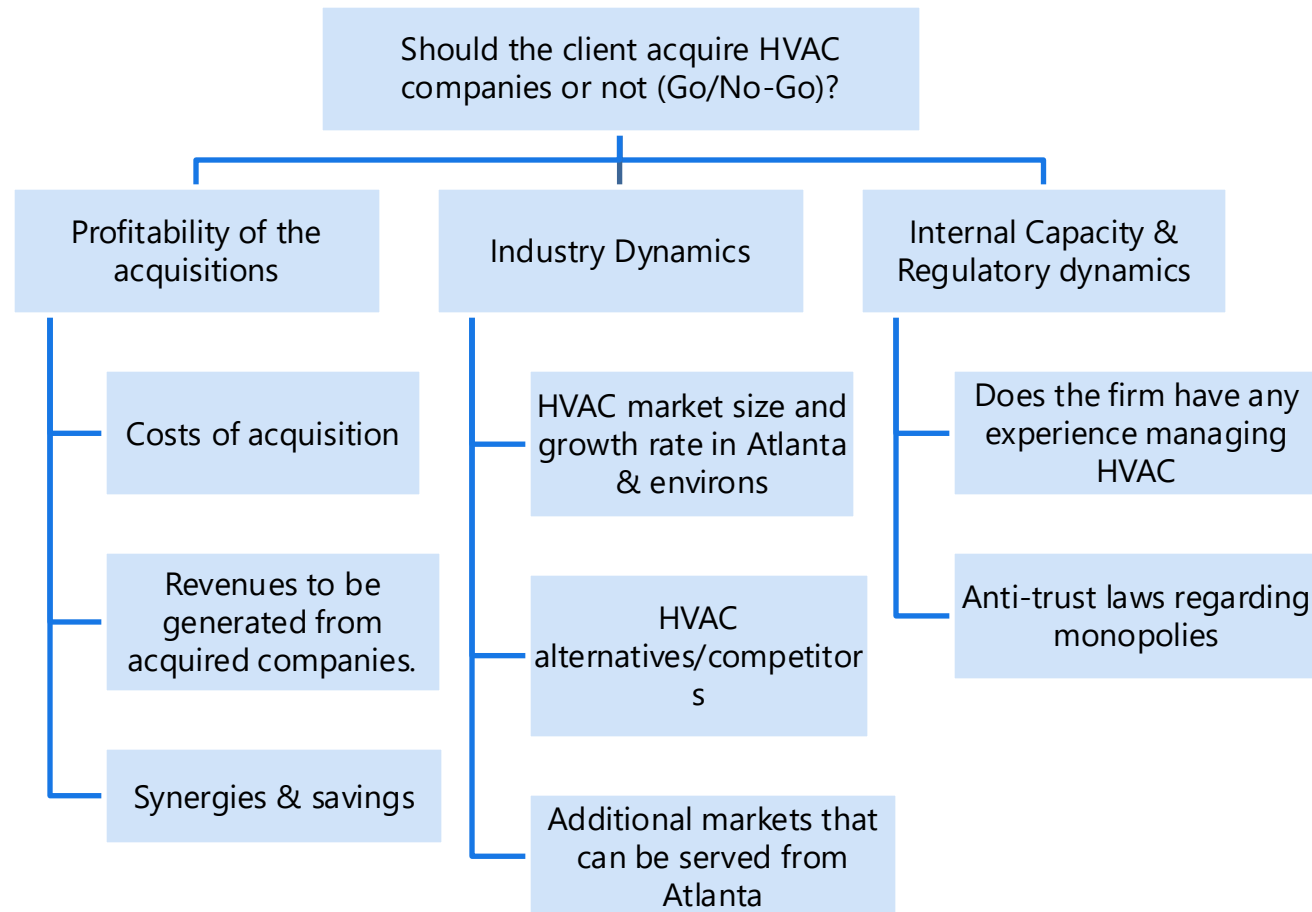
Tell me about an important role model in your life.
What do you do for fun outside of school?

Case Tracker		Case Stem
Industry	Energy	Your client is an energy firm that has a lot of extra cash and wants to know if they should buy up the 50 HVAC (heating, ventilation, and cooling) service firms in the Atlanta area and consolidate them into one larger firm. The client would like to know if this is a viable investment that they should consider. Clarifying Questions - How will the company evaluate the success of the acquisitions? <i>The company has set a three-year breakeven period.</i>
Type	Merger & Acquisition	
Leader	Interviewee	
Concepts	Break-even analysis, cost synergies	

Interview Guidance
• Ideally, the candidate should be able to break this case down into 3 parts: 1) Calculate cost of acquisition 2) Calculate new profits from cost savings due to buying up the 50 HVAC service firms 3) Break-even analysis • A strong candidate will ask up-front is there is a quantified objective or metric for “success” the client is looking to achieve through acquisition (in this case, 3 year break-even) • Candidate structure should ideally include a profitability analysis; if it does not, the interviewer should guide the candidate to brainstorm additional ways they could arrive at a go/no-go decision. • A good candidate should be able to calculate the acquisition costs quickly by recognizing that % of profit is the only relevant number from the cost bucket.

#12 – HVAC Service Provider (II of IV)

Sample Framework



#12 – HVAC Service Provider (III of IV)

Guidance for interviewer and information provided upon request

Only provide additional information after being specifically asked by the candidate.

- Atlanta market consists of 50 firms
 - Average annual revenue: \$10M
 - Revenue growth: 3%
 - Acquisition cost: Perpetuity cost of profits
 - Cost of capital: 13%
- Cost structure (% of revenues)
 - Labor: 50% (*Technicians are 100% utilized*)
 - Equipment: 25%
 - Administrative: 20%
 - Profit: remaining 5%
- Savings areas (for each firm purchased):
 - Labor dispatching efficiency: 5% decrease in labor
 - Equipment: 5% decrease through bargaining power
 - Admin: 1% decrease after IT and advertising investments
- Client's finance department requires a 3-year break-even
- Assume all costs savings occur immediately
- Assume revenues will remain stable for each target

#12 – HVAC Service Provider (IV of IV)

Encourage candidate to do the calculation for 1 unit instead of all 50 acquisitions.

Candidate should calculate implications of changing cost structure...

<u>Cost centre</u>	<u>Cost (% rev)</u>	<u>Cost (\$)</u>	<u>Savings (% cost)</u>	<u>Savings (\$)</u>	<u>New Cost (\$)</u>
Labor	50%	\$5M	5%	\$250K	\$4.75M
Equipment	25%	\$2.5M	5%	\$125K	\$2.375M
Admin	20%	\$2M	1%	\$20K	1.98M
Profits	5%	\$500K			\$895K PROFIT

...then conduct a break-even analysis

<u>Current profit</u>	<u>Discount Rate</u>	<u>Cost of Firm</u>	<u>Expected Profit</u>	<u>Undiscounted Break-even</u>
\$500K	10%	\$5M	\$895K	5.59 years
	(CoC - growth)			(stating "over 5 years" is fine)
				TOO LONG = NO GO

What are additional risks associated with the acquisition?

A solid interview will come up with a strong list of risks

- The energy firm has no particular experience in the HVAC industry
- Cultural issues (small operations purchased by large company)
- National entrants overpowering effort
- Antitrust issues may arise from the HVAC monopoly

Suggestions for improvement and final recommendation

A solid interviewer will suggest improvements for breakeven before heading into recommendation

- Based on the information we have today, *we do not recommend* that our client invest in consolidating HVAC companies as it would require a 5-year payback period which does not reach their goal of breaking even in 3 yrs
- *Potential ways to improve break-even:*
 - Reduce purchase price
 - Seek further cost improvements (IT systems, etc.)
 - Improve revenues through advertising efficiency, brand name, referrals, etc.

#13 – Vidi-Games

Analysis	Structure	Overall
Hard	Medium	Medium

Behavioural Prep

Tell me about a time you pushed back on a project timeline.
Describe a time you had to change your communication style to influence different stakeholders.

Case Tracker		Case Stem
Industry	Telecom, Media & Technology	Our client is Vidi-Games, a manufacturer and retailer of video games. They are planning to enter the Brazilian market. One of the factors that has prevented them from entering that market is that there is a tariff of 50% of production costs for importing products that can be produced in Brazil. However, the government is planning to lower the tariff down to 15% in the next years, decreasing it by 5% every year (from 50% to 45%, and so on). What variables should they consider? What is the market size? Should they enter the market? And if so, should they import the product from China, where they currently have a production facility or produce it in Brazil?
Type	Opportunity Assessment	
Leader	Interviewee	
Concepts	Market Entry	

Interview Guidance
- First, let the interviewee structure the problem. It is a profitability problem. Some things that should be mentioned, on top of industry analysis and profitability drivers, are repatriation costs, regulation, currency, inflation, culture issues, etc. <i>Upon request:</i> the client wants to break-even (recover upfront investment of \$18M) within 4 years after entry.

#13 – Vidi-Games (II of VII)

Sample Framework

Market Landscape of Brazil:

- Market Size
- Vidi-Games' potential market share
- Competitive landscape
- Customers preferences in Brazil (are they interested in these video games?)

Potential Financial Gain from entering Brazilian market:

- Revenue: Market Share / revenue calculation over 4 years
- *Cost of Production (comparing China and Brazil):*
 - Brazil: cost of setting up new facility (upfront investment costs) – do breakeven calculation
 - China: additional distribution costs (importing, tariffs, including changing tariff costs)

Other Considerations:

- Cultural barriers in starting production in Brazil
- Political considerations

#13 – Vidi-Games (III of VII)

Information to be provided to interviewee

The interviewee should try to estimate these numbers, but no real prior knowledge is required. Because it is a market very different from the US, these numbers should be provided after the person tries to estimate them.

- Brazil population = 200M
- Younger population – under 35 = 50% of total (it can be assumed that these are the only ones interested in video-games)
- Affluent population = 40% of total (Those people that can afford the game)
- Average number of people per household = 4
- Penetration of video games within the affluent and young segment in Brazil = 75%
- Games per year per household with affluent youngsters = 3
- Retail price \$50
- Expected market share = 20%
- Upfront investment = \$18M (only relevant later in the case)

Market Size Calculation

Total number of video-games that can be expected to be sold per year = (total number of households that will purchase video games) x (number of video games per year per such household) = $[(\text{total population} \times \% \text{ of young people} \times \% \text{ affluent people}) / (\text{average people per household})] \times (\text{number of video games per year per household})$

Total number of video-games that can be expected to be sold per year = $((200\text{M} \times 50\% \times 40\%) / 4) \times 75\% \times 3 = 22.5\text{M}$ video games.

With a retail price of \$50, the size of the video game market is \$1.125B. The savvy interviewee will point out that this is a sizeable market. Real insight from this number, however, can only be derived by comparing it with the client's current revenues. At a 20% market share, the company can expect to sell 4.5M video games and make \$225M in revenue.

#13 – Vidi-Games (IV of VII)

Exhibit/ Information to be provided upon request

Profitability drivers	Per unit value
Retail price	\$50
Production in China	\$14
Production in Brazil	\$21
Distribution (same)	\$6
Labor	\$8
Overhead	\$6
Other costs	\$10.40

Profitability analysis

Per unit profit = price – per unit cost

	Per unit value China	Per unit value Brazil
Production	\$14	\$21
Tariff	\$7	\$0
Distribution (same)	\$6	\$6
Labor	\$8	\$8
Overhead	\$6	\$6
Other costs	\$10.40	\$10.40
Total cost	\$51.40	\$51.40
Price	\$50.00	\$50.00
Per unit profit	-\$1.40	-\$1.40

A good interviewer will share the following insights:

- Because of the tariff, production cost are currently the same in China and Brazil
- Currently, entering the Brazil market it is not a profitable business, hence Vidi-Games should postpone entry, unless cost savings or price increases are possible; note producing in Brazil will never be profitable

#13 – Vidi-Games (V of VII)

But the tariff is going to decrease...

A good candidate will proactively move on to answer the following questions: "If at all, when will it make sense for the client to enter the Brazil market". Give the interviewer some time to think about next steps themselves but provide guidance if interviewee gets stuck or moves in another direction.

The candidate should mention but can ignore inflation.

Break-even point when cumulative profits = upfront investment (\$18M):

- Sum of all costs not influenced by tariff = \$30.40 (Distribution + Labor + Overhead + Other cost)

	Year 2	Year 3	Year 4	Year 5	Year 6
Price	\$50	\$50	\$50	\$50	\$50
Stable unit cost	\$30.40	\$30.40	\$30.40	\$30.40	\$30.40
Tariff	45%	40%	35%	30%	25%
Unit production cost (production + tariffs)	\$20.30	\$19.60	\$18.90	\$18.20	\$17.50
Total unit cost	\$50.70	\$50.00	\$49.30	\$48.60	\$47.90
Per unit profit	-\$0.70	\$0.00	\$0.70	\$1.40	\$2.10
Sales volume	4.5M	4.5M	4.5M	4.5M	4.5M
Yearly profit	-\$3.15M	\$0.00M	\$3.15M	\$6.3M	\$9.45M

Insight: Client should enter in 2 years (Year 3). This is the earliest time to enter for which cumulative profits over 4 years exceed upfront investment of \$18M. Entering later would increase the risk of competition taking over the market, thus limiting client's potential share.

#13 – Vidi-Games (VI of VII)

Bonus Question

What could Vidi-Games do if they needed to enter now?

Interviewer guidance:

This is a structure brainstorming question. Here are some ideas of potential answers:

ORGANIC OPPORTUNITIES:

Revenue side:

- Analyze how viable an increase in price is (price elasticity analysis)
 - Important: since per unit profits are negative, increasing volume won't solve the problem (a savvy interviewer will point this out)
- Selling video-games to retailers there (hence, savings in distribution, labor, overhead, etc., but selling at a reduced wholesale price)

Cost side:

- Negotiating cheaper labor
- Moving manufacturing sites to lower cost areas
- Optimizing distribution network
- To produce in Brazil: See possibility of reducing production costs

INORGANIC OPPORTUNITIES:

- Entering in partnership with a Brazilian partner
- Acquiring a competitor in Brazil

#13 – Vidi-Games (VII of VII)

What do you think of the current scenario? I have to call Vidi-Games now – what should I tell them?

Recommendation:

Vidi-games should enter the Brazilian market in two years.

Rationale:

This is the earliest time to enter for which cumulative profits over 4 years exceed upfront investment of \$18M.

Risks:

There could be additional competitors entering the market during this time.

We based the calculations from the current tariff projections; the government could change them.

Next Steps:

Do more thorough scan of competitive landscape to understand the main players in the market

Consider other potentially more profitable markets to enter

#14 – Molds R Us

Bain & Co.

Analysis	Structure	Overall
Hard	Medium	Medium

Behavioural Prep

Tell me about a time when you led a team through a challenge.
Give me an example of a time when you had to motivate others who did not report to you.

Case Tracker		Case Stem
Industry	Consumer Products	Our client is a private equity firm interested in Molds R Us, a small company that makes plastic moldings for houses in Russia. They want to know if we think investing in this company is a good idea. The firm also wants to understand what the 2011 market for moldings, particularly in plastics, will look like.
Type	Opportunity Assessment	
Leader	Interviewee	
Concepts	Market Sizing, Go no-go	

Interview Guidance

Once the candidate lays out a framework and asks the relevant questions you should give them exhibits A and B.

After the candidate analyzes the exhibits **ask them to calculate their estimate for the number of meters of plastic moldings being sold in 2011.**

#14 – Molds R Us (II of VIII)

Information provided upon request

The PE firm wants to see growth of 20% in the first year to justify this purchase.

This company only plays in the Russian market and the PE firm is not interested in expanding across borders.

This company is the only player in plastic moldings.

Moldings are used where walls meet the ceiling to add a decorative appeal to houses and are only used in residential buildings.

All housing starts require moldings in the year they are started and are all completed by the next year.

#14 – Molds R Us (III of VIII)

Sample Framework

- 1) Market Size and Growth
 - What is the size of the market for plastic moldings?
 - What are the key segments for plastic moldings?
 - What are the growth drivers?
 - What is the outlook for these drivers? (including for 2011)
- 2) Define Molds R Us as a business
 - Size and growth
 - Product offerings
 - Plastic moldings
 - Any other products? (*if this was not addressed by interviewee in the clarifying questions*)
 - Key segments of Molds R Us's business
 - Ex.: Customers, Geographies
- 3) Competitive Positioning
 - What is the competitive landscape for plastic moldings?
 - What is Molds R Us's market share?
 - How does Molds R Us perform across customers' key performance criteria?
 - How does this indicate the potential to gain market share?
 - How protected is Molds R Us from competition?
 - Ex.: Barriers to entry
- 4) Value Creation
 - How can the PE firm create value through an acquisition of Molds R Us?
 - Ex.: grow organically, tuck-in acquisitions
 - What is Molds R Us's management forecast vs. ours?

#14 – Molds R Us (IV of VIII)

Exhibit 1 – Types of Moldings

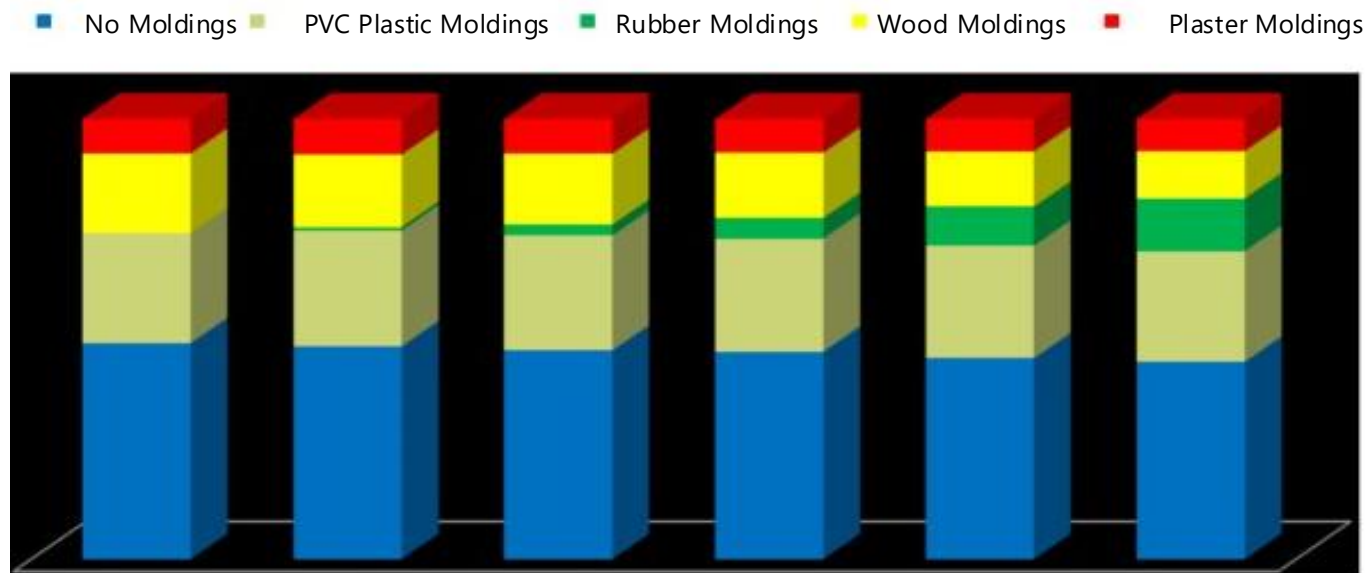
Molding Options	None	PVC Plastic	Rubber	Wood	Plaster
Price/10 Metres	0	1 Ruble	1.5 Rubles	5 Rubles	15 Rubles
Installation Requirements	None	Contractor	DIY	Contractor	Contractor
Replacement	None	Every 5 Years	Every 7 Years	Every 10 Years	Every 25 Years

Other important information:

- A contractor can lay down 1000 feet of molding per hour
- A contractor makes, on average, \$50 Rubles per hour
- DIY-Do it yourself or self installation
- The average house has 4000 metres of walls

#14 – Molds R Us (V of VIII)

Exhibit 2 – Moldings used in the Russian market



	2005	2006	2007	2008	2009	2010
Total Residences	29,689,297	30,145,394	30,696,939	31,375,374	32,170,804	33,122,149
Housing Starts	456,097	551,545	678,435	795,430	951,345	1,245,890

#14 – Molds R Us (VI of VIII)

Estimate # of meters of plastic moldings being sold in 2011

- This can be done by multiplying the market share of plastics for 2010 by the number of residences in 2011 (2010 residences + 2010 starts) plus the estimated housing starts in 2011 multiplied by the market share of plastics. This gives the expected number of houses using plastics in 2011. Given that plastic moldings are replaced every 5 years, the candidate should realize that only 1/5 of existing households will be replacing their moldings in 2011.

Math:

- Note: It is ok to round in the math portion of this case
- 34.3M Residences + ~1.4M Starts = Approx. 35.7M houses in 2011
 - *Interviewer guidance: Note that there is no information for 2011 on Exhibit B. Candidate should recognize that total residences + housing starts in a given year add up to total residences in the following year (i.e. total residences 2011 $\approx$ 33.1M + 1.2M = 34.3M). Also candidate should look at growth rate of housing starts to estimate # of house starts in 2011.*
- 34.3M Residences * 25% market share of plastic moldings = 8.6M houses
- 8.6M Residences * 0.2 (replacement rate for moldings) = 1.7M existing houses replacing moldings in 2011
- Estimated growth in housing starts = $(1,245,890/456,097)^{1/5} - 1 \sim 22\%$
- Estimated housing starts in 2011 = $0.2 * 1,245,890 = 1.5\text{M}$ housing starts
- Estimated 1.5M Housing starts in 2011 * 25% market share = 0.38M
- So, 1.7M existing homes + .38M starts = 2.08M Houses in 2011 using plastic moldings
- 2.08M * 4K meters of wall per house = **8.3B meters of plastic moldings** being sold in 2011.

#14 – Molds R Us (VII of VIII)

Case Progression (Guidance for Interviewer)

- After reviewing the charts and graphs, the candidate should notice the stagnant pace of the market share of plastic moldings
- A good candidate will begin to calculate the overall changes in market size to see if there is enough growth to make this deal worthwhile
- **Either way, have the candidate calculate the overall market growth rate from 2005-2010**
- This will begin to clue the candidate into the major issue: the growth will not be high enough for the PE firm to move forward with the investment
- Existing homes growth rate $(33.1/29.7)^{(1/5)} - 1 \sim 2\%$
- New homes growth rate $(1,245,890/456,097)^{(1/5)} - 1 \sim 22\%$
- The key here is for the candidate to recognize that the market of plastic moldings for existing homes (about 82% of market) far outweighs housing starts (about 18% of market – see calculations on previous slide) and thus recognize that overall market growth will fall well short of the 20% target. Actual growth rate is $<10\%$.

Once candidate recognizes low growth rate, ask them for their final recommendation to the PE firm.

#14 – Molds R Us (VIII of VIII)

Recommendation

The PE firm should not purchase Molds R Us.

- Plastic molding market share is stagnant among all moldings sold
- The overall growth in housing does not make up for the stagnant growth and they will not grow revenues by 20% in their first year

Risks:

- Lack of diversification in their portfolio (depending on other investments)

Next Steps

- The PE firm should look at rubber molding companies to see if there is an opportunity to purchase an organization because of high growth of market share in the market
- They should look at the sales and marketing of Molds R Us to see if there is opportunity to spur sales to increase growth by investing in marketing, distribution, or sales channels

#15 – The Coffee Grind

Analysis	Structure	Overall
Hard	Medium	Medium

Behavioural Prep

Describe a positive/negative experience working with a customer.
Tell me about a time you pushed back on a project timeline.

Case Tracker		Case Stem
Industry	Consumer Products	The CEO of a major client has requested a short-term study examining a small part of the client’s product portfolio. The company has a small division that manufactures automatic drip coffeemakers for the US and Canadian market. The division has been steadily producing coffeemakers for 20 years and has made few changes to the business over its history. The client has always enjoyed healthy margins for the coffeemaker division, and annual volumes have been steady. Recently, however, the coffeemaker division’s profits have been declining. The CEO wants to understand what is going on. What broader insights would you want to explore first to answer the CEO’s question?
Type	Profitability Improvement	
Leader	Interviewer	
Concepts	Distribution Channels	

Interview Guidance
Although this looks and feels like a profitability case (which it ultimately is), the point here is to push the interviewee to develop a framework beyond the standard profitability setup. Additionally, the case is meant to train the interviewee to listen to the question being poised: the interviewer is asking for “broader” insights, not basic profitability analysis. If the interviewee tries to go down the typical profit = price x volume – fixed + variable costs, push him or her a little harder to think bigger picture. This case was done at the partner level and thus involves a little more ambiguity.

#15 – The Coffee Grind (II of IX)

Information to be provided upon request

Changes in fixed or variable costs:

- The production lines and facilities are mature. The business has been steady, overall production is rather efficient given the advantages of a long-term steady state
- No major changes to fixed or variable cost inputs have occurred recently

Production:

- All production occurs at a facility in Michigan. The plant is operating at about 90-95% capacity

Product mix:

- The division produces 4-cup, 10-cup, and 12-cup coffeemakers. The overall mix between these categories has been fairly consistent.

Volume:

- Relatively constant

Price:

- On average, prices have been declining (but why? To answer this question a broader framework like the one on the next page is required)

#15 – The Coffee Grind (III of IX)

Sample Framework (w/o profitability portion) and follow-up information to be provided upon request

Market	Customers	Channel (Company)
• How are competitors performing? • Have new competitors entered? • Have new substitutes emerged? • Are we missing new technological developments? • Where are our prices relative to our competitors?	• Are customer preferences changing? • Has the product mix changed? • Are customers more or less price sensitive than before? • Have customers become more concentrated? • Are customers shopping in new channels?	• How do we reach our customers? • Has distribution changed? • Have we lost shelf space/visibility to other products? • Are retail outlets changing? • How are sales price established? • Are channels consolidating?
Follow-up and guidance	Follow-up and guidance	Follow-up and guidance
There have been some new entrants, primarily for very low-priced coffeemakers. The client's coffeemakers are mid-price range and there are several other competitors at this price point. New technology and substitutes are minimal. <i>Try not to let the candidate go too far down this area, the real issues are in distribution.</i>	Volumes by coffeemaker size have been relatively steady, indicating that customer preferences have not undergone dramatic changes. <i>Again, try not to let the candidate go too far down this path.</i>	Channel is the real issue. <i>When the candidate reaches this area, present them with the first exhibit:</i> <i>Opening: "We were able to get some quick shelf data on coffeemaker sales by channel."</i> If the candidate missed this bucket, thinking of a 4C structure helps remember.

#15 – The Coffee Grind (IV of IX)

Exhibit 1 – number of coffeemakers sold (in thousands of units) by retailer category

	2010				2009				2008				2007	2006	2005	2004	2003
	4-cup	10-cup	12-cup	Total	4-cup	10-cup	12-cup	Total	4-cup	10-cup	12-cup	Total	Total	Total	Total	Total	Total
National Chains																	
Best Buy	10	12	12	34	9	12	12	33	9	12	12	33	33	32	32	31	30
Target	29	38	17	85	28	38	17	83	28	38	17	84	84	82	80	78	79
Walgreens	1	1	1	4	1	2	2	4	1	2	2	5	5	5	5	5	5
Sears	1	6	7	14	1	6	7	14	1	6	7	14	15	15	16	18	36
Office Max	-	6	10	15	-	6	10	16	-	6	10	16	17	17	18	19	20
Kohl's	1	3	2	6	1	3	2	6	1	3	2	7	7	7	7	7	8
Bed, Bath, & Beyond	8	8	7	24	9	9	8	25	10	9	8	27	29	30	32	33	34
Costco	15	16	19	50	17	16	19	52	17	17	21	56	59	62	67	69	25
BJ's	12	14	47	73	11	13	42	66	10	12	38	60	57	52	48	44	20
Walmart	41	76	71	188	34	65	68	167	28	54	60	142	125	110	93	83	99
Total National	119	179	194	492	111	168	187	466	107	158	178	443	429	414	398	388	357
Memo: Percent of Sales	15%	23%	25%	63%	15%	22%	25%	61%	14%	20%	23%	57%	52%	52%	50%	46%	46%
Non-National Retailers																	
Grocery	19	25	36	80	23	30	37	90	26	32	39	97	116	109	110	120	104
Medium Retailers	36	24	25	85	45	25	20	90	45	31	33	108	101	109	120	114	102
Smaller Retailers	38	21	15	74	10	24	19	53	16	25	20	61	100	91	95	112	106
Other	10	20	21	51	12	24	25	61	12	29	29	70	76	70	75	104	101
Total Other	103	90	97	290	89	103	101	293	99	117	120	336	393	379	400	450	413
Memo: Percent of Sales	13%	12%	12%	37%	12%	14%	13%	39%	13%	15%	15%	43%	48%	48%	50%	54%	54%
Total Sales	222	269	291	782	200	272	288	760	206	275	298	779	822	793	798	838	770
Memo: Percent of Sales	28%	34%	37%	100%	26%	36%	38%	100%	26%	35%	38%	100%	100%	100%	100%	100%	100%

#15 – The Coffee Grind (V of IX)

Questions for Exhibit 1

Question 1a: What conclusion can you make from this data?

- *Key insight:* The candidate should identify that National chains increasingly make up a greater share of the client's sales
- *Bonus points:*
 - One retailer, Wal-Mart, buys 25% of their coffeemakers
 - Total sales have stagnated over the last 7 years
 - As might have been mentioned by this point, the product mix is fairly stable

Question 1b: What are the implication for this shift towards National retailers?

A great candidate proactively discusses the implication of the key insight about National chains increasing share of client's sales.

Sample answer:

- Larger chains have more bargaining power, and are putting more pressure on the division to provide discounts on its products based on volume, which is squeezing the company's margins

Math teaser

Question 2: What percent of the market do you think the client has?

As for every analysis during a case the candidate should lay out a structure or an approach to the answer first before diving into calculations. The candidate should make their own assumptions. Only correct if they are wildly off the mark.

Approach: Market share = Client sales volume / Market sales volume. Therefore, the candidate needs to start by estimating the number of coffeemakers sold in the US and Canada annually:

- Any logical approach is acceptable.
- A recommended approach is to start with the population of households in the US and Canada (~100M households in the US and ~10M in Canada)
- Estimate the percent that have electric coffeemakers (~65%)
- Replacement repurchases of coffeemakers every 5-10 years
- Bonus points if candidates also considers commercial and industrial places with coffeemakers (offices, hotels, etc.)
- In the end, the candidate should have an estimate in the range of 10-15M coffeemakers sold annually in the US and Canada

Answer: The client's market share should be around 5-8%. Again, a good candidate will gut check the result (there are a lot of coffeemakers so 5-8% seems reasonable) and point out the implication that the client seems to be a relatively small player with little bargaining leverage with retailers.

#15 – The Coffee Grind (VI of IX)

Follow-up question

Question 3: During a brief meeting with the CEO, you share your early insight that national retailers are squeezing the company's margins. He states that the dishwashers and cooking appliances division recently launched a website to sell its products. He wants to know if a website would also work for coffeemakers. Do you think it will work?

Interviewer guidance: Any answer here is acceptable if the candidate can create a logically structured argument to support his or her answer. One way to structure the answer is a simple cost benefit analysis. The candidate should at a minimum identify a couple of risks. Additionally, the candidate should want to know more about the possible profitability of a website.

Possible reasons for a website (sample)

- *Profitability:* allows client to capture the margin that presently goes to the retailer
- *Channel:* reduces client's dependency on national retailers, could add additional volume
- *Company:* the company already has an existing website for dishwashers and cooking appliances, so capabilities may exist

Possible reasons against a website (sample)

- *Profitability:* need to understand economics (how expensive are the fixed costs?)
- *Customers:* coffeemakers are more of an immediate purchase decision than dishwashers or ovens, customers may not want to wait for shipping times for online purchases
- *Competition:* may face increased competition online from websites as Amazon.com
- *Channel conflict:* customers may already be purchasing our coffeemakers from our retailers' websites, which creates risk that retailers will retaliate by pulling our products from their shelves
- *Company:* a coffeemaker division may not have the expertise or skills needed for online marketing, sales, shipping, etc.

#15 – The Coffee Grind (VII of IX)

Break-even analysis for website

Question 4: Given the following information we've collected, what percent of the company's current sales would the client need to achieve for the website to break-even?

The interviewer should tell the candidate the following:

- Annual website costs: \$500k
- Additional marketing costs: \$300k
- Average retail price of coffeemakers: \$60
- Handling website sales: \$4 per coffeemaker
- Shipping cost: paid by customers

The interviewer should only provide the following when asked (about cost or margins):

- Client's gross margins on coffeemakers 40%

Approach:

- Determine contribution margins per coffeemaker sold through website to calculate how many units are required to cover fixed costs
- Contribution margins = Price – (COGS + website sales)
- COGS (variable cost) approach: Gross margins = (Price – COGS)/Price → COGS = \$36 → contribution margin = \$60 - \$36 - \$4 = \$20
- Total fixed cost = \$500k + \$300k = \$800K
- Break-even volume = \$800k/\$20 = 40k coffeemakers

Answer: The client sells approximately 800k coffee makers annually, so about 5% of the client's volume.

#15 – The Coffee Grind (VIII of IX)

Other Solutions

Question 5: The CEO really appreciates all the work you've done so far on identifying the issue with national retailers squeezing margins and the breakeven analysis for the website. He would also like your insight on the other possible ideas. What other solutions might you suggest?

Interviewer guidance: This is another structured brainstorming exercise. Anything is acceptable as long as it's reasonable; the idea is to push the candidate's creativity. After the candidate has provided a few options, continue to ask "what else" until the candidate cannot produce any more ideas (McKinsey often uses this type of questioning during its interviews and structured brainstorming skills are generally appreciated across firms)

Sample answers (ideally structure in buckets such as revenue-side and cost-side improvements):

- Investigate selling the division to another coffeemaker
- Move production overseas to increase margins
- Investigate purchasing other coffeemaker manufacturers to gain leverage with retailers
- Consider exclusive retailing rights to one distributor
- Increase marketing efforts to create a pull-strategy from customers to increase leverage with suppliers
- Seek other channels (Starbucks, hotel chains, etc.)
- Consider international expansion
- Increase focus and attention on small to medium retailers
- Consider raising prices to offset margin loss (but with further investigation of customer price sensitivity)

#15 – The Coffee Grind (IX of IX)

Recommendation

Question 6: What final recommendation would you give the CEO on how to move forward?

The candidate should either provide an initial (“from what we have seen so far, it seems favorable to...”, if no concrete data to support the claim) recommendation the website or another solution identified above and provide supporting reasons.

Risks: In the candidate’s conclusion, the candidate should mention risks such as competition from other website, retaliation by retailers, risk of moving production overseas, etc. Prompt candidate if they forget to mention risks. E.g. Consider raising prices to offset margin loss (but with further investigation of customer price sensitivity)

Example Recommendation:

Recommendation: Go ahead with the website

Support: Only 5% of client’s volume is needed for website to break-even. This is very attainable. Further increases would provide more contribution to company profits than sales through other channels

Risks: Alienating retail partners, causing retaliation. Not getting enough traffic to website.

Mitigation: Negotiate contracts with major retailers pre-emptively to prevent costly retaliation. Online marketing to drive traffic.

Next Steps: Start negotiations with retailers to secure position. Perform market research to optimize marketing plan/strategy

#16 – Winter Olympics Bidding

Analysis	Structure	Overall
Medium	Medium	Medium

Behavioural Prep

Tell me about a time when you had to convince a difficult client to follow your recommendations
Tell me about a time you faced conflict in the workplace.

Case Tracker		Case Stem
Industry	Telecom, Media and Entertainment	Our client is a major US television network that is trying to figure out how much to bid for the 2022 Winter Olympics broadcasting and has brought you in to help us figure out the right bid. The Winter Olympics are a huge deal and will require a significant amount of capital to secure the rights. Before making a bid, our client wants to ensure that we have considered all the right things. <i>For this case consider that there are exactly 6 years to go to the Winter Olympics</i>
Type	Valuation	
Leader	Interviewee	
Concepts	NPV and breakeven analysis	

Interview Guidance

In this case, candidates are tasked with a quantitative-heavy evaluation of an Olympic bid. They need to crunch the numbers, considering potential ad revenue and cost data, while also calculating the Net Present Value (NPV) to determine an appropriate bid size.

Successful candidates will proactively seek additional information essential to solve the problem rather than relying solely on the interviewer for data.

It's important to note that some candidates, particularly those less inclined toward finance, may require gentle guidance through the mathematical aspects of the analysis.

#16 – Winter Olympics Bidding (II of VII)

Clarifying answers to provide

1. When will the payment be due?

The payment will be paid when the Winter Olympics starts, which is 6 years from now.

2. What is the Winter Olympics' schedule?

If this question is asked before candidate provided a structure, tell them you will show an exhibit later in the case.

If the candidate already presented the structure, show **Exhibit 1**

Revenues

- No subscription revenue, but can keep 100% of advertising revenue
- Ad rates are \$400K/30 second ad for prime time (M-F 7-11 PM, all weekend) and \$200K/30 second ad for non-prime time
- Market research has shown that you can include no more than 10 minutes of advertising per hour.

Costs

- Direct Costs: \$428M
- Opportunity cost (lost revenue from other programming): \$1M/hour
- Time value of money: 6-year lag for receipt of revenue
- WACC: 12%

Guide to case

Part 1 – Quantitative discussion

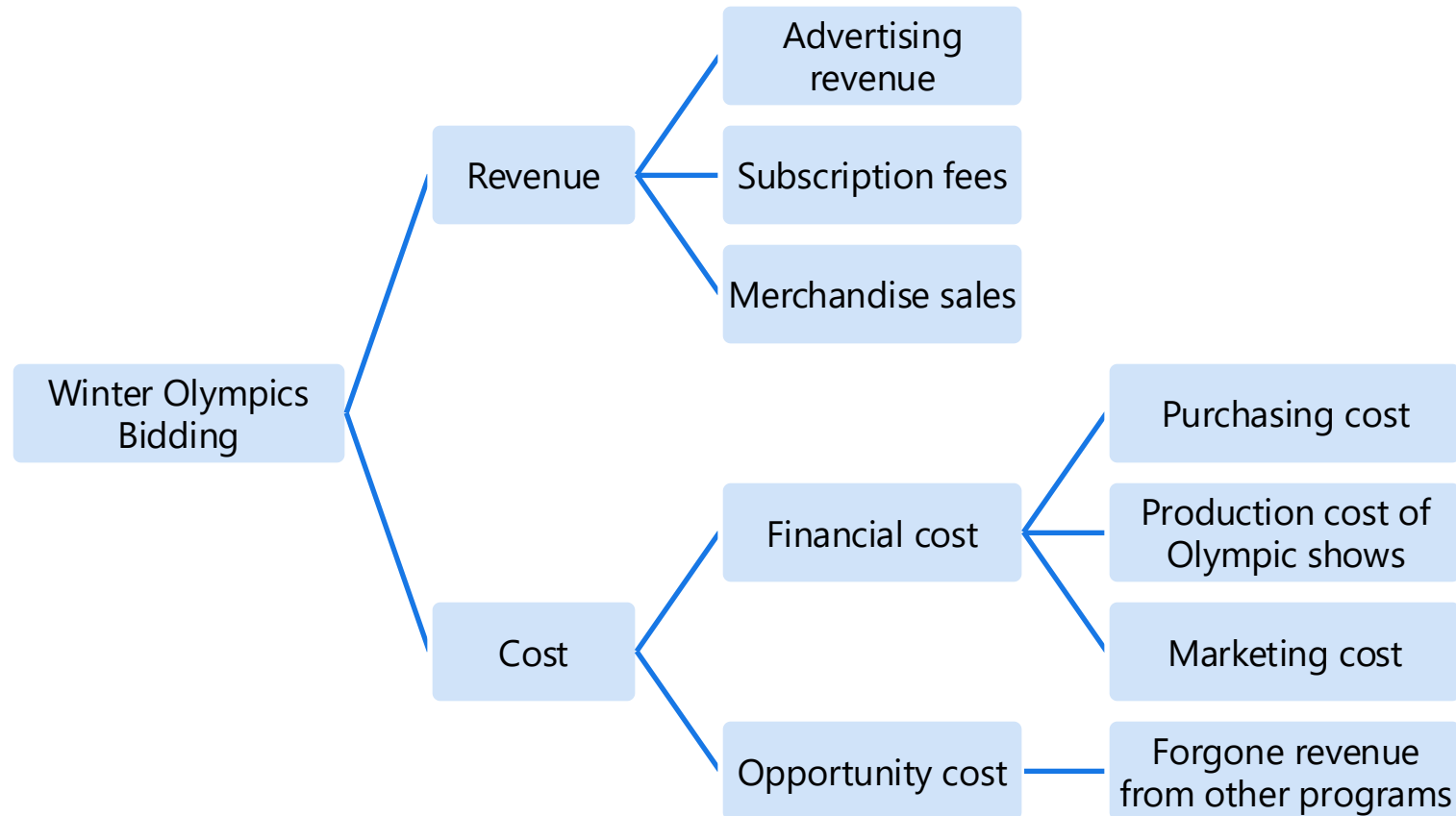
- Some of the numbers are assumptions here are difficult, so nudge the candidate along if necessary.
- First of all, the candidate should find the revenues from hosting the Olympics.
- Then, the candidate will have to figure out if this is a good investment. They should identify 3 costs (production costs, opportunity costs, and time value of money). By factoring in these costs, the candidate will find out if the Olympics are worth the investment.

Part 2 – Qualitative discussion

- After finding the NPV of \$177M, ask the candidate about intangible factors, benefits, and risks. Some critical factors:
 - Might give network access to new viewers
 - There is prestige associated with hosting this event
 - We can the airtime to promote other programming
 - Opportunities for product tie-ins, supplemental revenue
- After finishing the discussion, ask the candidate for a recommendation

#16 – Winter Olympics Bidding (III of VII)

Sample Framework



#16 – Winter Olympics Bidding (IV of VII)

Exhibit 1 – Winter Olympics Broadcasting Schedule

Winter Olympics broadcasting schedule (16 days)

- Day 1 – Friday - Opening ceremonies
 - 3 hours of programming from 8-11 PM
- Day 2-15 – Games – 10 hours/day
 - Weekday (9 – 12pm, 2 – 5pm, 7-11pm)
 - Weekend (11 AM – 9 PM)
- Day 16 – Saturday – Closing
 - 3 hours of programming from 8-11 PM

#16 – Winter Olympics Bidding (V of VII)

Quantitative questions

1. Calculate the revenue from broadcasting the Winter Olympics
2. Factoring in costs, is this a good investment? Find the NPV

Revenue calculation

Day	# of days	Prime Time	Broadcast time (hour)/day	Advertising Time (Minutes)	Advertising revenue / 30 sec	Advertising income
1	1	Y	3	$1 \times 3 \times 10 = 30$	400,000	24,000,000
2-3; 9-10	4	Y	10	$4 \times 10 \times 10 = 40$	400,000	320,000,000
4-8; 11-15	10	Y	4	$10 \times 4 \times 10 = 40$	400,000	320,000,000
4-8; 11-15	10	N	6	$10 \times 6 \times 10 = 600$	200,000	240,000,000
16	1	Y	3	$1 \times 3 \times 10 = 30$	400,000	24,000,000
					Total	928,000,000

#16 – Winter Olympics Bidding (VI of VII)

Quantitative questions

1. Calculate the revenue from broadcasting the Winter Olympics
2. Factoring in costs, is this a good investment? Find the NPV

Cost and profit calculation

- Direct purchase cost: 428,000,000
- Opportunity cost: Total broadcast hours * forgone per hour revenue = $146 \times 1,000,000 = 146,000,000$
- Profit = Total revenue – direct purchase cost – opportunity cost = $928,000,000 - 428,000,000 - 146,000,000 = 354,000,000$

Using the “Rule of 72,” we know that 72/rate of return means the number of years to double our money. With a six-year lag and a 12% WACC, we know that all future cash flows must be halved.

Note: Candidate may not know the Rule of 72. Provide clarification if candidate is unsure how to proceed from this point.

- $\$354/2 = \177M in present value (NPV)

#16 – Winter Olympics Bidding (VII of VII)

Solution and recommendation

- While the NPV of the project is \$177M, the fact that there are other intangibles (new viewers, plugging our programs, and prestige) the bid should not just be \$177M. While there is no one correct answer, most answers should be in the range of \$200M. If there is significant fluctuation from \$200M, the candidate will have to provide in-depth justifications and make a concrete argument.

Bonus/ Guide to an excellent case

- This case tests the interviewee's comfort with numbers and understanding of how intangible factors may influence financial value. The bid process requires another level of understanding around game theory and what dynamics will ultimately determine the value of the bid beyond NPV.
- Ultimately, the best interviewees will make a very strong argument using the facts provided and support their bid and explain why they moved their bid from the NPV figure.
- There is also a lot of room for creativity for the interviewee to discuss other factors, including supplemental streams of revenue, intangible factors, and things to consider during the bid process.

#17 – Butcher Shop

Analysis	Structure	Overall
Medium	Medium	Medium

Behavioural Prep

Tell me about a time you faced opposition in the workplace. What did you do and what was the outcome?
What is your proudest accomplishment?

Case Tracker	
Industry	Manufacturing
Type	Operations/Supply Chain
Leader	Interviewee
Concepts	Matching Supply with Demand; Market Sizing; Operation Cost

Case Stem
Your client is a fast-food chain that has recently purchased a bovine meat-processing outlet to supply it with fresh hamburgers and other meats. The shop process is: cows enter from one end of the shop, meat gets processed in the middle, and then the meat gets packaged and delivered at the other end. Enter → Processing → Distribution However, the manager of the butcher shop can't decide whether to have the cows walk or run into the meat processing room and has engaged you to advise him.

Interview Guidance
The candidate may have questions regarding the products that the processing facility produces. For the purposes of this case, assume that we are only concerned with hamburger production. The candidate may ask about the criteria on which the manager is basing his decision. You might ask the candidate to generate some ideas of their own. Ultimately the manager will consider three things: 1. Cost implications 2. Meeting demand from the outlets 3. Any other qualitative considerations (<i>don't reveal these, but some might include quality, freshness, regulatory, animal rights/impact on company brand</i>) <i>At this time, we are not assessing options for mixed speed operations, renting out additional capacity to other vendors, or selling excess inventory to other chains, or via wholesale or grocery, but these are options that the candidate might take into consideration in their risks/next steps.</i>

#17 – Butcher Shop (II of III)

Structure

Structure could include assessment of: (a) Demand, distinguishing between present demand (current customer base, purchasing patterns, and volumes) and future demand, including growth potential and evolving consumer trends; (b) Supply, including sourcing reliability, input costs, processing efficiency, and distribution capacity; (c) Financials, focusing on revenues, costs, and profitability

Shop capacity

Overall Shop Hours

(only reveal if asked: 1 cow = 20 hamburgers)

- The meat processing facility is operational 10 hours per day, 5 days per week
 $10 \times 5 = 50 \text{ hours per week}$

Walking

- If cows are processed at the walking rate, the facility can process 10 cows per hour
 $10 \times 10 \times 5 = 500 \text{ cows per week}$
or
 $500 \times 20 = 10,000 \text{ hamburgers per week}$

Running

- If cows are processed at the running rate, the facility can process 25 cows per hour
 $25 \times 10 \times 5 = 1,250 \text{ cows per week}$
or
 $1,250 \times 20 = 25,000 \text{ hamburgers per week}$

Assume the supply of cows is unlimited for the purposes of this case

Demand

Demand

Note: we have no information on future demand

Assume that there are 10 fast food outlets. All outlets are supplied by the meat processing facility, and each serves an area with a population of 30,000 people.

- In each area, there are three other competitors, and all competitors have equal market share (25%)

$30,000 \times 10 \times 25\% = 75,000 \text{ potential customers}$

(For an advanced candidate, you may now ask them to complete a hypothetical market sizing exercise to estimate demand before revealing additional segmentation information)

- Of the fast food chain's potential customers, only 40% of those individuals fall within the chain's target demographic
- Given healthy eating trends of the chain's target consumers, only one third will frequent the restaurant
- Of the chain's consumers, each individual will visit an outlet twice per week, on average, and consumer a hamburger half the time

$75,000 \times 40\% = 30,000 \text{ target customers}$

$30,000 \times 33\% = 10,000 \text{ consumers}$

$10,000 \times (2/2) = 10,000 \text{ hamburgers per week}$

#17 – Butcher Shop (III of III)

Cost		
	Cows Walk	Cows Run
Overhead	\$5,000	\$10,000
Labour	\$1,000	\$2,500
Total	\$6,000	\$12,500
Burgers/Week	10,000	25,000
Cost/Burger	\$0.60	\$0.50

- Running, although more costly in absolute terms, reduces per burger cost by \$0.10
- We do not have information on the price of a hamburger, but assume that prices are the same across all outlets, and that they are the same regardless of whether the hamburgers are produced via walking or running (meaning that \$0.10 is pure profit)

Recommendation

Recommendation:

- Have the cows walk
 - We are able to meet demand
 - This ensures fresh burgers, despite a higher per-burger cost
 - Having the cows walk circumvents issues related to animal cruelty

Risks:

- An increase in demand could result in a stock-out (no safety stock if cows walk)

Next Steps/Additional Considerations (might include)

- Forecast future demand for hamburgers to determine if walking cows will be a sustainable long-term strategy
 - Determine if mixed-speed processing is a cost-effective solution for potential demand fluctuations
- Assess opportunity to have the cows run and sell excess inventory through wholesale, grocery, or other food service businesses
- Assess opportunity to have cows run into processing until demand from the franchises is met, and then stop processing, or rent out additional capacity

#18 – Everlasting Light Bulb

Analysis	Structure	Overall
Medium	Hard	Medium

Behavioural Prep

Tell me about a time when you exhibited leadership
Tell me about a time you faced opposition in the workplace. What did you do and what was the outcome?

Case Tracker		Case Stem
Industry	Consumer Goods	Your client is a manufacturer of standard light bulbs for domestic use. Assume your client enjoys a perfect monopoly in the domestic light bulb market. Recently, a scientist invented the world’s first everlasting light bulb and has been granted a rock-solid patent for it. Your client is, not surprisingly, concerned about the impact this new innovation might pose for their business, and wants to know how much the patent is worth so they can begin negotiations with the scientist.
Type	Valuation	
Leader	Interviewee	
Concepts	Market Sizing	

Interview Guidance

Note to Interviewer: This case requires a thorough pre-read by the interviewer in order to make it work.

They often panic at the start as no real framework exists to answer this beyond the simple market sizing at the start. It is a great conceptual thinking, decision flow-chart, what-if style question that rewards candidates for confident thinking on their feet. It can be a very hard case. The best candidates remain composed while bombing certain sections. The test of this case is to show grace under fire.

#18 – Everlasting Light Bulb (II of IV)

Clarifying answers to provide

Info to be given if asked

Patent:

- Assume that the patent is for eternity, meaning that no one else can manufacture an everlasting light bulb other than the patent owner.
- You are valuing the PATENT not an individual everlasting light bulb.

Product:

- This is a domestic light bulb only so has no commercial applications such as offices or cars
- Conventional light bulbs price: \$2. Everlasting light bulbs will be priced at \$5.
- You can assume that conventional bulbs last 2 years.
- Assume a profit margin of 20% on conventional and everlasting bulbs.

Market (probe the interviewee to come up with the approach to market sizing; give this information if asked)

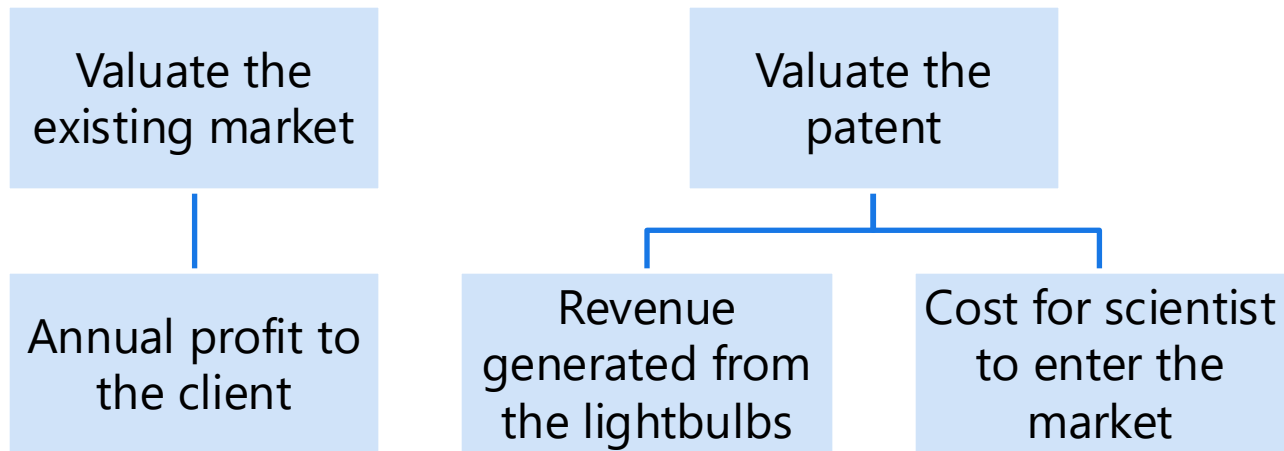
- There are 6B people in the world. You can assume that only 4B people have access to electricity.
- You can assume that there are 8 people on average per household (HH).
- Assume that there are, on average, 5 rooms per household
- Assume that there are, on average, 2 light bulbs per room
- Assume everyone switches to everlasting in year one. The phase of the transition will have little impact on the eternal patent value (imagine the transition is immediate)

Other:

- Can assume a 5% discount rate.
- The scientist has no sales or marketing infrastructure at present. To enter the market, the scientist would need to spend an additional \$500m upfront in sales, marketing, and distribution.

#18 – Everlasting Light Bulb (III of IV)

Sample structure



#18 – Everlasting Light Bulb (IV of IV)

Valuing the industry

Valuing the current industry:

2.5B domestic rooms * 2 bulbs per room = 5B bulbs every 2 years

5B bulbs * \$2 per bulb = \$10B every 2 years or \$5B every year

\$5B per year in sales * 0.20 margin = \$1B per year profits

\$1B yearly profits / 5% discount rate = \$20B value in perpetuity

Valuing the new industry:

5B bulbs * \$5 = \$25B * 0.20 margin = \$5B profit (** remember, this is ONE TIME profit – these are everlasting bulbs!)

Valuing the patent

Value of the existing industry: \$20B total

Value of the new industry: \$5B total

Capital investment required (by scientist) to achieve \$5B: \$500M

Value of the patent to the scientist: \$4.5B (\$5B - \$500M)

Value of the patent to the client: \$20B

Recommendation

Recommendation:

The client should negotiate a price of the patent close to \$4.5B, but not exceeding \$20B.

Risks:

The scientist might recognize that financial impact to the client company of commercializing the patented innovation is greater than \$4.5B and one cent, and refuse the offer.

The scientist may glean that by selling the patent, it is extremely likely that his work will be squashed and never commercialized, and so demand a higher price.

Additional Considerations:

There may be some additional incentives the client can emphasize to keep the scientists' asking price low when they enter negotiations, for ex.:

- Effort that would be required to establish a sales and distribution network

#19 – Vie Tire

Analysis

Medium

Structure

Medium

Overall

Medium

Behavioural Prep

Tell me about a time when you had to convince a difficult client to follow your recommendations
Give me an example of a time when you had to motivate others who did not report to you.

Case Tracker

Industry	Consumer Goods
Type	Valuation
Leader	Interviewee
Concepts	Market Sizing

Case Stem

Your client is a tire manufacturer based in Vietnam called VieTire. Due to high tariffs, VieTire has historically been the only player in the market and as such, they dominate the tire industry. The Vietnamese government has plans to lower the tariff. VieTire is very concerned about this change, as it will radically alter the landscape of the industry in Vietnam. They hire you to determine when they can expect the tariff reduction to impact their business as it stands, and to generate some ideas around what they should do to minimize the impact.

Interview Guidance

Clarifying information to be provided if asked:

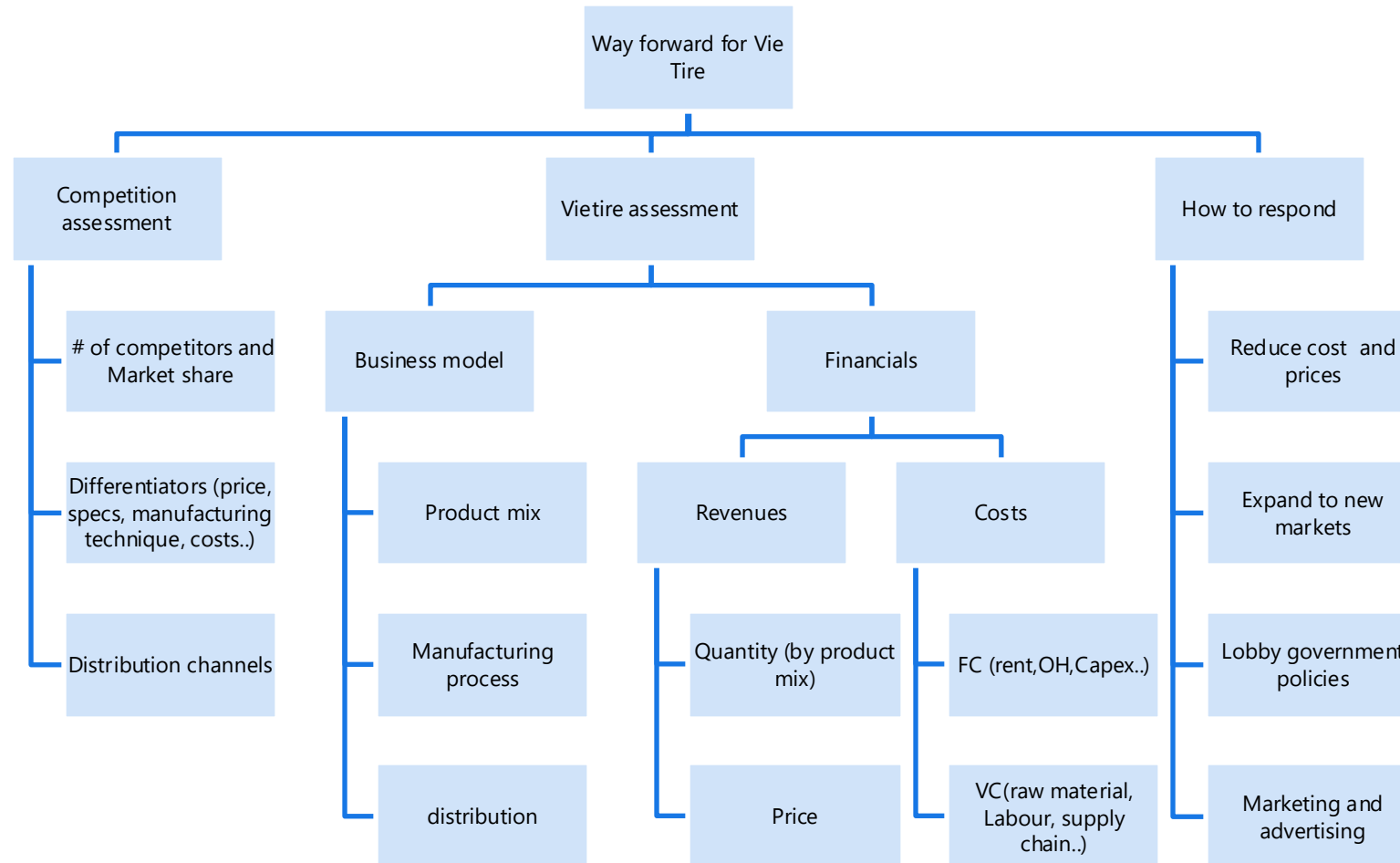
- **What is a tariff:**
 - A tariff is a tax imposed by the government of a country on imports or exports of goods. Government might impose a tariff on imports of foreign goods to protect local manufacturers
- **What is the current & proposed tariffs:**
 - Tariff applies to all foreign tire manufacturers. As it stands, the tariff is 50% of the total cost to produce a tire. Because of the forces of globalization and lower consumer prices, the Vietnamese government decided to lower the tariff by 5% a year for the next ten years (straight line, not compounded, i.e., in year 1 the tariff will be 45% and in year 2 the tariff will be 40, and so on).

A high-level plan of attack might include:

- Understanding the current cost structure of VieTire's product
- Determining the impending competitive situation
- Calculating the impact the reduction of tariff will have for VieTire
- Recommending specific steps that VieTire can take to protect themselves from increased competition

#19 – Vie Tire (II of IV)

Sample structure



#19 – Vie Tire (III of IV)

Cost information

(to provide when asked for)

Cost structure in Vietnam

- Raw material comprises about 20% of the cost, labor 40%, and all other costs such as overhead 40%. The average tire cost is about \$40 to manufacture.

It seems that labor is a major cost at \$16 per tire. Why?

- Things are done more manually. Most of the technological advances in the industry have not yet been implemented in Vietnam.

Cost structure in U.S.

- An average tire manufacturer in the U.S. produces tires at a cost of \$30 each.
- Shipping cost to Vietnam is \$4 each tire

Prompt the candidate to calculate current cost for U.S. tire manufacturer to sell in Vietnam

Solution:

Current cost of an imported tire is \$49

- \$30 manufacturing cost + \$15 tariff (50% of manufacturing cost) + \$4 shipping
- It's important to remember that the tariff is applied to the cost of the item (before shipping).

So currently, even though the cost to produce a tire in the U.S. is much cheaper due to technological advances, foreign competitors are out of luck because of the tariff.

Calculations

Prompt the candidate to calculate when will U.S. competition wins the Vietnam market

Solution:

Year	Tariff	Cost	Result
Now	50%	\$49	Competition will not enter
1	45%	\$47.5	Competition will not enter
2	40%	\$46	Competition will not enter
3	35%	\$44.5	Competition will not enter
4	30%	\$43	Competition will not enter
5	25%	\$41.5	Competition will not enter
6	20%	\$40	Competition enters market – Competition enters & splits market
7	15%	\$38.5	In competition – Competition wins market

- Candidate should recognize that every year the tariff is reduced by 5%, which is \$1.5 in cost for U.S. producers ($\$30 \times 5\%$)
- The current price difference is \$9 ($\$49 - \40), which means in year 6 the costs would be the same. In year 7, U.S. tires will be cheaper.

#19 – Vie Tire (IV of IV)

Key findings	Recommendation
- Depending on what price they are willing to set, the competition will start to think about entering the market in year four. In year seven, the competition will surely enter as their prices become lower than domestically produced tires. - The cost of imported tires only becomes cheaper than domestic production in year 7. All other forces being equal, and with no product differentiation, local producers might completely exit the market. - This analysis assumes that the cost structure for the competition will remain constant. It is important to note that because of the rapid advances in technology, chances are that the costs of producing tires will decrease resulting in competitors entering the market even sooner. - It also assumes that the production costs for local consumers remain constant. Technology advances could also drive down the cost of local production and prevent competition from entering the market. - Finally, it assumes that all tires are not differentiated.	VieTire needs to benchmark against word class tire manufacturers and reengineer production methods and cost structures. They must invest in the latest advances in order to reduce their labor/operations costs. Need to develop loyalty from their customers/consumers in order to lock in a certain percentage of the market share. (improve customer service, bundle some services..) Our analysis suggests that competition will surely enter in year six – however, this assumes that the level of technological advancement/ sophistication/ capability of our competitors remains constant. Advances in technology may expedite the market entry timeline as costs decrease – VieTire may need to act more swiftly than this analysis suggests.

Note: This case does not contain a brainstorming question; however, most cases will include a qualitative brainstorming component.

#20 – Megabank Under-Penetration

McKinsey & Company

Analysis	Structure	Overall
Medium	Medium	Medium

Behavioural Prep

Describe a time you had to change your communication style to influence different stakeholders.
Tell me about a time you pushed back on a project schedule

Case Tracker		Case Stem
Industry	Financial Services	Megabank is a bank that issues credit cards. New cards are sold in three main ways: 1. Cross-selling to existing banking customers 2. Sales to new customers via direct mail campaigns 3. Distribution via private label partnerships with retailers and airline Megabank is looking for new card member growth areas in the United States. Its Hispanic market penetration is low relative to comparable banks' penetration rates. That group is a fast-growing demographic, and the bank wants to capitalize on it. What is the current problem? How should the bank move forward?
Type	Marketing	
Leader	Interviewer	
Concepts	Marketing, market sizing, customer value drivers	

Interview Guidance

This is an interviewer-led case in which the interviewer asks the interviewee a series of questions about the client. As the interviewer you should be aware of the time being spent on each question and try to keep the candidate moving along at a pace that will allow him/her to complete as many of the questions as possible.

#20 – Megabank Under-Penetration (II of IV)

Question 1 and solution

Question 1: Let's start by thinking about what might be wrong.

Solution:

The candidate should discuss briefly the extent to which each of the following could affect penetration rate:

Product definition:

- Does the card, as it currently exists, meet the needs of the customer?

Pricing of Credit Card Terms:

- Are the fees and rates on par with other comparable cards?

Marketing / Advertising:

- Are the messages properly directed to the audience (in terms of both content and distribution)?

Channel Partners:

- Does our target audience shop at / eat at / buy from the channels the client is present in?

Internal Sales Messages / Incentive Structure:

- Are the sales messages correctly structured to entice the client's potential customers?
- Is the client's internal sales force (i.e., teller and desk personnel) trained and incented properly to promote the card? Discuss in detail what might be misaligned and how it might affect adoption rates

Question 2 and solution

Question 2: Calculate the number of additional members Megabank wants to add based on the following information:

- There are 40M Hispanic people in the US (candidate can assume no growth over relevant time period)
- 3/8ths of them are too young to have credit cards
- The average customer is worth \$180 to the bank over the course of his/her life
- Due to decreased acquisition costs, the average Hispanic customer is worth 10% more than that
- Currently the bank's penetration rate in the Hispanic market is 10% (of valid customer prospects)
- The bank wants to get to 30% penetration over 5 years

Solution:

3/8ths are too young, so 5/8ths of 40M are valid customer prospects.
That's 25M people

They currently have 10% of 25M = 2.5M

They want 30% of 25M = 7.5M

They need 5M additional members over 5 years

#20 – Megabank Under-Penetration (III of IV)

Question 3 and solution

Question 3: How much is that additional market share worth to Megabank? In other words, how much would Megabank be willing to spend on that additional market share?

Note: Stronger candidates will have already performed this analysis during the previous section

Solution:

If the average customer's lifetime value is \$180, but the average Hispanic customer is worth 10% more, each Hispanic customer is worth \$198

If you round that to \$200, the total value of the 5M extra members is \$1B

Question 4 and solution

Question 4: Megabank's cross-selling to branch customers is significantly below the industry norms (Average = 15,000 to 20,000 per month; Megabank = 5,000 per month). What might be the reason for this?

Solution:

The candidate should brainstorm potential reasons including:

Product

- Is the Megabank product different from competitors' products?
- Interviewer: NO

Pricing

- Are the fees and rates different than other comparable cards?
- Interview: NO

Customer

- Is the client targeting a fundamentally different audience?
- Interviewer: NO

Channel partners

- Are the distribution channels misaligned?
- Interviewer: NO

Marketing/Advertising

- Is there something wrong with the client's sales mechanism?
- Interviewer: Let's investigate.... (see Question 5)

#20 – Megabank Under-Penetration (IV of IV)

Question 5 and solution

Megabank uses direct mail campaigns to solicit new cards members.

- Megabank's historical response rate from Hispanic prospects is 3%
- The bank is planning to target 15M Hispanic potential customers this year, with each customer getting 3 mailings
- It expects that after the first mailing, the response rate will drop by one-third in each of the subsequent mailings
- The bank has a conversion rate of 45% of respondents

Question 5: How many Hispanic customers should the bank expect after the third mailing?

Solution:

$3\% \times 15M = 450,000$ from first mailer

$3\% - (1/3) \times 3\% = 2\% \times 15M = 300,000$ from second mailer

$2\% - (1/3) \times 2\% = 2\% \times 15M = 200,000$ from third mailer

Total new customers
= $950,000 \times 45\%$ conversion
= 427,500 customers

Recommendation

You bump into the SVP of Sales for the credit card business in the hallway, and he asks: "How does it look?"

How do you respond?

Possible answer

Based on your current market position, your goal of increasing penetration to 30%, and your historical conversion rates for direct mail campaigns, our initial estimates suggest that you will fall well short of your goal by mainly relying on that method of acquisition. In fact, it will only get you about half-way to your goal ($5 \times 428K = \sim 2.5M$) (or less than half-way given that the response rate for the same customers would likely go down over time)

We need to discuss other measures to increase penetration of the Hispanic market. Specifically, we need to look at your sales force compensation structure, training and specific sales and marketing messages. Let's plan to review our formal recommendation later in the week.

Other possible approaches

- Target a greater portion (more than the 15M planned)
- Look at other channels than direct mail

#21 – Trains in Spain

McKinsey & Company

Analysis	Structure	Overall
Easy	Easy	Medium

Behavioural Prep

1. Tell me about the best piece of feedback you ever received.
2. Why do you think you'd be a good consultant?

Case Tracker		Case Stem
Industry	Transportation	The railway company of Spain, a former monopoly, is now opening to competition. The monopoly has been divided into two separate companies: Train Espana (TE), a train operation and Rail Espana (RE), a company that manages the tracks, train stations and traffic control. A friend of yours has €90M and has approached you for advice on starting a new train operation company to compete against the new Train Espana. In particular, he wants to start by opening a new high speed train route between Madrid and Barcelona. Should your friend go ahead and launch the business?
Type	Profitability; industry analysis	
Leader	Interviewer	
Concepts	Break-even analysis	

Interview Guidance

A good candidate will try to gain a solid understanding of the industry and the market conditions before asking for the relevant financial information to complete a profitability assessment. The candidate should also determine the break-even point and offer a reasonable explanation as to whether or not the break-even timeline would be acceptable to the friend.

Clarifying information:

Objective: payback in 10 years

New operation: focus on passenger train only. Main revenue stream is ticket sales.

#21 – Trains in Spain (II of IV)

Phase 1: Industry overview

Interviewer: What kind of information do you need to assess market attractiveness?

Let the candidate lay out the framework and start brainstorming about competitors and competitive situation.

Give the information below when asked the appropriate questions. Let the candidate focus on competition, and the differences between this new venture and the competition.

Main competitors: Train Espana and air travel. Evaluate price, time and convenience.

New venture compared to TE

- Less expensive
- More efficiently run (as a former monopoly, is very inefficient)
- Equal travel time
- Equally convenient (same stations and tracks)

New venture compared to Air Travel

- Less expensive
- Run equally as efficiently
- Longer travel time
- Convenience (stations downtown while airports are outside city)

KEY INSIGHT: This information suggests there may be an opportunity to succeed in this market. Now that we understand market conditions, we should evaluate financials (profitability)

Phase 2: Business Overview

Interviewer: Now that we've identified the main issues in the market, it's time to see whether the venture can be profitable. How would you evaluate the viability of your friend's business? What do you think the main costs and revenue streams are?

Possible answers:

Costs

- Marketing
- Operations and maintenance
- Trains
- Personnel
- IT

Revenue streams

- Tickets (main revenue driver)
- Ads in trains
- Food and drinks
- Wi-Fi / entertainment on trains

****** For purpose of profitability calculations later in the case, candidate should only consider tickets, advertising, and food/drinks

Interviewer: *Once the interviewee has identified the revenue/cost sources, you can provide them with the following operational info:*

- We can run two round-trips per train per day
- Will need 3 operating trains and 1 train for backup (in case of breakdown)
- Trains operate 365 days a year from 8am to 10pm

#21 – Trains in Spain (III of IV)

Phase 3: Break-even calculations

Question 1: What information do you need to calculate expected profits? Keep in mind that the company only has 10 years to pay back the investment.

The candidate should ask specifically for all the revenue, cost, and capacity information.

Capacity: Each train has a capacity of 500 passengers

Possible Ticket Price (Per single direction trip)	Expected Train Occupancy	Costs	
€40	95%	Trains	€30M per train
€50	80%	Marketing	€15M annually per train
€60	60%	Operations/Maintenance	€10M annually per train
		Corporate Overhead	€5M annually

Question 2: Given this data, how much do you think the tickets should be priced?

Answer:

- $€40 \times 95\% \times 500 = €19,000$ per trip
 - $€50 \times 80\% \times 500 = €20,000$ per trip
 - $€60 \times 60\% \times 500 = €18,000$ per trip
- It's most profitable to sell tickets for €50 each

Question 3: Now figure out the profits per year

Answer:

- Revenue per train: $365 \text{ days} \times 2 \text{ round-trips per day} \times 2 \text{ directions} \times €20,000 \text{ per single trip} = €29.2\text{M}$ per year
- Profit per train: $€29.2\text{M} - €25\text{M} = €4.2\text{M}$ operating profit per year
- Total annual profit: $€4.2\text{M} \times 3 \text{ active trains} - €5\text{M} = €7.6\text{M}$

Key insight: This does not look great for your friend – it will take more than 10 years to recoup the €120M investment in 4 trains

#21 – Trains in Spain (IV of IV)

Recommendations

Interviewer: What would you recommend for your friend? What options could you consider to make the business profitable?

Possible Answers: He should not invest under the current circumstances, as it would take over 10 years to breakeven his investment. That said, he may also explore options such as leasing trains, additional revenue from onboard sales and advertising, a joint venture with a foreign operator that has excess capacity, etc.

The candidate should brainstorm several ideas for how to make this a profitable venture.

#22 – Money Bank Call Centre

Analysis	Structure	Overall
Medium	Medium	Medium

Behavioural Prep

What do you do for fun?
What skills can you bring to this position?

Case Tracker		Case Stem
Industry	Financial Services	Our client is a large financial services firm with multiple locations around the world. Part of their service offering includes a 24-hour help line. The client has their call centres in New York and Paris. The client has recently acquired a small firm (Firm B) in order to expand its reach in a particular geography. Firm B provides a subset of the services and has its call centre located in The Philippines. The client has asked us to evaluate its strategy going forward for handling customer calls. They want us to investigate call centre operations.
Type	Operations	
Leader	Interviewee	
Concepts	Financial analysis; organizational fit	

Interview Guidance

- This is a typical outsourcing case that requires the candidate to:
 - Use a framework that covers the most important areas of M&A and cost-cutting
 - Read the exhibits to assess the cost-effectiveness and efficiency of the 3 locations provided
 - Discuss qualitative information on the acquired company (i.e., products, culture, customers, etc.)
- There is no right or wrong recommendation, as it will depend on the interviewee's assessment of the qualitative concerns
- Key steps in the case:
 - Evaluate existing and potential cost structure
 - Explore alternatives / ideas for implementation
 - Make a recommendation based on the data

#22 – Money Bank Call Centre (II of V)

Additional information to be provided

The following information can be provided to the interviewee if asked:

Client Characteristics

- Provides full range of financial services for individuals and small organizations
- Acquired firm was started 5 years ago and is still run by its original founders

Nature of Call Centre

- New employees are college graduates with basic knowledge of financial services and products
- Fluency in English and several European languages is required

Regulatory Environment

- Very difficult to lay off employees in the Paris call centre – significant costs would be incurred

Sample structure

Case structure: The interviewee should take a couple of minutes to draft a framework that covers the majority of the important considerations in this case

Sample Framework:

Options for Client

- No changes – maintain all three call centres
- Close Firm B's call centre in The Philippines and route calls to one of the existing locations
- Consolidate to a single location

Key Factors to Consider

- Services offered each of the three call centres (product mix)
- Call centre performance, including identification of relevant performance metrics (cost per call, calls per agent, customer satisfaction, etc.)
- Employment regulations
- Shut-down costs
- Infrastructure capabilities
- Tax benefits

#22 – Money Bank Call Centre (III of V)

Determining cost-efficiency

If the interviewee doesn't immediately ask for cost data for each of the three call centres, guide them towards that branch of their framework and provide them with Exhibit 1

Key Takeaways from Exhibit 1

- Total cost incurred across all three locations is \$9.6M
- The Philippines can process 20 calls/employee/day at a cost of \$6 per call (assume 200 days per year)
- To process the *total* call volume, The Philippines will need 325 employees – leading to a new total cost (including overhead) of \$7.7M (assumption: overhead grows to 4x current cost)
= $325 \times \$20K + \$1.2M$ overhead
- The interviewee should realize that the \$7.7M does not account for all costs. When prompted, tell the candidate that there's a one-time cost of \$5M to cover severance of NY & Paris, and the expansion of The Philippines facility)
- CONCLUSION: Moving all calls to The Philippines not profitable

	Calls/Employee	Cost/Call
New York	40	\$7
Paris	33.3	\$9
The Philippines	20	\$6

Notes to interviewer

- Prompt the interviewee to explore ways to make outsourcing to The Philippines more feasible
- Inform the interviewee that the CEO requires any solution she pursues to be, at a minimum, cost-neutral by Year 2
- The interviewee needs to identify the difference in calls/employee between The Philippines and New York locations
 - Ask the interviewee to assume that the best practices can be transferred from NY to The Philippines and be implemented within 3-6 months
- Potential cost savings if The Philippines achieves the same call efficiency as New York
 - # employees required
= $(1.3M \text{ calls}) / (40 \text{ calls/employee}) \times (200 \text{ days/year}) = 163$
 - OR, doubling the number of calls/employee reduces the number of employees needed by half ($325/2 = 162.5$)
 - New employee cost = \$3.25M
- Other questions to consider:
 - Does the Philippines have sufficient space/capacity to handle expansion?
 - Can client find sufficient talent within short timeframe?
 - Customer satisfaction in new location?
 - Legal constraints in NY & Paris: how quickly can staff be laid off?

#22 – Money Bank Call Centre (IV of V)

Exhibit 1

	Calls/Year	Cost/Employee/Year	Overhead Cost/Year	# Employees
New York	600,000	\$50,000	\$450,000	75
Paris	400,000	\$50,000	\$600,000	60
The Philippines	300,000	\$20,000	\$300,000	75

Note: On average, an employee works 200 days/year

#22 – Money Bank Call Centre (V of V)

Recommendations

This is an open-ended case. The interviewee needs to justify their recommendations based on the qualitative and quantitative considerations

A good recommendation would include three sections:

1. **Recommendation:** If the recommendation is to outsource, the interviewee needs to highlight the risks associated with outsourcing and nature of the acquired firm. If the recommendation uses any other approach, sufficient justification needs to be given to overcome the cost savings
2. **Risks:**
 - Risks associated with increasing capacity ~100% (people, infrastructure, service quality, gaps in knowledge transfer, organizational changes, etc.
 - Reputational impact: Do customers notice difference in service? Can the competitor leverage this to steal customers?
3. **Next Steps:** If outsourcing, some of the next steps would be analyzing the infrastructure requirements and capabilities, finding the right talent, ensuring smooth transfer and implementation of best practices, etc.

Bonus/ Guide to an excellent case

Excellent interviewees need to address the qualitative information provided in the case (i.e. nature of merger, nature of markets being served, etc.)

The interviewee should explore the option of improving effectiveness of The Philippines location without being prompted to do so

#23 – Upscale Restaurant

McKinsey & Co.

Analysis	Structure	Overall
Hard	Medium	Hard

Behavioural Prep

Tell me about an important role model in your life.
What is an important issue facing your previous industry?

Case Tracker		Case Stem
Industry	Travel & Leisure	Our Client is an upscale restaurant in Tianjin, China, serving government officials and high-level business customers. Its monthly revenue is 1.2 Million Yuan. The CEO recently hired McKinsey to figure out why revenues are lower than expected, and to help them increase profits.
Type	Profitability Improvement	
Leader	Interviewer	
Concepts	Incremental profit	

Interview Guidance
• This is an interviewer-led case that is quite quant heavy and tests a candidates business acumen. • During opening structure, once candidate mentions revenue streams and capacity issues, interviewer can provide Exhibit 1 • After candidate has derived the key insights from Exhibit 1, interviewer should push candidate to brainstorm potential solutions • Interviewer will proceed with follow-up questions pertaining to two potential avenues – raising prices and turning big room tables into individual rooms

#23 – Upscale Restaurant (II of V)

Information provided upon request

- As China's economy is booming, the upscale dining market is growing at 20% every year.
- Customers for high-end dining are generally price insensitive.
- All competitors are earning money. Competitors' price and value proposition are similar.
- Variable costs across industry are 50% of revenue.
- Assume no fixed costs

Sample Framework

Current Assessment: Why are revenues declining?

- Revenue Streams:
 - Spend per customer (are customers buying cheaper items, not buying alcohol)?
 - Capacity (are we at capacity? Are we declining capacity)
- Market
 - Competition: Have there been any new restaurants opening in the area, taking our customers? *(if didn't ask about competition in clarifying questions)*
 - Are we in an economic downturn, causing customers to want to save, eat at home? *(if didn't ask about customer price sensitivity in clarifying questions)*

How can we improve Profits?

Increase Revenue

- Expanding hours
- Is there a way to increase capacity? (e.g., adding more tables)
- Marketing, promotions (e.g., deals)

Decrease costs:

- Raw ingredients (reneg supplier contracts), labour

#23 – Upscale Restaurant (III of V)

Exhibit 1

Individual Rooms (20 total)			Main Room (20 tables)		
	Week Day	Weekend		Week Day	Weekend
Lunch	Occupancy: 80% Price per person: 150 Party size per table: 4	Occupancy: 30% Price per person: 100 Party size per table: 4	Lunch	Occupancy: 20% Price per person: 100 Party size per table: 4	Occupancy: 30% Price per person: 100 Party size per table: 4
Dinner	Occupancy: 100% Price per person: 300 Party size per table: 6	Occupancy: 50% Price per person: 200 Party size per table: 6	Dinner	Occupancy: 30% Price per person: 200 Party size per table: 4	Occupancy: 30% Price per person: 200 Party size per table: 4

NOTE: During weekday dinners, there is always a line for individual rooms.
As a result, the restaurant has to turn away half its potential customers due to capacity constraints

#23 – Upscale Restaurant (IV of V)

Expected initial insight

Revenues are low because currently our client’s offering is not meeting customer demand. Government officials and business customers prefer individual rooms to big rooms because of their requirement for privacy.

Follow-up Questions

What are potential solutions for this situation?

Candidate should identify two main options:

- Raising price.
- Turning big room tables into individual rooms.

Tell candidate you would like to explore those options separately.

Price: Through market research, we have determined that if we raise weekday individual room price by 33%, we will lose 10% of customers. How will it change our profitability on a given weekday?

Solution #1

For weekday lunch, changing the price will result in 10% customer loss.

	Previous	Now
Customers	$4 \times 20 \times 80\% = 64$	$64 \times (1 - 10\%) = \sim 58$
Price	150	$150 \times (1 + 33\%) = 200$
Revenue	$64 \times 150 = 9600$	$58 \times 200 = 11600$
Profit	$9600 \times 50\% = 4800$	$11600 \times 50\% = 5800$
Incremental Profit		$5800 - 4800 = 1000$

For weekday dinner, underlying demand is 200% of capacity, so raising price WON'T reduce volume.

	Previous	Now
Customer	$6 \times 20 = 120$	120
Price	300	$300 \times (1 + 33\%) = 400$
Revenue	$120 \times 300 = 36K$	$120 \times 400 = 48K$
Profit	$36K \times 50\% = 18K$	$48K \times 50\% = 24K$
Incremental Profit		$24K - 18K = 6K$

Incremental Profit Per Weekday: $1K + 6K = 7K$

#23 – Upscale Restaurant (V of V)

Follow-up Question #2

The second option is converting half of big room tables into 5 individual rooms. It will take 2 weeks for the restaurant to finish the decoration, during which the time the restaurant has to be completely shut down. The decoration will cost 100K Yuan. What is the total cost of this project? How much incremental profit will it add per weekday?

Cost

- Capital investment: 100K
- Opportunity Cost: ~300K (2 weeks of profits, based on given revenues of 1.2M Yuan/month – no need for candidate to do detailed calculations)
- Total cost = 400K Yuan

Incremental profit per weekday:

- The candidate should realize that all current main room demand can be served even with just 10 tables, so there is no change to profits. There is also no change in individual lunch profits, because capacity is not at 100%. Adding 5 individual rooms will allow the restaurant to capture 30 new customers at \$300 each, for revenue of \$9,000 and profit of \$4,500.

Which choice would you recommend?

Sample Recommendation:

Raising price offers greater incremental profit without any cost/opportunity cost, so it is preferred to renovating the restaurant. However – there may be risks associated with this!

This will result in an incremental profit per weekday of \$7K, compared to \$4500 for the converting big room tables to individual rooms.

Risks:

Customers may be more price-sensitive than we think, possibly resulting in greater customer loss.

Backlash from government officials on the new higher prices could lead to unfavorable political treatment, e.g., licensing (remember, the restaurant is in China)

Next Steps:

Soft-launch: increase prices in two waves to cause less potential backlash from customers, reducing possible customer loss

Explore opinions of government officials

#24 – Jamaican Land Investment

Analysis	Structure	Overall
Hard	Medium	Hard

Behavioural Prep

Describe a recent unpopular decision you made. What was the result?
What is your proudest accomplishment?

Case Tracker	
Industry	Agriculture and Real Estate
Type	Opportunity Assessment
Leader	Interviewee
Concepts	Contribution Margins, Capacity & Demand

Case Stem
Your friend is thinking about buying a piece of land in Jamaica for \$3,000 and has asked you to determine whether it's a good idea.

Interview Guidance

This case offers great practice in organizing analysis. Consider pointing out to the candidate that organization of information in tables is very important for this case, to ensure that they get the most out of this exercise. Candidates will also benefit from knowing that calculations are on a per unit basis, and contribution margins ($CM = P - VC$) are sometimes critical. In addition, this case tests an understanding of capacity and demand constraints ($Volume = \min\{Demand, Capacity\}$).

Don't just give away that the use case is agricultural if the candidate asks for it. Ask the candidate what their friend could do with the land. A great candidate will lay out a structured (MECE) answer grouping potential use cases in some way (e.g., use, renting out, sell OR commercial, residential, industrial, agricultural).

#24 – Jamaican Land Investment (II of VII)

Information provided upon request

Answers to potential clarifying questions:

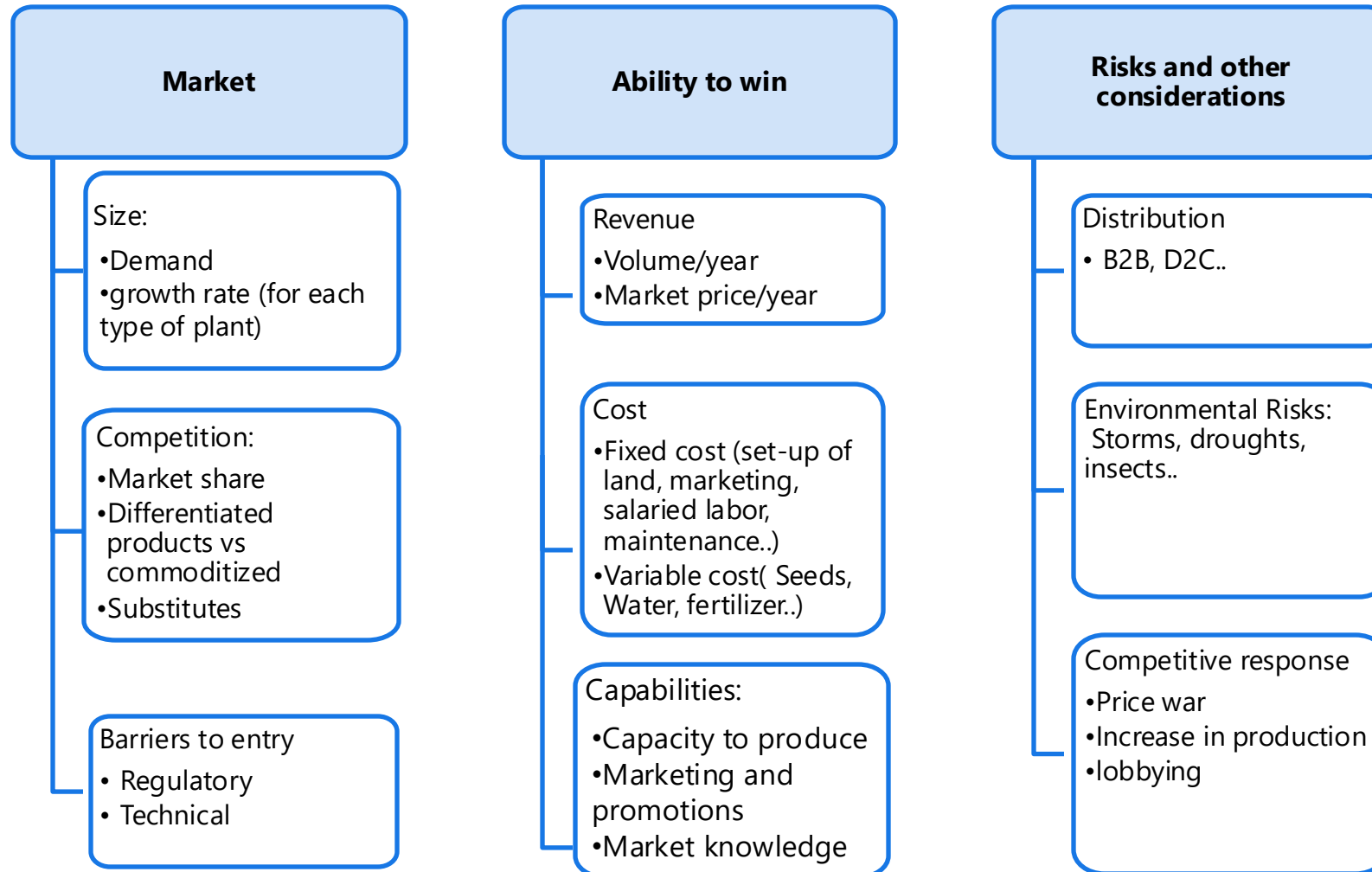
- Purchase price of the land: \$3,000
- Total acreage: 10 acres
- Financial target: \$4,500 profit in the first two years combined, excluding the purchase price (\$1,500 after purchase price considered)
- When prompted about the use of land, ask the interviewee to brainstorm possibilities before giving the answer: real estate development, agriculture, hold and sell once it appreciates, and other options you can think of
- The land will be used for agriculture to plant trees, shrubs, fruit, and flowers
- Cannot mix products (e.g., trees and shrubs) on the same acre; only one type of plant allowed per acre

Information to be provided during the case:

- Price per plant: trees \$50, shrubs \$35, fruits \$15, flowers \$25
- Variable cost per plant: trees \$30, shrubs \$25, fruits \$11, flowers \$17
- Fixed cost: \$500 initial set-up (first year only), \$350 per year for salaried labour
- Market demand per year: 5,000 trees, 1,000 shrubs, 1,000 fruit, 2,500 flowers
- Penetration rate: competitors cannot meet demand, assume that your friend can capture all the leftover demand
 - Competitor share of demand: trees 60%, shrubs 20%, fruits 85%, flowers 90%
- Number of plants per acre (before giving the answer, ask the interviewee what they think about which plants would have the least or most per acre)
 - 10 trees per acre, 25 shrubs per acre, 75 fruits per acre, 50 flowers per acre

#24 – Jamaican Land Investment (III of VII)

Sample Framework



#24 – Jamaican Land Investment (IV of VII)

Sample Approach

Clarify the problem

Interviewee should always spend time clarifying the objective(s) upfront. Find out what the land will be used for, how many acres, financial target, etc., in order to produce a more precise structure and to avoid stumbling later in the case.

Determine demand and assign plants to number of acres based on profitability per acre

Interviewee should determine total demand of each product and the friend’s potential market share. After finding estimated demand, interviewee should look to assign each type of plant to a specific number of acres. This involves figuring out the margins per unit, the profitability per acre of each plant (most profitable plants get first priority in acre assignments), and how many plants can fit onto an acre.

Determine if investment would meet the friend’s financial target

Once interviewee assigns plants to a specific number of acres, they should proactively calculate the total profitability for the first two years. Interviewee should take into account the fixed costs for each year. Running the numbers, they’ll find that the investment will exceed the client’s financial target of \$4,500.

Use tables to organize information in a clear and easy-to-calculate manner

This particular case tests how organized an interviewee will be as they go through a lot of quantitative information across revenues, costs, demand, etc., for four different products at the same time. The interviewee should come up with a table that is similar to the one provided on the following page.

#24 – Jamaican Land Investment (V of VII)

Sample Analysis									
Product	(1) Price per unit (\$)	(2) VC per unit (\$)	(3) Margin per unit (\$) [(1) – (2)]	(4) Units per acre	(5) Profit per acre (\$) [(3) x (4)]	(6) Market size	(7) Market share	(8) Client demand [(6) x (7)]	(9) Acres to be dedicated to product
Tree	50	30	20	10	200 (rank 4)	5,000	40%	2,000	-
Shrub	35	25	10	25	250 (rank 3)	1,000	80%	800	3
Fruit	15	11	4	75	300 (rank 2)	1,000	15%	150	2
Flower	25	17	8	50	400 (rank 1)	2,500	10%	250	5

#24 – Jamaican Land Investment (VI of VII)

Profitability analysis

If the table is done correctly, the interviewee should see that they should rank the products by profitability and assign acres based on demand. For example, while flowers are the most profitable, only 5 acres should be used to meet the demand. Use this same logic moving down the profitability ranking.

Once this has been completed, the final quantitative task is to calculate the estimated profits from the first two years.

Total Revenue

5 acres of flowers at \$400 per acre = \$2,000

2 acres of fruits at \$300 per acre = \$600

3 acres of shrubs at \$250 per acre = \$750

Total revenue = \$3,350 (\$6,700 over two years)

Total Costs

One-time set-up cost = \$500

Labour cost over two years = \$350 x 2 = \$700

Total costs = \$1,200 (remember to exclude purchase price!)

Total Profit = \$6,700 - \$1,200 = \$5,500

#24 – Jamaican Land Investment (VII of VII)

Recommendation

Recommendation

The interviewee should recommend that their friend invest in the land. A sample recommendation may be: "My recommendation to my friend is to invest in the property. My calculations show that we would achieve \$5,500 profit in the first two years, which exceeds their target by \$1,000."

Risks

These can include the growth rate and expected demand of the products, as they may change over time; we assume in this case that these are constant. Also, consider the nature of the case: it is an agricultural development; therefore, risks may include plant disease and drought, and your friend is investing in Jamaica, which experiences hurricanes that can destroy property.

Next Steps

Considering how attractive this investment is financially, the next step would be buying the land, etc.!

#25 – Grape Control Systems

Bain & Co.

Analysis	Structure	Overall
Hard	Easy	Hard

Behavioural Prep

Tell me about a time you had a conflict with a teammate.
Describe the last time you disagreed with a supervisor

Case Tracker		Case Stem
Industry	Industrial Goods	Grape Control Systems (GCS) produces, installs, and services air temperature control systems. The CEO is concerned about the company’s profitability. GCS has, therefore, asked you to help them understand what’s going on and how to increase profitability by about 2 percent points.
Type	Profitability Improvement	
Leader	Interviewee	
Concepts	Product Mix, Customer Value Drivers	
Interview Guidance		
The case is in its core a straight-forward profitability case. What makes it harder is that to really understand why profits are lower than for competitors and how the product mix can be improved, it is critical for the candidate to drill a little bit deeper in the client’s product/service offering and into customer’s preferences. Since this is an advanced case, try not to provided too much guidance for the candidate to uncover these specific insights but let them know that they are missing a piece of the puzzle if they are not there yet.		

#25 – Grape Control Systems (II of V)

To be provided for clarification upon request

- The air temperature control systems consist of two primary components: digital readers and remotely controlled flaps that are placed in air ducts
- GCS bundles their air temperature control systems with installation
- GCS sells service plans (e.g., warranties, maintenance, and repair) separately from the monitoring system
 - There are break-fix services (when product breaks, customer calls GCS, which sends an employee to fix the ATC system on-demand) and long-term care services (preventive service, usually once per year → ATC systems never break)
- So, the two revenue streams for GCS are ATC systems including installation and service plans

#25 – Grape Control Systems (III of V)

Solution guide – Competition

The candidate should provide an approach that includes at least the following: competition, customers, company (product & services capabilities), profitability (incl. product mix)

Competition

Information provided if asked:

- There is little differentiation among manufacturers of air temperature control systems. The market is evenly split among several national players
- The competition tends to be more profitable than GCS
- GCS's air temperature control systems can be serviced by GCS, by customers (e.g., in-house servicing by the client's personnel), or by a third party
- GCS cannot service a competitor's air temperature control system; the opposite is also true

Solution guide – Profitability

- At this point, the candidate should try to find out why the competition tends to be more profitable, despite there not being a great deal of differentiation among air temperature monitoring systems. The candidate should think about:
 - Mix of manufacturing/installing (sales) vs. servicing
 - Mix of customers
 - Relative profitability or costs for each
- Product mix:
 - GCS mix: 70% product, 30% service
 - Competition mix: 60% product, 40% service
- Candidate should recognize that the product mix MAY be a driver for lower profitability, but there's no way of knowing without more information; a superior candidate will hypothesize that services probably have higher profit margins
- Profit margins
 - GCS product margin: 4%
 - GCS service margin: 15%
 - Competition product margin: 5%
 - Competition service margin: 15%

[see next slide for more profitability analysis]

#25 – Grape Control Systems (IV of V)

Solution guide – Profitability (continued)

Now that the candidate has this information, ask them to calculate the overall profitability of GCS relative to the competition assuming that the margin percentages reflect the full costs of their businesses (an excellent candidate would do this proactively to quantify if GCS would reach its 2% profit margin improvement goal by replicating competitors' mix and product margins)

- GCS profit margin: $4\% \times 70\% + 15\% \times 30\% = 7.3\%$
- Competition: $5\% \times 60\% + 15\% \times 40\% = 9\%$

Key insights:

1. GCS's profit margins are lower than competitors' because of it has unfavourable mix between product sales and services and because profit margin of its ATC systems is lower
2. By replicating competitors' mix and margins GCS could reach its profitability targets

Solution Guide – 3C deep dive

Once the candidate identifies service as the area of focus, they should try to understand the services offered, the company's capabilities and customers' preferences a bit better:

Services:

- Break/Fix: customers just wait until their air temperature control systems break before calling GCS
- Long-term service contracts: customers have a contract with GCS for annual preventative maintenance

GCS / company:

- GCS has traditionally sold more break/fix service contracts
- Therefore, sales team is more versed selling break/fix service contracts and mainly targets customers that may require break/fix service contracts

Customers:

- The interviewee should find out by themselves that some customers prefer one contract over the other and why (try to push them a little before providing this)
- While a breakdown of the air temperature control is acceptable (low cost associated with breakdown) for some customers (e.g., in an office building) for a short period of time, it is unacceptable (high cost associated with down-time) for some customers in hospitals especially in areas where a constant temperature must be maintained

#25 – Grape Control Systems (V of V)

Solution guide – Customers (continued)

- Once the candidate gets to this point, let them know that majority of GCS service business is Break/Fix. Ask candidate why this might be the case.
 - The ideal answer will reference the fact that customers will be drawn to service contracts depending on their business model. Businesses for who continuous temperature control is critical will likely gravitate toward long-term service contracts that prevent downtime. Businesses willing to endure temporary outages will likely opt for break/fix agreements. GCS does a much better job of selling service to customers with a high-risk tolerance (such as office buildings) as opposed to those that have a low risk tolerance (such as hospitals and pharmaceutical companies) due to the tremendous cost that the temperatures variations could present for their business

Key insight:

- GCS is not currently targeting effectively a sub-segment (long-term contract type clients that are sensitive to downtimes) of the market

Solution Guide – Potential solutions

- Ask the candidate to think about how GCS could address this other group of customers that is more averse to risk (downtime)
- Answers can include:
 - Retrain sales personnel to help push service contracts (educate them on value proposition)
 - Retrain service personnel to ensure they know how to service new group of customers
 - Reallocate sales personnel and/or hire additional sales employees to have more staff targeting long-term service contracts
 - Hire or build relationship with thought leaders in the industry
 - Advertising

Recommendation

Profitability can be increased by 2% by improving the product mix towards services and by increasing margins for its ATC systems, while selling more services is the bigger lever.

Reasons: 2% increase is reasonable as competition has 2% higher margins; increasing services contribution to revenues to 40% will increase overall profit margins by 1.5% (percentage points); sales team currently focused on break/fix-type customers leading to a low share for GCS with long-term service contract type customers.

#26 – New Rubber Plant Investment

Boston Consulting Group

Analysis	Structure	Overall
Hard	Hard	Hard

Behavioural Prep

Tell me about a time you had a conflict with a teammate.
Describe the last time you disagreed with a supervisor

Case Tracker		Case Stem
Industry	Industrial Goods	The federal government of a country in a certain part of the world is investigating whether to restart a rubber factory. The factory was operational in the past but has not been used for 7 years. The plant was closed due to terrorism in the area which has now come down significantly though there are still issues and skirmishes reported in the area. All the equipment is considered usable but the government still estimates to spend \$12M to rejuvenate the plant which would enable the plant to produce up to 10M lbs of rubber per month. The demand of rubber worldwide is strong but rubber must be exported via trains. Due to current high traffic on the route, only up to 2 trains per day can be used for this.
Type	Value Chain	
Leader	Interviewee	
Concepts	Operations Process Analysis, Bottleneck	
Interview Guidance		
Since this is an advanced case, try to provide as little guidance as possible as the interviewer. Push the candidate to include non-profitability elements into their structure if they don't naturally do so. It may prove beneficial to use the discussion around risk as a brainstorming question to be conducted either before the recommendation was given or after. The candidate should be able to provide at least 5 risks and ideally put them into a MECE structure (this is not straightforward and will require some time).		

#26 – New Rubber Plant Investment (II of VII)

Information provided upon request

Clarifying Questions:

How many suppliers are there?

We have identified one supplier.

Who are our customers?

We would be selling the rubber in the commodity market to the entire world.

What is the demand of rubber in the market?

The market demand is strong. We can assume that whatever the plant produces can be sold.

What is the payback period / timeline they are considering?

The government expects a payback period of 1 year due to high risks.

What are the raw materials?

Gum resin, 3 lb. of gum resin needed to produce 1 lb. of rubber.

Where are the resins coming from?

They need to be transported from the capital. Up to 6 trains can be used for to transport them **(this is a key question, a candidate not asking the question misses out a key element of the case).**

Information to be provided upon request during the case:

What price can rubber be sold at?

Sell the rubber at \$20 per lb., gum resin costs \$5 per lb.

#26 – New Rubber Plant Investment (III of VII)

Sample Framework

This is an operations case mixed with cost-benefit analysis. The analysis may include, but is not limited to, the following areas:

Basic Structure:

Analyze the financial benefits of the investment:

- Analyze ROI [revenues vs costs (upfront investment, recurring costs such as transportation, materials, etc.)]
- Other non-financial considerations:
 - Risks: threat of terrorism, timely delivery, labour shortage, etc.
 - Non-financial benefits: economic development, employment

Advanced Structure:

- Analyze the **financial benefits** of the investment:
 - Analyze ROI [revenues vs costs (upfront investment, recurring costs such as transportation, materials, etc.)]
- Examine **value chain** for the rubber plant and identify the bottleneck:
 - Raw Materials → Manufacturing → Distribution
 - Analyze production capacity of plant. Given the equipment capacity is 10M lb. per month, production is probably limited by supply and distribution
 - After drawing the value chain, the candidate should clearly identify that there is a transportation element
- Identify the **other benefits** associated with this investment considering this is a government investment:
 - Employment
 - Economic development
- Identification of **risks**
 - Assess risks in the investment (timely delivery, terrorism, labour shortage, etc.)

*After the candidate lays out the plan, ask the candidate to calculate the **payback period**, if not already identified.*

#26 – New Rubber Plant Investment (IV of VII)

First Step: Identify Bottleneck

A common mistake is to assume 10M lb per month as production level (10M lb per month is the maximum production capacity, but not necessarily the plant's production level)

Ask the candidate to calculate on monthly basis. Assume 25 days in a month. Show exhibit #1 upon request for train information

Outgoing Train Calculation

- Daily capacity: $2 \times 8 \times 25 \times 500 = 200\text{K lb}$
- Monthly capacity: $200\text{K} \times 25 = 5\text{M lb}$

Incoming Train Calculation

- Daily capacity: $6 \times 10 \times 25 \times 640 = 960\text{K lb}$
- Monthly capacity: $960\text{K} \times 25 = 24\text{M lb}$
- Conversion to rubber: $24\text{M} / 3 = 8\text{M lb}$ (resin:rubber ratio)

Expected key insights

- Bottleneck is the **outgoing** train capacity; production level is limited by distribution capacity
- **Therefore, monthly rubber production is 5M lb**
- Quantity of resin needed is 15M lb ($5\text{M lb} \times 3$)

Second step: Identify revenue and cost

Information to be provided upon request

- Labour: \$8M per month
- Fixed overhead costs: \$10M per month
- Cost per train round trip: \$40K (both inbound and outbound)
- Sell the rubber at \$20 per lb.
- Gum resin costs \$5 per lb.

Expected calculation

- Transportation cost per month: $\$40\text{K} \times 6 \text{ trains} \times 25 \text{ days} = \6M per month
 - **Important information:** we need all 2 outbound trains, but only 4 inbound trains per day to transport 15M lb of resin ($2+4 = 6$ trains / day)
- Revenue: $5\text{M lb} \times \$20 / \text{lb} = \100M
- Material costs: $15\text{M} \times \$15 = \75M

#26 – New Rubber Plant Investment (V of VII)

Third step: Identify payback period

Rough income statement

• Revenue:	\$100M
• Material costs:	(\$75M)
• Labour costs:	(\$8M)
• Transportation cost:	(\$6M)
• Fixed overhead costs:	(\$10M)
• Profit:	\$1M per month
	\$12M per year

Expected key insight

The candidate is expected to state that the payback period is one year (\$12M investment, \$12M profit in year 1). However the profit margin is very thin at 1%

A great candidate will note that small fluctuations in prices and cost and incorrect assumptions can make the project unprofitable which is a significant risk and should prompt further analysis into input price and market price analyses, hedging opportunities.

Non-Financial Considerations

A strong candidate will immediately offer to brainstorm non-financial considerations (both benefits and risks). If they don't, interviewer should prompt them.

If the candidate asks any of the questions below, provide answers given.

Q: We are limited by our outgoing capacity. Do we have market demand for 5M lbs. of rubber?

A: Yes, we have demand for a lot more.

Q: Can we increase the number of trains incoming or outgoing? We have additional production capacity.

A: Not right now, but a good thing to explore in the future.

Expected key insights

The candidate should identify non-financial benefits:

- Increasing employment in the area (\$8M per month in labour costs indicates a labour intensive process)
- Increasing economic development in the area

[see following page for risks]

#26 – New Rubber Plant Investment (VI of VII)

Non-Financial Considerations: Risks and Mitigation

- Low margin, exposure to price fluctuations of resin and rubber
 - Mitigation: forward contract purchases of resin
 - Forward contract sale of rubber
- Labour risks
 - Introduce automation
 - Facilitate migration of labour from other areas
- Terrorism risk
 - Involve community leaders in the process
 - Look at getting government or private security for the plant
- Supply chain risk - sole dependency on trains
 - Maintain enough safety stock of resins and rubber
 - Ensure the rail tracks are protected sufficiently in sensitive areas
 - Crucial to diversify into other modes of transportation, invest in building roads and if applicable pursue waterways
- Single supplier
 - Diversify supplier base

Recommendation

- **Recommendation:** go ahead with investment in the plant; it is profitable
- **Rationale/Analytical Support:** With production at 5M lbs. of rubber, we make a profit of \$12M per year. Production is limited by transportation, an area that can be looked at and addressed. This should further increase our profits in the future.
- **Risks and considerations:** We have highlighted a lot of the risks (candidate can summarize their brainstorming), key is that the government takes steps to mitigate the risks, the government can take some steps based on our analysis of the mitigation. Some of the steps could mean a long-term investment.
 - E.g., terrorism risk, supply chain risk, single supplier
- **Next steps:** Assess how plant can be staffed, whether transportation bottleneck can be alleviated, the level of terrorism threat and steps to mitigate.
 - E.g., should explore finding security for the plant, diversify transportation modes, diversify supplier base

#26 – New Rubber Plant Investment (VII of VII)

Exhibit – Train Information

	Outgoing	Incoming
# of trains available per day	2	6
Bogies/train	8	10
Containers/bogie	25	25
Lb/container	500	640

#27 – Wealth Management

Analysis	Structure	Overall
Hard	Hard	Hard

Behavioural Prep

What is your leadership style?
Tell me about a time when you struggled to contribute.

Case Tracker		Case Stem
Industry	Financial Services	Your client is a Canadian Bank. Within their retail wealth management business. They are experiencing higher than industry average turnover, specifically among their Financial Advisors. They have brought you in to understand the impact this is having on their business and to come up with a way to reduce this turnover moving forward.
Type	Human Capital	
Leader	Interviewee	
Concepts	Segmentation	

Interview Guidance

This case is about a Canadian Bank that is experiencing higher than average industry turnover, and the candidate will have to analyze the financial impact this is having on the firm.

The candidate will have to consider segmentation of advisor groups (segmented by years of tenure) in order to drive the case forward to deliver a solution.

#27 – Wealth Management (II of IX)

Clarifying answers to provide if asked

- 'Impact' here is referring to financial impact

About Retail Wealth Management

- Financial advisors invest money on behalf of individuals (such as you or I wanting to invest) and not companies

About Financial Advisors

- Entrepreneurial set up, advisors are responsible for finding their own clients. Their compensation is 100% commission based
- Are motivated by financial incentives

Competition

- Advisors in the 0 – 10 year tenure are leaving the industry – this is common across the competition
- Advisors in the 20 – 25 year tenure are leaving to the competition (think another big bank) – we have more advisors in this tenure than our competitors

Give above information only after showing exhibit 1

- Advisors receive 'recruiting bonus' when they join another firm
 - 2% on assets for all clients they manage to bring over
 - We do not pay this but just offering this is not a solution – we want to prevent them from leaving!
- Competition's pay scale is the same

Note: There are 3 exhibits. Provide only when asked for.

#27 – Wealth Management (III of IX)

Sample structures

This is a non-traditional case so candidates can be creative with how they structure the case. Below are two examples:

Basic structure:

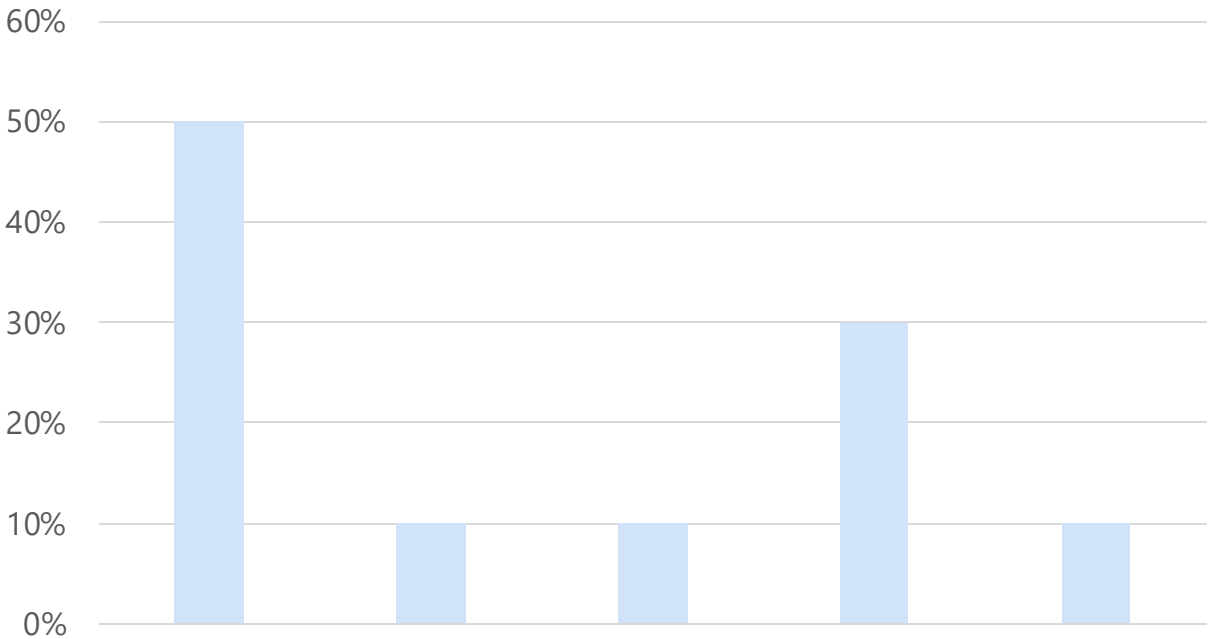
- Understanding financial impact
 - Financial advisor turnover
 - Number of advisors
 - Turnover rate & trend
 - Revenue/profit impact
 - Commission
- Reducing turnover
 - Industry
 - Competitors
 - Compensation
 - Financial incentives
 - Commission

Advanced structure:

- Understanding financial impact
 - Financial advisor turnover
 - Number of advisors
 - Grouping by tenure or by total asset managed
 - Turnover rate & trend
 - Revenue/profit impact
 - Commission
- Reducing turnover
 - Industry
 - Competitors
 - Compensation
 - Financial incentives
 - Base salary
 - Commission
 - Bonus and other benefits (retirement plans, insurance, etc)
 - Other incentives
 - Career progression
 - WFH
- Other considerations
 - Cultural impact of high turnover
 - Losing talents to competitors

#27 – Wealth Management (IV of IX)

Exhibit 1: Advisor turnover by tenure



Tenure	0 – 10 yrs.	10 – 15 yrs.	15 – 20 yrs.	20 – 25 yrs.	25+ yrs.
Avg. Advisor Age	35	40	45	50	65
No. of Advisors	600	300	300	350	50

#27 – Wealth Management (V of IX)

Exhibit 2: Business size by tenure

Tenure	Assets Under Management (per average advisor)	Return on Assets (RoA) (per average advisor)
0 – 10 years	10 million	1.5%
10 – 15 years	40 million	1.25%
15 – 20 years	60 million	1%
20 – 25 years	100 million	0.80%
25+ years	80 million	0.80%

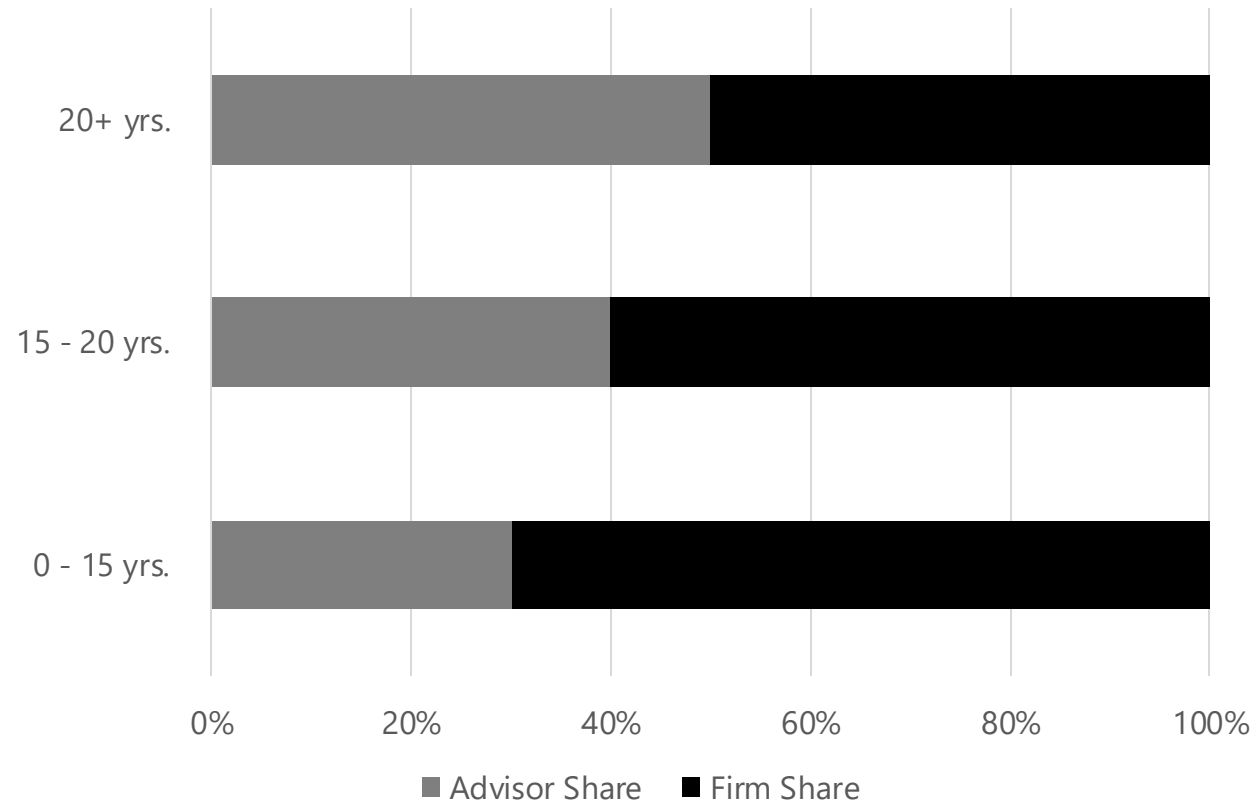
#27 – Wealth Management (VI of IX)

Note for Exhibit 2 (for INTERVIEWER ONLY)

RoA is an industry term, if the candidate does not ask for clarification ensure that they understand that this is the fee that is charged to the client based on the assets the advisor manages (from here the fee gets split between advisor payout and firm share)

#27 – Wealth Management (VII of IX)

Exhibit 3: revenue sharing model



#27 – Wealth Management (VIII of IX)

Calculating financial impact

Candidate should notice from Exhibit 1 that there are two main areas of concern 0 – 10 years and 20 – 25 years. In terms of the financial impact, they should just look at these areas (note: if the candidate is practicing quants you can have them calculate for all tenures but give them extra time).

- When asked how many clients leave with the advisor – tell them to assume 100% leave in all tenures.

Financial Impact Tenure 0 – 10 years

of advisors who leave = 300 ($600 * 50\%$)

Revenue loss per advisor = 150k ($10M * 1.5\%$)

Total Revenue Loss = 45M ($300 * 150k$)

Firm Share = 31.5M ($45M * 70\%$)

Financial Impact Tenure 20 – 25 years

of advisors who leave = 105 ($350 * 30\%$)

Revenue loss per advisor = 800k ($100M * 0.8\%$)

Total Revenue Loss = 84M ($105 * 800k$)

Firm Share = 42M ($84M * 50\%$)

Total Financial Impact

= **73.5M** ($31.5M + 42M$)

Note: Candidate has nothing to use as a baseline but even still this is a large number which they should point out.

#27 – Wealth Management (IX of IX)

Analysis and recommendation

Brainstorming Question – What could be the reasons behind the high attrition rate in the two identified groups?

Candidates should realize that the 0 – 10 year tenured advisors are leaving the industry because they are not making enough money (only 45k per year and they are 35 years old). **The firm doesn't make a considerable amount of money off of them so if the candidate doesn't do it on their own, nudge them to move away from investigating this further.**

In terms of the 20 – 25 year tenured advisors nudge candidate to think about what they know about this group. Average age of 50 should suggest that they are thinking of retirement.

When asked, provide to candidates that the client does not offer pension so what the advisors make is what they retire off of. This is why the recruiting bonus of competitors is so alluring.

Recommendation:

The final summary and recommendation should highlight the \$73.5M financial impact of the current turnover rate. A good candidate will recognize that the 0-10 year tenured advisors are leaving because of low compensation and will make recommendations around increasing the pay. For the 20-25 year tenured advisors, the client should consider offering a pension plan or other incentives to retain the advisors.

For example, the client can offer a loyalty bonus to Financial Advisors once they have been at the firm for 25 years. Bonus should be approx. 1.5% of assets which is lower than the recruiting bonus because advisors do not need to go through the extra effort of transitioning clients over.

#28 – A+ Airline Co.

Analysis	Structure	Overall
Hard	Hard	Hard

Behavioural Prep

What is one difficult and one easy thing that you consider for yourself in getting this job?
Why do you want to firm XYZ?

Case Tracker		Case Stem
Industry	Airline	Our client is A+Airline Co., the third largest airline in the United States by the number of passengers carried. This week, we have been flying on our primary competitor, Gamma airline, and we noticed something interesting; they stopped accepting cash for in-flight food and beverage services and they now only accept major credit cards. The CEO of A+Airline Co. wants to know, why did Gamma Airline switch from a Cash & Card system to a credit card only system, and should we follow them?
Type	Opportunity assessment	
Leader	Interviewee	
Concepts	NPV, Operations, Market sizing	

Interview Guidance

This case will force an interviewee to rely on logic, business acumen, and structure

Commonly, the interviewee will ask a lot of questions about historical costs, revenues, etc. This is not the correct approach to this question.

The savvy interviewee will realize that this is a BEFORE vs. AFTER comparison of switching from a CASH & CARD system to a CARD ONLY system. Therefore, the questions asked should focus on the DELTA, or the cash changes that occur when the switch is made.

#28 – A+ Airline Co. (II of VIII)

Clarifying answers to provide if asked

Industry Characteristics – Market Economics

- Card Use: Roughly 99% of all consumers purchase their airline tickets using a credit card, i.e., all consumers on an airplane have a credit card available to them.

Client Characteristics

- Items Sold: Only food and alcoholic beverage items are sold on A+ Airline flights
- Locations: This is a US Domestic decision only. Ignore international.

Competitive Dynamics

- Gamma is the only airline that has made the switch; however, all other airlines are evaluating the switch.
- For the purposes of this case, Gamma and A+ Airline should be considered to be exactly the same in all regards.

Interviewer guide to case

Case Structure – Interviewee’s structure should be a BEFORE vs. AFTER comparison of the switch from CASH & CARD to CARD only and should include:

- Revenue Changes: Loss of Cash Only customers vs. Increase in Credit Card customers
- Cost Changes: Benefit of Cash Management Cost Removed vs. Incremental Cost (Fee) of Credit Card
- Cash Flow Changes: Interest and Time Value of Money (TVM), and Working Capital impacts due to an increase in collection speed

Exhibit 1 – After Interviewee walks through structure, they will likely ask questions about consumer purchase behavior on airplanes. Once you feel that they have identified the need to do a market sizing, hand out Exhibit 1. (**DO NOT ALLOW ROUNDING**)

- Ask the interviewee to determine the Total Market Size in (\$) for food and beverage purchases on an average flight
 - The interviewee should calculate the CURRENT allocation of purchases (Cash vs. Card) for an average flight.
 - The interviewee should correctly identify that some of the current “Cash” customers will not convert to Card. **Tell them we will lose 1/3.**

Exhibit 2 – The interviewee should recognize that there might be a cost savings due to the change. Most interviewees know that there is a Credit Card processing fee but do not realize that there are many costs associated with cash management. Ask them about the types of costs A+Airline might face under both processes before handing out the exhibit. When asked, explain that the **total, per flight, savings from eliminating overhead due to Cash Management Operations is \$35/flight**

#28 – A+ Airline Co. (III of VIII)

Sample structure

1-Why Gamma changed

- Changing Customer purchasing behavior
- Higher margins
- Competitive advantage

2-Impact of potential switch

- Revenue
 - Loss of cash only customers
 - Increase in # of Credit card customers
 - Increase in spending/transaction
- Cost
 - Cash Management Cost saving
 - Costs associated with switch to credit card only
- Cash Flow
 - Interest and Time value of Money
 - Working capital impact
- Qualitative considerations
 - Customer satisfaction
 - Airline reputation

3-Implementation of new method:

- Vendor selection
- Pilot program
- Feedback and evaluation

#28 – A+ Airline Co. (IV of VIII)

1- Change in revenue

Interviewee will likely begin with the correct assumption that revenue will be lost due to unhappy, cash-only passengers

Once asked about how many are lost, show Exhibit 1 to do a market sizing

Calculations

1-Total number of passengers per plane

- Number of seats:
 - First class : $200 \times 25\% = 50$
 - Economy: $200 \times 75\%$ or $200 - 50 = 150$
- Occupants: multiply # of seats calculated above by load factor from exhibit 1
- Split between Business and leisure: use percentages provided in exhibit 1

2-Total number that purchase & total spend:

- Business: $110 \times 75\% = 45$ | $45 \times 10 = 450\$$
- Leisure: $60 \times 25\% = 15$ | $15 \times 5 = 75\$$

3-Cash vs Card Spend:

- Cash = $20\% \times 525\$ = 105\$$
- Card = $525 - 125 = 420\$$

Lose 1/3 of cash customers: $105 \times 1/3 = 35\$$

New total:

70\$ is revenue from Cash and we get it from $105\$ - 35\$$

Notes on Exhibit 1

Using exhibit 1, interviewee should calculate:

1. Total Number of Passengers per Plane

	Seats	Occupants	Business	Leisure
First Class	50	50	50	0
Economy	150	120	60	60
Total	200	170	110	60

2. Total Number that Purchase & Total Spend (\$)

	# that Purchase	Total Purchase \$
Business	45	\$450
Leisure	15	\$75
Total	60	\$525

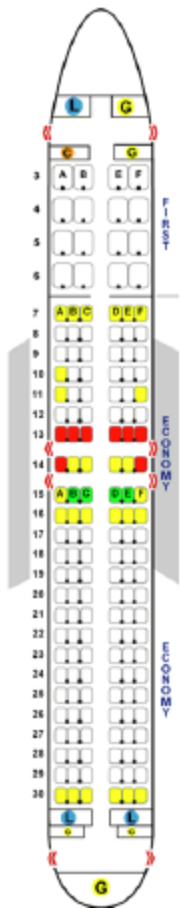
3. Cash vs. Card Spend + LOSS

	Cash	Card	Total
Inflight \$	\$105	\$420	\$525

Lose 1/3 of Cash Customers			
	#	\$	New Total
Business	3	(\$30)	\$420
Leisure	1	(\$5)	\$70
Total	4	(\$35)	\$490

#28 – A+ Airline Co. (V of VIII)

Exhibit 1: An average A+ Airline flight



A+ Airline Boeing 737-800 Vers. 2 (738)

Total seats	200			
	% of Seats	Load Factor	% Business	% Leisure
First Class	25%	100%	100%	0%
Economy	75%	80%	50%	50%
	% that Purchase		Avg. Spend (\$)	
Business	75%		\$10	
Leisure	25%		\$5	
*Note: First Class Passengers Receive free food and beverage				
	% of in-flight purchases			
Cash	20%			
Card	80%			

*Note: First Class Passengers Receive free food and beverage

#28 – A+ Airline Co. (VI of VIII)

2- Change in costs

Interviewee should make mention of Cost changes due to the shift to a Card Only strategy.

Ask the interviewee to detail the types of costs that might be involved before handing out Exhibit 2

Notes on Exhibit 2

- Exhibit 2 shows the current operations for Cash Management and Card Management at A+ Airline.
- The interviewee should notice the following information:
 - Cash Management & Card Management both have 2% fees associated, so this is a “wash” in terms of savings.
 - Cash Management requires additional overhead, 7 total employees per airport, that could be eliminated for further savings.
- **When asked, explain that the total, per flight, savings from eliminating overhead due to Cash Management Operations is \$35/flight.** Interviewee should notice that the \$35 savings offsets the \$35 loss in Revenue (Slightly more because this impacts bottom line, but ignore margin)

3- TVM & working capital

Exhibit 2 also shows that there is a 30-day time benefit to collecting payment via Credit Card vs. Cash

Prompt 1: Ask the interviewee to qualitatively explain the impact his time savings will have

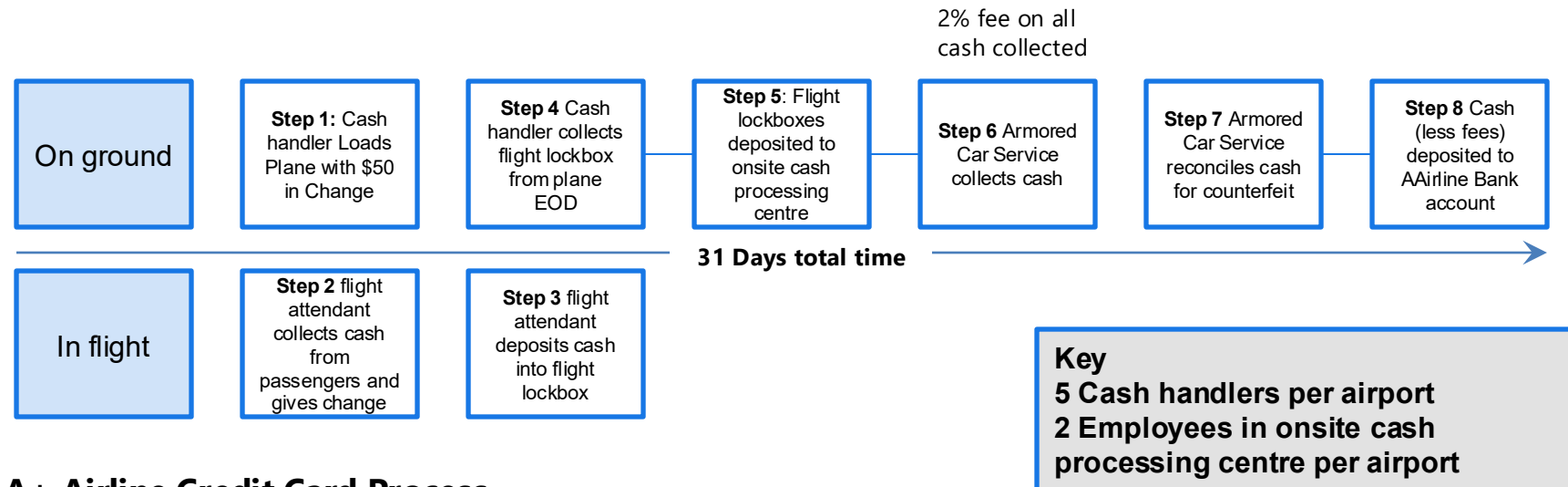
Possible Answer:

The interviewee should qualitatively mention that there are benefits to eliminating the longer cash management process:

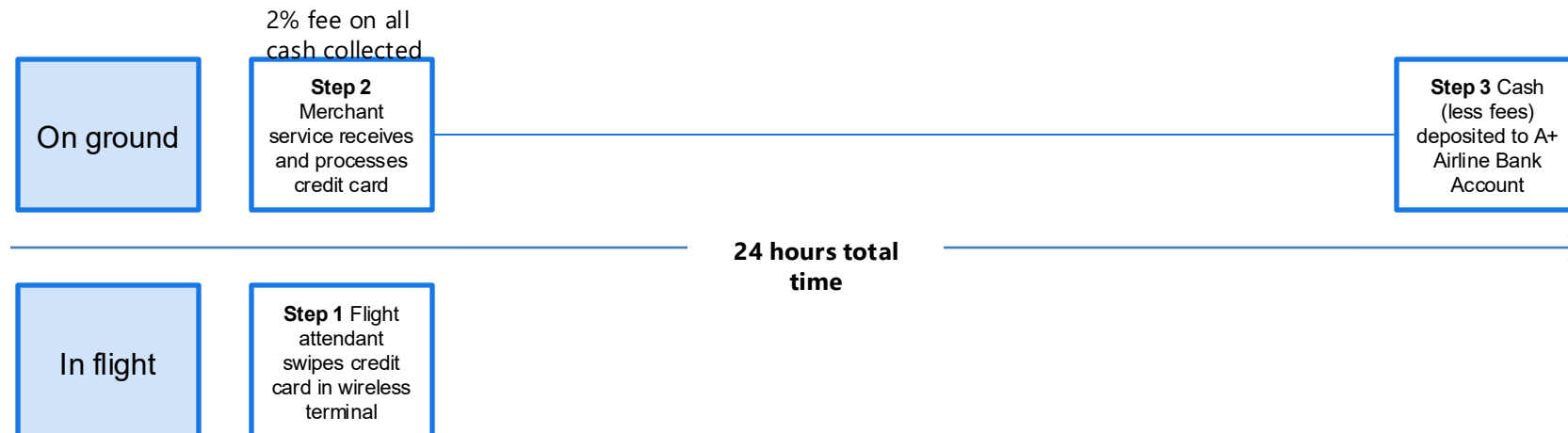
- **Time Value of Money:** A+ Airline will receive their money 30 days sooner, and this money could be used to:
 - Invest in interest earning accounts or growth projects
 - Pay off suppliers early and take advantage of discounts
 - Pay down lines of credit faster
- **Working Capital Improvement:** \$50 per flight in change can be eliminated, thus freeing up cash flow. Additionally, there will be a reduction in loss of cash due to theft and damage
- **Happier Customers:** The majority of customers will be able to place and receive their order faster on the plane which will also increase sales

#28 – A+ Airline Co. (VII of VIII)

Exhibit 2: A+ Airline cash and card operations



A+ Airline Credit Card Process



#28 – A+ Airline Co. (VIII of VIII)

Solution and recommendations

Recommendation:

Overall, our client, A+ Airline, should switch to a Credit Card only system for in-flight food & beverage because:

Quantitative Benefits:

- We calculated a Revenue loss of \$35. We also found that we could save \$35 in overhead expenses by eliminating the Cash Management process. There are no net benefits or losses

Qualitative Benefits:

- We also found that we will receive payment 30 days sooner by only accepting credit cards. This improves our cash flow and could allow us to earn interest, pay down creditors, or invest in projects. Additionally, by reducing cash losses and eliminating "Change" tied up on airplanes, we can improve our Working Capital and also put this money to work for us. Finally, it appears as though customers, in general, might actually be happier because the speed of transactions on the airplane will improve.

Risks:

- Customer Resistance: Passengers who prefer cash/ have no access to credit cards may be dissatisfied.
- Technical Issues: System glitches or outages could disrupt transactions.
- Security Concerns: Handling credit card data may pose security risks.

Next steps:

- Communicate the change to passengers effectively.
- Ensure robust and secure payment infrastructure.
- Provide comprehensive training for cabin crew.

Bonus

An excellent interviewee will note:

- Due to the increased transaction speed, probability of purchase for both Business and Economy passengers in the AFTER state should go up due to a reduction in frustration. The people on the back of the airplane often abandon a purchase if it takes too long to place an order. Making change takes a lot of time!
- Average purchase amount should also increase. There is a proven psychological phenomenon that shows how consumers who do not carry cash purchase less when a cash option is offered because they feel guilty using their card. A card-only option eliminates this guilt and consumers don't mind using the card.
- Business Acumen: The fact that Gamma Airline has already switched is a clear indication that this is a logical idea.

#29 – Old Pharma

McKinsey (LBS)

Analysis	Structure	Overall
Hard	Medium	Hard

Behavioural Prep

What is your proudest accomplishment?
Tell me about a problem in your professional life and how you were able to overcome it.

Case Tracker	
Industry	Pharmaceuticals
Type	M&A
Leader	Interviewer
Concepts	Cost synergies, market entry, sizing, profitability

Case Stem
Our client is OldPharma, a major pharmaceutical company with \$10B in annual revenue. They are headquartered in Germany and have regional sales offices worldwide. OldPharma has a long history of researching, developing, and selling small molecule (SM) drugs, a segment that comprises the vast majority of drugs today. They are interested in entering a new, rapidly growing segment, called Biologicals, which are more complex drugs that can treat conditions not typically addressable by SM drugs. R&D for Biological is vastly different than for SM and OldPharma is years behind its competitors in attaining the R&D capabilities in biologicals. They are considering jump-starting its biological program by acquiring BioFuture. BioFuture is a San Francisco-based start-up founded 12 years ago by prominent scientists that now employs 200 people. It is publicly traded and worth \$1B at its current share price. Old Pharma has engaged us to evaluate the BioFuture acquisition and advise on its strategic fit with OldPharma’s biologicals strategy.

Interview Guidance
This case asks the question, should OldPharma acquire BioFuture. As an interviewer led case, the candidate should not spend too much time doing upfront structuring. There will be opportunity for analysis at each question. Warning: This is a long case

#29 – Old Pharma (II of IX)

Question 1

What factors should the team consider when evaluating whether OldPharma should acquire BioFuture?

Candidate should take the time to organize his/her thoughts in order to develop an overall approach.

A good answer would include consideration of:

Value of BioFuture's current pipeline: Number of drugs, probability of success, potential revenues and profits.

BioFuture's R&D capabilities: scientific talent, intellectual property, building, equipment. All factors that can help assess BioFuture's future drug pipeline.

BioFuture's marketing / sales capabilities: Relationships with key opinion leaders, marketing history.

A very strong answer would also address the below points:

Acquisition Price: What premium over the \$1B market price would be required.

OldPharma's capability gaps: Does BioFuture help fill these gaps in sales/marketing, etc.

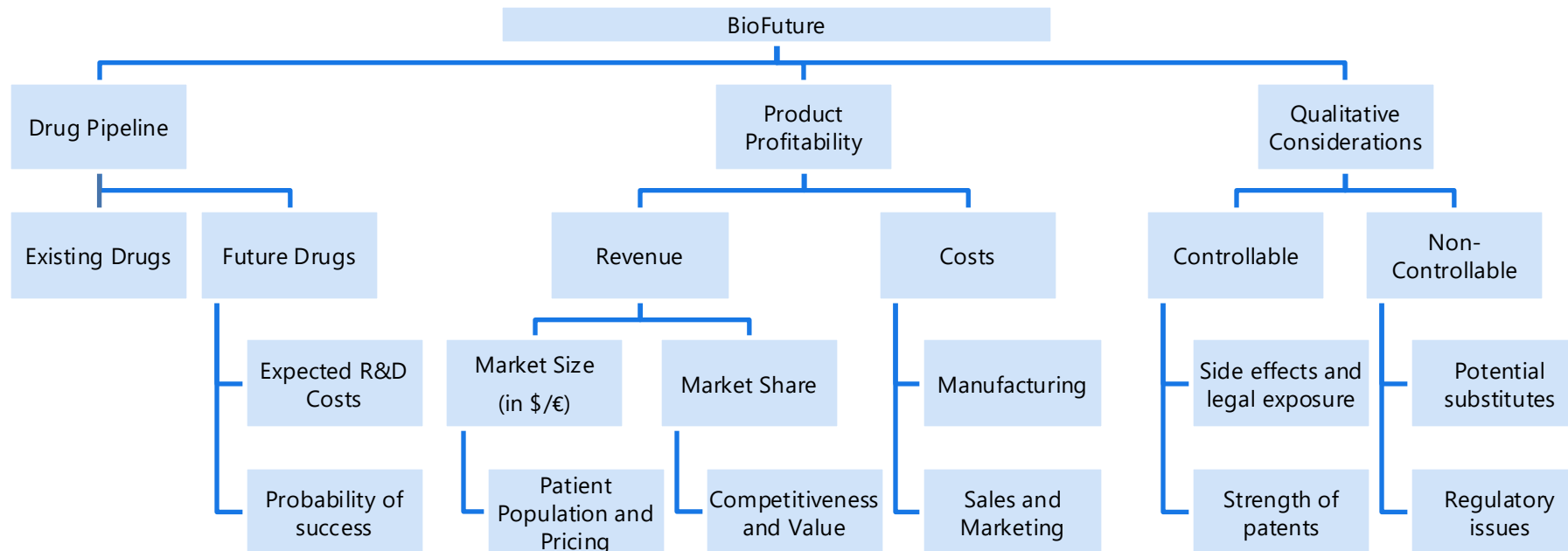
OldPharma's alternatives: Alternative companies to be acquired, alternative strategies (joint venture, going at it alone, not entering the biologicals segment).

#29 – Old Pharma (III of IX)

Question 2

The team wants to explore BioFuture's current drug pipeline. The team decides to focus first on evaluating the value of BioFuture's drug pipeline – both its current portfolio, as well as its ability to generate drugs on an ongoing basis. What issues should the team consider when evaluating the value of BioFuture's existing drug pipeline?

A possible consideration set is shown below. The interviewee should draw out their thought process first and then deep dive into specific areas that need further explanation or details.



#29 – Old Pharma (IV of IX)

Question 3

OldPharma believes that the likelihood of success of BioFuture's primary candidate drug can be improved by investing an additional \$150M in a larger phase II trial. The hope is that the investment would raise the success rate in Phase II, meaning a higher probability that the drug makes it to Phase III and beyond. By how much would phase II success rate need to increase for the investment to break even?

See Exhibit 1 for a description of the expected probability of success, by stage, in the Pharma R&D pipeline.

After handing out the exhibit to the interviewer, let them know that if the drug were to be successfully marketed and sold, it is expected to be worth \$1.2B (ie: the NPV of the future expected profits from selling the drugs).

Recommended Solution:

Currently, the maximum amount that BioFuture should be willing to invest in the drug would be the expected value of the drug times the probability of success: $(70\% \times 40\% \times 50\% \times 90\%) \times \$1.2B = 151.2M$ (the Probability of success is obtained by deducting from 100% the % of drugs that won't make it to the next phase, which is found in Exhibit 1 below the circles)

Investing \$150M would mean that:

$$(151.2M + 150M) = (70\% \times (\text{New Phase II}\%) \times 50\% \times 90\%) \times \$1.2B$$

$$\text{New Phase II \%} = (301.2M) / (378M) = \mathbf{80\%}$$

Doubling the probability of success from 40% to 80% will be extremely difficult and likely not possible.

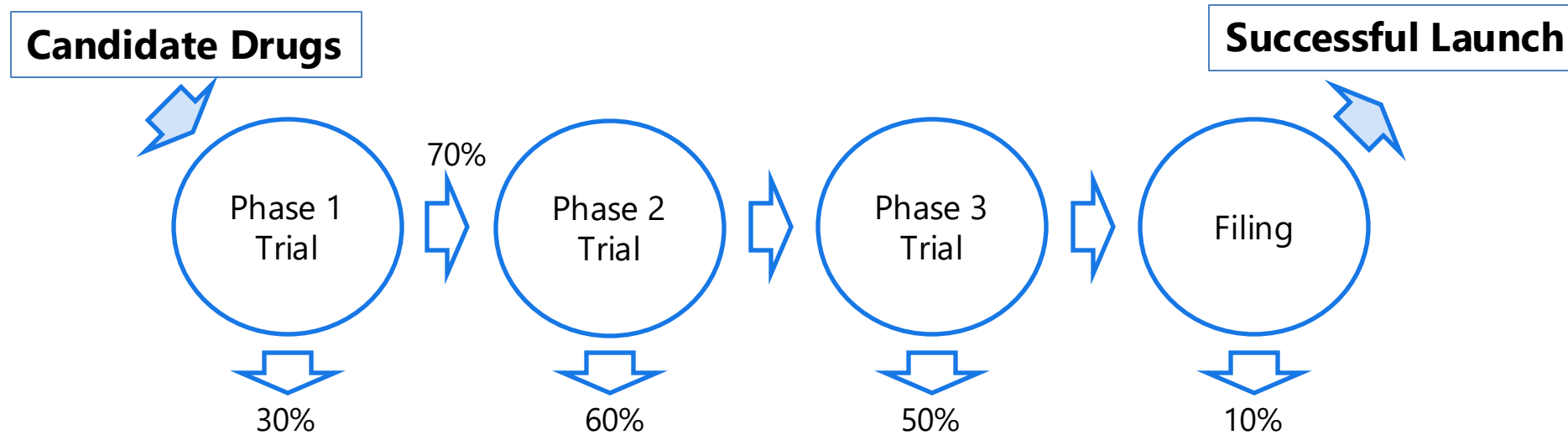
Note: Filing is the process of submitting the all of the clinical and safety evidence and asking for regulatory approval from the authorities to be able to sell the drug

#29 – Old Pharma (V of IX)

Exhibit 1: expected probability of success, new drugs

Expected Probability of success, by stage of research and development

Percent



#29 – Old Pharma (VI of IX)

Question 4

Next, the team explores the potential setup of BioFuture after the acquisition. Although BioFuture's existing drug pipeline is relatively limited, *OldPharma* is highly interested in its ability to serve as a biological research "engine" that, when combined with *OldPharma's* existing R&D assets, will produce many candidate drugs over the next 10 years. What are your hypotheses on the major risks of integrating the R&D functions of BioFuture and *OldPharma*?

Some potential risks include:

- Scientists do not have overlapping disease (therapeutic area) interests or expertise and are unable to materially collaborate.
- Integration into the process driven *OldPharma* culture kills the entrepreneurial culture at BioFuture that has been key to its success
- Language barriers severely hinder ability to communicate and share information
- Poor management and a sense of community as a result of R&D operations that might come with a time difference of 9 hours
- Key scientific talent leaving BioFuture after the acquisition – either because the acquisition makes them wealthy, or they do not want to be part of the new larger *OldPharma* organization

#29 – Old Pharma (VII of IX)

Question 5

Post-acquisition, *OldPharma* believes that it will be necessary to consolidate all biologicals R&D into one centre. There are two logical choices: (1) *OldPharma*'s existing headquarters in Germany and (2) BioFuture's headquarters in San Francisco. *OldPharma* does not currently have any biologicals facilities or operations in Germany, so new facilities would have to be built. How would you think about this situation?

Reasons to consolidate in Germany:

- Better coordination with non-Biologicals
- Better coordination with other business units (marketing, sales, manufacturing)
- Easier to intermix scientists between biologicals and traditional R&D to transfer knowledge and capabilities

Reasons to consolidate in San Francisco:

- Less likely that they will see a flight of talent
- Easier to recruit top talent in San Francisco (region is a centre for new technology, research, etc)
- Ability to retain the entrepreneurial spirit and culture of BioFuture
- No need to rebuild facilities or infrastructure

#29 – Old Pharma (VIII of IX)

Question 6

While researching the integration barriers, the team learns that one of OldPharma's competitors, DrugMax, has already partnered with BioFuture on their lead drug candidate essentially agreeing to split the development costs and future profits 50/50.

OldPharma is considering buying out DrugMax's 50% share in the lead drug candidate. As a first step in the valuation, they have asked your team to estimate the potential peak sales of the drug candidate – this is another way to verify potential future profits of a drug. The drug candidate is intended to treat non-Hodgkin's Lymphoma. New cases are diagnosed each year in 25 out of every 100,000 US men and 15 out of every 100,000 US women.

What is the estimated US peak sales of the drug?

Information to provide upon request and answer

US population is 300M

Full course of therapy takes 90 days

OldPharma believes the drug can be sold at a price of \$500 per day

Estimated market share is 25% (% of eligible patients who are treated with this drug).

Answer

A very good answer would include the following:

- Expected peak sales of this drug candidate are USD \$675 million
- Assuming a U.S. population of 150 million men and 150 million women, there would be 37,500 (150M/100k, then multiplied by 25) estimated diagnoses among men, and 22,500 diagnoses (150M/100K, then multiplied by 15) among women, or 60,000 new cases of non- Hodgkin's lymphoma per year
- Each course of therapy will yield \$45,000 in revenue (90 days at \$500 per day). Therefore total U.S. market potential is \$2.7 billion(45K*60K). Estimated market capture is 25%, leading to an estimated U.S. peak sales of \$675 million. (25%*2.7B)

#29 – Old Pharma (IX of IX)

Question 7

On the third day of the engagement, you run into the Vice President of Business Development for *OldPharma* in the cafeteria. He asks what the team's current perspective is on the BioFuture acquisition and what next steps you are planning to take. How would you respond?

Possible answer

There is no right or wrong answer regarding whether to buy or not buy, and there are various valid ways to build an argument. One possible very good answer would be:

An acquisition of BioFuture can bring two major sources of value to *OldPharma*: the value of its existing compounds and the potential value of integrating its research capabilities into *OldPharma*

In terms of BioFuture's existing pipeline there are a couple of challenges: firstly, the proposed idea of investing heavily in Phase II trials is not likely to be a profitable investment; secondly, one of your competitors, DrugMax, currently has a partnership with BioFuture for its lead drug candidate. This needs to be taken into account when trying to acquire BioFuture. We are still looking into other potential synergies, but it appears unlikely that *OldPharma* can justify the cost of an acquisition purely based on BioFuture's existing pipeline

The greater source of upside is likely to be the long-term benefits of integrating BioFuture's research capabilities with *OldPharma*. There are significant risks to this as well, given the "two worlds" nature of their organizational cultures. As next steps we therefore want to better understand the feasibility of bridging the cultural gap and better understand pros and cons of different consolidation options; estimate the cost of this research integration; get a better understanding of the value of BioFuture's future potential to develop drugs

#30 – Grocer Co.

Analysis	Structure	Overall
Medium	Medium	Medium

Behavioural Prep

Tell me about a time you upset someone at work. How did you handle the situation?
Tell me about a time you took a risk at work.

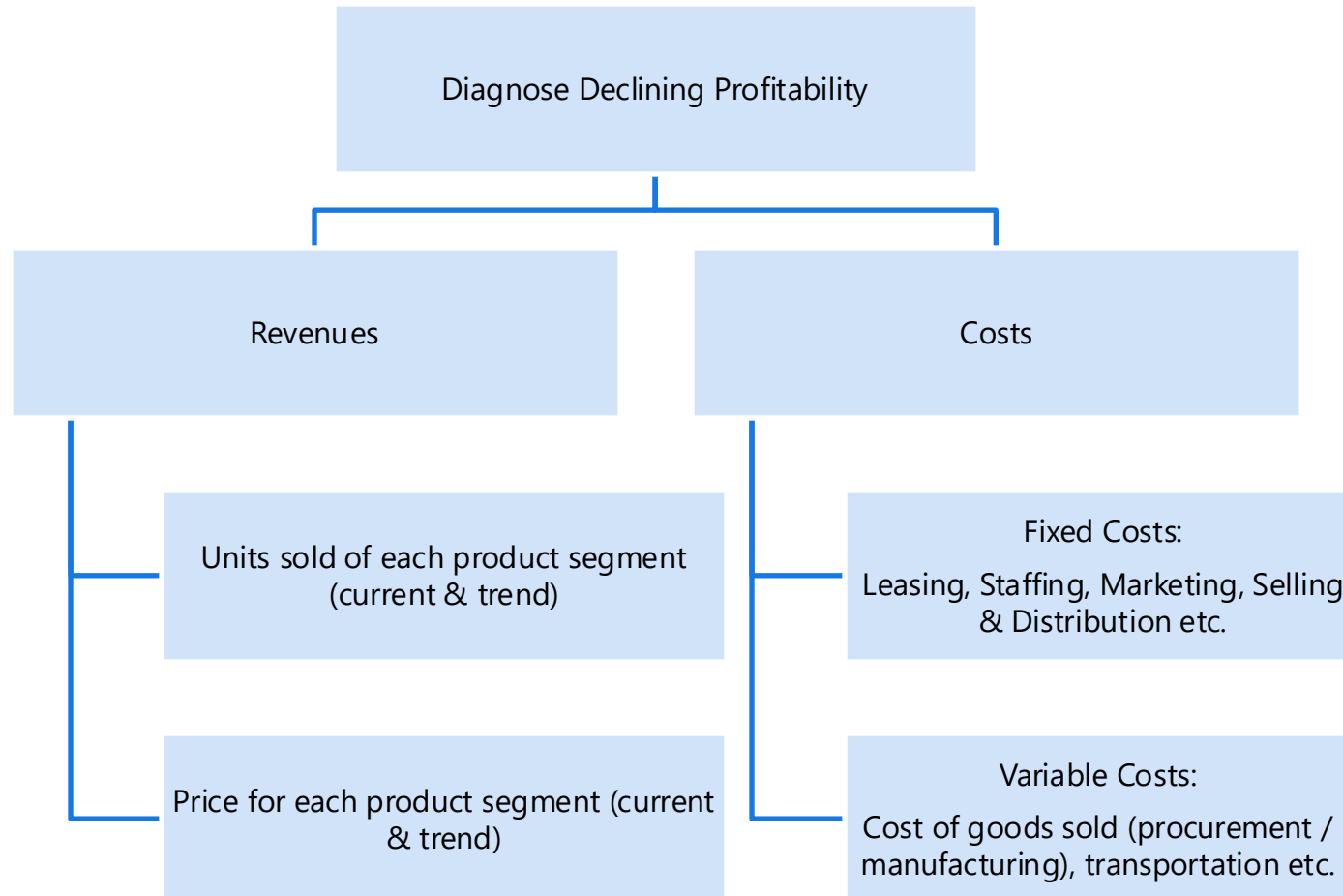
Case Tracker		Case Stem
Industry	Consumer Goods	GrocerCo. is one of the top 3 grocery chains in US and Canada, with a diverse portfolio of products. They have 500 stores across the regions they operate in. They have recently seen a decline in performance in their Asian food section. What could be causing that decline and how can GrocerCo improve it?
Type	Profitability	
Leader	Interviewee	
Concept	Profitability improvement, product portfolio optimization	

Additional Information
Answers to clarifying questions: • What does performance imply? • Performance here is concerned with profitability or bottom line • Who are the competitors and are they facing the same issue? • Large grocery chains like Walmart, CostCo, 7/11 etc • No competitors are doing fine or even better in some regions. • Is this a company wide issue or does it pertain to certain stores or region? • It is a company-wide issue • What products does the Asian food segment include? • It includes products like noodles, sauces and other condiments.

Case adapted from a 1-on-1 coaching session with a consulting preparation firm. Industry details and numbers modified for educational purposes.

#30 – Grocer Co. (II of VIII)

Simple Sample Structure



#30 – Grocer Co. (III of VIII)

Exhibit 1

To be given when asked about revenues from various products sold in the Asian foods segment over the years. (Encourage the candidate to keep the data at the store level)

Store-level Data	2023		2024	
	Avg Units (K)	Avg Price	Avg Units (K)	Avg Price
Noodles	30	\$4	31	\$3.5
Sauces	21	\$2	20	\$2.2
Other Condiments	22	\$2	20	\$2.5

#30 – Grocer Co. (IV of VIII)

Case Math – Exhibit 1

Key Insights:

1. Prices for noodles have fallen and average units for noodles have increased
2. Prices for other 2 categories have increased and average units have fallen.
3. A good candidate will identify the need to calculate revenues for each segment for 2023 and 2024 to identify the biggest problem area

Store-level Data	2023			2024			
	Products	Avg Units (K)	Avg Price	Revenue (K)	Avg Units (K)	Avg Price	Revenue (K)
	Noodles	30	\$4	\$120	31	\$3.5	\$108.5
	Sauces	21	\$2	\$42	20	\$2.2	\$44
	Other Condiments	22	\$2	\$44	20	\$2.5	\$50

After realizing that the revenue from noodles is falling, the candidate should ask for the specific products that are being sold within this segment and how have their purchasing patterns changed to understand what is the best opportunity to increase revenues.

#30 – Grocer Co. (V of VIII)

Exhibit 2

To be given when asked about further segmentation of products within noodles. (Encourage the candidate to round off)

Store-level Data	2023		2024	
	Revenue (K)	Avg Units (K)	Revenue (K)	Avg Units (K)
Brand A	\$48.2	12	\$27.9	7
Brand B	\$31.8	8	\$24	6
Value Brand	\$40	9.9	\$56.6	18

#30 – Grocer Co. (VI of VIII)

Case Math – Exhibit 2

Key Insights:

1. Revenue from Brands A and B have fallen while the Revenue from Value Brand is rising
2. Units seem to fall and rise in tandem with revenue across products. A good candidate will realize that they should calculate the price to understand the driver behind change in units and revenues

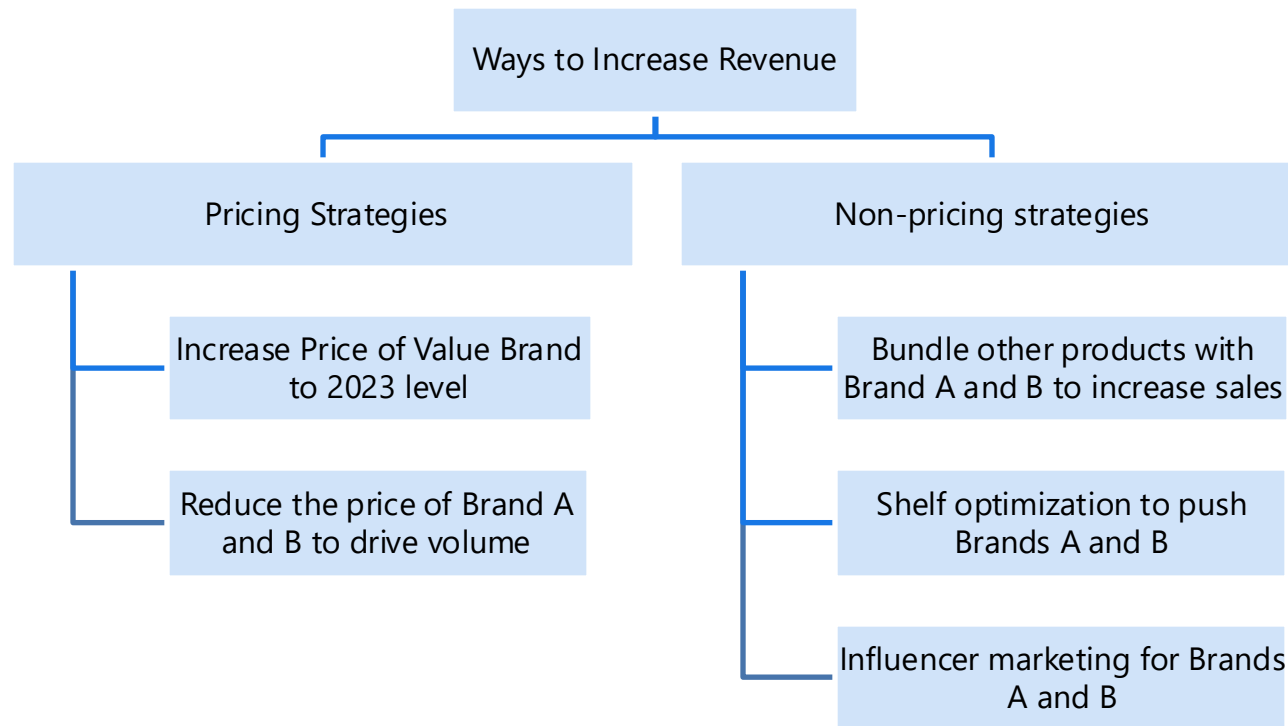
Store-level Data	2023			2024		
	Revenue (K)	Avg Units (K)	Price	Revenue (K)	Avg Units (K)	Price
Brand A	\$48.2	12	~\$4	\$27.9	7	~\$4
Brand B	\$31.8	8	~\$4	\$24	6	\$4
Value Brand	\$40	9.9	~\$4	\$56.6	18	~\$3

- The calculations will show that the price of Brand A and B hasn't changed, but the price of Value Brand has reduced further.
- This means that the value brand is cannibalizing sales for Brands A and B, driving reduced profits
- **A great candidate will also notice that of the 8K increased units for Value Brand in 2024, 7k come from Brand A and B, but 1K are net new sales which can come from either existing customers purchasing more or new customers coming into the stores**

#30 – Grocer Co. (VII of VIII)

Brainstorming

Once the candidate realizes that Value Brand is cannibalizing sales for Brand A and B, push them to brainstorm ideas on how to increase the revenue to previous levels. **A great candidate will arrive on it on their own.**



This is a non-exhaustive list of ideas. **A great candidate will also give a recommendation for which way the client should go and also a few top-of-mind risks for their recommendation.**

#30 – Grocer Co. (VIII of VIII)

1. Recommendation

- GrocerCo. revenues are declining due to cannibalization of Brand A and B sales by price reduction in value brand and they should increase the price of Value Brand back to \$4

2. Supporting Data

- It will mitigate cannibalization of Brands A and B
- It could increase revenue by \$11.5K from the noodles segment in each store and bring it back to baseline levels
- This will also boost profitability from this segment as costs to produce noodles will probably remain unchanged.

3. Risks and Next Steps

- **Risks:** -Price-sensitive customers may switch to competitors or reduce purchases eventually hindering baseline revenue achievement
- Other low-cost brands may undercut, forcing a reactive strategy later
- The client risks losing foot traffic and the incremental sales from other products typically purchased by Value Brand customers
- **Next Steps:** - Hire a customer insights (CI) team to understand consumer price sensitivity to gauge impact a price increase for Value Brand
- The CI team could also assess the aligned sales that new Value Brand customers bring (e.g., from other segments of the grocery stores) to see if indirect benefits of reduced price are higher than direct losses
- The client should assess potential competitor reaction before making a change to their pricing strategy.

#31 – Electronics Retailer

Boston Consulting Group

Analysis	Structure	Overall
Easy	Medium	Medium

Behavioural Prep

Tell me about a successful business relationship you built with a client, boss, or peer in your previous job.
Tell me about a time you faced opposition in the workplace. What did you do and what was the outcome?

Case Tracker	
Industry	Retail
Type	Opportunity Assessment
Leader	Interviewee
Concepts	Incremental revenues/profits

Case Stem
Your client, Circuit Co. is a national consumer electronics retailer similar to Best Buy. For the past 5 years, Circuit Co. has grown its revenues and earnings primarily through new store openings. However, Circuit Co. knows that this type of growth can not be maintained forever. For the past year, the company has focused on several initiatives aimed at improving their same-store sales and earnings. One of these initiatives has fallen by the wayside and you have been hired to analyze the situation. Specifically, in the 3rd quarter of 2003, Circuit Co. ran a pilot program in the digital camera departments of its Southwest Region stores. The CEO wants to know: 1. Was the program a success? 2. What improvements can be made to the program? 3. Should Circuit Co. roll the program out to the rest of the country? ADDITIONAL INFORMATION ON FOLLOWING PAGE

Interview Guidance

- Candidate should recognize that they need to inquire about the financials of the pilot program as well as the quantified objective or metric for “success” of the pilot project
- After candidate recognizes that while incremental revenues increased during the pilot project, digital camera profits actually declined:
 - Interviewer should push candidate to brainstorm why this might be the case. A strong candidate will notice that shift in compensation structure between “generalists” and “specialists” may have an impact

#31 – Electronics Retailer (II of V)

Case Prompt (continued) and Clarifying Information

Details about the pilot program (read verbatim):

Traditionally, all of Circuit Co.'s ground level employees were "generalists" in the sense that every one of the them did all of the jobs that needed to be done: stocking shelves, answering customer questions, running the cash register, etc. The pilot program involved setting up a group of "specialists" in the digital camera area who were solely responsible for answering customer questions and selling digital cameras. The remaining employees remained "generalists". Generalists maintained their previous wages. Specialists were paid a small draw plus commissions based on digital camera revenues.

Information to be provided upon request:

Question: How does the CEO measure trial success?

Answer:

- Incremental revenues exceeded incremental costs
- Program did not create significant problems for the general store operations

#31 – Electronics Retailer (III of V)

Sample Framework

- 1) Incremental Revenues of Pilot Program
 - Revenues without the program
 - Timeframe
 - Segments of digital camera sales
 - Prices of cameras in segments
 - Number of cameras sold
 - Revenues with the program
 - Timeframe
 - Segments of digital camera sales
 - Prices of cameras in segments
 - Number of cameras sold
- 2) Incremental Costs of Pilot Program
 - Costs without the program
 - Salaries
 - COGS for different types of digital cameras
 - Costs with the program
 - Salaries, commissions
 - COGS for different types of digital cameras
- 3) Effects of program on other departments
 - How do generalists react to the change in compensation for specialists?
 - Ex. lower motivation, degrading quality of customer service?
 - Should the specialist model be rolled out across other departments?

#31 – Electronics Retailer (IV of V)

Pilot Program Financials

NOTE: Only provide each item when asked. Company is still growing by adding stores. A strong candidate should know to ask about this, as it was included in the prompt.

	3Q 2002	3Q 2003
Total Digital Camera Rev (\$M)	500	900
Total Per Store Revenue (\$M)	7.75	12
# of Stores	200	300
Digital Camera COGS (\$M)	250	675

Additional Information

- SKU portfolio did not change
- Pilot did not affect other operations
- Q3 runs from June to September
- Generalist wage changes were insignificant

For the interviewer (quant analysis answer):

	2002	2003
Digital Camera (Per Store Revenue \$M)	2.5	3
Digital Camera (Per Store COGS \$M)	1.25	2.25
Per Store Profitability (\$M)	1.25	0.75

#31 – Electronics Retailer (V of V)

Analysis of the financial performance...

- Digital camera revenues increased by about 80%
- Stores increase by 50%. Camera revenue per store increased by 20%
- However, COGS increased by 170%. COGS per store increased by 80%
- Digital camera profits actually declined during the pilot program, even though revenues increased dramatically
 - Before, profit = \$250
 - Now, profit = \$225

...leads to clients' answer

A strong candidate will offer to brainstorm ideas on why the revenue increased, but profitability decreased. If not, interviewer should prompt candidate to do so:

E.g.:

- Hiring new specialists at higher wage rate reduced profitability
- It appears specialists were focusing on selling lower margin cameras in order to earn revenue and drive commissions
- In terms of revenues, the program was a success, however, the client suffered in terms of profits

Recommendation: re-run the program after determining a profit-driven commission plan, based on results, determine rollout possibility for system.

#32 – Frontier Adventure Racing Inc. Bain & Co.

Analysis	Structure	Overall
Hard	Hard	Hard

Behavioural Prep

Give me an example of a time when you had to motivate others who did not report to you.
Describe a recent unpopular decision you made. What was the result?

Case Tracker	
Industry	Travel & Leisure
Type	Profitability Improvement
Leader	Interviewee
Concepts	Incremental profitability, capacity and demand

Case Stem
A friend of mine has his own event management business organizing adventure races across Canada. An adventure race involves mountain biking, paddling and trekking through a wilderness location and is done in teams of 3 or 4 people. It is a relatively new sport in Canada that is growing wildly with the television success of an international adventure race called Eco-Challenge. His business, Frontier Adventure Racing (FAR) Inc., is the leading adventure race company in Canada with a total of 15 different events per season and three different types of races. Competition in the Canadian market is also growing rapidly although no company rivals FAR Inc.'s reputation and quality of events yet. All three types of races offered by FAR are well managed and both variable and fixed costs are near optimal. However, the business is not currently very profitable. With such a strong brand in a growing market there must be ways to increase profitability. Where should he focus his energy?

Interview Guidance

This case has a lot of details, both provided by the interviewer, as well as in the Exhibit. Although much of it is important, the interviewee should not spend too much time going through all the details and instead find the important points to build the recommendation.

This case first requires the candidate to break down FAR's business into its three different race types and determine the profitability and success of each.

#32 – Frontier Adventure Racing Inc. (II of VII)

Information Provided Upon Request

Race Information:

This should be provided early on as it will lead the interviewee on the path of segmenting the business by race type

FAR offers 3 types of events per year:

- 1) *Adventure Challenges* are the easiest and most accessible events and average 80 teams per race with an entry fee of \$350 per team. There are 10 Adventure Challenges across the country.
- 2) *Raid the North* races are more serious and competitive; they are the foundation of FAR Inc.'s reputation. 3 of the 4 national events are selling out at 50 teams per event with an entry fee of \$1,500 per team.
- 3) *Raid the North Extreme* is an internationally competitive event and qualifier for the AR World Championships. It typically attracts 40 teams per year (1 event) with an entry fee of \$5,000 per team.

Competition:

- There are main competitors in the Adventure Challenge races with only a handful of competitors offering races similar to Raid the North
- These competitors are similarly priced and are perceived to offer events of equivalent quality and challenge

Company Financial Information

Annual Sponsorship Rev. = \$70,000

Annual Business O/H = \$300,000

Race Specific Financial Information:

See Exhibit 1

#32 – Frontier Adventure Racing Inc. (III of VII)

Sample Framework

Current profitability of each race type:

- Revenue (for each race): price, number of participants, revenue from merch, sponsorships
- Costs:
 - Fixed costs
 - Variable costs
- A strong candidate may suggest a breakeven assessment

Characteristics and differences of each race type (e.g., what are the differences of each race):

- How do the demographics and preferences/interests of participants vary by race type?
- Style of race (more intense, competitive, more fun (i.e., "fun run"))
- Where is each race located?

Potential ways to increase profitability:

- **Growth from Existing Races**
- **Growth from providing new races:**
 - What new locations/cities could they expand to? (e.g., in areas where demographics of people match current customers)
 - Are there certain types of races they could offer? (e.g., IronMan, partner with a charity)
- **Reducing costs:**
 - Fixed costs
 - Variables costs

Competitive Assessment:

- Other racing companies: where are they located, how many participants, demographics of participants

#32 – Frontier Adventure Racing Inc. (IV of VII)

Financial Solution

The candidate should start by understanding the financials and calculating contribution per race for each race type (shown below). This will require first identifying the different cost types (variable, fixed per event, overhead) and calculating the revenue per event.

The conclusion this should lead to is that fixed costs are a major factor in this business.

Profitability can be increased by better leveraging either fixed costs per event or O/H costs. Strong candidates will be able to do the math in their head by looking at the contribution per team. Struggling candidates can be hurried through the calculations by giving them the contribution per team.

Profit and Break Even per race

Event Cont.	AC	RtN	RtNE
Number of Events	10	4	1
Revenue per Event	28,000	75,000	200,000
Costs per Event	(22,000)	(50,000)	(100,000)
Profit per Event	6,000	25,000	100,000
Total Profit	60,000	100,000	100,000
B/E # of Teams	50	25	15
Contribution per team	200	1000	4000

Overall company profitability

EVENT	Overall Corp.	FAR
	Revenue	780,000
	Fixed Costs	(260,000)
	Variable Costs	(260,000)
	Event Contribution	260,000
CORP.	Sponsorships	70,000
	Fixed Overhead	(300,000)
	Net Profit	30,000

#32 – Frontier Adventure Racing Inc. (V of VII)

Exhibit #1: Race Type Details

Provide info:	Adventure Challenge (AC)	Raid the North (RtN)	Raid the North Extreme (RtNE)
Length	8 hour event	36 hours per event	6 days
Entry Price	350	1500	5000
Number of Events	10 per year (5 Ont., 2 Qc., 3 BC)	4 per year (2 On., 1 Qc., 1 BC)	1 per year
Avg. # of Teams / event	80	50	40
Variable Costs per team	150	500	1,000
Event Fixed Costs	10,000	25,000	60,000
Details	• Fun & accessible • Attracts broad audience • Up to 100 teams per event possible • Costs are well managed and entry fee is on par with competition • Quality of events between key competitors is not widely different and brand loyalty is low	• Wilderness-based events • Attract both athletes and non-athletic outdoors people • Selling out ON and QC, but not BC; no other company is selling out • Brand loyalty strong • Setup costs are high because of safety and infrastructure requirements	• Requires advanced skills and certifications • Less than half of competing teams are Canadian • Most prestigious race in Canada and only event of its length and competitiveness in Canada • Although brand loyalty is very strong it is unlikely that international teams would travel to Canada for more than one event per year

#32 – Frontier Adventure Racing Inc. (VI of VII)

Opportunity Solution:

Give a few ideas on how the client can increase profitability.

Leverage the Brand

- Expand the number of races in the profitable segments (ie: RtN in Ontario).
- Move into new segments – (ie: US, Winter Events)

Average candidates will get the idea of adding events. There is a limit to the number of events that can be added (only so many weekends in summer season) so more is required.

Good candidates will choose to add Raid the North events and understand the trade-offs between RtN and AC events.

Strong candidates will also talk about expanding into US or extending season by offering winter races.

Leverage the Fixed Costs

- Increase revenue per event by:
- Increasing Price in sold out events
 - Increase marketing to have more teams
 - Increase the max teams per event

Average candidates will get increased marketing to fill races that are not full and potential to increase price on RtN.

Good candidates will question the maximum number of teams per race.

Strong candidates will suggest running an AC and RtN on the same weekend to share fixed costs and the possibility of adding extra services or “features” to get more revenue per team at each event.

Leverage the Core

- Move into similar product areas:
- Corporate events
 - Leadership training

For strong candidates that finish early, we can discuss whether other products are close enough to the core to not just be distractions. What is similar (content, branding)?

What is different (audience/marketing, suppliers, cost structure, competitors)? Is there really profit potential here anyway?

#32 – Frontier Adventure Racing Inc. (VII of VII)

Sample Recommendation

**Note: Recommendation will vary, but will ideally be based around one of the three profitability levers presented on the previous page.*

Recommendation: After our discussion today, FAR should add more runs, through

1. providing more runs, particularly Raid the North
2. Expand into the US
3. Offer winter races

Supporting Data:

- Raid the North has higher profitability per team than AC, and in similar regions, which is why we propose offering more RtN
- RtN is selling out, so there is likely excess demand

Risks:

- Possible lack of brand recognition in US
- Possible increased risks or costs offering winter races (need to plow snow)

Next Steps:

- Competitive scan of races in US to explore where to start
- Further analysis to decide which Canadian provinces to expand RtN to

#33 – Competitive Threat in Cookies

Kearney

Analysis	Structure	Overall
Medium	Medium	Medium

Behavioural Prep

Tell me about a piece of feedback you received that you disagreed with. How did you handle it?
Tell me about a time when you led a team through a challenge.

Case Tracker		Case Stem
Industry	Consumer Products	Your client is a US based manufacturer of branded cookies (cookies that carry the name of the manufacturer). Recently, private label cookies (those carrying the name of the retailer or a retail-owned label) have emerged and threatened branded cookies. Private labels cookies are made by the same manufacturers that make branded cookies. We are going to evaluate the impact of this changing market on our client.
Type	Profitability Improvement, Supply Chain	
Leader	Interviewer	
Concepts	Market Sizing, Profitability, Marketing	

Interview Guidance
This is an interviewer-led case that follows the following key steps: 1) Market sizing 2) Competitive assessment 3) Strategic insight

#33 – Competitive Threat in Cookies (II of VII)

Sample structure

What is the impact of the emergence of private label cookies threatened branded cookies?

Sample solution:

What is the impact on Market landscape?

- How has the market size increased, and how does this impact our clients' percentage market share?
- Competitors: respective market shares, pricing strategies, branding, product offerings
- Do consumers have a preference for private label cookies or branded cookies?
 - Need to assess the customer segments, their respective price sensitivities, customer perceptions regarding quality, brand loyalty, etc.
 - Changes in economy that may have impacted customer preferences? Are these temporary or will they continue in the long-term?

What is the impact on the supply chain?

- Value Chain:
 - ingredients cost and availability (e.g., does the emergence of private label increase supplier power if there is now more competition for raw materials?)
 - production costs
 - production capacity (can capacity meet demands for both branded and private label?)
 - distribution (e.g., shelf space, distribution channels)

#33 – Competitive Threat in Cookies (III of VII)

Question 1

How large would you estimate the overall US cookie market in terms of dollars?

There is no right or wrong answer, but the interviewee should follow a logical structure and make reasonable assumptions. Push the interviewee to estimate values for each variable.

Sample response

US Population	330M
/ (3 ppl / household)	110M
X (10 boxes a year / household)	\$1.1B
X (\$5 average price per box)	\$5.5B

Regardless of what number is arrived at, tell the interviewee to use \$10B for the rest of the interview for simplicity.

Question 2

How large of a threat to you believe the trend in private label cookies is to your client?

Provide additional information given on the following page and Exhibit 1 when asked by the interviewee.

This question requires the interviewee to (1) figure out what is causing the trend towards private labels (2) decide if that trend will continue and (3) determine what extent the client is threatened by the increasing percent of the overall cookie market represented by private label.

They should get there by analyzing the information that they acquire by probing the interviewer for additional information (*see next pages for information*).

Key insights to be generated:

The trend towards private branded has historically been caused by:

- Retailers interest to utilize spare capacity
- Grocers' desire to sell products with their own brand
- Consumers' concerns about the economy (driving price vs. quality)

(2) The interviewee can decide if that trend is continuing based on his/her own interpretation of the information. The lack of industry capacity would suggest it is not, but the economic concerns suggest that the trend might continue.

(3) The private label segment originally cut into the competitors market share, but not the client's market share. The competitor reacted by entering the segment and they are now stealing overall market share from the client.

#33 – Competitive Threat in Cookies (IV of VII)

Information provided upon request

Private Label information

- Private label cookies emerged 10 years ago
- 5 years ago they made up 10% of the overall cookie market
- Now they make up 20% of the overall market

Pricing and Costs

Current market costs and prices per bag of cookies

	Private Label	Branded
Manufacturing Costs	\$1.50	\$2.00
Wholesale Price	\$2.00	\$2.75
Retail Price	\$2.50	\$3.50

Competitive Landscape

- The client only makes branded cookies
- Main competitor makes both branded and private labels
- A collection of small players make both

Quality

- Consumer studies have shown a noticeable difference in taste, texture, and quality in favour of branded cookies

Information provided upon request

Industry Capacity

- There has been some excess capacity at the smaller retailers and at the main competitor in the past (5 years ago)
- Today there is very little excess capacity in the industry

Retail Competition

- Competition amongst grocery stores is increasing
- Overall grocery stores are pushing private label branding across other food categories to populate shelves

Distribution Strategy

- Grocery stores: All sell branded, but also usually carry their own private label that accounts for 90% of sales
- Mass merchants: only sell branded

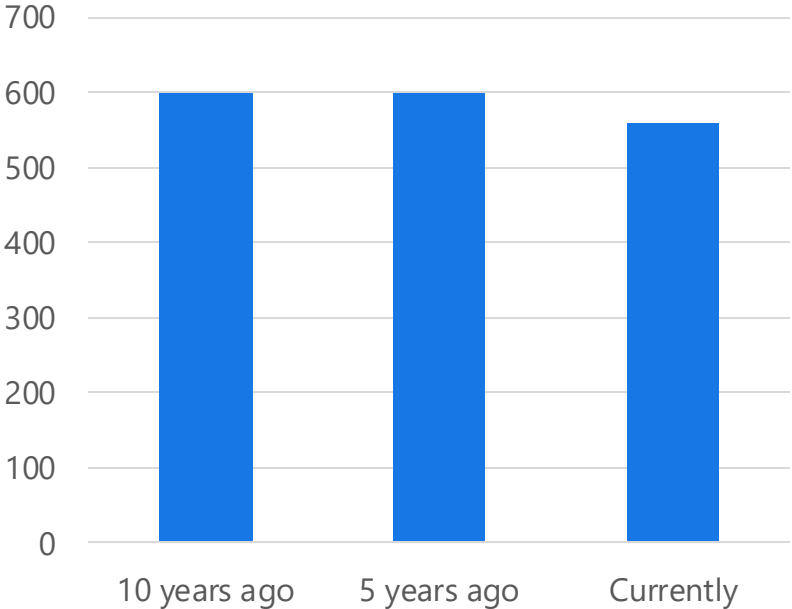
Economic Environment

- Economy has been slowing and there is concern about a recession

#33 – Competitive Threat in Cookies (V of VII)

Exhibit 1

Client Sales (\$M)



Market Share

Branded Cookie Segment of the Market

Years Ago	Client	Primary Compet.	Small Players
10 years ago	60%	30%	10%
5 years ago	67%	25%	8%
Currently	70%	23%	7%

Private Brand Cookie Segment of the Market

Years Ago	Client	Primary Compet.	Small Players
10 years ago	N/A	N/A	N/A
5 years ago	0%	0%	100%
Currently	0%	40%	60%

#33 – Competitive Threat in Cookies (VI of VII)

Question 3

What is the magnitude of the threat of private label to the client (High, Low, or None)? Please explain your reasoning.

The candidate should state the basis of their assessment in the previous question.

If the interviewee sees the risk as HIGH:

- slowing economy prompting customers to seek cost-effective options
- growing private label segment from 10% to 20% of overall cookie market over past 5 years
- grocery strategies pushing private label over branded
- retail competition increasing
- competitive pricing (lower manufacturing costs and lower retail prices compared to branded cookies)

If the interviewee sees the risk as LOW

- lack of excess capacity
- Customers prefer quality/taste of branded cookies

#33 – Competitive Threat in Cookies (VII of VII)

Question 4

Based on your assessment of the level of threat, what is your final recommendation for the appropriate strategy the client should follow?

Sample recommendation for HIGH risk:

Recommendation & Rationale: the threat of competitor private label cookies is HIGH: due to slowing economy, the fact that they make up 20% of the market, groceries prefer private label:

- Recommend client to begin supplying private label products. The basis of the private label competition is primarily done on cost (ie. Cost is the main factor that influences competition or purchasing decisions).
- *Risks:* supplying private label products could cannibalize sales of client's branded product.
- *Next Steps:* client's branded product should be protected: undertake overhead cost reduction efforts, seek partnerships with retailers, gain maximum product knowledge in the private branded segment, or explore deals with mass merchants.

Sample recommendation for LOW risk:

Recommendation & Rationale: the threat of competitor private label cookies is LOW (due to lack of excess capacity, product quality differentiation):

- Recommend client to stay with branded cookies only. The basis of competition is differentiation. Additionally, your client should do all it can to halt or reverse the momentum of the private label segment.
- *Risks:* there is a chance that private label cookies are a larger threat than we think, customers may be more price-sensitive than expected
- *Next Steps:* invest in brand image to support premium prices, innovate through line extensions, consider alternative distribution channels, manage price gap, and explore price sensitivity.

#34 – Bell Computer Co

Analysis	Structure	Overall
Hard	Hard	Hard

Behavioural Prep

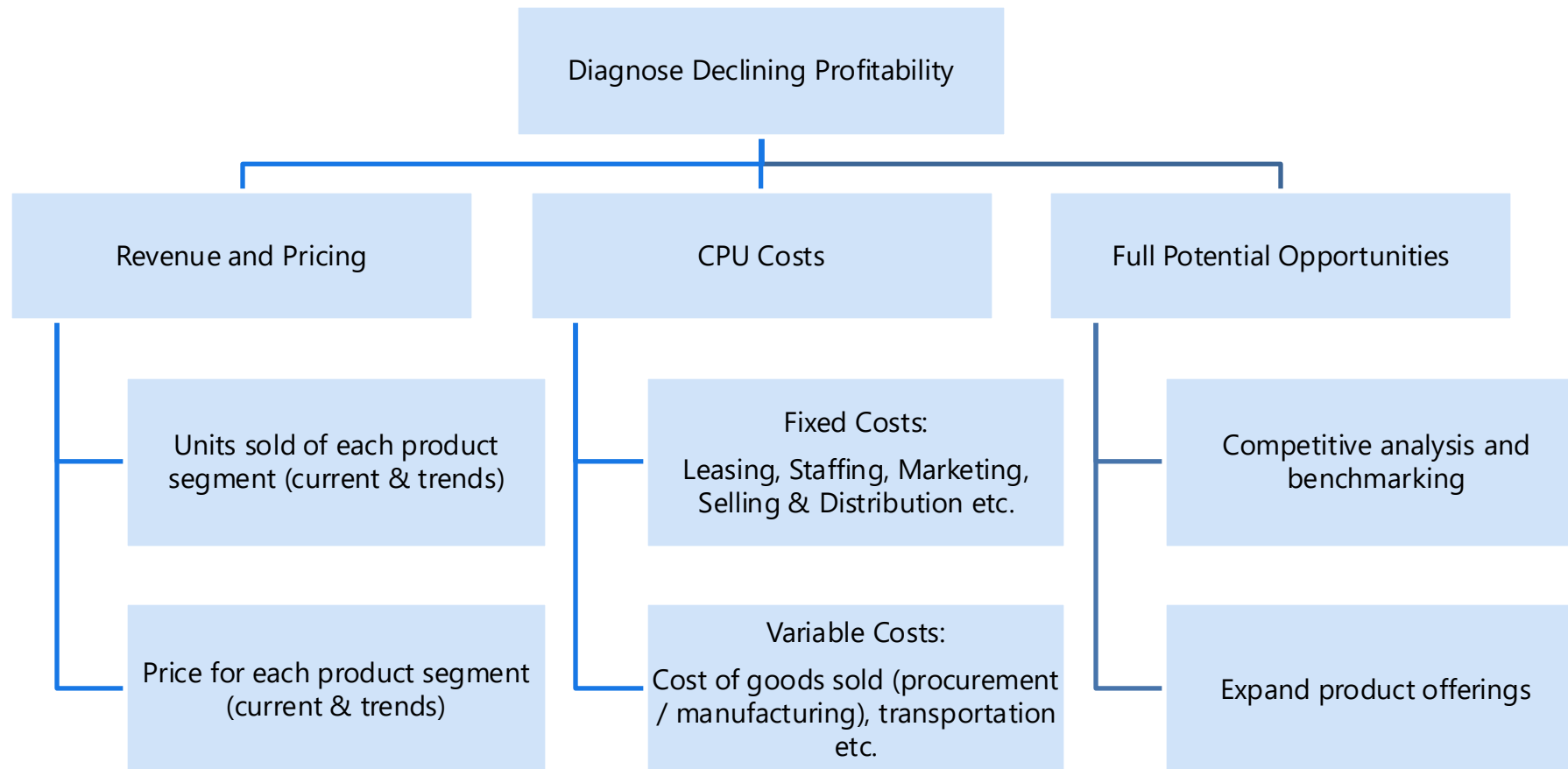
Why consulting?
Give me an example of a business problem your company faced and tell me how you solved it?

Case Tracker		Case Stem
Industry	Technology	Our client, Bell Computer Co, is the second largest PC manufacturer, by unit sales, in the United States. Over the past 5 years, Bell has been gaining market share and growing revenue, but at the same time, their net income is eroding. The founder of Bell has returned to the company and taken over as CEO. He has hired us to determine: • Why have our profit margins declined? • What can we do to improve our profitability and reach our “Full Potential”?
Type	Profitability	
Leader	Interviewee	
Concept	Marketing Strategy, Competitive Analysis, Supply/Value Chain	

Additional Information
Answers to clarifying questions: Industry Definitions • ASP = Average Selling Price, the average price sold by the company for a computer. Multiplying units sold × ASP gives total revenue. • Client Segment = PCs designed for, and sold to, retail consumers either directly or through retailers like Best Buy. Client Characteristics • Items Sold: Assume only hardware (as in Exhibit 1). Interviewee could suggest add-ons like printers, software, warranty services to improve profit. • Locations: U.S. domestic decision only; ignore international. Competitive Dynamics • All competitors face the same ASPs and sell comparable products.

#34 – Bell Computer Co (II of X)

Sample Structure



#34 – Bell Computer Co (III of X)

Interviewer Guide to case and handouts

Case Structure – Interviewee’s structure should be framed as a comparison of Bell Computer’s financials from Before (2020) vs. After (2025). The interviewee should note the “Profit Equation” and make the following inferences from the setup:

- Revenues are going up and Market Share is increasing AND Net Income is going down.
- Thus: Prices (ASPs) are being reduced to buy market share. This is causing more units to be sold, but with Net Income going down, it is clear that **costs have not reduced to keep up with the reduction in ASPs.**

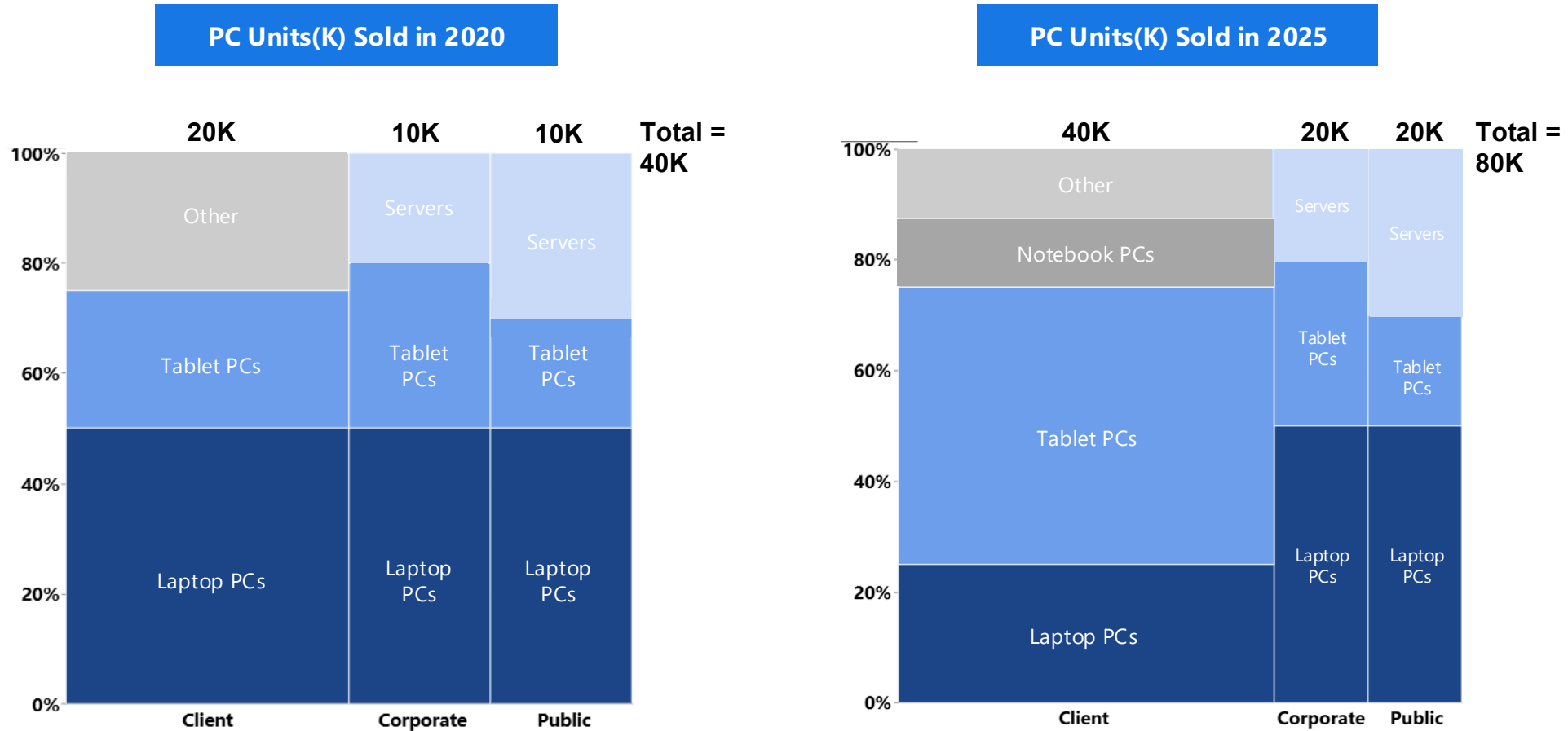
Exhibit 1–3 – After the interviewee lays out structure, they should ask questions about the components of Net Income (Revenue and Costs).

- Have a conversation with the interviewee to force them to talk through the essential components of the Profit Equation needed to answer the question.
- Let the interviewee drive the case. When they’ve asked enough about the following topics, hand them the corresponding exhibit
- Sales Units → Exhibit 1
- Prices → Exhibit 2
- Costs → Exhibit 3

Answer – The interviewee cannot solve this case without all three exhibits. Once they’ve given sufficient rationale to see each exhibit, they should drive through the case to answer both questions.

#34 – Bell Computer Co (IV of X)

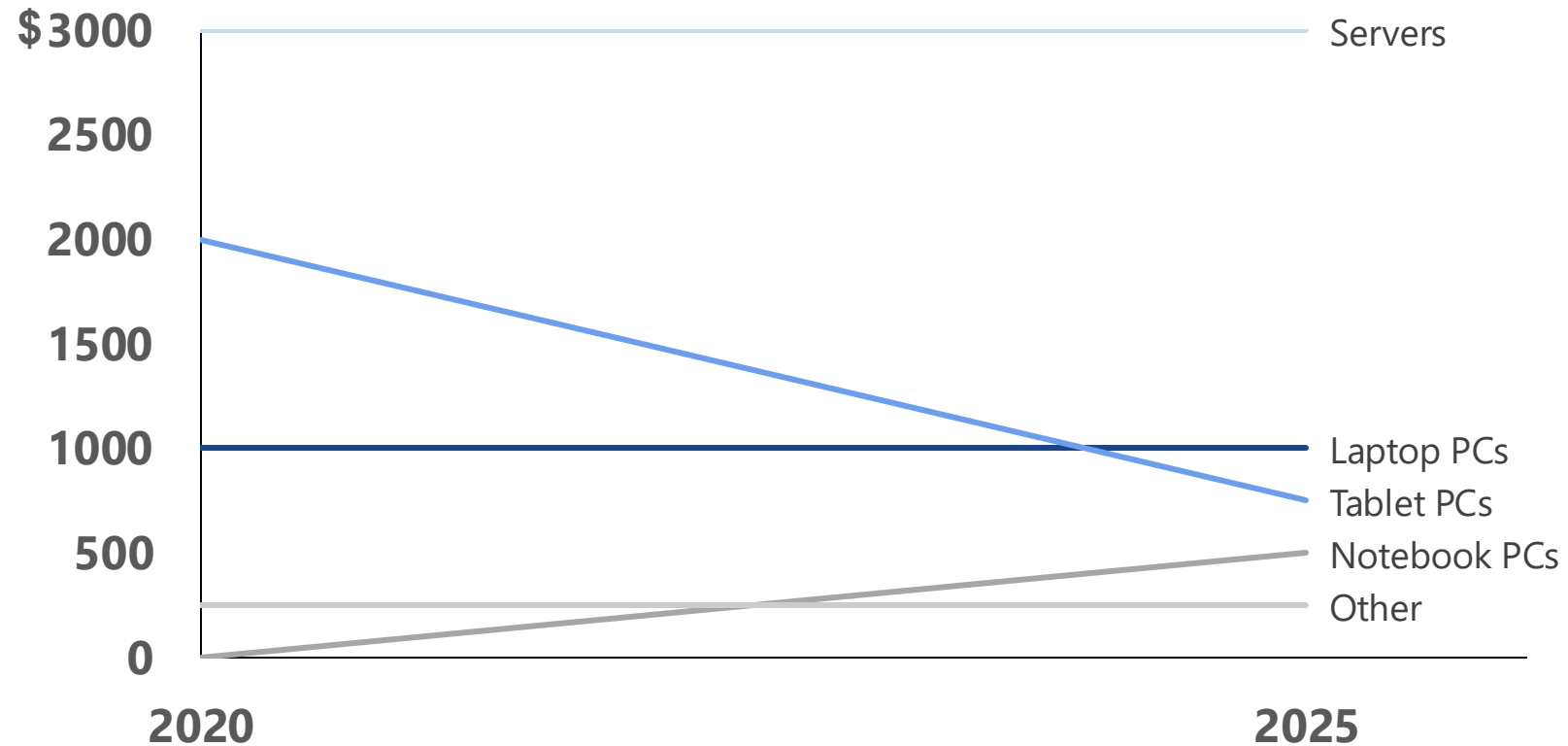
Exhibit 1: Product Mix breakdown by Business Segment for 2020 and 2025



*Note: Bell Computer Co did not introduce a Notebook PC model until after 2020; “Other” products include Pocket PCs and Calculators

#34 – Bell Computer Co (V of X)

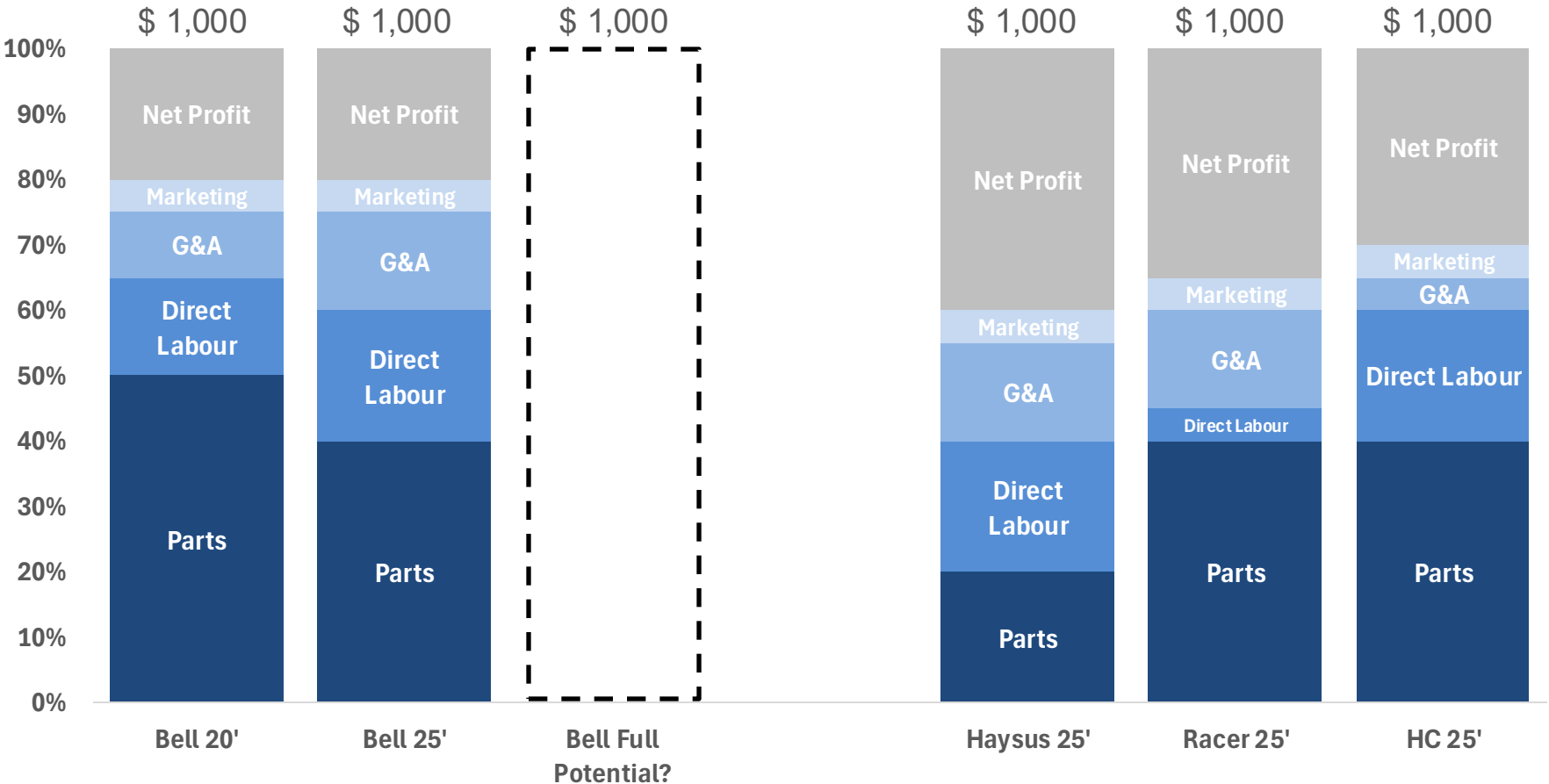
Exhibit 2: ASPs by Business Segment 2020 and 2025



*Note: Bell Computer Co did not introduce a Notebook PC model until after 2020; “Other” products include Pocket PCs and Calculators

#34 – Bell Computer Co (VI of X)

Exhibit 3: Avg. Per Unit Cost Breakdown of a Laptop PC for Bell (20' and 25') and Top 3 Competitors



*Note: Servers, Tablets, and Notebook Product types have the same costs per unit; however, Net Income will vary based on ASP. "Other" products have a total CPU \$200

#34 – Bell Computer Co (VII of X)

Case Math – Part I

Marketing Strategy

Using Exhibits 1–3, the interviewee should be able to determine that margins are falling due to product mix shifting toward Tablet PCs and price declines

The interviewee should be able to answer the question, “**Why have our profit margins declined?**” with the following rationale:

- Exhibit 1: Shift in Client preference towards Laptops & Netbooks
- Exhibit 2: Declining ASPs for Laptops and Low ASPs for Netbooks
- Exhibit 3: Cost position has remained unchanged

The interviewee could answer the question qualitatively by interpreting Exhibits 1–3. To make this case more quantitatively focused, ask the interviewee to calculate various metrics for 2020 and 2025

The interviewee should be able to calculate the following using Exhibits 1-3:

- Net Income per Unit by product type ①
- Revenue by product type and Total Revenue ②
- Cost by product type and Total Cost ③
- Net Income by product type and Total Net Income ④

Bell FY2020

	Laptop PCs	Tablet PCs	Notebook PCs	Servers	Other	TOTAL
Units Sold (K)	20.0	10.0	0.0	5.0	5.0	40.0
ASP	\$1,000.0	\$2,000.0	\$0.0	\$3,000.0	\$250.0	
CPU	800.0	800.0	0.0	800.0	200.0	
NI/Unit	\$200.0	\$1,200.0	\$0.0	\$2,200.0	\$50.0	①
Revenue (\$M)	\$20.0	\$20.0	\$0.0	\$15.0	\$1.3	\$56.3 ②
Total Cost (\$M)	16.0	8.0	0.0	4.0	1.0	\$29.0 ③
NI (\$M)	\$4.0	\$12.0	\$0.0	\$11.0	\$0.3	\$27.3 ④
Profit Margin	20.0%	60.0%		73.3%	20.0%	48.4%

Bell FY2025

	Laptop PCs	Tablet PCs	Notebook PCs	Servers	Other	TOTAL
Units Sold (K)	30.0	30.0	5.0	10.0	5.0	80.0
ASP	\$1,000.0	\$750.0	\$500.0	\$3,000.0	\$250.0	
CPU	800.0	800.0	800.0	800.0	200.0	
NI/Unit	\$200.0	(\$50.0)	(\$300.0)	\$2,200.0	\$50.0	①
Revenue (\$M)	\$30.0	\$22.5	\$2.5	\$30.0	\$1.3	\$86.3 ②
Total Cost (\$M)	24.0	24.0	4.0	8.0	1.0	\$61.0 ③
NI (\$M)	\$6.0	(\$1.5)	(\$1.5)	\$22.0	\$0.3	\$25.3 ④
Profit Margin	20.0%	(6.7%)	(60.0%)	73.3%	20.0%	29.3%

#34 – Bell Computer Co (VIII of X)

Case Math – Part II

Competitive Analysis + Supply Chain

Using Exhibit 3, the interviewee should identify a “Full Potential” improvement for Bell to improve its margins by copying the strategy of its direct competitors

The interviewee should be able to answer the question, “**What can we do to improve our profitability and reach our ‘Full Potential’?**” with the following **COST REDUCTION** strategy:

- Copy Haysus’ strategy for part procurement to reduce the build cost of PCs
- Copy Racer’s strategy for Direct Labor to reduce the labor cost of producing PCs
- Copy HC’s strategy for G&A to reduce the total company overhead

This will reduce the CPU from \$800 in 2020 to \$350 in 2025 (see table on right)

The interviewee could answer the question qualitatively by interpreting Exhibit 3. To make this case more quantitatively focused, ask the interviewee to calculate various metrics for “Full Potential” with the Units Sold (K) from 2025:

- Net Income per Unit by product type ①
- Revenue by product type and Total Revenue ②
- Cost by product type and Total Cost ③
- Net Income by product type and Total Net Income ④

	BELL Full Potential					TOTAL
	Laptop PCs	Tablet PCs	Notebook PCs	Servers	Other	
Units Sold (K)	30.0	30.0	5.0	10.0	5.0	80.0
ASP	\$1,000.0	\$750.0	\$500.0	\$3,000.0	\$250.0	
CPU	350.0	350.0	350.0	350.0	200.0	
NI/Unit	\$650.0	\$400.0	\$150.0	\$2,650.0	\$50.0	①
Revenue (\$M)	\$30.0	\$22.5	\$2.5	\$30.0	\$1.3	\$86.3 ②
Total Cost (\$M)	10.5	10.5	1.8	3.5	1.0	\$27.3 ③
NI (\$M)	\$19.5	\$12.0	\$0.8	\$26.5	\$0.3	\$59.0 ④
Profit Margin	65.0%	53.3%	30.0%	88.3%	20.0%	68.4%

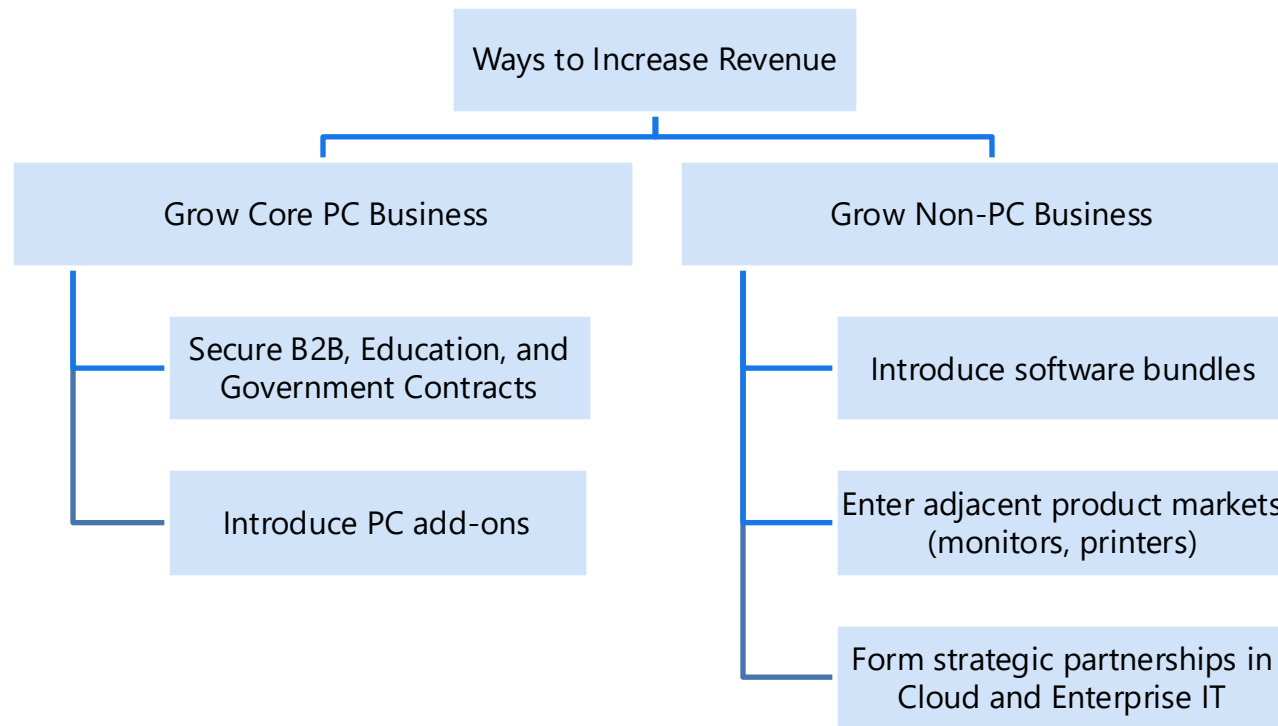
Results:

	Laptop PCs	Tablet PCs	Notebook PCs	Servers	Other	TOTAL
NI (\$M) Improved	\$13.5	\$13.5	\$2.3	\$4.5	\$0.0	\$33.8
Cost Per Box Reduced	\$450.0	\$450.0	\$450.0	\$450.0	\$0.0	

#34 – Bell Computer Co (IX of X)

Brainstorming

Once the interviewee completes the Full Potential cost reductions, push them to brainstorm ideas on how to increase the revenue with the following question: **“In addition to reducing costs, what are potential growth levers Bell Computer Co could explore to drive revenue growth in the next 3–5 years?”**



This is a non-exhaustive list of ideas. **A great candidate will also give a recommendation for which way the client should go and also a few top-of-mind risks for their recommendation.**

#34 – Bell Computer Co (X of X)

1. Recommendation

- Bell Computer Co's profit margins have declined due to decreasing average price decreases in the Tablet PC product segment and shifting client preferences to Notebook PCs
- To return to profitability, they must reduce their cost per unit sold through "Full-Potential" cost reductions

2. Supporting Data

- Bell Computer Co's ASP for Tablet PC's has decreased from \$2000 in 2020 to \$750 in 2025 despite a constant cost structure; profit margins and net income have deteriorated despite increases in revenue
- Using our competitor's line-item costs as a benchmark, we could potentially reduce our cost per box by \$450, amounting to a total Net Income improvement of roughly \$34M

3. Risks and Next Steps

- **Risks:** Cutting costs could negatively impact product quality
- Workforce reduction or labor changes could affect productivity and morale
- Competitors may respond with further price cuts, intensifying commoditization
- **Next Steps:** Phase cost reductions to protect product quality; prioritize operational efficiencies before supplier changes
- Pair cost cuts with selective growth plays (e.g. services, warranties) to avoid a race to the bottom
- Conduct market research to track impact of cost, quality, and customer metrics to ensure sustainable execution

#35 - Sheep Of Wall Street Co.

Analysis	Structure	Overall
Medium	Medium	Medium

Behavioural Prep

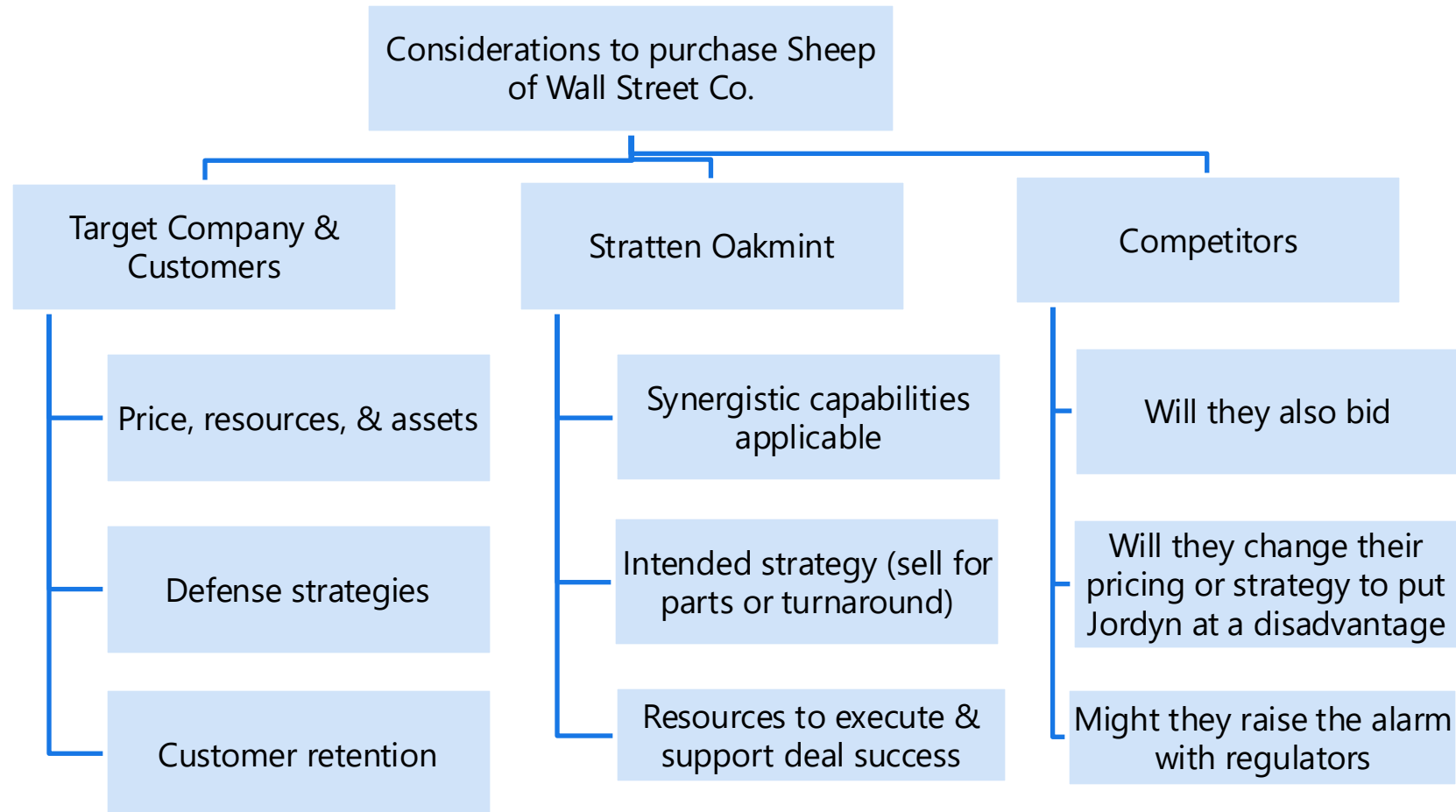
Tell me about a time you had a conflict with a teammate.
Tell me about the best piece of feedback you ever received.

Case Tracker		Case Stem
Industry	Financial services	You have been hired by Stratten Oakmint, a private equity firm. Currently, the CEO Jordyn Belfert has his eye on purchasing a publicly traded company, Sheep of Wallstreet Co., via a hostile takeover. What should Jordyn consider before he does this?
Type	M&A – Financial Buyer (PE)	
Leader	Interviewer	
Concept	Opportunity Assessment	

Additional Information
Answers to clarifying questions: • How does Jordyn’s private equity firm make money? • His firm purchases companies and makes a profit from selling them for parts or through bringing in a new management team that executes a turnaround strategy to make the business more profitable. • What is a hostile takeover? • A type of acquisition where an acquirer takes control of a target company without consent from the target company’s board of directors. • Why is Jordyn considering purchasing this target company? • To generate alpha (outperformance of stock market benchmarks) for his investors.

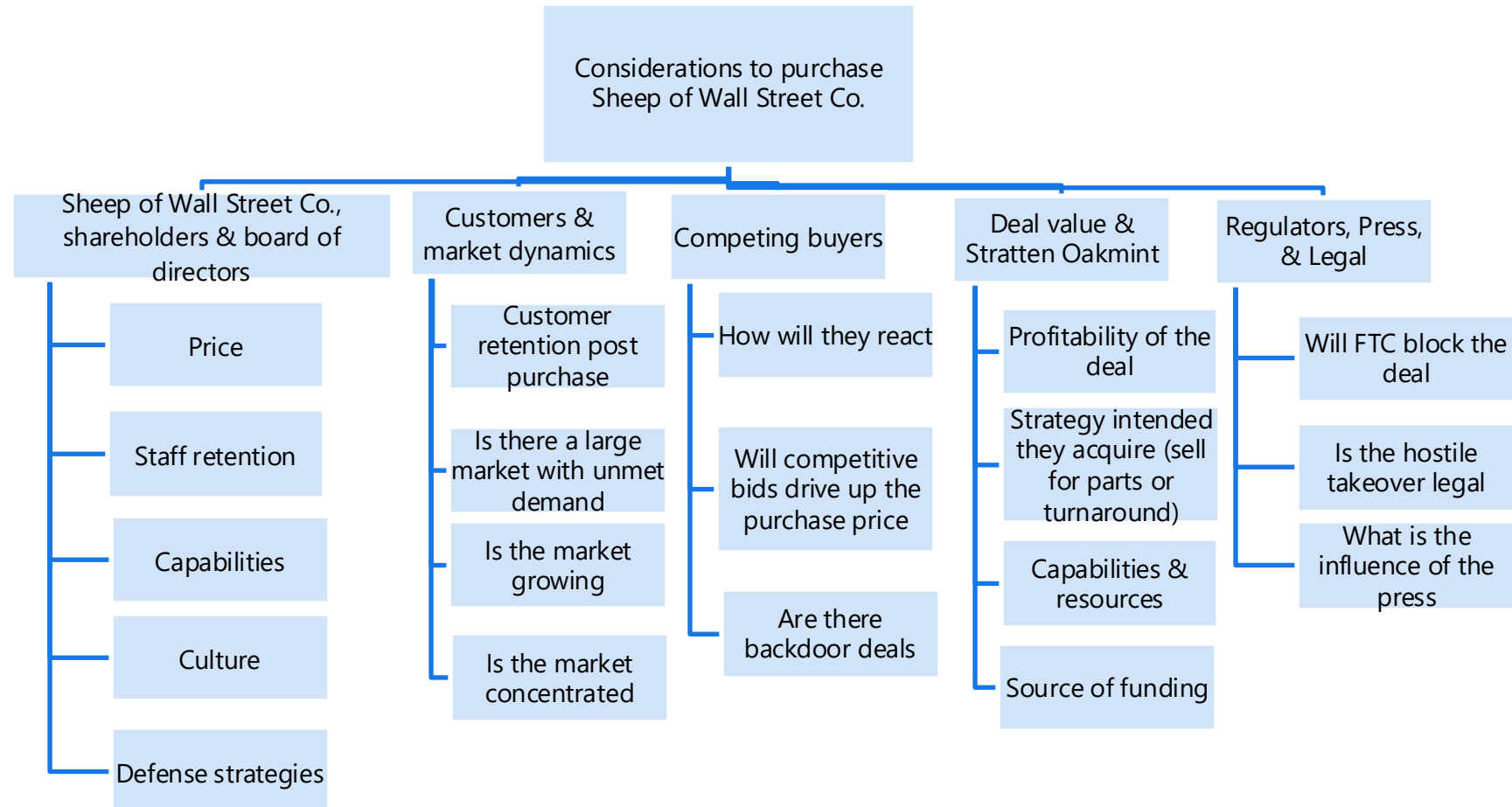
#35 - Sheep Of Wall Street Co (II of IX)

Sample Structure - Basic



#35 - Sheep Of Wall Street Co (III of IX)

Sample Structure - Advanced



#35 - Sheep Of Wall Street Co (IV of IX)

Case creative question and answer

Jordyn is unsure of how he wants to fund this deal. What are funding options Jordyn should consider?

Level 1 Answer	Level 2 Answer
1.Cash. 2.Debt. 3.Partnerships.	1.Stratten Oakmint assets fund the deal. a) Cash. b) Debt. 2.Other peoples' money funds the deal. a) Partnerships. •Sheep of Wall Street Co. competitors. •Stratten Oakmint corporate peers. •Others. b) Investor money in the PE fund finances the deal. •High-net-worth individual investors. •Pension funds. •Endowments. •Insurance companies.

#35 - Sheep Of Wall Street Co (V of IX)

Case Math Question

Jordyn believes he can purchase the company for \$500,000,000. If he purchases Sheep of Wall Street Co., what should he do with it?

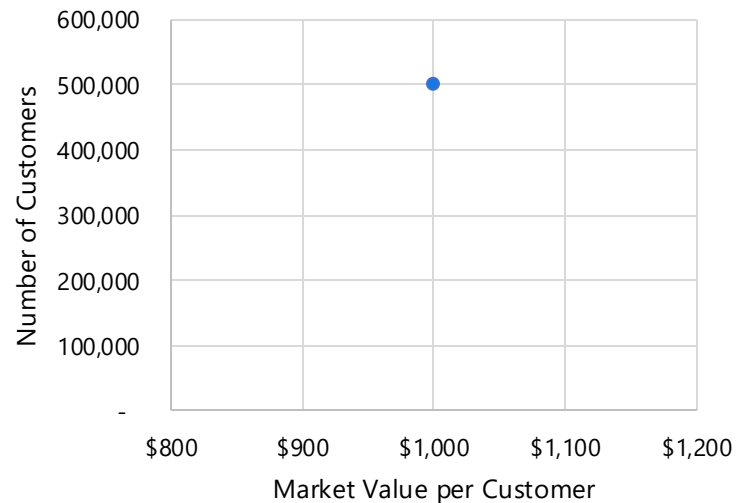
Should he sell it for parts or execute a turnaround strategy? He is interested in the option that generates the greatest value. He shares the following data with you. Provide a final recommendation.

#35 - Sheep Of Wall Street Co (VI of IX)

Exhibit 1 and 2

Turnaround Strategy

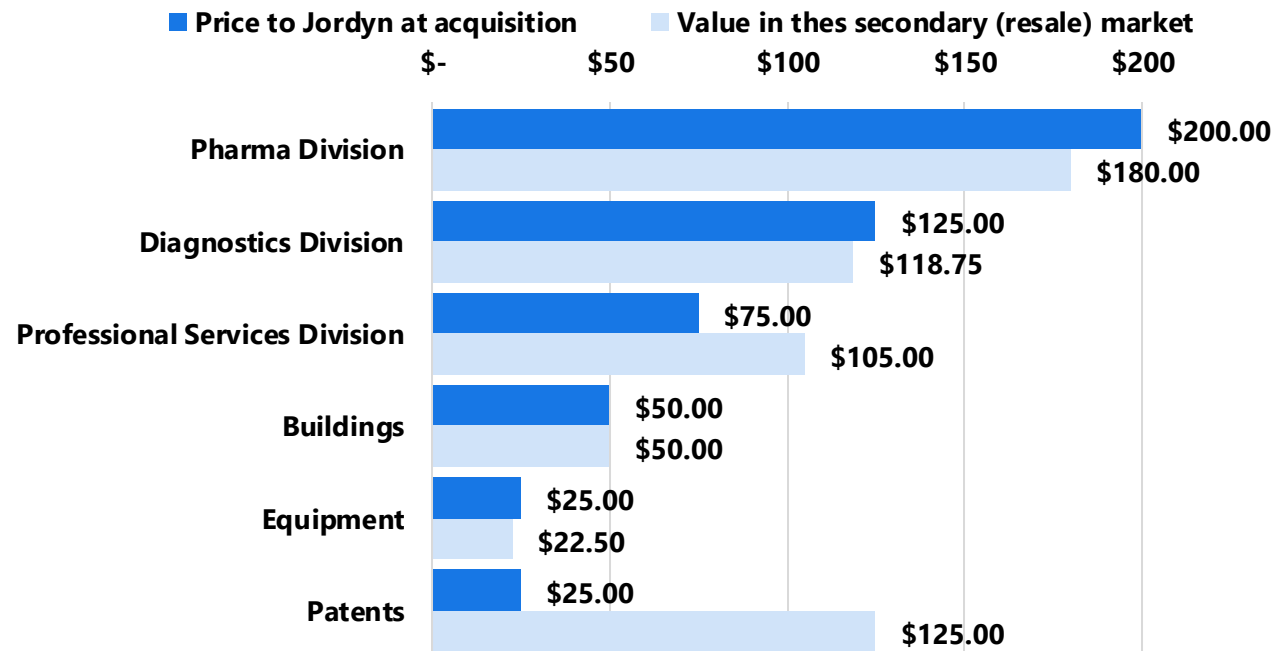
Market Value per Customer vs
Number of Customers



Note: Jordyn believes that he can add 100,000 new customers, if he selects the turnaround strategy option.

Sell for Parts Strategy

Value of Sheep of Wallstreet Co. Assets (\$MM)



#35 - Sheep Of Wall Street Co (VII of IX)

Case Math Answer

1. Evaluate the value generated from purchasing the company for \$500,000,000 and selling the sum of its parts in the secondary market. Examine the bar chart. The candidate can ignore the bar charts displaying the **Assets price to Jordyn at acquisition** because from the prompt we know he is purchasing the company for \$500,000,000. Sum up the value of the **Price in the secondary Market** bars to arrive at secondary market value of \$601,250,000 ($\$180,000,000 + \$118,750,000 + \$105,000,000 + \$50,000,000 + \$22,500,000 + \$125,000,000 = \$601,250,000$). Compare the price Jordyn purchased the Sheep of Wall Street Co. vs what he can sell the sum of its parts for to unlock \$101,250,000 of value ($\$601,250,000 - \$500,000,000 = \$101,250,000$).
2. Evaluate the Turnaround Strategy option where the current value of the company is \$500,000,000 (from the prompt & the candidate can also get this by multiplying the market value per customer x # of customers). The candidate can assess how much value is unlocked by this strategy simply by multiplying the number of net new customers by the market value per customer to realize the value unlocked by the Turnaround Strategy is \$100,000,000.
3. Compare the value unlocked by both options to realize that the selling for parts option unlocks \$1,250,000 more value than the turnaround option, and hence should be selected ($\$101,250,000 - \$100,000,000 = \$1,250,000$).

#35 - Sheep Of Wall Street Co (VIII of IX)

Case Math Answer

Level 1 Answer	Level 2 Answer	Level 3 Answer
-Jordyn should purchase Sheep of Wall Street Co. & sell it for parts.	-Jordyn should purchase Sheep of Wall Street Co. & sell it for parts since it unlocks maximum value (\$101,250,000) versus a turnaround strategy.	-Jordyn should purchase Sheep of Wall Street Co. & sell it for parts since it unlocks maximum value (\$101,250,000) versus a turnaround strategy. Next, I'd like to examine if there are any other considerations besides maximum financial return we should consider – for example, the turnaround strategy offers only \$1.25M less value but could offer other longer-term or intrinsic benefits.

#35 - Sheep Of Wall Street Co (IX of IX)

1. Recommendation

- Jordyn should purchase Sheep of Wall Street Co. & sell it for parts.

2. Supporting Data

- It unlocks \$101,250,000 of value.
- It unlocks \$1,250,000 more value than a turnaround strategy.
- The professional services division & patent value are significantly undervalued that drive an opportunity in the secondary (resale) market.

3. Risks and Next Steps

- Jordyn should perform competitive intelligence on competitors to ensure no other deals are in motion that could impact the purchase price of the company.
- Jordyn should consult with legal to evaluate if regulators will block the deal.
- Jordan should consider customer retention post the acquisition to validate the post-purchase value of the professional services division.
- Jordyn should examine what is driving the value of the patent to see if there are any alternative and higher value uses for it than selling it in the secondary (resale) market.

#36 - WEBFLIX

Analysis

Medium

Structure

Medium

Overall

Medium

Behavioural Prep

Tell me about yourself.
Tell me about a time when you exhibited leadership.

Case Tracker

Industry	Technology/Media
Type	Pricing Strategy
Leader	Interviewee
Concept	Marketing strategy, consumer research

Case Stem

Webflix is a U.S.-based OTT streaming company that is over 3 decades old.

Webflix is the incumbent in the streaming company and has built strong brand equity and loyalty amongst its subscriber base.

However, over the last few years, competition in the streaming space has intensified significantly and there have been a lot of new entrants into the market. As a result of the increase in options, consumers are becoming increasingly price sensitive to subscription fees.

The CEO of Webflix is currently evaluating their annual pricing strategy and wants your help in deciding the best path forward.

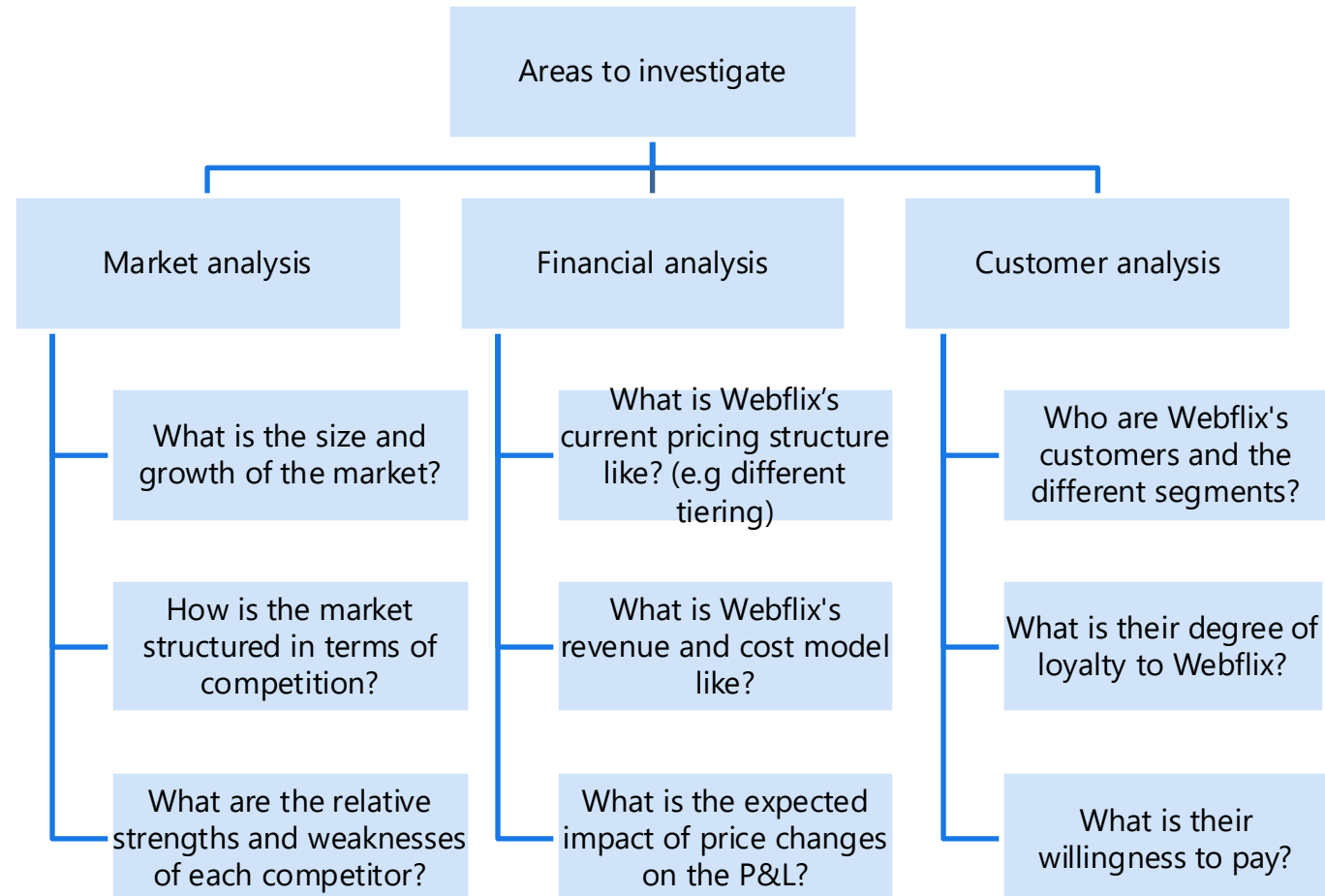
Additional information

Answers to clarifying questions:

- Webflix is U.S. based but its content is streamed globally
- For this case, let's only focus on the U.S. market because it forms the bulk of Webflix's revenue
- Webflix only has one pricing tier
- The main differentiator between streaming providers is the type and quality of content on the platform. Streaming providers have two types of content – (1) self produced and (2) third-party (third party content is content that streaming providers "rent" from production houses like Warner Brothers

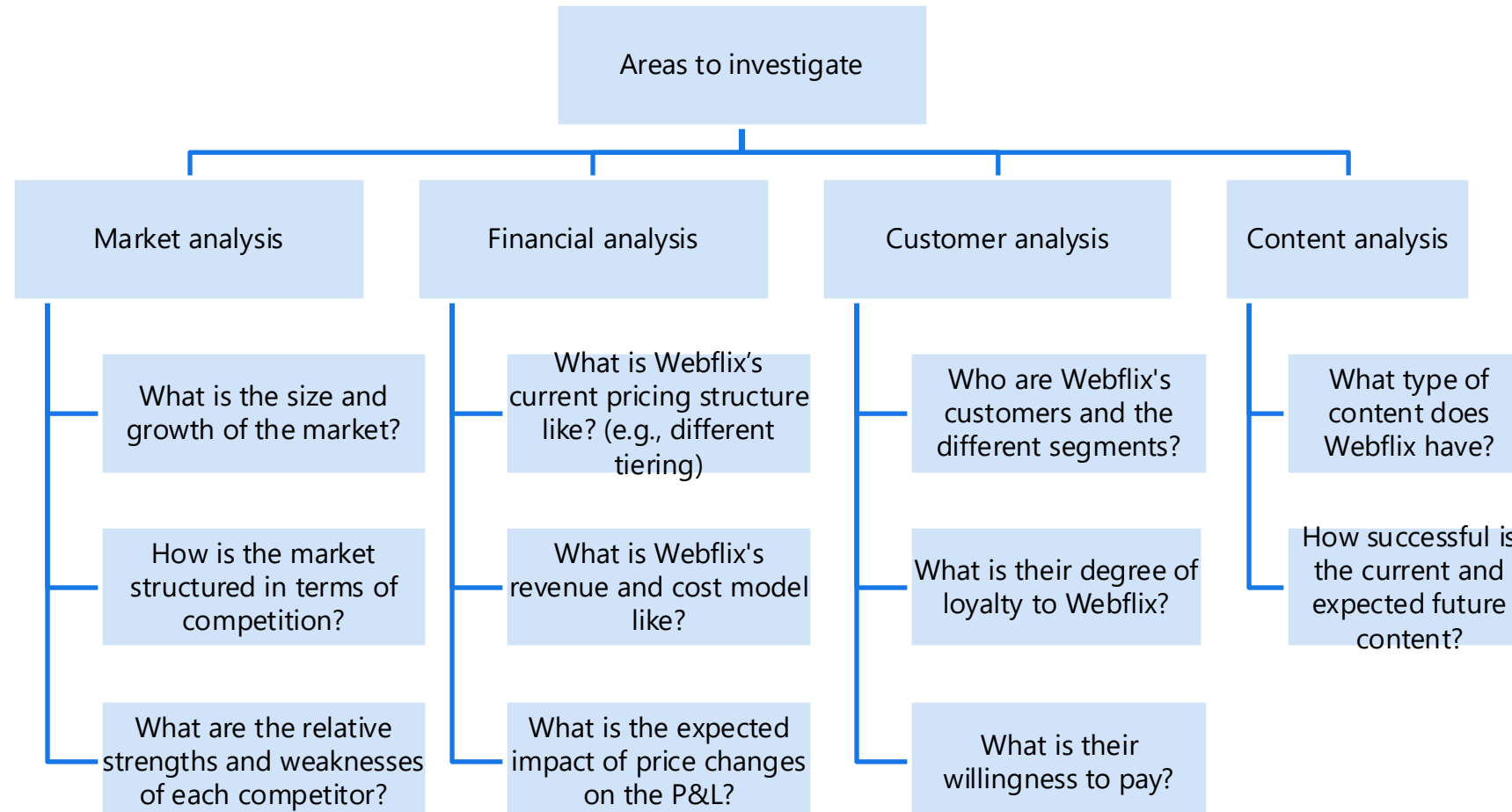
#36 - WEBFLIX (II of IX)

Sample Structure - Basic



#36 - WEBFLIX (III of IX)

Sample Structure - Advanced



#36 - WEBFLIX (IV of IX)

Case creative question and answer

During your first meeting with the CEO, both of you discuss the possibility of a subscription pricing increase over the next year.

One obvious concern of implementing a price increase is the potential loss of subscribers.

The CEO asks you how would you approach this problem and what ideas you have to minimize this risk.

Level 1 Answer	Level 2 Answer
- Less creative with no structured framing: - Understand our subscribers' willingness to pay (WTP) - Analyze the impact of different price points on revenue and profit (i.e., elasticity of demand) - Producing higher quality content and offerings that can justify the price changes	- More creative ideas with structured framing (non-exhaustive examples below): - Segmentation and analysis - Segment the base into different segments and understand WTP of each segment - Perform data analysis to quantify impact on retention rates and P&L - Implementation approach - Implement a gradual increment over time with higher WTP segments getting an accelerated increase over lower WTP segments - Introduce different subscription tiers with differentiated experiences/ offerings - Content library and value-added services - Increasing the quality of content in Webflix's library - Differentiating the type of content available under each subscription tier - Adding value-added services for higher tiers (e.g., 4K streaming, watch parties etc.) - Communication and transparency - Be extremely clear on comms to customers on the rationale for these increases - Implement feedback loops for Webflix to understand the impact and feedback from customers

#36 - WEBFLIX (V of IX)

Case Math Question

After your initial discussion with the CEO, the strategy team of Webflix has done some research and shared data on two different subscription price points and the associated impact on the P&L.

Please look at the exhibit on the next page and let me know what Webflix's optimum pricing is.

#36 - WEBFLIX (VI of IX)

Exhibit 1

U.S. streaming industry data:

# of streaming subscribers in whole of U.S (in M)	200
Average cost of third-party content titles per subscriber (in \$)	2.0
Other variable costs per subscriber (in \$)	1.0

Webflix's data:

Current pricing model metrics:

Current subscription price per month (\$)	15
Current market share of subscribers (%)	50%
Own produced content cost per month (in \$ M)	800
Other fixed costs per month (in M)	200

Pricing model #1 (moderate increase):

Subscription price per month (\$)	20
Estimated market share of subscribers (%)	40%

Pricing model #2 (aggressive increase):

Subscription price per month (\$)	25
Estimated market share of subscribers (%)	30%

#36 - WEBFLIX (VII of IX)

Case Math Answer

Current P&L per month:

# of Weblifx's subscribers	100,000,000
(*) Subscription per month	15

Total revenue per month	1,500,000,000
--------------------------------	----------------------

Content costs:

Own produced content	800,000,000
Third-party content	200,000,000

Other variable costs	100,000,000
-----------------------------	-------------

Other fixed costs per month	200,000,000
------------------------------------	-------------

Total costs per month	1,300,000,000
------------------------------	----------------------

Total profit per month	200,000,000
-------------------------------	--------------------

Pricing model #1 P&L per month:

# of Weblifx's subscribers	80,000,000
(*) Subscription per month	20

Total revenue per month	1,600,000,000
--------------------------------	----------------------

Content costs:

Own produced content	800,000,000
Third-party content	160,000,000

Other variable costs	80,000,000
-----------------------------	------------

Other fixed costs per month	200,000,000
------------------------------------	-------------

Total costs per month	1,240,000,000
------------------------------	----------------------

Total profit per month	360,000,000
-------------------------------	--------------------

Pricing model #2 P&L per month:

# of Weblifx's subscribers	60,000,000
(*) Subscription per month	25

Total revenue per month	1,500,000,000
--------------------------------	----------------------

Content costs:

Own produced content	800,000,000
Third-party content	120,000,000

Other variable costs	60,000,000
-----------------------------	------------

Other fixed costs per month	200,000,000
------------------------------------	-------------

Total costs per month	1,180,000,000
------------------------------	----------------------

Total profit per month	320,000,000
-------------------------------	--------------------

#36 - WEBFLIX (VIII of IX)

Case Math Answer

Level 1 Answer	Level 2 Answer	Level 3 Answer
- Webflix's revenue model is straightforward – it's based on the number of subscriptions per month multiplied by the subscription fee. - Webflix's cost model has both fixed and variable components - Fixed costs are own produced content and other fixed costs. - Variable costs are third-party content titles and other variable costs. - Webflix's market share goes down by 10% with each \$5 increase in subscription price.	- The third-party content cost that we pay to other production houses is likely charged based on a per-view basis (i.e., for each subscriber that views a third-party title on our platform, we will need to pay the production house a fee). That cost averages out to be \$2 per subscriber. - Webflix's own produced content costs \$800M per month to produce. This is likely their average budget per month across all productions. - Webflix's market share goes down by 10% with each \$5 increase in subscription price but need to weigh the price increase impact against the loss in market share.	- Based on the data shown to us, pricing model #1 (\$20 per month) is the optimum approach for us. This not only gives us a higher revenue per month of \$1.6B but also a higher profit per month of \$360M. - The reason for the increase in revenue is the positive impact from the price increase more than offsets the loss in subscribers. - The reason for the increase in profit is because variable costs (i.e., third party content and other variable costs) go down with the loss in subscribers.

#36 - WEBFLIX (IX of IX)

Recommendation

- I recommend Webflix increase its subscription price from \$15 per month to \$20 per month.

Supporting Data

- Increasing its subscription price to \$20 per month will increase revenue by \$100M per month and profit by \$160M per month.

Risks and Next Steps

- Risks:
 - Market share loss – Even though doing this yields great financial benefit, we will be losing 10% in market share to competition. We need to balance financial gains with platform health in the long-run.
- Next steps:
 - Understand in detail impact of loss in market share on platform health (e.g., future growth potential) and competitive dynamics (e.g., will competition be significantly stronger with greater share).

#37 - Death Valley Energy

Analysis	Structure	Overall
Medium	Medium	Medium

Behavioural Prep

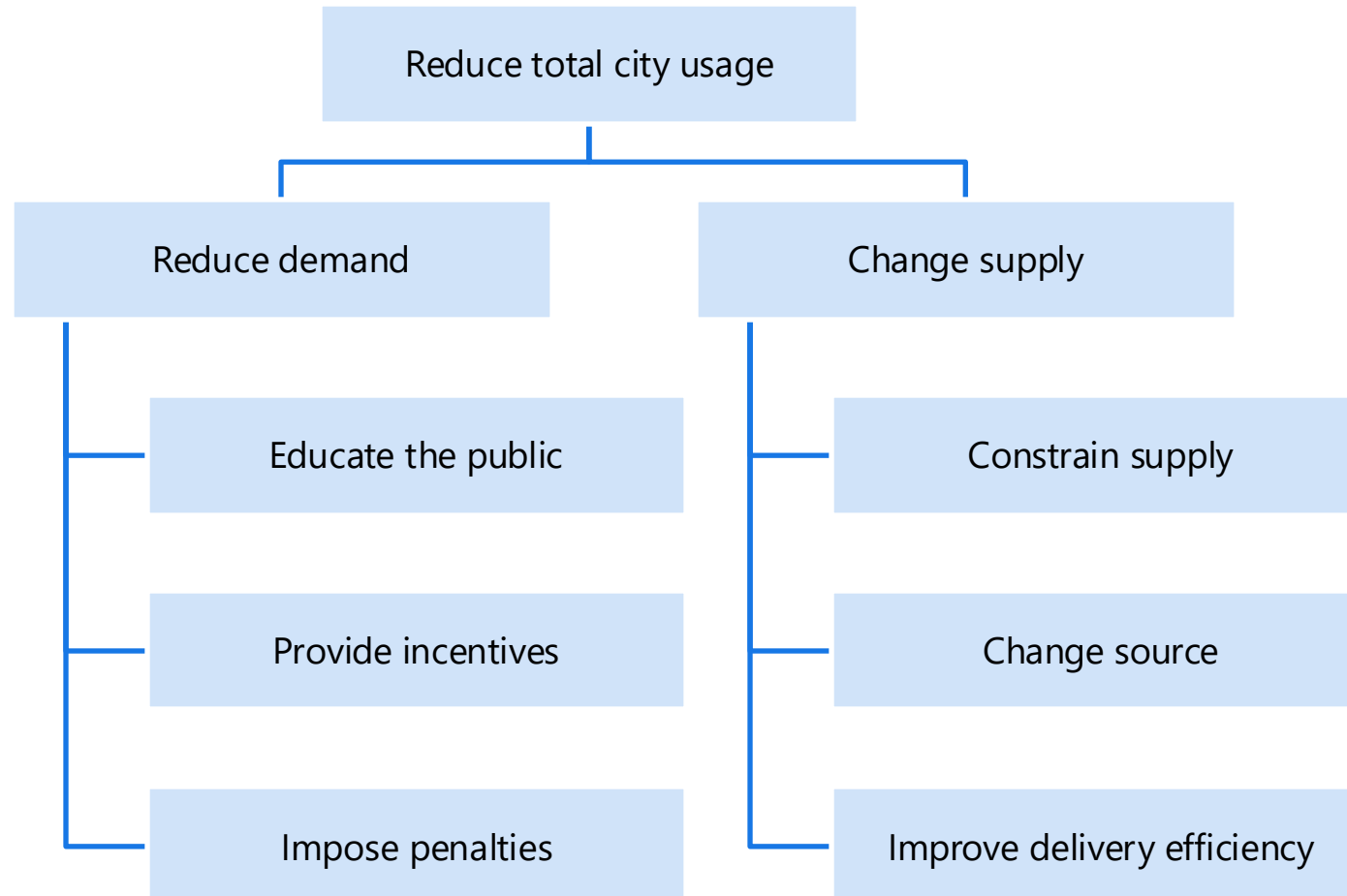
Tell me about a time when you had to convince a difficult client to follow your recommendations.
What is your proudest accomplishment?

Case Tracker		Case Stem
Industry	ESG	The Mayor of Death Valley, Mayor Metalika, wants to help the city reduce its total energy usage by 10% in 5 years (by 2028) to reach its ESG goals. Death Valley is a city located in the California desert and is thought to be the hottest place on Earth during the summer. How might the Mayor Metalika reduce energy usage?
Type	Market Study - Strategy	
Leader	Interviewee	
Concept	Market Strategy, Sustainability, Cost-Benefit Analysis	

Additional Information
Answers to clarifying questions: • Why does Mayor Metalika want to reduce energy usage? • To comply with a state law that goes into effect in 2028 and which may impose a financial penalty on the city for non-compliance. • What is meant by “energy usage?” • Energy usage is electricity that is delivered to any type of real estate within the jurisdiction of Death Valley. • What is current energy usage? • That is a good question. We don’t yet have that data but may be able to gather it later.

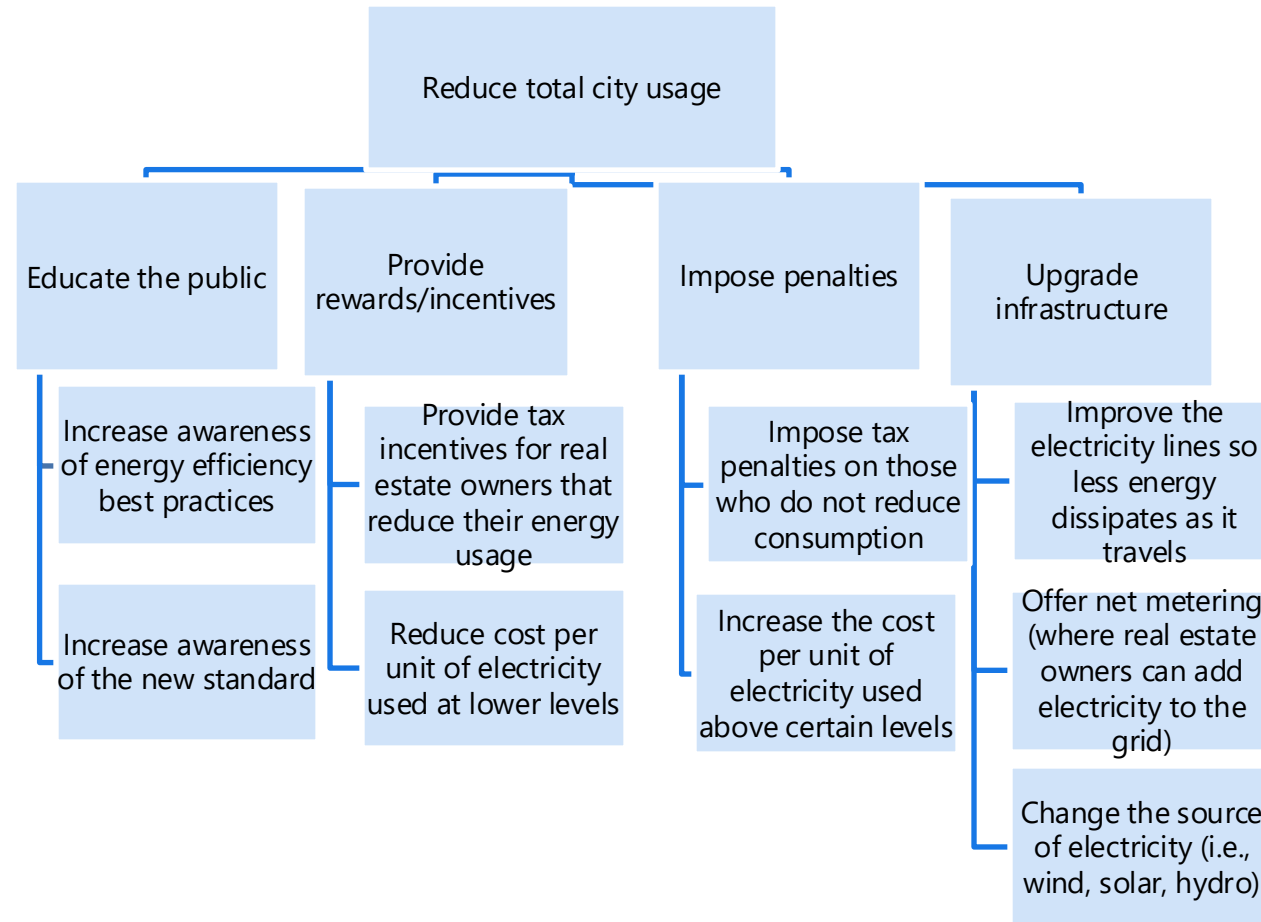
#37 - Death Valley Energy (II of IX)

Sample Structure - Basic



#37 - Death Valley Energy (III of IX)

Sample Structure - Advanced



#37 - Death Valley Energy (IV of IX)

Case creative question and answer

If the mayor invests in an advertising campaign intended to motivate the public to reduce their energy consumption, what types of messages might he consider including?

Level 1 Answer	Level 2 Answer
1. Instill perceived value of compliance. 2. Instill perceived fear of non-compliance.	1. Instill perceived value of compliance. - Financial benefit to audience. - Non-financial/emotional benefit to audience. 2. Instill perceived fear of non-compliance. - Financial impact. - Environmental impact. 3. Increase perceived ease of compliance. - Easy changes in behavior. (i.e., unplug devices from outlets when not in use) - Easy changes in systems (i.e., LED lighting)

#37 - Death Valley Energy (V of IX)

Case Math Question

After an expensive advertising campaign educating the public on energy efficient best practices failed to motivate residents to reduce their energy usage, the Mayor is starting to become desperate.

He needs the city to reduce its total energy usage by 10% by 2028.

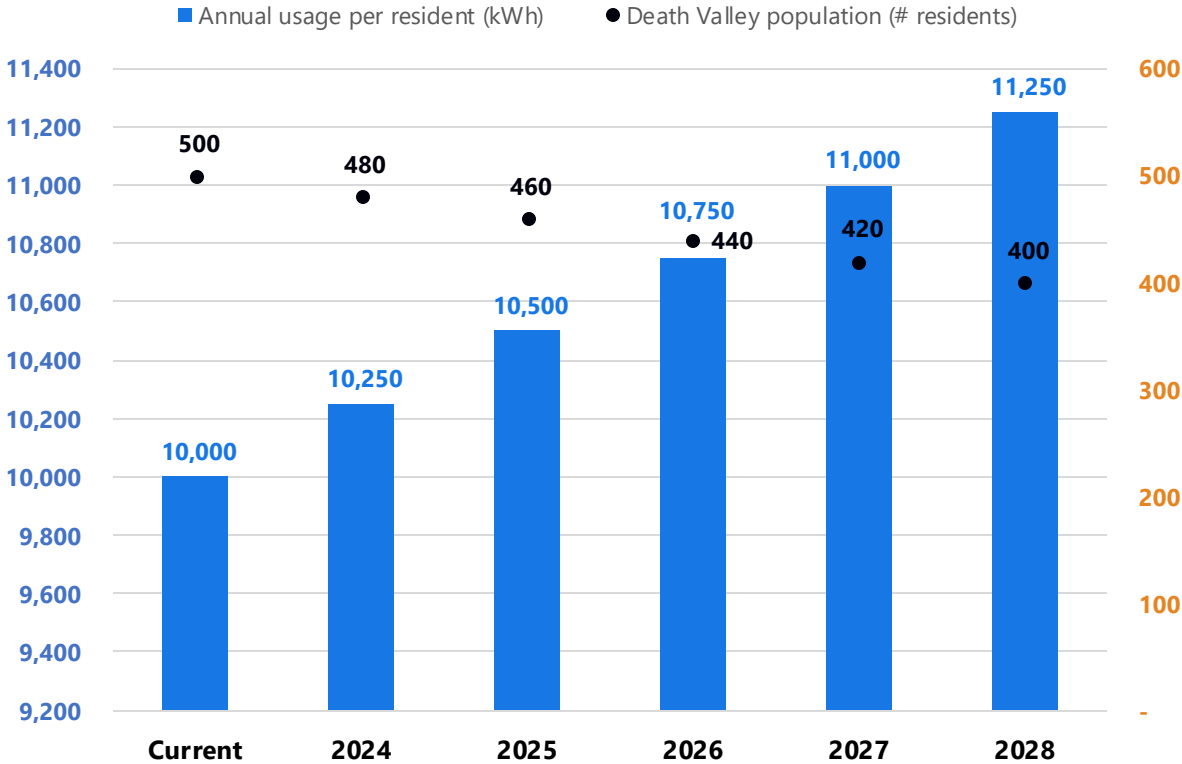
He provides you with the following information and wants to know if he should invest in upgrading powerlines, build a wind farm, or do nothing.

Provide a recommendation both before and after examining the following data.

#37 - Death Valley Energy (VI of IX)

Exhibit 1 and 2

Energy Usage vs Population



Wind Farm

Price	\$2,000,000
Incremental energy saving/day (kWh)	1,369.86

Upgrade Power Lines

Price per square mile replaced	\$667
Square miles in Death Valley	3,000
Incremental energy savings/year (kWh)	500,000

#37 - Death Valley Energy (VII of IX)

Case Math Answer

1. Examine the Energy Usage vs Population chart to calculate the current level of total consumption and the forecasted consumption in 2028. Realize that since the city population is declining, it will naturally reduce total energy consumption by 10% in 2028 compared to current levels. The mayor doesn't need to take any action.
 - **Current total annual energy consumption = 500 residents x 10,000 kWh annual usage = 5,000,000 kWh**
2. Calculate 2028 total annual energy usage goal = $5,000,000 \text{ kWh} \times 90\% = \mathbf{4,500,000 \text{ kWh}}$. This is a savings of 500,000 kWh per year.
3. Calculate the projected 2028 energy usage of the city = $400 \text{ residents} \times 11,250 \text{ kWh per resident} = \mathbf{4,500,000 \text{ kWh}}$. Notice that this projected 2028 usage hits the 10% goal naturally.
4. The powerline & windmill options are red herring distractions. If the candidate performs math on them, they have the same cost (\$2,000,000) and generate the same level of annual energy consumption savings (500,000 kWh per year).

#37 - Death Valley Energy (VIII of IX)

Case Math Answer

Level 1 Answer	Level 2 Answer	Level 3 Answer
The Mayor should do nothing.	The Mayor should do nothing. The population decline will offset the energy usage per resident to achieve a 10% reduction in total energy usage in 2028.	The Mayor should do nothing. The population decline will offset the energy usage per resident to achieve a 10% reduction in total energy usage in 2028. The windmill & electric cable options each cost \$2,000,000 and would reduce energy usage, but they are not necessary. Next, I'd like to ensure that are population projections are accurate and build a bear, base, and best-case scenario for population decline.

#37 - Death Valley Energy (IX of IX)

Recommendation

- The mayor should do nothing.

Supporting Data

- The population decline will offset the energy usage per resident to achieve a 10% reduction in total energy usage in 2028.
- The city will be using 500,000 kWh less energy per year in 2028 compared to now.
- The windmill & electric cable options each cost \$2,000,000 and are not necessary.

Risks and Next Steps

- The mayor should validate that the city population projections are accurate.
- The mayor should validate that all energy usage comes from residents (and not from people who commute to and through Death Valley).
- The mayor should calculate what the actual penalty would be for non-compliance. It is possible that the penalty amount is insignificant where it costs less to pay a penalty than to comply.

#38 - Beer Burger

Analysis	Structure	Overall
Medium	Medium	Medium

Behavioural Prep

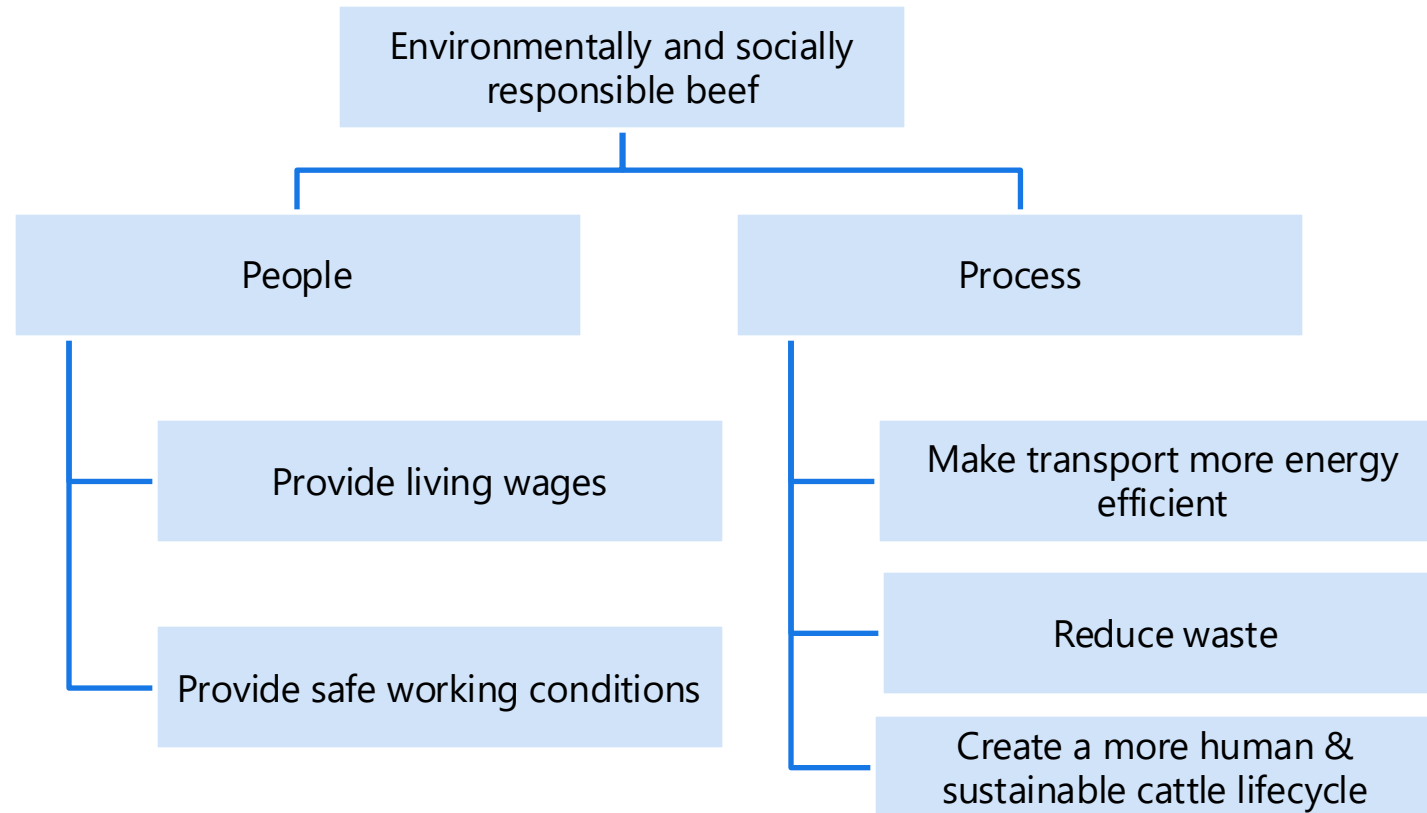
How would your teammates describe working with you?
Tell me about a time you had a conflict with a teammate.

Case Tracker		Case Stem
Industry	ESG	Beer Burger is a publicly traded fast food chain that serves beef burgers along with tall cold mugs of beer to its many hungry and loyal customers in Canada. The company recently appointed a new CEO, Vicky Burgess. Vicky is a big proponent of ESG policies and is seeking out ways to make the supply chain more environmentally & socially responsible. By examining the end-to-end supply chain of Beer Burger’s meat, Vicky wants your help hypothesizing the areas where the company could become more environmentally & socially responsible.
Type	Market Study - Strategy	
Leader	Interviewee	
Concept	Sustainability, supply chain	

Additional Information
Answers to clarifying questions: • What do you mean by beef supply chain? • We mean the process from which the cattle are born on cattle farms all the way through to when their meat (in the form of a burger) is served to customers. • What is meant by environmentally and socially responsible? • It means the company operates in a way that may be better for entire ecosystem in which it operates. How can it become better for all stakeholders involved (human, animal, and planet). • How does beer burger currently source their beef? • Beer Burger sources its beef from cattle farms in Guatemala

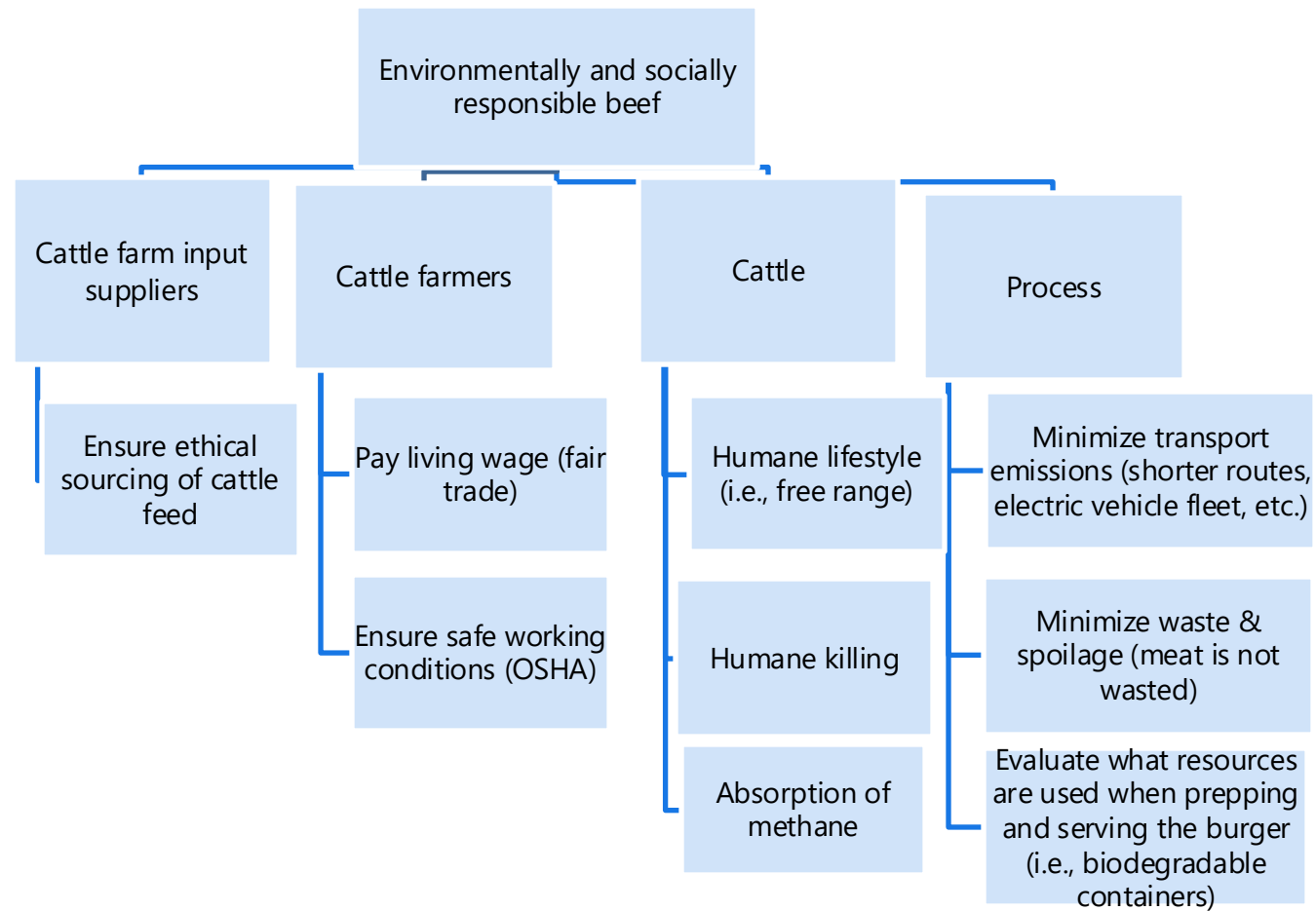
#38 - Beer Burger (II of IX)

Sample Structure - Basic



#38 - Beer Burger (III of IX)

Sample Structure - Advanced



#38 - Beer Burger (IV of IX)

Case creative question and answer

After performing an environmental impact assessment, Vicki is upset by the size of the carbon footprint the company creates from the beef supply chain.

From the methane produced by the cows to the fossil fuels burned during the transporting of the meat from Guatemala to Canada, Vicki feels a dramatic change may be warranted.

She has researched lab grown meat and feels that it might be good alternative to utilizing cattle-raised meat. Prior to bringing this recommendation to the board (that the company forgo cattle sourced beef and transition to serve lab grown beef), what should Vicki consider?

Level 1 Answer	Level 2 Answer
1. Company. 2. Customers. 3. Competition.	1. Lab grown meat. • Flavor profile (how does it compare to real beef). • Scalability (can it be supplied all restaurants). • ESG considerations (would it reduce the carbon footprint). 2. Beer Burger. • Alignment with value proposition. • Investor/BOD considerations. • Operating model impact. • Financial impact. 3. Market dynamics & customers. • Is there a demand for this meat & how does the flavor profile compare to customer tastes & preferences? • Are there competitors that already do this better than Beer Burger can?

#38 - Beer Burger (V of IX)

Case Math Question

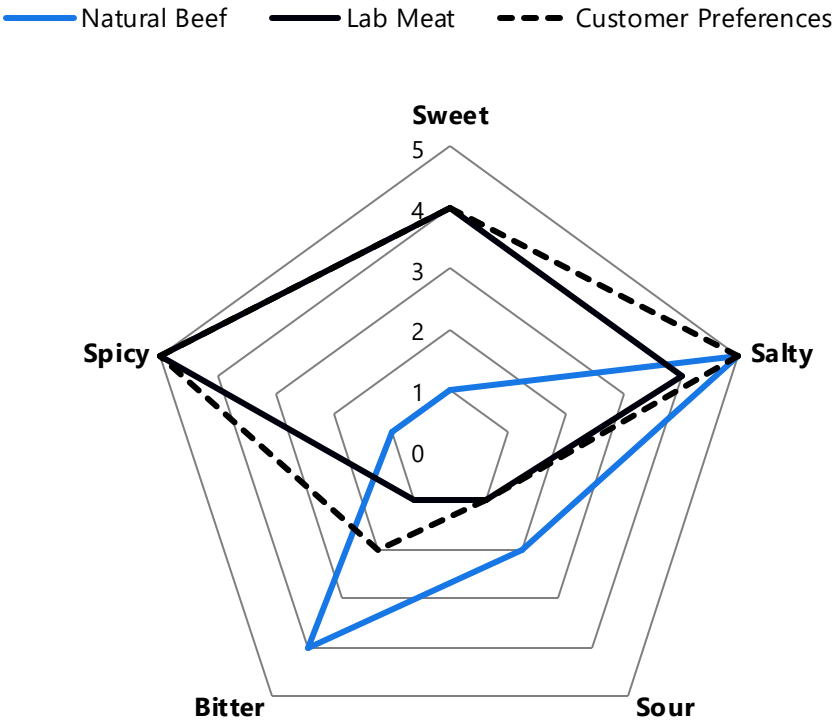
The board likes Vicki's idea. However, they are concerned about how it might impact profitability of the company. They will only give Vicki the green light if it results in greater profitability.

Vicki provides you with the following data. Should the company transition to lab grown meat? Provide a final recommendation.

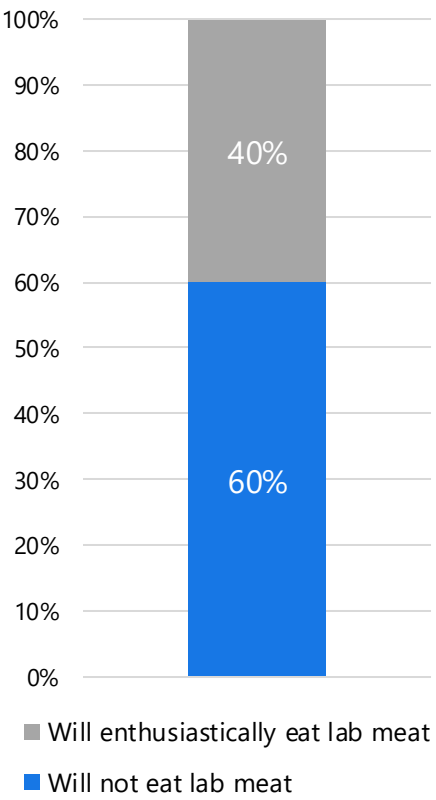
#38 - Beer Burger (VI of IX)

Exhibit 1,2 and 3

Flavor profiles vs Customer preference



Current customer segmentation



Current Beer Burger Snapshot

Total customers	10,000,000
Current profitability	\$ 1,000,000,000

Note: Customers classified as those who will enthusiastically eat lab meat will generate \$300 in annual profitability each should all burgers at Beer Burger become lab meat burgers.

#38 - Beer Burger (VII of IX)

Case Math Answer

1. Examine the radar chart. Visually, observe that the flavor profile of lab meat better matches the flavor preferences of customers. Mathematically, the candidate can turn each data point into a chart and then compare the total absolute deviation. If a candidate builds this table, consider it a bonus.

Attribute	Natural Beef	Lab Meat	Customer Preferences	Lab Meat vs Customer preferences (absolute deviation)	Natural Beef vs Customer Preferences (absolute deviation)
Sweet	1	4	4	0	3
Salty	5	4	5	1	0
Sour	2	1	1	0	1
Bitter	4	1	2	1	2
Spicy	1	5	5	0	4
Total Deviation (sum of absolute values)				2	10

2. Explore profitability impact. Note that current profitability is \$1,000,000,000. If the company switches to lab meat, then only 40% of the customer base will survive. $40\% \times 10,000,000 \text{ customers} = 4,000,000 \text{ customers}$ will remain. Referring to the callout at the bottom of the exhibit, each of these surviving customers will generate \$300 in annual profitability. New company profitability will become $4,000,000 \text{ customers} \times \$300 = \$1,200,000,000$.
3. Compare new profitability with lab meat to old profitability with natural meat. $\$1,200,000,000 - \$1,000,000,000 = \$200,000,000$. Lab grown meat makes the company 20% more profitable.

#38 - Beer Burger (VIII of IX)

Case Math Answer

Level 1 Answer	Level 2 Answer	Level 3 Answer
Beer Burger should transition its beef supply chain to lab grown meat.	Beer Burger should transition its beef supply chain to lab grown meat. It will unlock \$200,000,000 in additional annual profitability. Although only 40% of our pre-existing customer base will remain, their impact is substantially more significant than the loss of the other 60%.	Beer Burger should transition its beef supply chain to lab grown meat. It will unlock \$200,000,000 in additional annual profitability. Although only 40% of our pre-existing customer base will remain, their impact is substantially more significant than the loss of the other 60%. Lab grown meat better matches the flavor profile desired by our customers and reduces the sum of absolute deviation from 10 to 2. Next, I'd like to examine if lab meat suppliers can scale to meet our demand, and I'd like to test lab grown meat in a pilot market.

#38 - Beer Burger (IX of IX)

Recommendation

- Beer Burger should transition to lab grown meat.

Supporting Data

- It will unlock \$200,000,000 in additional annual profitability.
- Although only 40% of our pre-existing customer base will remain, their impact is substantially more significant than the loss of the other 60%.
- Lab grown meat better matches the flavor profile desired by our customers and reduces the sum of absolute deviation from 10 to 2.
- Lab grown meat will eliminate the carbon footprint generated by natural cattle methane & from fossil fuels utilized to transport the meat from Guatemala to Canada.

Risks and Next Steps

- Vicki needs to assess if the lab meat suppliers have capacity to scale because Beer Burger is a huge company that will place large orders.
- Vicki should validate with legal that there are no department of health issues associated in all the provinces Beer Burger operates where licensing must be attained to serve this type of meat.
- Vicki should consider impact to the Guatemalan cattle farmers who may lose their livelihoods when Beer Burger pulls out & should be ready to counter any unintended public backlash.

#39 – Mayor Of Maine

Analysis

Structure

Overall

Hard

Hard

Hard

Behavioural Prep

Why are you interested in working at Firm X?
Tell me about your biggest strength/weakness?

Case Tracker

Industry	Public Sector
Type	Market Study - growth
Leader	Interviewee
Concept	Policy, profitability

Case Stem

Marine biologists have warned the Mayor of Maine, Mayor Crabbycakes, that the wild lobster population off the coast of Maine has experienced a dramatic decline over the past year.

Maine's economy is heavily reliant upon lobster fishing. Mayor Crabbycakes has hired our firm to help figure out what to do. What should Mayor Crabbycakes consider before authoring new regulations that protect the wild lobster population off the coast of Maine?

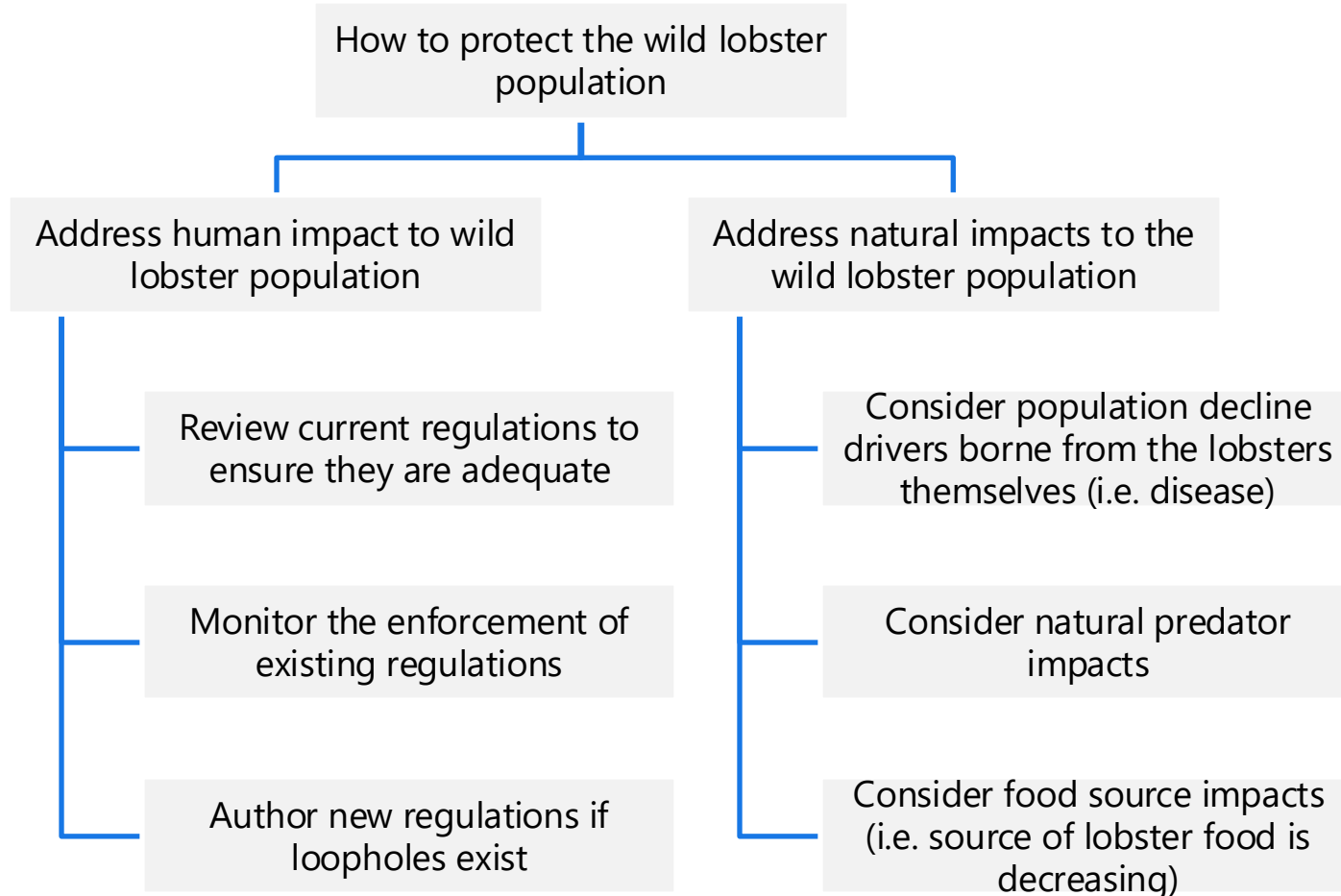
Additional Information

Answers to clarifying questions:

- Are Maine-based fishermen the only human parties that fish for wild lobster off the coast of Maine?
 - Yes.
- Is all Maine lobster production from harvesting lobster fished from the wild exclusively?
 - Yes.
- Do regulations already exist?
 - Yes, wild lobster fishing regulations already exist.

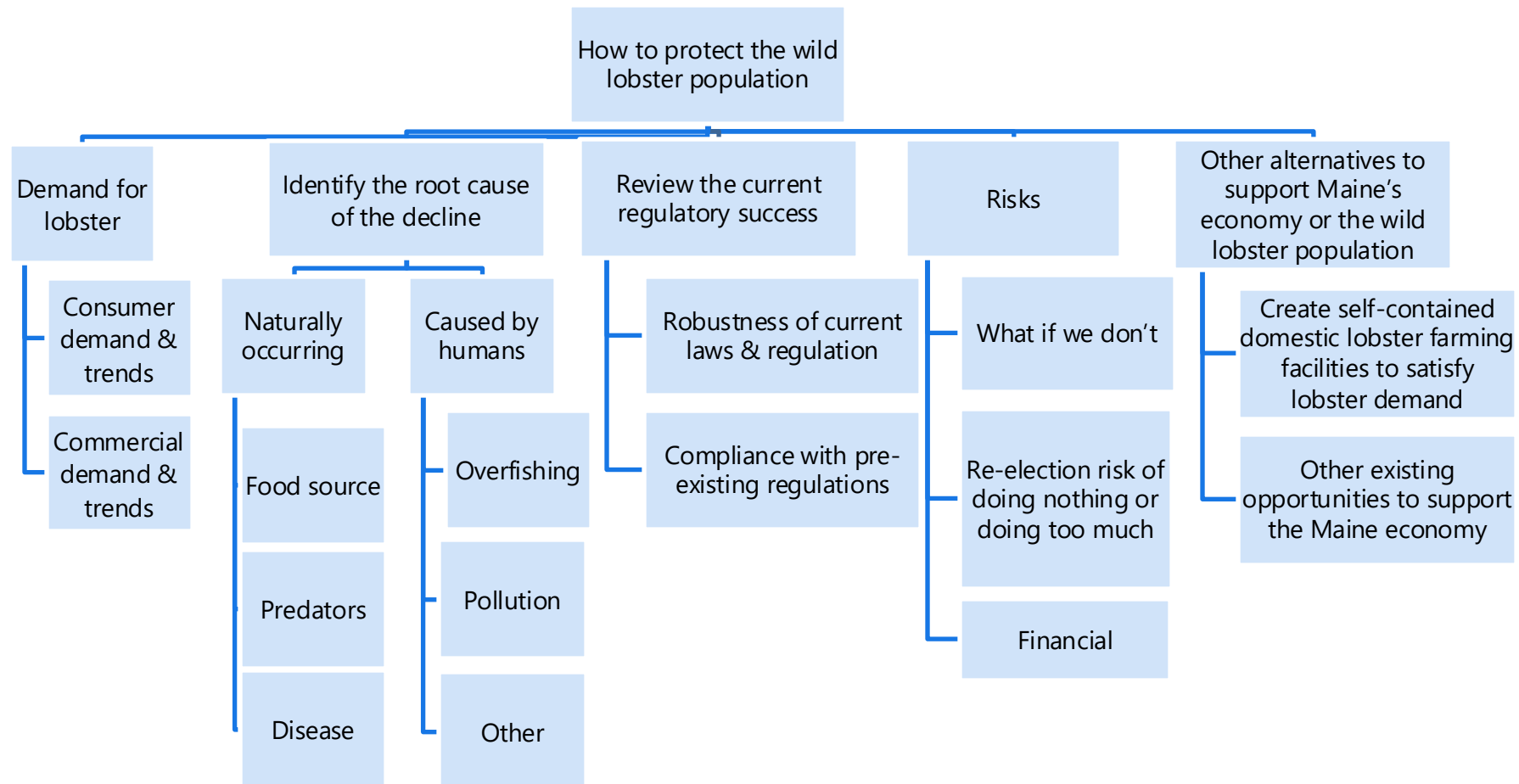
#39 – Mayor Of Maine (II Of X)

Sample Structure - Basic



#39 – Mayor Of Maine (III of X)

Sample Structure - Advanced



#39 – Mayor Of Maine (IV of X)

Case creative question

After attending a marine hatchery expo in Las Vegas, Mayor Crabbycakes learned about a new alternative to harvesting wild lobster. Technology now exists where domesticated lobster can now be sustainably farmed in a self-contained system. Setting up such a system in Maine would cost \$400M in 2023.

What should Mayor Crabbycakes consider before investing taxpayer funds into developing Maine-based lobster farms?

#39 – Mayor Of Maine (V of X)

Case creative answer

Level 1 Answer

1. Financial considerations.
2. How fishermen will be impacted.
3. How will the wild lobster population be impacted.
4. Lobster demand.

Level 2 Answer

- 1.Lobster demand.
 - Will lobster demand hold firm?
- 2.Farming capacity & logistics.
 - Who would own and run these facilities?
 - Is there real estate available to locate these facilities?
 - Who would benefit?
 - How much lobster could be produced from these facilities?
 - Does the captive lobster flavor profile match that of wild lobster?
- 3.Financial considerations.
 - Would this investment pay off financially for Maine?
 - How might funding to other parts of the state be impacted?
 - Are there better investment options available?
- 4.Wild lobster population impact.
 - Would this enable the natural lobster population to re-grow to a healthy volume?

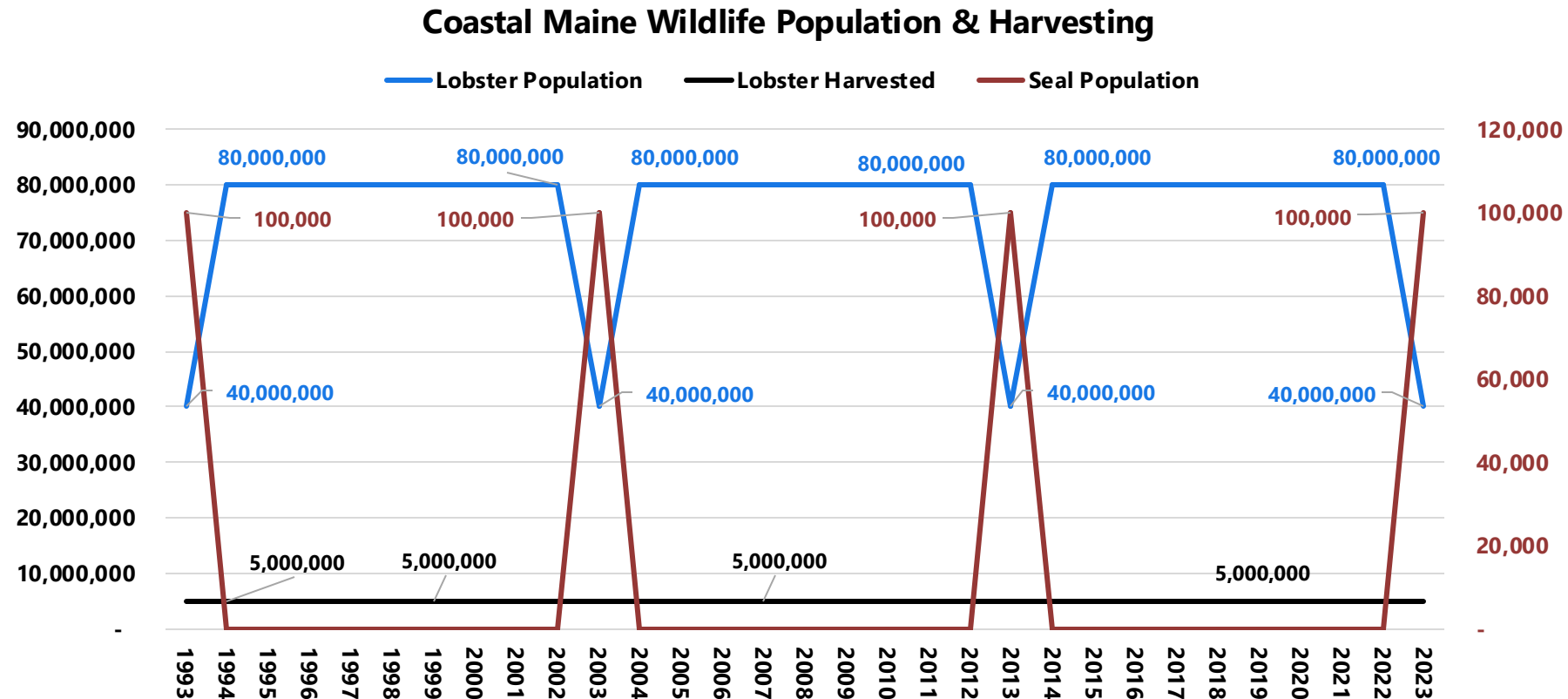
#39 – Mayor Of Maine (VI of X)

Case Math Question

Mayor Crabbycakes has provided you with the following data on Exhibits 1, 2, and 3. Should she invest \$400M of Maine's budget into lobster farms? What should she consider? Provide a recommendation.

#39 – Mayor Of Maine (VII of X)

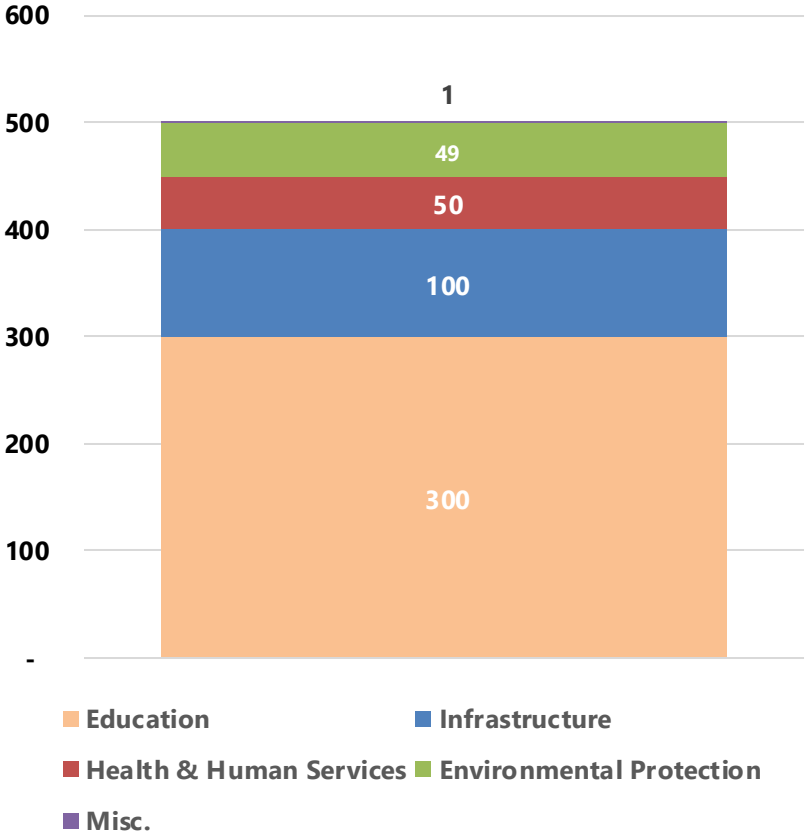
Exhibit 1



#39 – Mayor Of Maine (VIII of X)

Exhibit 2 and 3

2023 State of Maine Budget (\$MM)



Lobster Farm Forecast

Year (t)	Profit	Lobster Hatched
Initial Outlay (0)	\$ (400,000,000)	N/A
1	\$ 10,000,000	200,000
2	\$ 10,000,000	200,000
3	\$ 20,000,000	200,000
4	\$ 20,000,000	200,000
5	\$ 20,000,000	200,000
i	15%	

#39 – Mayor Of Maine (IX of X)

Case Math Answer

1. Exhibit 2: Examine the state of Maine budget to realize it is \$500M total. Compare that to the \$400M required investment in 2023 ($\$400\text{M}/\$500\text{M}=80\%$), to realize that this investment would torpedo the existing budget and cause shortages in almost all other departments that require state funding (i.e., education).
2. Exhibit 1: Examine the Coastal Maine Wildlife Population & Harvesting chart to observe that the lobster population dips every 10 years as migrating seal populations increase to 100,000. Observe that human harvesting has stayed consistent over the past 30 years typically harvesting no more than 6% of the wild lobster population ($5\text{M harvested}/80\text{M wild population}=6\%$)
3. Exhibit 3: Examine the financial forecast table. Although it is set up for an NPV calculation, the candidate should abstain from performing the NPV calculation because the sum of the yearly profit from years 1-5 is \$80M. \$80M, is clearly less than the \$400M outlay required in 2023. Discounting cash flows would only make this \$80M worth less in present value terms (today's dollars).
4. Exhibit 3: From the financial forecast table, examine the amount of annual lobster hatched 200,000 and compare it to the 80,000,000 wild population to realize that these farms would only generate 0.25% of the overall lobster population. It is a moot effort.

Level 1 Answer

This investment would take 80% of the state budget; the Mayor should not move forward with the investment.

Level 2 Answer

This investment would take 80% of the state budget; we should not move forward with the investment. Not only is this solution too expensive, but it would generate less than 1% of the lobster population, not enough to make a dent in the problem.

Level 3 Answer

This investment would take 80% of the state budget; we should not move forward with the investment. Not only is this solution too expensive, but it would generate less than 1% of the lobster population, not enough to make a dent in the problem. Next, I'd like to examine diversifying our economy away from lobster fishing.

#39 – Mayor Of Maine (X of X)

Recommendation

- Mayor Crabbycakes should not invest \$400M in lobster farms.

Supporting Data

- It is not a viable investment as it is NPV negative.
- It would require 80% of the 2023 state budget which would devastate essential state programs (i.e., education).
- It would generate less than 1% of the overall lobster population.
- Every 10 years, seals migrate to Maine and deplete the lobster population. However, the past 30 years have shown a rebound to natural levels the following year.

Risks and Next Steps

- What if the seals do not migrate away from Maine and continue to deplete the lobster down extinction? We should put into place a seal monitoring & tracking program.
- What if there is no causation between the seals and lobsters, where something else is causing lobster population decline (i.e., disease is killing off the lobster)? We should take samples of lobsters and test them for disease to rule this out.
- What other industries should Maine consider to fortify its economy aside from lobster fishing? The Mayor should validate that lobster demand is likely to stay strong in the near & long term and seek out ways to diversify the state economy to manage risk.

#40 - Doctors Within US Borders

Analysis	Structure	Overall
Hard	Hard	Hard

Behavioural Prep

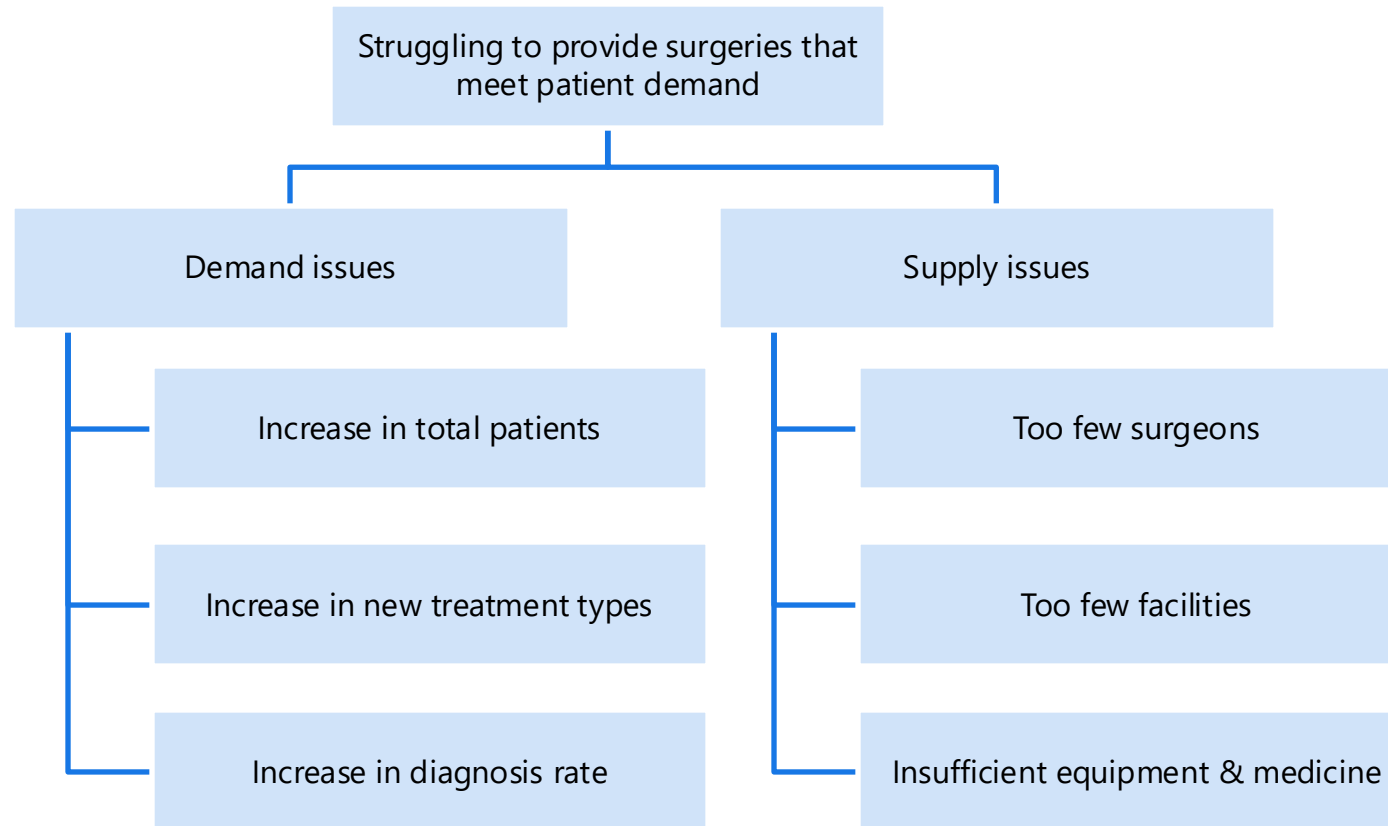
Why did you choose to do an MBA?
Tell me about a time when you led a team through a challenge.

Case Tracker		Case Stem
Industry	Non-Profit	The non-profit Doctors within US Borders is a domestic medical humanitarian organization that delivers emergency aid to people affected by epidemics, natural disasters, and exclusion from healthcare. Their activities span a wide range of medical scenarios, from primary care to secondary hospital services. Doctors within US Borders operates independently from any political, military, or religious agendas. They make their own decisions about where to provide medical assistance. In addition, the organization aids people based on need, irrespective of race, religion, or political beliefs. The organization can quickly respond to emergencies because of its operational flexibility. They can initiate medical programs within hours after a disaster strikes. Due to this, Doctors Within US Borders is known to experiment and innovate, especially in challenging situations where standard procedures might not work. The non-profit is struggling to deliver services due to outsized demand. Why might this be and how can they solve this problem?
Type	Market Study – Growth	
Leader	Interviewee	
Concepts	Operations	

Additional Information
Answers to clarifying questions: - Are there specific services that we're most under capacity in terms of delivering? - Our biggest issue is delivering surgeries – that's where we'd like to focus our problem-solving efforts. - How are they funded? - A combination of private and public donations.

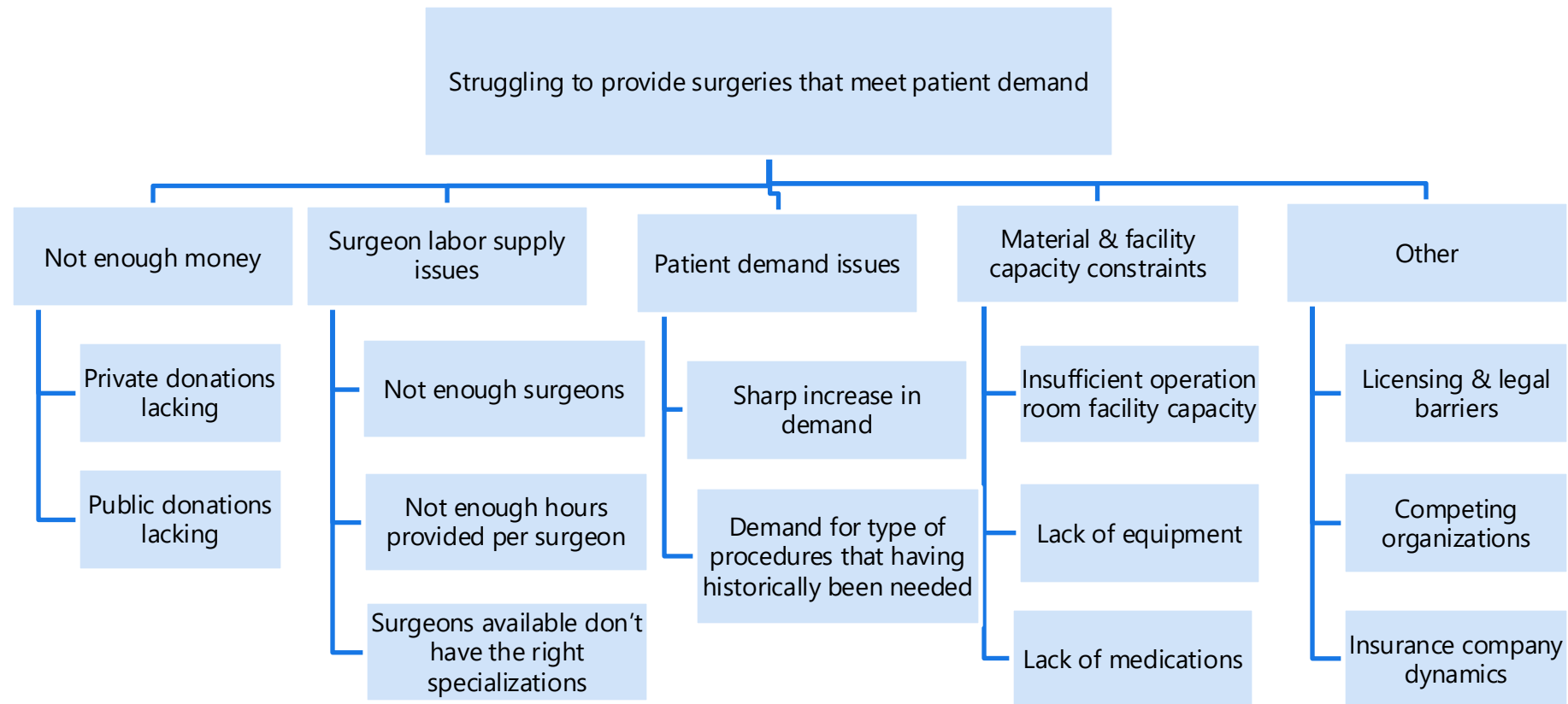
#40 - Doctors Within US Borders (II of IX)

Sample Structure - Basic



#40 - Doctors Within US Borders (III of IX)

Sample Structure - Advanced



#40 - Doctors Within US Borders (IV of IX)

Case creative question and answer

If supplying in-demand surgeries is the issue (not a sharp increase in demand), what could be the root cause of this supply shortage?

Level 1 Answer

1. Lack of people.
2. Lack of materials.

Level 2 Answer

1. Surgeon supply.
 - Not enough surgeons.
 - Not enough available hours per surgeon.
 - Not enough surgeons with the "right" specializations.
2. Facility constraints.
 - Not enough operating room capacity.
 - Operating rooms are not the right type.
3. Equipment constraints.
 - Don't have the required machinery.
 - Don't have the sufficient surgical tools.
4. Medicine constraints.
 - Insufficient access to medicines enabling the procedure.
5. Post-operation care constraints.
 - Lack of patient care support.
 - Lacking a venue for patients to recover.

#40 - Doctors Within US Borders (V of IX)

Case Math Question

The CEO believes that the main issue is a lack of available operating rooms. To increase operating room capacity, he is considering either investing in a fleet of mobile operating room RVs or constructing medical buildings. He can only select one option.

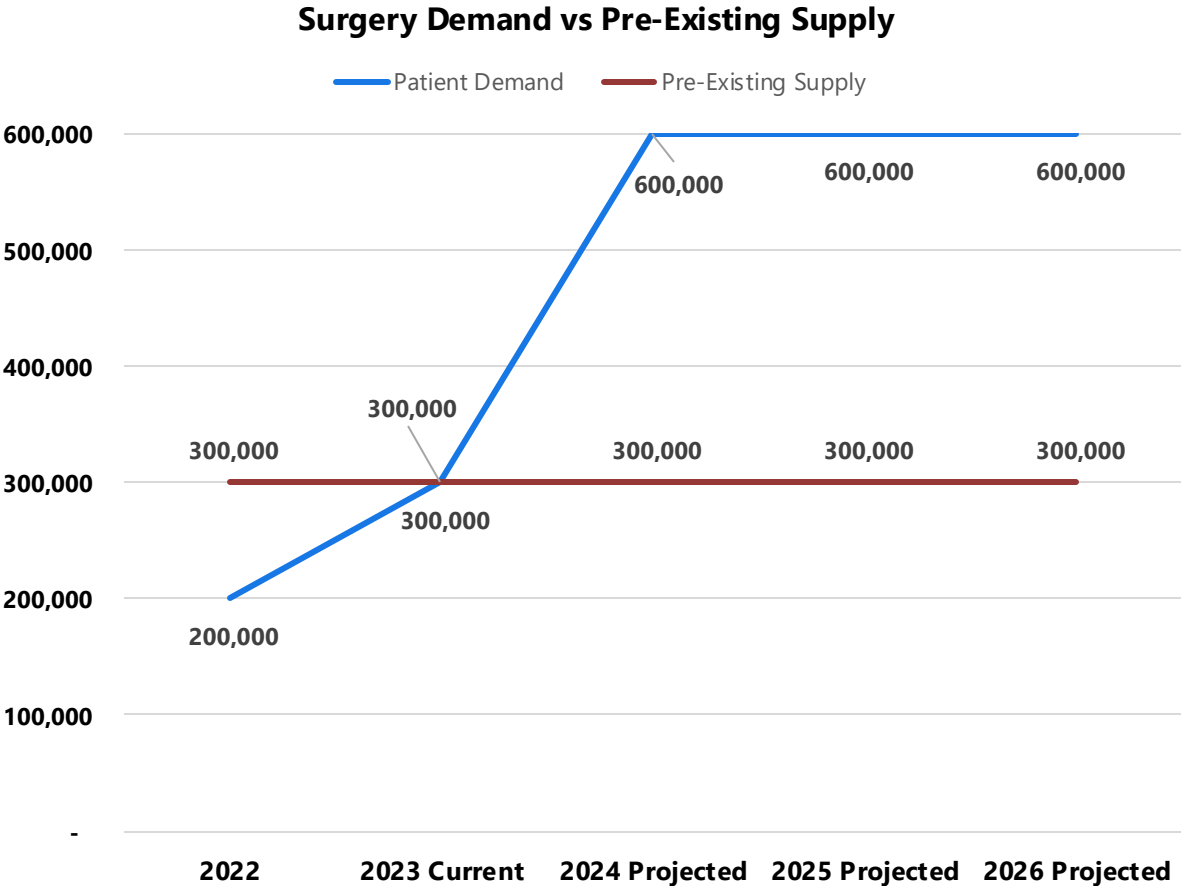
Which option should he pursue & why? Provide a final recommendation after examining Exhibits 1 and 2.

#40 - Doctors Within US Borders (VI of IX)

Exhibit 1 and 2

Investment Options

	RV Fleet	Medical Building
Surgery capacity/month	1,000	10,000
Price per each	\$250,000	\$2,000,000



#40 - Doctors Within US Borders (VII of IX)

Case Math Answer

1. Examine the Surgery Demand vs Pre-Existing supply to realize an annual shortage of 300,000 surgeries per year. Hence, the investment option must reach or exceed 300,000 surgeries per year. **600,000 demanded - 300,000 supplied = 300,000 shortage.**
2. Notice that the Investment Options table displays capacity monthly. Convert the 300,000 shortage into monthly shortage to arrive at 25,000 surgeries per month are required to hit the goal. **300,000 shortage per year / 12 months = 25,000 surgeries per month.**
3. Find out how many RVs are required to achieve 25,000 surgeries per month. It is 25 RVs. **25,000 monthly surgery shortage / 1,000 surgeries per RV = 25 RVs required.**
4. Find out how many buildings are required to achieve 25,000 surgeries per month. It is 3 buildings (because you cannot construct a partial building). Note this will provide 5,000 of additional capacity per month. **25,000 monthly surgery shortage / 10,000 surgeries per building = 3 buildings.**
5. Calculate the cost of each option. **RV option: 25 RVs x \$250,000 per RV = \$6,250,000. Building option: 3 buildings x \$2,000,000 = \$6,000,000.**
6. Subtract the difference, to inform that buildings cost \$250,000 less than the RV option.

#40 - Doctors Within US Borders (VIII of IX)

Case Math Answer

Level 1 Answer

The CEO should invest in constructing 3 medical buildings.

Level 2 Answer

The CEO should invest in constructing 3 medical buildings. This will provide sufficient capacity to cover the 300,000 per year shortage with additional capacity available for future growth.

Level 3 Answer

The CEO should invest in constructing 3 medical buildings. This will provide sufficient capacity to cover the 300,000 per year shortage with additional capacity available for future growth. This option also costs \$250,000 less than the RV option. Next, I'd like to examine the benefits and risks of having fixed vs. mobile surgery locations.

#40 - Doctors Within US Borders (IX of IX)

Recommendation

- The CEO should invest in constructing 3 medical buildings.

Supporting Data

- This will provide sufficient capacity to cover the 300,000 per year shortage.
- It will provide an additional 60,000 in capacity beyond the shortage, positioning Doctors within US Borders for future growth.
- The building option costs \$250,000 less than the RV option.

Risks and Next Steps

- The CEO should consider that having fixed buildings means that doctors & patients must travel to the site. He should survey patients & doctors to confirm they can travel to these sites.
- The CEO should consider where to strategically locate each building. The CEO should look at patient & provider population heat maps & local market pricing to identify optimal locations.
- The CEO should consider how quickly the buildings can be constructed as next year a 300,000-surgery shortage may occur if they are not open. He should consult with a builder & architect. He should also consider if there are pre-existing buildings he can purchase instead of construct to shorten this time horizon and meet demand.

#41 - TIKTAC CO

Analysis	Structure	Overall
Hard	Hard	Hard

Behavioural Prep

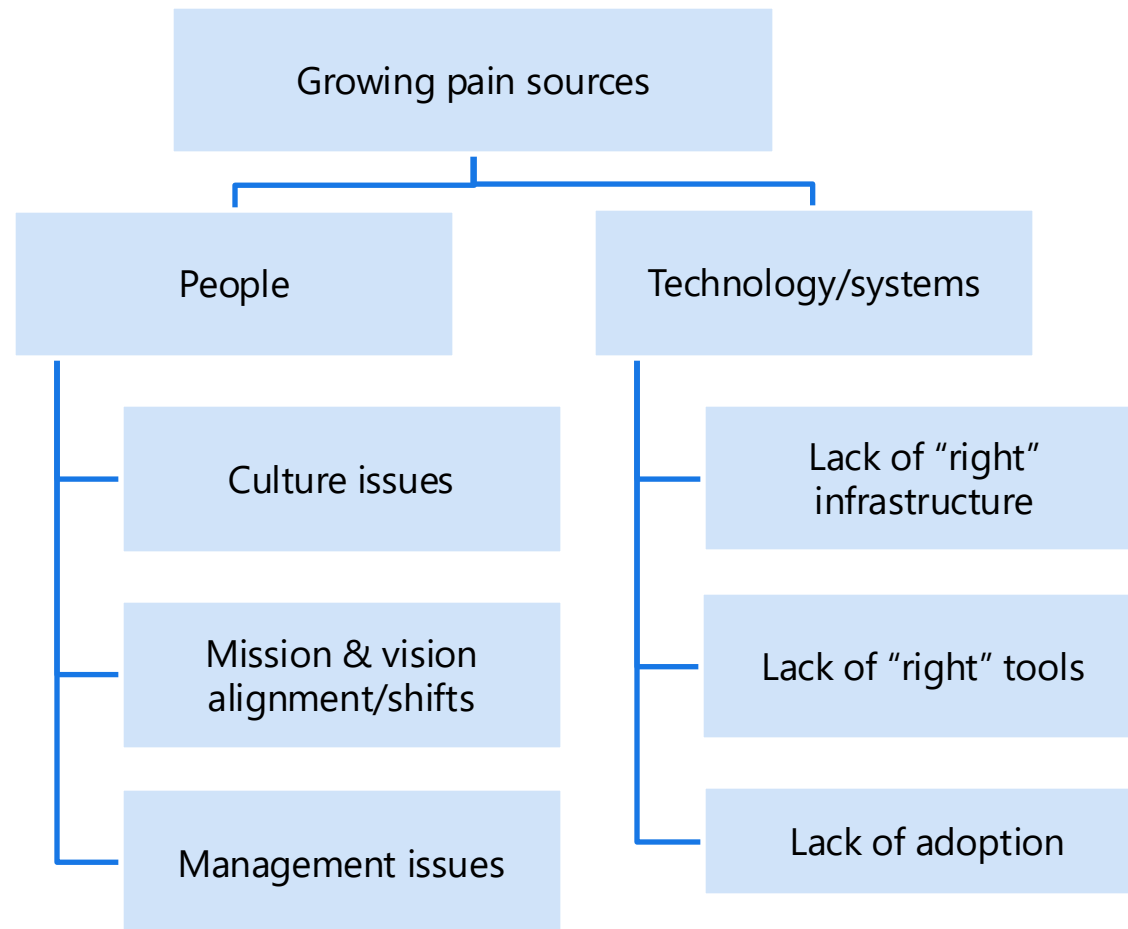
What do you do for fun?
Give me an example of a tough business problem you previously solved.

Case Tracker		Case Stem
Industry	Tech	A once small social media tech company, TikTac Co., has experienced explosive growth. Over the past 5 years, the company has grown from 50 employees to 5,000. As a result, they are experiencing growing pains that are starting to interfere with the day-to-day operations of the company. You have been hired by their CEO, Zoey Toey. What might be the sources of these growing pains?
Type	Market Study - growth	
Leader	Interviewee	
Concept	Operations, human capital	

Additional Information
Answers to clarifying questions: - What do you mean by growing pains? - Problems that have resulted by the scaling in size of the company. - What does this company do? - They are a social media platform where users post video content. - Where do they operate? - Users of the platform are global, the company (and all employees) are in Canada.

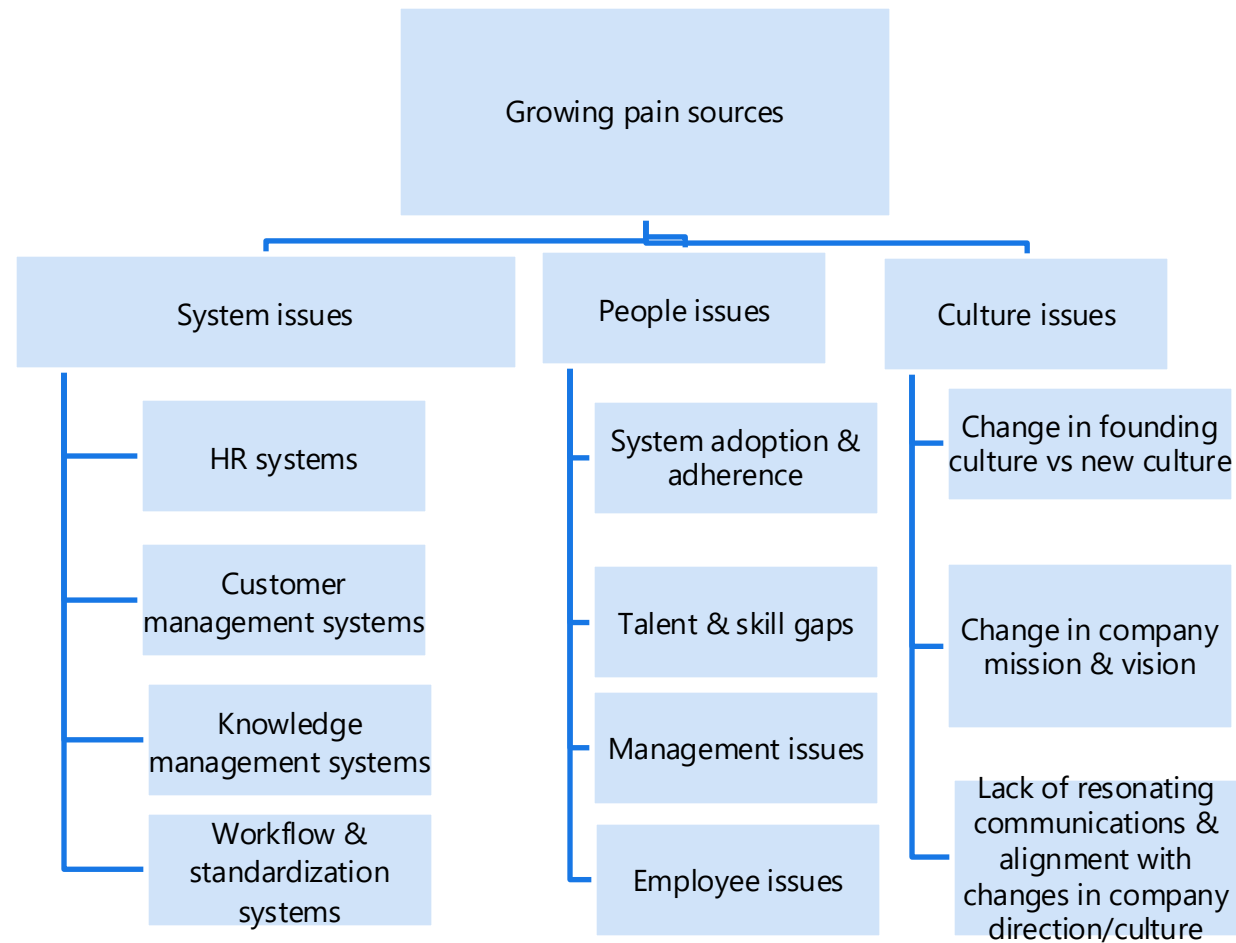
#41 - TIKTAC CO (II of IX)

Sample Structure - Basic



#41 - TIKTAC CO (III of IX)

Sample Structure - Advanced



#41 - TIKTAC CO (IV of IX)

Case creative question and answer

Zoey suspects that her employees may be struggling because they lack the necessary skills & capabilities required to operate the now-large and evolving company.

How might Zoey effectively assess the skills gap of her employees to triangulate where specific gaps exist?

Level 1 Answer

1. Ask employees.
2. Ask customers.
3. Perform desk research

Level 2 Answer

1. Survey employees.
 - Qualitative surveys.
 - Quantitative surveys.
 - Skills-based assessments (test their skills to inform aptitude)
2. Survey customers.
 - Qualitative surveys.
 - Quantitative surveys.
3. Perform industry benchmarking.
 - Assess against peers in the same industry.
 - Assess against latest academic/subject matter expertise literature.

#41 - TIKTAC CO (V of IX)

Case Math Question

After performing a 360-degree survey and benchmarking assessment, it was uncovered that employees lack spreadsheet analysis skills, which is causing significant bottlenecks in the organization.

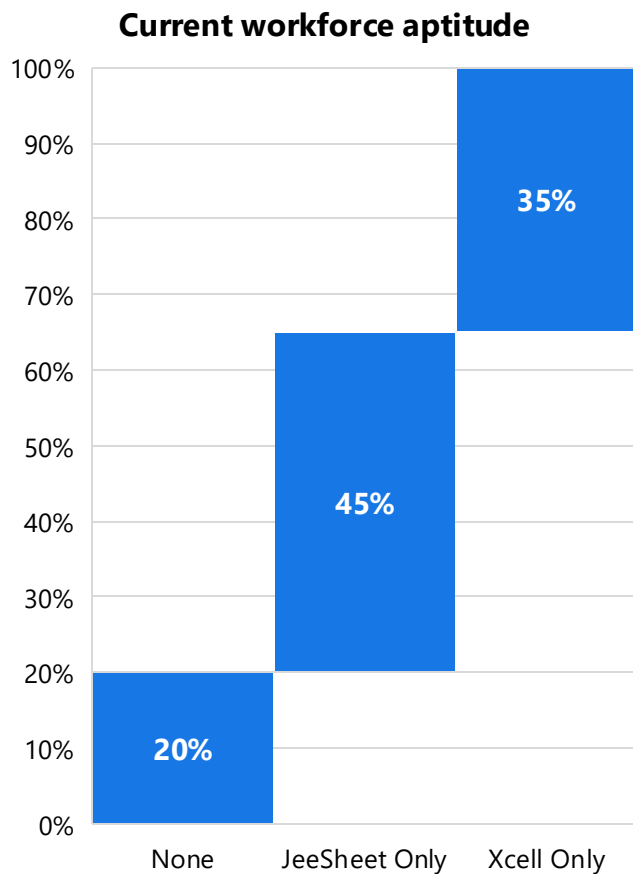
Zoey has contacted an upskilling vendor who has offered two spreadsheet training options, a JeeSheet training or a Xcell training. Zoey's budget can only afford one of the options.

The application selected for the training will become the company standard moving forward (i.e., if JeeSheet is selected, then employees can only use JeeSheet moving forward and vice versa for Xcell).

Zoey wants to invest in the spreadsheet option that will result in the greatest amount of employee productivity. Which should she select? Provide a final recommendation after reviewing Exhibit 1.

#41 - TIKTAC CO (VI of IX)

Exhibit 1 and 2



Annual productivity by aptitude (productivity units)				
Spreadsheet Application	None	Novice User	Intermediate User	Advanced User
Xcell Only	10	50	80	250
JeeSheet Only	10	40	75	200

Training outcomes key: If Xcell training selected				
Current aptitude		Aptitude achieved post-training		
None		Novice Xcell		
JeeSheet User		Intermediate Xcell		
Xcell User		Advanced Xcell		
If JeeSheet training selected				
Current aptitude		Aptitude achieved post-training		
None		Novice JeeSheet		
Xcell User		Intermediate JeeSheet		
JeeSheet User		Advanced JeeSheet		

#41 - TIKTAC CO (VII of IX)

Case Math Answer

1. Calculate the composition of aptitude of the entire current workforce. Examine the Current workforce spreadsheet aptitude chart and multiple each category by the total population of the workforce.
Non-users: $5,000 \times 20\% = 1,000$.
JeeSheet only users: $5,000 \times 45\% = 2,250$.
Xcell only users: $5,000 \times 35\% = 1,750$.
2. **If Xcell training selected**, calculate the number of users in each aptitude category, then multiple these populations by the productivity associated with each aptitude classification, and then sum it up to arrive at total workforce productivity.
Native Non-users: $1,000 \rightarrow 1,000$ Novice Xcell $\times 50$ annual productivity units = 50,000 productivity units.
Native JeeSheet only users: $2,250 \rightarrow 2,250$ Intermediate Xcell $\times 80$ annual productivity units = 180,000 productivity units.
Native Xcell only users: $1,750 \rightarrow 1,750$ Advanced Xcell $\times 250$ annual productivity units = 437,500 productivity units.
Total Xcell workforce productivity: $50,000 + 180,000 + 437,500 = \mathbf{667,500 \text{ productivity units}}$.
3. **If JeeSheet training selected**, calculate the number of users in each aptitude category, then multiple these populations by the productivity associated with each aptitude classification, and then sum it up to arrive at total workforce productivity.
Native Non-users: $1,000 \rightarrow 1,000$ Novice JeeSheet $\times 40$ annual productivity units = 40,000 productivity units.
Native Xcell only users: $1,750 \rightarrow 1,750$ Intermediate JeeSheet $\times 75$ annual productivity units = 131,250 productivity units.
Native JeeSheet only users: $2,250 \rightarrow 2,250$ Advanced JeeSheet $\times 200$ annual productivity units = 450,000 productivity units.
Total Xcell workforce productivity: $40,000 + 131,250 + 450,000 = \mathbf{621,250 \text{ productivity units}}$.
4. Compare the productivity of both training options to find that Xcell training results in 46,250 more annual productivity units than JeeSheet training.
Xcell training annual productivity 667,500 productivity units – JeeSheet training annual productivity 621,250 productivity units = **46,250 annual productivity**.

#41 - TIKTAC CO (VIII of IX)

Case Math Answer

Level 1 Answer	Level 2 Answer	Level 3 Answer
Zoey should upskill her workforce with Xcell.	Zoey should upskill her workforce with Xcell, as it results in annual workforce productivity of 621,250.	Zoey should upskill her workforce with Xcell, as it results in annual workforce productivity of 621,250. This generates 46,250 more productivity than JeeSheet training. Next, I'd want to verify that Xcell is the best solution for our business needs.

#41 - TIKTAC CO (IX of IX)

Recommendation

- Zoey should upskill her workforce with Xcell.

Supporting Data

- It results in annual workforce productivity of 621,250.
- It generates 46,250 more units of productivity than JeeSheet training.
- Excel has the highest productivity per user output of 250, which means that as novice and intermediate users gain more experience over time, that they will eventually arrive at optimized productivity levels (vs. JeeSheet that maxes out at 200).

Risks and Next Steps

- Since 45% of the workforce currently uses JeeSheet, Zoey should anticipate how to communicate this standard application change to Xcell as there could be employee resistance to this change.
- Zoey should examine the cost of the trainings & timelines of the trainings.
- Zoey should examine other growth pain drivers to ensure this is the most impactful one.

#42 - Whatcha Say Research

Analysis	Structure	Overall
Hard	Hard	Hard

Behavioural Prep

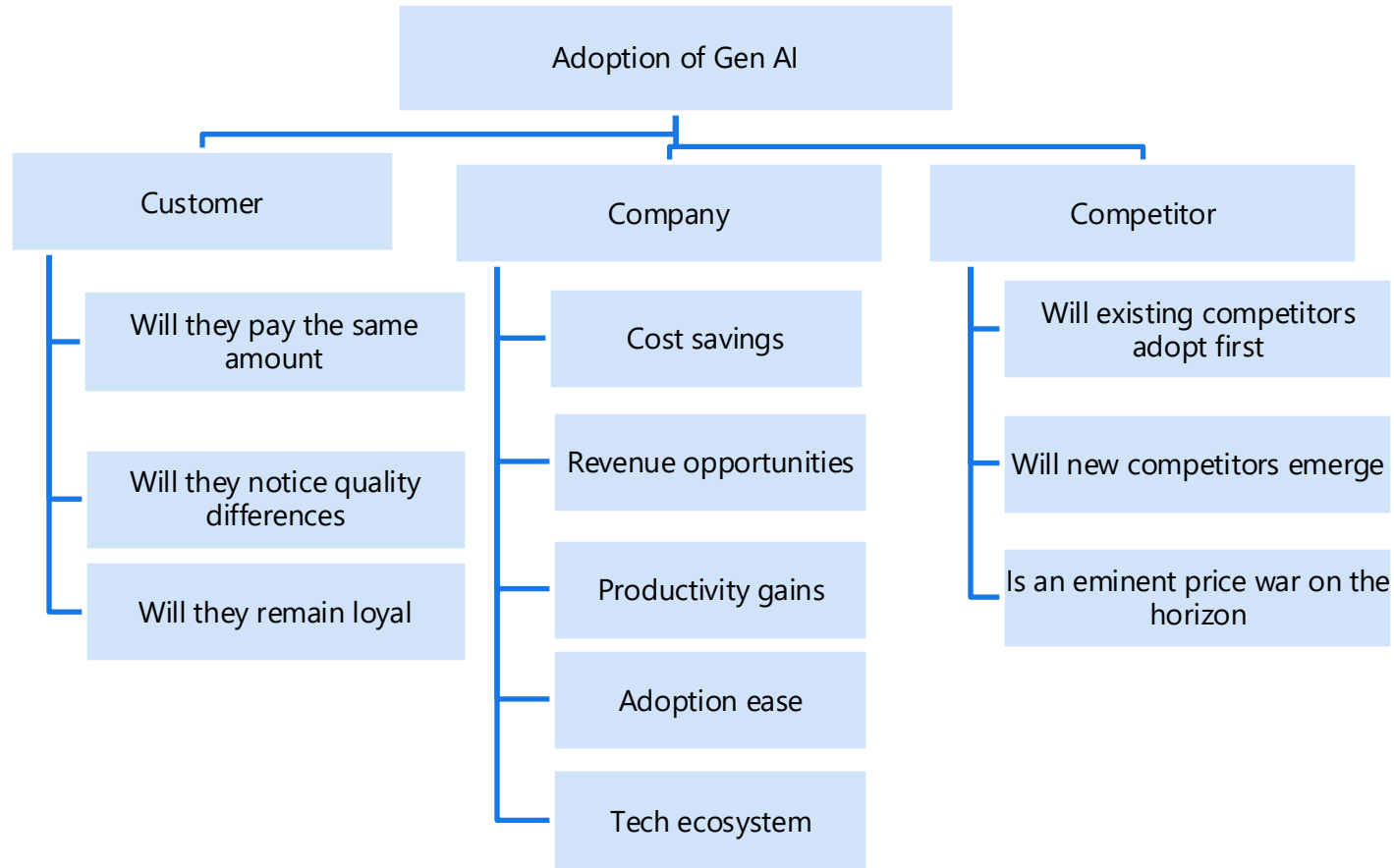
What do you see as the most challenging aspect of this job?
Describe a time you had to change your communication style to influence different stakeholders.

Case Tracker		Case Stem
Industry	Tech	You have been hired by the CEO, Jordan Derulo, of Whatcha Say Research. Jordan’s company provides primary market research service to clients. Jordan recently read about how generative artificial intelligence has the capability to synthesize long strings of text & distill it into key takeaways. What should Jordan consider prior to adopting this technology?
Type	System transformation	
Leader	Interviewee	
Concept	Profitability, human capital	

Additional Information
Answers to clarifying questions: • What is primary market research? • It is researching a population (usually a target customer sample) and conducting in-depth interviews to understand how to better serve their wants and needs for products and services. • What is generative AI? • “Generative artificial intelligence is artificial intelligence capable of generating text, images, or other media, using generative models. Generative AI models learn the patterns and structure of their input training data and then generate new data that has similar characteristics (Wikipedia, 2023).” • What is Jordan’s business model & process? • Jordan charges clients a project fee (informed by hourly labor costs plus a margin). Jordan’s firm authors interview guides, procures interview subjects, facilitates in-depth interviews with the subjects, manually captures verbatim notes, synthesizes those notes into key takeaways, and delivers a deck with recommendations to his clients (informed by key takeaways uncovered).

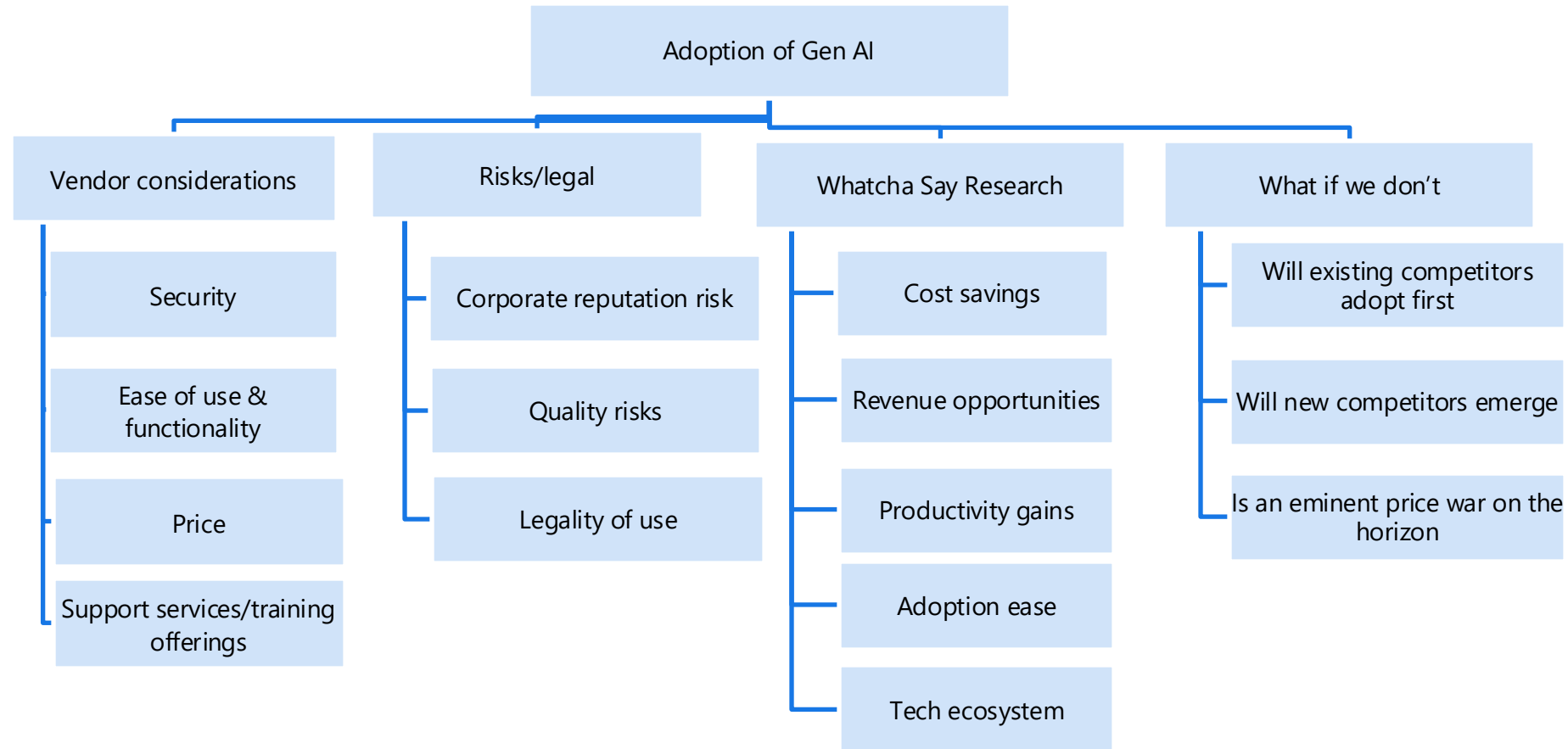
#42 - Whatcha Say Research (II of IX)

Sample Structure - Basic



#42 - Whatcha Say Research (III of IX)

Sample Structure - Advanced



#42 - Whatcha Say Research (IV of IX)

Case creative question and answer

Jordan believes that generative AI could significantly reduce the thruput time and the number of human-staffed hours that are required to complete client projects.

What are some ways Jordan can capitalize on this opportunity (if this is true) to improve the strategic positioning of the company?

Level 1 Answer	Level 2 Answer
1. Improve margins. 2. Pass on savings to customers (reduce price).	1. Improve project margins. a) Keep price the same & reduce the human-staffed hours per project. 2. Increase speed of project delivery to open capacity to take on more projects. 3. Increase market share through reducing pricing and passing (some or all) savings to customers with the intent of garnering more loyalty & undercutting competitors on price. 4. Lean up the company. a) Layoff much of the workforce to reduce costs and keep only the top performers (do the same amount of work, but with fewer people).

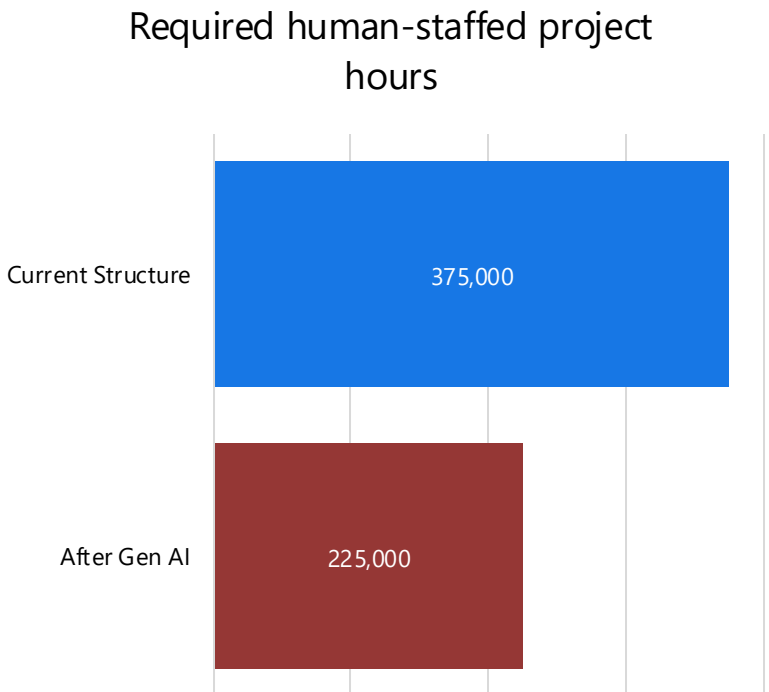
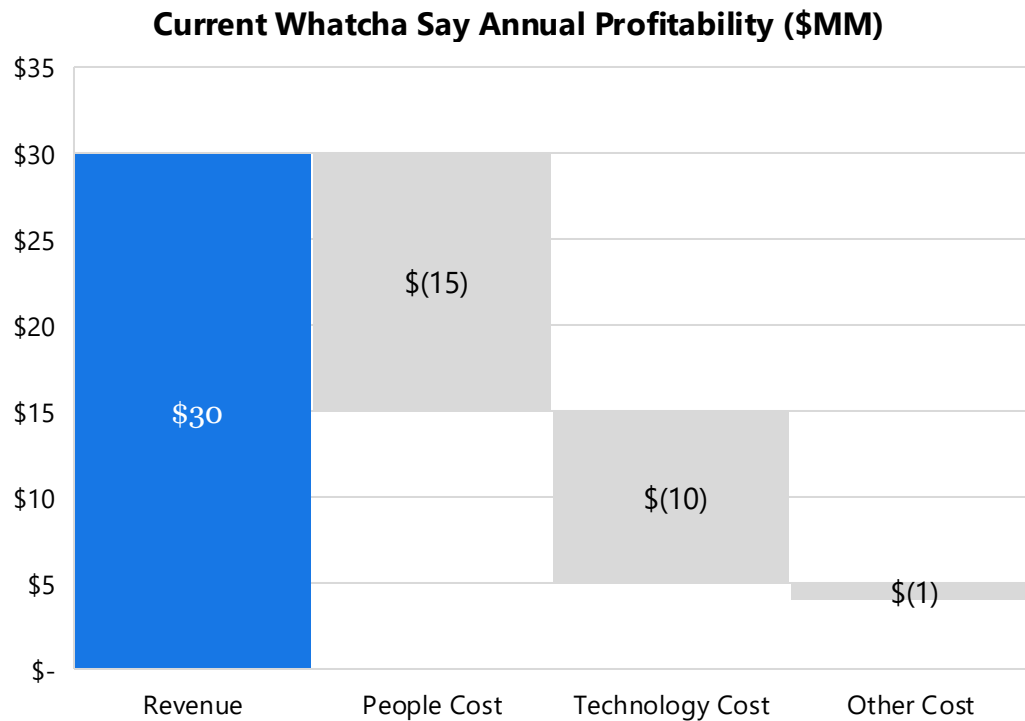
#42 - Whatcha Say Research (V of IX)

Case Math Question

Jordan wants to invest in this tech only if it increases firm profitability by 50% (or more). Should he invest in generative AI? Provide a final recommendation after examining Exhibit 1.

#42 - Whatcha Say Research (VI of IX)

Exhibit 1 and 2



Note: All employees earn \$40/hour and work exactly 40 hours a week. All people costs are for staffing human employees on market research projects. Gen AI creates an additional technology cost of \$2,000,000 per year.

#42 - Whatcha Say Research (VII of IX)

Case Math Answer

1. Calculate current company profitability by examining the waterfall chart to arrive at **current Whatcha Say profit of \$4,000,000**.
2. Calculate the profitability that must be achieved to hit the target. $\$4,000,000 \times 150\% = \textbf{\$6,000,000 (or more) is the required level of profitability after generative AI is implemented to warrant its investment.}$
3. Calculate the new profitability by adding \$2,000,000 more in Technology costs (from the footnote) and subtracting the human-staffed hour costs saved of \$6,000,000 to arrive at **\$8,000,000 profitability after generative AI is implemented**.
 - A. Savings in human-staffed hours is calculated by taking the following steps:
 - a) Examine the Required human-staffed project hours bar chart and subtract After Gen AI project hours from the current structure hours to discover **150,000 human hours are saved**. $375,000 \text{ hours} - 225,000 \text{ hours} = 150,000 \text{ human-staffed hours saved}$
 - b) Multiple the 150,000 human-staffed hours saved by the \$40 hourly rate **to realize a human-staffed hour cost savings (people cost savings) of \$6,000,000 per year**.
4. Realize that an investment in generative AI is warranted because **the new profitability of \$8,000,000 is greater than** the requirement of **\$6,000,000**.

#42 - Whatcha Say Research (VIII of IX)

Case Math Answer

Level 1 Answer	Level 2 Answer	Level 3 Answer
Jordan should invest in generative AI.	Jordan should invest in generative AI, as it saves 150,000 human hours (\$6M) per year.	Jordan should invest in generative AI, as it saves 150,000-man hours (\$6M) per year. This generates profitability of \$8M, which is greater than the \$6M target. Next, I'd want to ensure generative AI upholds our quality standards.

#42 - Whatcha Say Research (IX of IX)

Recommendation

- Jordan should invest in generative AI.

Supporting Data

- It saves 150,000 human hours of annual people costs.
- It generates a profitability of \$8,000,000 which is greater than the \$6,000,000 target.
- It will provide Jordan many strategic positioning options including (but not limited to) the ability to increase margins or price more competitively in the market.

Risks and Next Steps

- Jordan should rigorously test that quality standards can still be met.
- Jordan should survey his customers to understand their perceptions & trust in generative AI to ensure they will remain loyal.
- Jordan should validate that his employees would embrace adoption of this technology.

#43 - Phuture

Analysis	Structure	Overall
Hard	Hard	Hard

Behavioural Prep

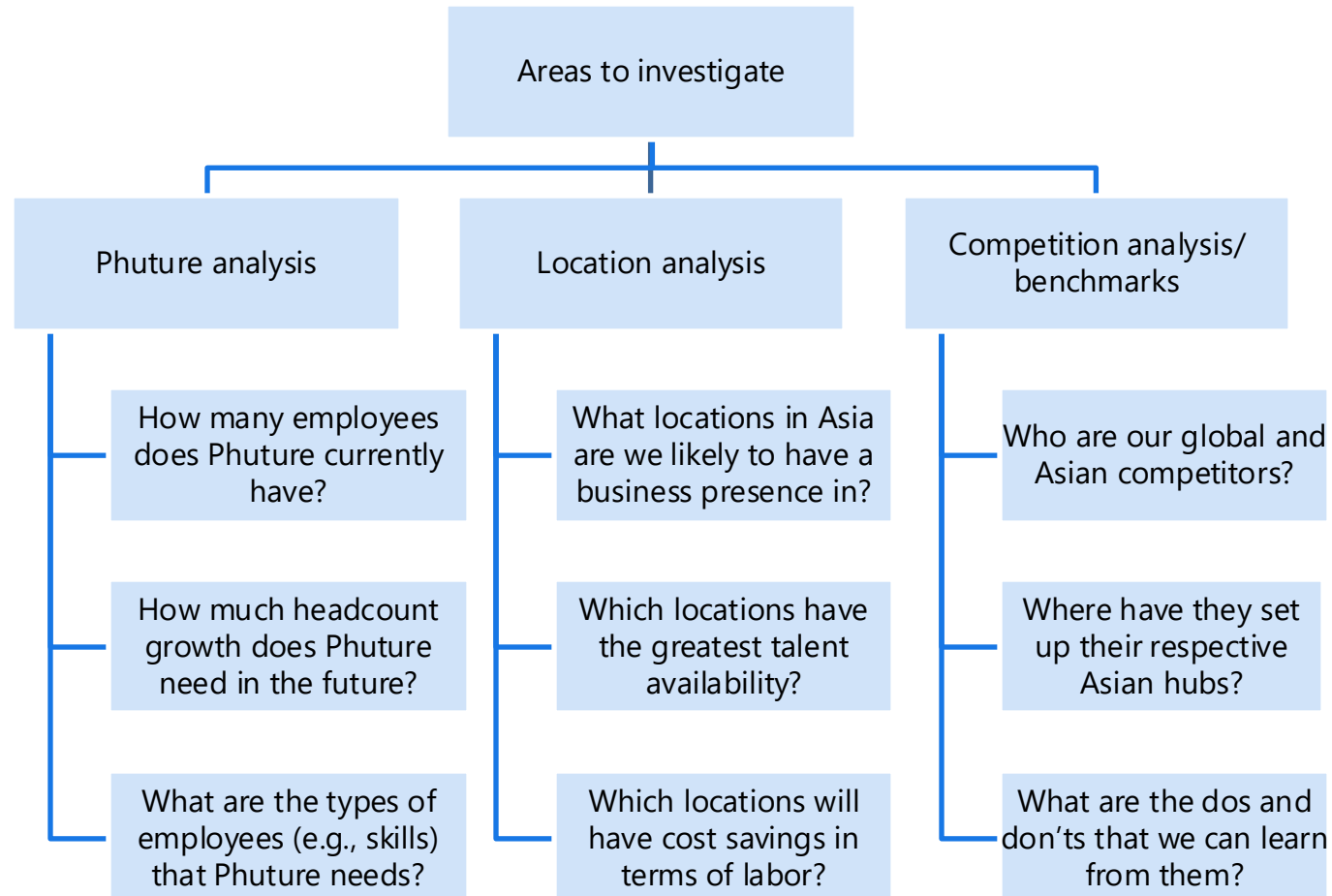
What is an important issue facing your previous industry?
Tell me about a piece of feedback you received that you disagreed with. How did you handle it?

Case Tracker		Case Stem
Industry	Tech	You are the Chief Talent Officer of Phuture, a rapidly growing Software-as-a-Solution (SaaS) company that offers a comprehensive Enterprise Resource Planning (ERP) solution to a variety of clients. Phuture is around a decade old and has a relatively mature and successful business. Phuture currently only has operations in the U.S. but is looking to expand its operations and talent footprint to Asia and set up its Asian hub in the next year. As the Chief Talent Officer of Phuture, you have been tasked by the CEO to think about this problem and chart a path forward.
Type	Internal Strategy	
Leader	Interviewee	
Concept	Market Entry, human capital	

Additional Information
Answers to clarifying questions: - Enterprise Resource Planning (ERP) is a platform companies use to manage and integrate essential parts of their businesses such as inventory management, finance systems, human resources systems, etc. - Phuture charges its clients a subscription fee per month to use its ERP solution.

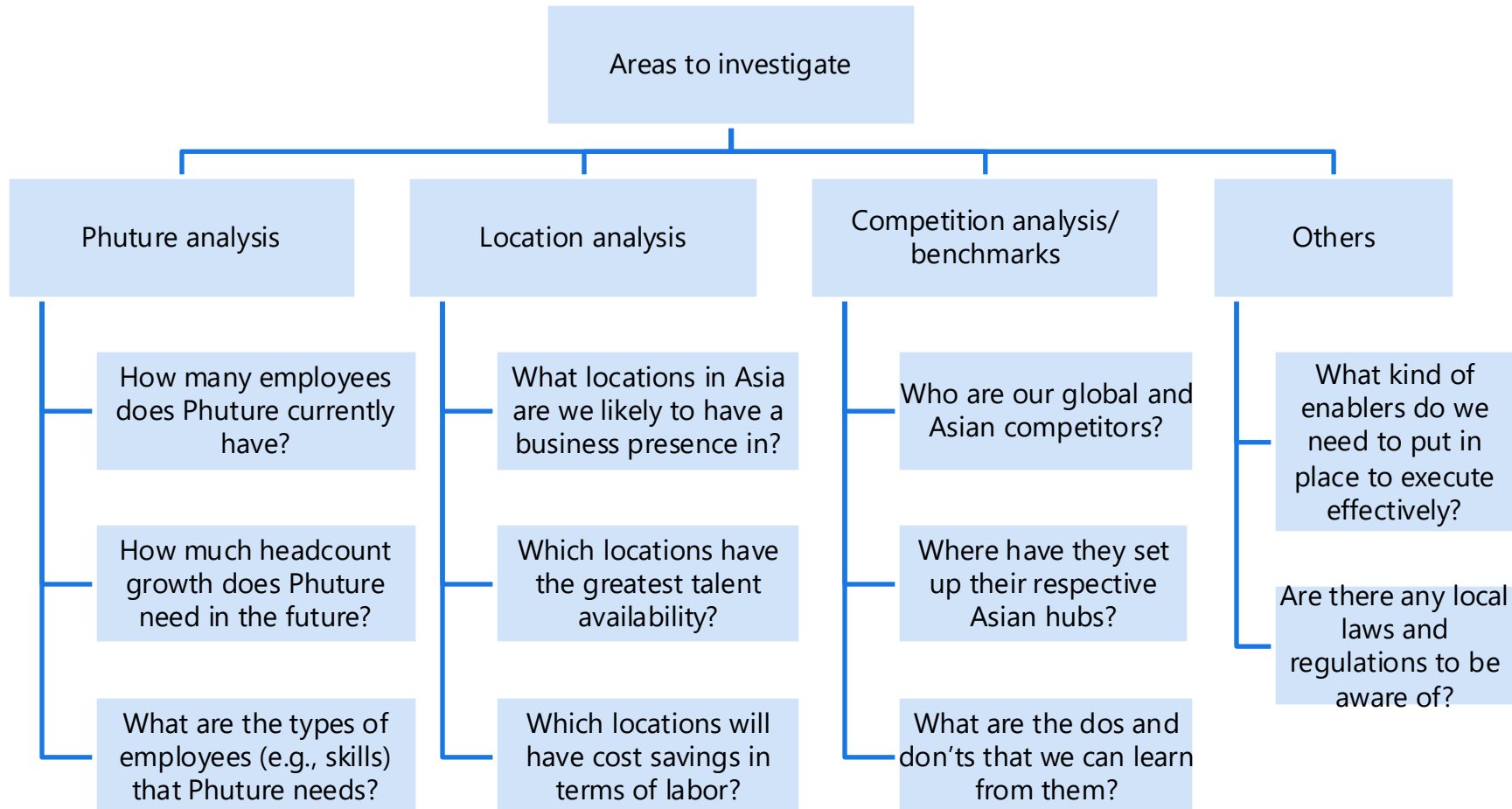
#43 - Phuture (II of X)

Sample Structure - Basic



#43 - Phuture (III of X)

Sample Structure - Advanced



#43 - Phuture (IV of X)

Case creative question and answer

After an intense couple of months working with the rest of Phuture's management team, Phuture has decided that the location of their Asian hub will be in Kuala Lumpur, Malaysia.

However, Phuture as an employer and brand is relatively unknown in Southeast Asia and the general sense is that it may be difficult to effectively recruit in Malaysia without putting in place critical enablers.

Could you please think of what enablers would be needed to recruit effectively in Kuala Lumpur?

Level 1 Answer

Less creative ideas with no thematic grouping of ideas

- Establish strategic partnerships
- Build an attractive office premise and infrastructure
- Execute a marketing plan to grow awareness
- Have competitive compensation and benefits

Level 2 Answer

More creative answers with thematic grouping (non-exhaustive)

- Candidate attraction
 - Build a strong understanding of the talent landscape and competition
 - Define and execute a comprehensive Employer Value Proposition (EVP) program to uplift the attractiveness of Phuture as an employer
 - Partner with local recruiters such as headhunters
 - Build and establish a local recruitment team in KL
 - Researching local salary benchmarks and ensuring compensation is competitive
 - Partner with local universities and career departments
- Candidate selection
 - Tailor job advertisements and descriptions to fit local standards and expectations
 - Define an interview selection process tweaked for local nuances that is reflective of the type of talent Phuture wants to hire
- Candidate integration
 - Develop a comprehensive onboarding and training program to ensure new hires are integrated well

#43 - Phuture (V of X)

Case Math Question

After defining the enablers, the next thing we need is a headcount projection over the next 3 years to understand what our workforce distribution will look like.

Please examine the exhibits on the next page and let me know what the workforce distribution between U.S. and Malaysia looks like over the next 3 years.

#43 - Phuture (VI of X)

Exhibit 1

Exhibit 1: Headcount formula

Start of year headcount

−

Attrition
(i.e., employees leaving)

+

Backfill
(i.e., how many replacement hires we do)

+

Growth hiring
(i.e., additional new hiring)

=

End of year headcount

#43 - Phuture (VII of X)

Exhibit 2

Exhibit 2: Headcount projections

Current headcount (all in US)	1000
Expected yearly attrition	10%
Backfill ratio	100%
Expected headcount YoY growth over next 3 years	20%
% of hiring to be done in KL	80%

#43 - Phuture (VIII of X)

Case Math Answer

	Year 1	Year 2	Year 3
Beginning headcount:	1000	1200	1440
US	1000	960	936
Malaysia	0	240	504
Attrition:	100	120	144
US	100	96	94
Malaysia	0	24	50
Backfills required in total	100	120	144
Growth hiring	200	240	288
Total number of hires needing to be done	300	360	432
Total hires by location:	300	360	432
US	60	72	86
Malaysia	240	288	346
End headcount:	1200	1440	1728
US	960	936	929
Malaysia	240	504	799
% of headcount:			
US	80%	65%	54%
Malaysia	20%	35%	46%

#43 - Phuture (IX of X)

Case Math Answer

Level 1 Answer

The expected attrition is 10% meaning 1 in every 10 employees will leave in a year and we expect to backfill all attrition (i.e., 100% backfill ratio).

Phuture's expected future headcount will grow significantly at ~20% per year.

Phuture is going to strongly prioritize hiring in Malaysia as 80% of all hires will be executed in Malaysia.

Level 2 Answer

Level 1 plus:

Given the talent projections, Malaysia is expected to grow from 0 to a size of ~800 in just 3 years. At the end of 3 years, the Malaysian hub and U.S. hub are almost identical in size.

Without doing calculations, strong candidates will pick up that given the 80% hiring focus in Malaysia and the significant headcount growth in 20% that the U.S. headcount base is likely to decrease in size over time and Malaysia will grow exponentially over the next 3 years.

Level 3 Answer

Level 2 plus:

We will need to consider the implications of such an aggressive Malaysian-centric strategy and whether that will impact morale of the U.S.-based employees.

In addition, we need to be certain that the quality and depth of the talent pool in Malaysia is comparable to U.S. and it can support the amount of hiring we want to do in KL (especially for specialized roles that require niche skills).

#43 - Phuture (X of X)

Recommendation

I recommend that Phuture expands its presence to Asia by building a second talent hub in Malaysia.

Supporting Data

Given Phuture's talent projections, the Malaysian hub will see exponential growth over the next 3 years and will be comparable in size to the U.S. hub.

Risks and Next Steps

Risks:

- Talent pool depth and availability
- Impact on morale of employees in the U.S. hub

Next steps:

- Start defining plan to put in place enablers to ensure successful execution of the strategy
- Analyze the Malaysian talent market and landscape to gain confidence that the talent pool can support our hiring ambitions

#44 - Salesforce

Analysis	Structure	Overall
Hard	Hard	Hard

Behavioural Prep

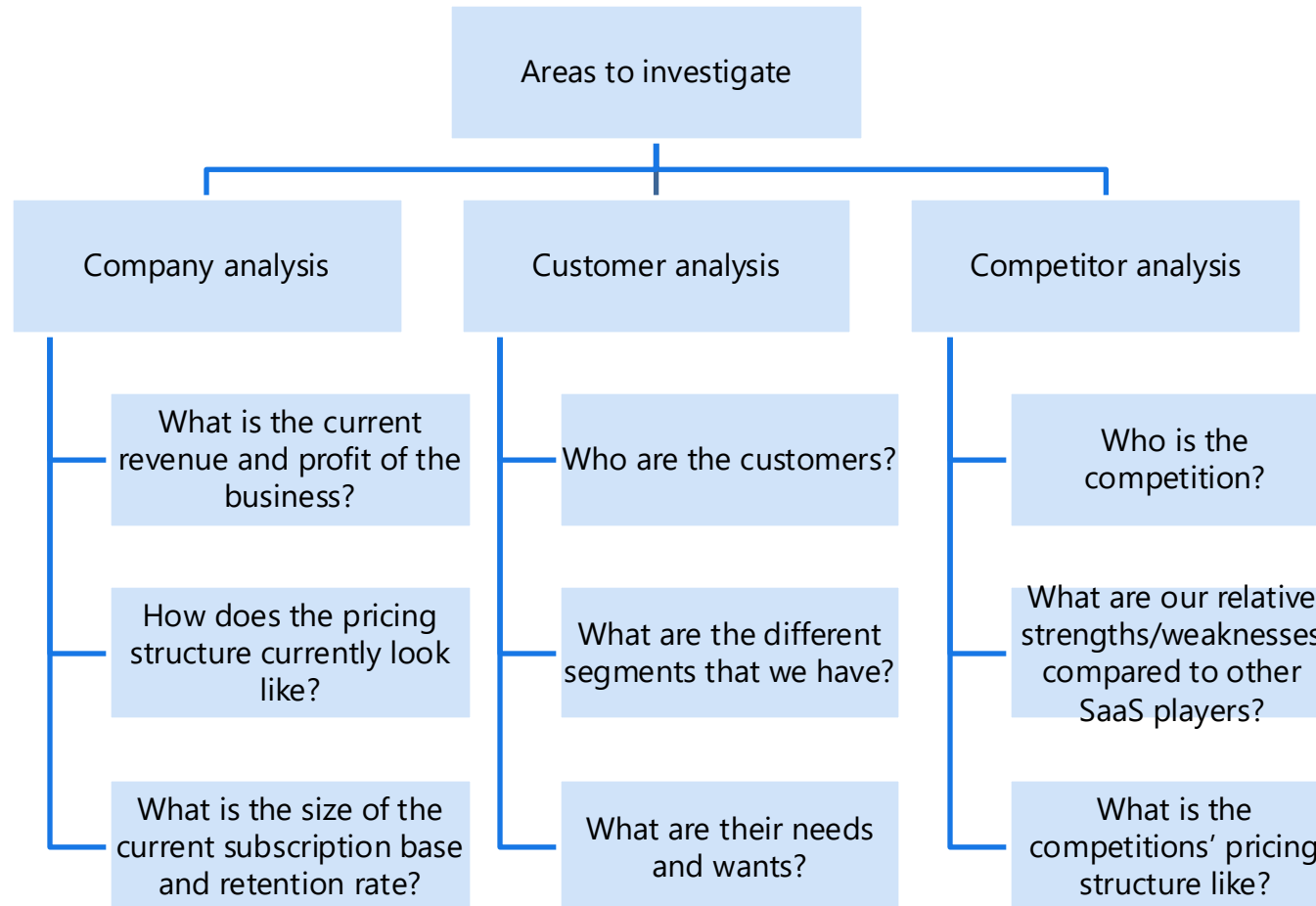
Tell me about yourself.
Tell me about a time when you exhibited leadership.

Case Tracker		Case Stem
Industry	Tech	You work for CloudForce, a Software-as-a-Solution (SaaS) company that offers a comprehensive project management platform to businesses of various sizes. The project management platform has different features that customers can choose from in exchange for a subscription fee. The company has achieved significant market penetration but is struggling to find the optimal pricing strategy to maximize revenue and profitability while maintaining customer satisfaction and market competitiveness. The CEO has come to your team to help revamp its pricing strategy so that it can find the right balance between maximizing financials while maintaining customers and being competitive against the market.
Type	Pricing Strategy	
Leader	Interviewee	
Concept	Market research, customer analysis	

Additional Information
Answers to clarifying questions: - CloudForce provides a cloud-based project management solution that enables teams to collaborate, plan, execute, and track projects efficiently. The platform offers various features such as task management, Gantt charts, team collaboration tools, integrations with third-party apps, and analytics. - CloudForce has a pricing structure with three tiers with each increased tier providing enhanced level of features. - CloudForce operates in a competitive market with both established players and emerging startups

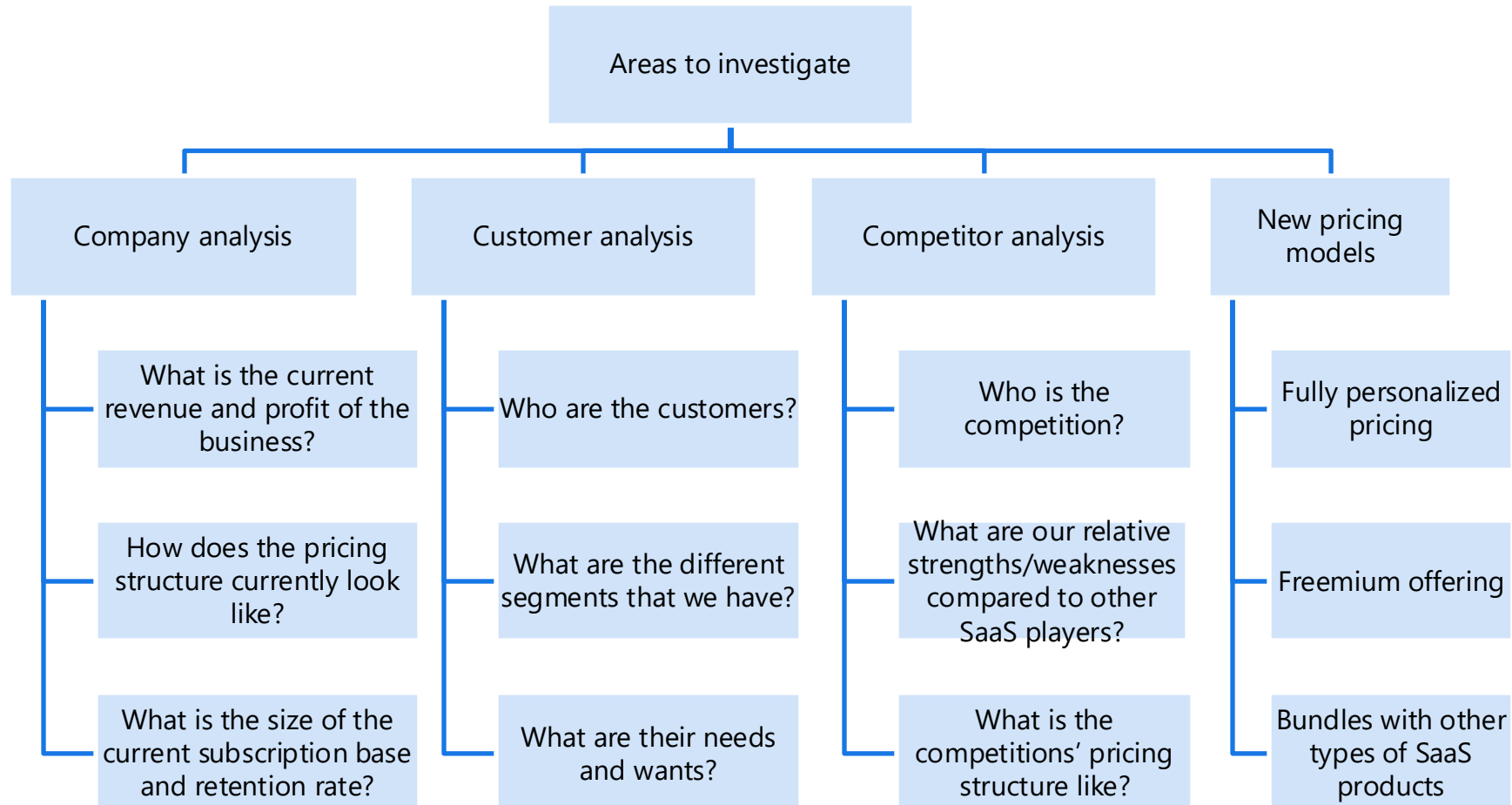
#44 - Cloudforce (II of IX)

Sample Structure - Basic



#44 - Cloudforce (III of IX)

Sample Structure - Advanced



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Case creative question and answer

The CEO of CloudForce has just invited you to a brainstorming session tomorrow to brainstorm different ideas and initiatives to revamp its pricing strategy. The CEO is hoping that you can bring some creative and innovative ideas to the table so that he can take some of it and test it out in CloudForce.

Given this context, can you please brainstorm some ideas and initiatives on what CloudForce can do to revamp its pricing strategy?

Level 1 Answer	Level 2 Answer
Less creative ideas with insufficient description of the ideas	More creative answers with further explanation
1.General pricing optimization	1.Personalized pricing: Move beyond tiered pricing structures and utilize data analytics to drive unique pricing for each customer based on needs and wants
2.Data-driven capabilities that help with pricing optimization	2.Freemium model: Introduce a free layer for new customers such as smaller businesses that can increase new user acquisition. These new users can be eventually nurtured to future high usage, high revenue customers
3.Creating new pricing tiers and structures	3.Value-added services: Elevate CloudForce's proposition by offering additional services like adjacent software (e.g., team communication such as Slack) or enhanced dedicated support
4.Rolling out of incentives or discounts	4.Ala-carte pricing: Move away from tiered pricing and allowing customers to customize their subscription by adding specific features ala carte
	5.Subscription bundles: Create subscription bundles that cater to specific industries or product types

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Case Math Question

The CEO appreciates your innovative suggestions and will have your Strategy team evaluate those ideas in more detail over the coming months.

In the meantime, CloudForce is evaluating a price increase across its three pricing tiers but needs your help to understand the potential impact of the price increase.

Please look Exhibit 1 on the next page and calculate:

- The current average monthly revenue per customer (across all segments)
- The **maximum percentage drop in customer retention** that CloudForce can have without losing net revenue (i.e., how many customers can CloudForce lose due to the price increase before experiencing a net negative revenue impact)

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Exhibit 1

Current distribution of customers:

Startups	20%
Small and Medium Enterprises (SMEs)	20%
Corporates	60%

Expected future distribution of customers:

Startups	25%
Small and Medium Enterprises (SMEs)	25%
Corporates	50%

Current subscription tiers (\$ per month):

Basic	\$30
Standard	\$40
Premium	\$60

New subscription tiers (\$ per month):

Basic	\$50
Standard	\$70
Premium	\$100

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Case Math Answer

Current case:

Weighted Startups revenue (20%)	\$8
Weighted SME revenue (20%)	\$8
Weighted Corporate revenue (60%)	\$48
Current average monthly revenue per customer	\$64

Future case:

Weighted Startups revenue (25%)	\$13
Weighted SME revenue (25%)	\$18
Weighted Corporate revenue (50%)	\$50
Future average monthly revenue per customer	\$80

Increase in average monthly revenue / cushion for revenue reduction due to churn (\$)	\$16
Buffer for decrease in current retention rate (%) (Increase in avg rev / current avg rev)	25%
Minimum retention rate required (%)	
[Current retention rate * (1- Buffer for decrease in retention rate)]	53%

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Case Math Answer

Level 1 Answer	Level 2 Answer	Level 3 Answer
CloudForce currently has three types of customers in its subscriber base – Startups, SMEs and Corporates. Most of the customer base is Corporates With the new price increases, there will be price increases across the board, but the price increase will be greatest in the Standard tier. After the price increase, we expect to lose some Corporates and Startups and Enterprises will make up a bigger portion of our customer mix.	Level 1 plus: CloudForce currently generates \$64 on average per customer across all segments. With the price increase, that is expected to increase by 25% to \$80 on average	Level 2 plus: The maximum decrease in retention rate that CloudForce can register before registering any net revenue decrease is 17% (70% retention rate now currently vs 53% retention rate required post-price increases).

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Recommendation

I recommend that CloudForce explore new pricing models in the future such as personalized pricing for each subscriber

However, in the short-term, CloudForce should implement the analyzed price increase assuming the retention rate does not drop below 53%

Supporting Data

After the price increase, the average monthly revenue per customer will increase by 25% from \$64 currently to \$80.

Risks and Next Steps

Risks:

- Customer churn and retention – no clear analysis done yet on the impact of the price increase on churn
- Competitive pressure – unclear how our pricing compares to competition and if we will lose significant market position

Next steps:

- Analyze in detail impact on churn
- Define timeline and plan for implementation
- Craft education and communications to customers behind the price increase